

A Geometric Framework for Identification in Characteristic-Based Factor Models

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Abstract

This paper develops a geometric theory of identification for characteristic-based factor models under no-arbitrage restrictions. The central premise is that identification is governed not only by the statistical model but also by the metric that determines which directions in characteristic space are economically distinguishable. Existing approaches implicitly treat the Euclidean metric as the underlying ruler of characteristic space. We show instead that the long-run cross-sectional distribution of firm characteristics induces an empirical geometry that serves as a hidden ruler governing latent pricing directions. Within this geometry, latent and observable factors become structured subspaces, allowing identification, normalization, and no-arbitrage restrictions to be characterized as intrinsic geometric properties rather than algebraic constraints imposed on an estimator. Building on this framework, we derive a no-arbitrage formulation of Instrumented Principal Component Analysis that admits a projected weighted least-squares representation together with rotation-invariant asymptotic inference. The resulting information geometry separates latent, observable, and non-systematic components, establishing a unified statistical foundation for estimation and inference in high-dimensional characteristic spaces. Geometry-calibrated simulations demonstrate that identification depends fundamentally on the underlying metric: Euclidean representations become unstable in anisotropic characteristic spaces, whereas the proposed geometry consistently recovers economically meaningful pricing spans. The paper therefore develops a general geometric framework for identification and statistical inference under no-arbitrage, of which No-Arbitrage Instrumented Principal Component Analysis (NA-IPCA) constitutes one econometric realization.

Keywords: Identification; Information Geometry; High-Dimensional Factor Models; Characteristic-Based Asset Pricing; No-Arbitrage; Instrumented Principal Component Analysis.

JEL Classification: C13, C14, C18, G12.

*Imperial Business School. Email:YL7919@ic.ac.uk. This paper develops the theoretical framework underlying the author's doctoral research on no-arbitrage characteristic-based asset pricing. An expanded version of the theoretical framework, incorporating updated datasets, extended empirical validation, and additional simulation studies, is currently under development. Companion papers on U.S. equity and corporate bond asset pricing are also in preparation. All errors are my own.

1 Introduction

1.1 Motivation: Systematic Risk in Characteristic-Rich Environments

Empirical asset pricing now operates in a characteristic-rich environment. Modern panels combine thousands of stocks with dozens of cross-sectional predictors—size, value, profitability, investment, momentum, and a growing range of accounting and market variables. In such settings, expected returns are no longer naturally described by a small handpicked set of factors. Instead, they are tied to a high-dimensional vector of firm characteristics that acts as an instrument set for latent risk exposures. Approximate factor models and characteristic-based estimators such as Instrumented Principal Component Analysis (IPCA) provide a natural statistical starting point for this environment (Bai and Ng, 2002; Bai, 2003; Bai and Ng, 2008; Kelly et al., 2020, 2019). Factor dimension still matters, but it is only part of the problem. Conditional on a chosen low-dimensional representation, the central question of this paper is which characteristic directions form the economically meaningful priced span, and which directions are merely redundant or residual.

Standard procedures often treat characteristic space as if it were approximately Euclidean. Principal component analysis (PCA), standard IPCA, and many machine-learning stochastic discount factor (SDF) estimators define scale and orthogonality using the identity metric. Under that geometry, any high-variance linear combination of characteristics can be treated as a candidate factor. But in actual data, firm characteristics are strongly correlated, unevenly scaled, and economically clustered. When the wrong geometry is imposed, estimators can rotate toward directions that look large under ordinary variance calculations but are weak from a pricing perspective, fragment one economically meaningful priced direction across several factors, or allow latent and observable blocks to load on overlapping risk directions.

The evidence developed later in Section 6 shows that this Euclidean view is misleading in the data studied here. Let N_t denote the number of assets observed at date t , and let $Z_t \in \mathbb{R}^{N_t \times L}$ collect the L firm characteristics. Define the period- t Gram matrix

$$W_t = \frac{1}{N_t} Z_t^\top Z_t.$$

Its time average,

$$\hat{Q}_Z = \frac{1}{T} \sum_{t=1}^T W_t,$$

converges to a positive-definite population limit Q_Z , which serves throughout the paper as the relevant long-run metric on characteristic space. In the U.S. equity data, the eigenvalues of this metric are highly uneven and the implied characteristic directions are strongly clustered. Two loading vectors that appear orthogonal under the identity metric can therefore be close to collinear once the empirical distribution of characteristics is taken seriously.

Economically, curvature in this metric can be read as the risk-bearing capacity of a characteristic direction. A direction with high curvature is one along which the cross section has more scope to support systematic risk compensation; a direction with low curvature may still display noticeable raw dispersion under ordinary Euclidean scaling yet contribute little to economically meaningful pricing once the characteristic geometry

is respected. This interpretation is empirical rather than metaphysical. The object Q_Z is not treated here as an immutable deep primitive of the economy; it is an estimated long-run description of the characteristic environment prevailing in the sample. If the distribution of characteristics changes materially across regimes, the relevant geometry should in principle be re-estimated rather than assumed fixed once and for all.

This distinction matters for both interpretation and inference. If factor extraction ignores the anisotropy of characteristic space, it becomes easy to confuse priced structure with mere statistical dispersion. Latent factors may duplicate observable benchmark exposures, while intercept terms may absorb directions that are large in-sample but difficult to reconcile with basic no-arbitrage restrictions such as Hansen–Jagannathan bounds (Hansen and Jagannathan, 1991; Cochrane, 2005). In a characteristic-rich environment, the geometric problem is therefore inseparable from the economic one: systematic risk should be identified relative to the characteristic directions that the data actually make meaningful.

For clarity, throughout the paper $r_{t+1} \in \mathbb{R}^{N_t}$ denotes the vector of excess returns, where N_t is the cross-sectional asset count at date t ; in the empirical chapter, by contrast, N denotes the total number of distinct firms in the sample summary. The characteristic-weighted return moment is

$$X_t = \frac{1}{N_t} Z_t^\top r_{t+1}.$$

Latent factors are denoted by f_t , with $F = [f_1, \dots, f_T]$ collecting their realizations over time. Observable factor proxies are denoted by F_t^{obs} , and the associated loading blocks Γ_β , Γ_δ , and Γ_α represent, respectively, latent characteristic-based exposures, observable factor exposures, and an intercept block capturing residual pricing structure. Appendix A provides the complete symbol map, but the central objects are introduced here because they are used from the outset.

The rest of the paper develops a no-arbitrage version of IPCA (NA-IPCA) designed for this setting. The guiding idea is simple: if the empirical metric Q_Z determines the relevant notion of scale, orthogonality, and curvature in characteristic space, then factor extraction, no-arbitrage discipline, and large-sample inference should all be built on that geometry rather than imposed only after estimation.

1.2 Literature Gap and Positioning

This thesis sits at the intersection of three strands of work. The first is the econometric literature on approximate factor models and their characteristic-based extensions. Classical large-panel factor methods identify low-dimensional common components under weak dependence and large cross sections (Geweke, 1977; Sargent and Sims, 1977; Ross, 1976; Chamberlain and Rothschild, 1983; Bai and Ng, 2002; Bai, 2003; Bai and Ng, 2006a,b, 2008; Bai and Li, 2016). In asset pricing, this logic extends naturally to panels of returns and firm characteristics, culminating in IPCA, where characteristics instrument latent exposures (Connor and Korajczyk, 1986, 1988; Kelly et al., 2019). What these methods typically leave implicit, however, is the geometry of characteristic space itself. Orthogonality and normalization are usually posed in Euclidean coordinates, even when the data display strong anisotropy and clustering.

The second strand imposes economic discipline more directly. No-arbitrage restrictions, Hansen–Jagannathan bounds, pricing-restricted latent factor estimators, and recent work on weak or spurious factors

all sharpen the distinction between statistically important directions and economically admissible ones (Hansen and Jagannathan, 1991; Lettau and Pelger, 2020b,a; Bryzgalova, 2015; Bryzgalova et al., 2023; Dello Preite et al., 2024; Zaffaroni, 2025; Zaffaroni and Zhou, 2024). This literature is closest in spirit to the present thesis. But even here, pricing restrictions are often imposed on top of a factor extraction problem whose underlying geometry has already been fixed. The empirical characteristic metric Q_Z is rarely elevated to the organizing object that determines which directions should count as distinct, which should be treated as redundant, and how residual pricing content should be bounded.

The third strand includes shrinkage and flexible SDF estimation. Characteristic portfolio methods and shrinkage-based models regularize exposures or expected returns using economically motivated priors or penalties (Kozak et al., 2018, 2020; Freyberger et al., 2020). Machine-learning SDF estimators allow for rich nonlinear structure and often perform well predictively (Gu et al., 2020; Chen et al., 2024; Chernozhukov et al., 2018; Belloni et al., 2014). These approaches are valuable, but they solve a different problem from the one emphasized here. Shrinkage methods typically regularize exposures, portfolio weights, or expected returns within a pre-specified representation rather than redefining the underlying characteristic geometry. Many flexible SDF methods, by contrast, prioritize predictive fit and flexibility over a transparent blockwise decomposition. Neither approach typically yields the same geometric decomposition of latent, observable, and residual pricing directions together with rotation-invariant asymptotic theory.

The gap motivating this paper is therefore specific. What is missing is a framework that treats the empirically estimated metric Q_Z as the relevant long-run geometry of characteristic space; uses that geometry to organize the decomposition between latent structure, observable factor proxies, and residual pricing components; and embeds no-arbitrage discipline directly into the feasible parameter space rather than only into ex post interpretation. This thesis proposes NA-IPCA as such a framework.

1.3 Contributions of the Paper

This thesis makes three distinct contributions: one theoretical, one methodological, and one empirical.

The first contribution is theoretical. The thesis recasts characteristic-based factor pricing on a no-arbitrage manifold built from the empirical metric Q_Z . In that formulation, the latent loading block is normalized in Q_Z geometry, observable factor proxies are separated from the latent span, and the intercept block is orthogonal to the latent span and disciplined by a curvature restriction that limits the pricing content of non-systematic directions. The economic point is that factor discovery should concern subspaces that are meaningful in the observed characteristic environment, not arbitrary Euclidean coordinates.

Under the regularity, identification, and centering conditions developed in Sections 2–4, this geometric structure delivers rotation-invariant local asymptotic normality (LAN) inference and a block-structured information matrix for the economically identifiable directions of the problem. The theory therefore links geometric discipline, no-arbitrage restrictions, and inference in a single framework. Conditional on a chosen low-dimensional representation, the analysis asks which characteristic directions form the economically meaningful priced span once the empirical geometry of the data is taken into account.

The second contribution is methodological. I develop a projected weighted least squares (WLS) estimator, implemented as feasible generalized least squares (GLS), that alternates between updating factor realizations

and loading blocks and then projects those updates back onto the feasible no–arbitrage set. In this way, the geometry is enforced during estimation rather than imposed only after the fact.

The resulting algorithm solves the projected Karush–Kuhn–Tucker (KKT) conditions of the constrained problem and remains operational in high-dimensional panels. This matters because the objective of the paper is not merely to propose a new interpretation of existing IPCA output. It is to provide an estimator that is numerically implementable, stable under the empirical anisotropy of characteristics, and faithful to the economic restrictions used in the theory. Appendix C develops the computational details, but the key methodological message is simple: the no–arbitrage geometry can be estimated directly.

The third contribution is empirical. Using CRSP–Compustat U.S. equities from 1964 to 2014, I show that the geometry of characteristic space is strongly anisotropic and that this fact materially changes factor extraction. Relative to standard IPCA and PCA, NA–IPCA recovers cleaner latent directions, sharply reduces overlap between latent structure and the Fama–French six-factor block (FF_6), and yields pricing errors that are at least as small as those of IPCA and uniformly smaller than those of PCA.

The empirical contribution is not that the framework produces an entirely different predictive universe. Rather, it reorganizes priced structure into directions that are more coherent economically and more disciplined statistically. The factor labels used later in the paper—for example, investment–profitability or size–momentum—should therefore be read as descriptive summaries of average loading patterns rather than as immutable structural names. A companion empirical study (the U.S. equity application) examines how stable these patterns are across subsamples and how they relate to observable benchmark factors.

1.4 Overview of Main Results

The main theoretical result is that once factor extraction is written in the Q_Z geometry, identification becomes a question about economically meaningful tangent directions rather than arbitrary Euclidean rotations. The feasible parameter space separates the latent span from observable factor proxies and disciplines residual pricing curvature, so the estimator behaves as a constrained WLS problem on a geometrically meaningful domain. This is what makes rotation–invariant inference possible and why the geometry matters for interpretation rather than only for normalization.

Geometry–calibrated simulations in Section 7 show why this distinction is first order. When the data-generating process inherits the anisotropy observed in real characteristic panels, Euclidean estimators tend to rotate toward high–variance but weakly priced directions and can allocate the same pricing content across multiple blocks. NA–IPCA instead tracks the economically relevant span more closely, keeps the residual component disciplined, and avoids the most important forms of misallocation in finite samples.

The U.S. equity application developed in a companion study reaches the same qualitative conclusion. Much of the long–horizon predictive content remains in the observable block, but the latent NA–IPCA block reorganizes priced structure into cleaner and more interpretable directions. For empirical researchers, the benefit is a more disciplined workflow for factor discovery and model comparison. For applied users such as portfolio managers, the benefit is a sharper distinction between genuine priced structure, benchmark factor exposure, and residual noise.

1.5 Practical Relevance and Broader Scope

The practical relevance of the framework is straightforward. In characteristic-rich environments, researchers often face a trade-off between interpretable but rigid benchmark factor models and flexible but less interpretable statistical estimators. NA-IPCA offers an intermediate route. It preserves the characteristic-based logic of IPCA, but it adds a geometric discipline that prevents latent factors from merely reproducing observable benchmark portfolios or absorbing residual misspecification. In that sense, the method changes not only estimation, but also how one should read the output of a factor model.

Although the empirical focus of the paper is U.S. equities, the same logic should be relevant in other asset classes where characteristic collinearity is at least as severe. Corporate bonds are a natural example: duration, spread, rating, liquidity, and covenant-related characteristics often overlap substantially, so the underlying characteristic geometry is unlikely to be close to Euclidean. More generally, the framework should be interpreted as sample dependent. If the cross-sectional distribution of characteristics shifts materially across regimes, then the relevant metric and the resulting factor geometry should be re-estimated rather than treated as permanently fixed.

1.6 Roadmap of the Paper

The remainder of the paper proceeds as follows. Section 2 defines the economic environment and no-arbitrage geometry, Section 3 introduces the estimator, and Section 4 develops the asymptotic theory and rotation-invariant inference. Section 5 positions the framework relative to adjacent factor-model approaches, while Sections 6 and 7 document the empirical characteristic geometry and the geometry-calibrated simulation evidence. Section 8 concludes. A companion empirical study develops the full U.S. equity application. The appendices collect notation, proofs, implementation details, and inference procedures: Appendices A–D contain the formal assumptions, the identification and efficiency results, the estimation algorithm, and the inference theory, while Appendices E–H provide structural comparisons with statistical IPCA and the curvature-projection details for the intercept and observable blocks.

2 Economic Environment and No-Arbitrage Geometry

We consider a panel of excess returns $r_{t+1} \in \mathbb{R}^{N_t}$ and firm characteristics $Z_t \in \mathbb{R}^{N_t \times L}$ observed at a regular frequency for dates $t = 1, \dots, T$. For each t , define the characteristic-weighted return moment and Gram matrix

$$X_t = \frac{1}{N_t} Z_t' r_{t+1}, \quad W_t = \frac{1}{N_t} Z_t' Z_t.$$

The moment X_t records return variation along characteristic directions, while W_t records the cross-sectional geometry of the characteristics themselves. The corresponding long-run characteristic metric is

$$Q_Z = \text{plim}_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T W_t.$$

This object aggregates the persistent scale and correlation structure of the characteristic panel and supplies the metric on which pricing directions are compared. Throughout, Q_Z is treated as the estimated long-run metric for the prevailing sample environment rather than as an immutable structural primitive; if the cross-sectional distribution of characteristics shifts materially across regimes, the relevant geometry must be re-estimated.

To keep the main text self-contained, Table 1 lists the notation used most repeatedly in this paper. Appendix A provides the complete symbol map and the full formal conventions.

Table 1: Main notation used throughout this paper

Symbol	Dimension	Meaning in this paper
r_{t+1}	$\mathbb{R}^{N_t}$	Vector of excess returns at date $t+1$ for the N_t assets observed at date t .
Z_t	$\mathbb{R}^{N_t \times L}$	Matrix of firm characteristics used to instrument latent exposures and define characteristic portfolios.
X_t, W_t	$\mathbb{R}^L, \mathbb{R}^{L \times L}$	Characteristic-weighted return moment and Gram matrix, $X_t = N_t^{-1} Z_t^\top r_{t+1}$ and $W_t = N_t^{-1} Z_t^\top Z_t$.
Q_Z	$\mathbb{R}^{L \times L}$	Long-run characteristic metric, defined as the probability limit of the time average of W_t .
$f_t, F = [f_1, \dots, f_T]$	$\mathbb{R}^K, \mathbb{R}^{K \times T}$	Latent factor realization at date t and the matrix collecting all latent factor realizations.
F_t^{obs}	$\mathbb{R}^{K_{\text{add}}}$	Prespecified observable factor vector (PSF), such as traded factors or macroeconomic proxies.
Γ_β	$\mathbb{R}^{L \times K}$	Loading matrix for the latent characteristic-based span.
$\Gamma_\delta, \Gamma_\alpha$	$\mathbb{R}^{L \times K_{\text{add}}}, \mathbb{R}^L$	Loading matrix for observable factors and intercept block capturing residual pricing structure.
$\theta = (\Gamma_\beta, F, \Gamma_\delta, \Gamma_\alpha)$	—	Parameter tuple collecting the latent block, observable block, and intercept block.
$\psi(\Gamma_\alpha)$	scalar	Curvature functional $\Gamma_\alpha^\top Q_Z \Gamma_\alpha$ used to bound the pricing content of the intercept block.

The metric Q_Z induces the inner product and norm

$$\langle a, b \rangle_{Q_Z} = a^\top Q_Z b, \quad \|a\|_{Q_Z} = (a^\top Q_Z a)^{1/2}.$$

Orthogonality in this metric corresponds to the absence of long-run co-movement between the associated characteristic-sorted portfolios and differs from Euclidean orthogonality because it respects persistent differences in characteristic scale and dependence. In the same metric, curvature measures the amount of priced variation carried by a characteristic direction: directions with large $a^\top Q_Z a$ support broader systematic variation than directions with comparable Euclidean length but little mass under Q_Z .

Expected excess returns are represented through a three-block decomposition in characteristic space,

$$\mathbb{E}_t[r_{t+1} \mid Z_t, F_t^{\text{obs}}] = Z_t (\Gamma_\beta f_t + \Gamma_\delta F_t^{\text{obs}} + \Gamma_\alpha),$$

where (Γ_β, f_t) is the latent characteristic-based block, $(\Gamma_\delta, F_t^{\text{obs}})$ is the block of prespecified observable factors (PSF), and Γ_α is the remaining intercept component.¹ This representation generalizes the characteristic-based formulation of Kelly et al. (2019) while remaining consistent with the approximate-factor environment

¹Throughout this paper, the terms *residual*, *non-systematic*, *asset-specific*, and *idiosyncratic* are used interchangeably for the component represented by Γ_α . Under the baseline specification, this block is separated from the latent span by $\Gamma_\beta^\top Q_Z \Gamma_\alpha = 0$ and bounded by $\psi(\Gamma_\alpha) \leq \delta$. An optional strengthening is to impose $\Gamma_\delta^\top Q_Z \Gamma_\alpha = 0$, so the intercept block is also orthogonal to the observable PSF span.

studied by [Bai and Ng \(2002, 2006b,a\)](#). Its economic content lies in the priced spans in characteristic space rather than in any particular basis used to represent those spans.

The feasible set collects normalization, orthogonality, and curvature restrictions in the same metric. The latent block is normalized in Q_Z , the observable block is restricted to the Q_Z -orthogonal complement of the latent span, and the intercept block is both orthogonal to the latent span and bounded in curvature. The admissible parameter space is

$$\Theta(\delta) = \left\{ (\Gamma_\beta, F, \Gamma_\delta, \Gamma_\alpha) : \Gamma_\beta^\top Q_Z \Gamma_\beta = I_K, \Gamma_\beta^\top Q_Z \Gamma_\delta = 0, \Gamma_\beta^\top Q_Z \Gamma_\alpha = 0, \psi(\Gamma_\alpha) \leq \delta \right\},$$

with curvature functional

$$\psi(\Gamma_\alpha) = \Gamma_\alpha^\top Q_Z \Gamma_\alpha.$$

The restriction $\psi(\Gamma_\alpha) \leq \delta$ confines the intercept block to a Q_Z -ellipsoid of radius $\sqrt{\delta}$; small values of δ approach strict no-arbitrage, while positive values allow limited departures associated with frictions, measurement error, or nontraded components of risk. Writing

$$P_\beta = \Gamma_\beta (\Gamma_\beta^\top Q_Z \Gamma_\beta)^{-1} \Gamma_\beta^\top Q_Z, \quad M_\beta = I - P_\beta,$$

the decomposition consists of the latent span $\text{col}(\Gamma_\beta)$, the effective observable span $\text{col}(M_\beta \Gamma_\delta)$, and the bounded intercept direction Γ_α . The baseline specification does not impose $\Gamma_\delta^\top Q_Z \Gamma_\alpha = 0$ because, under PSF centering ($\mathbb{E}[F_t^{\text{obs}}] = 0$) and sufficient time variation, the intercept direction cannot be replicated by $\Gamma_\delta F_t^{\text{obs}}$; in the baseline model, separation from the latent span and the curvature bound are the maintained restrictions. We therefore keep $\Gamma_\delta^\top Q_Z \Gamma_\alpha = 0$ as an optional strengthening rather than a baseline restriction: it sharpens separation between the observable and intercept blocks but is not needed for the baseline identification and asymptotic arguments developed below. It can nevertheless be useful when PSF centering is not imposed or when the observable PSF block contains strong low-frequency components.

Formal regularity conditions are collected in Assumptions 1–8 of Appendix A. They impose weak temporal dependence, finite $(2 + \tau)$ moments, existence and stability of Q_Z , smoothness of the moment map, compactness of the feasible set, and standard constraint–qualification conditions. Under these conditions, $\Theta(\delta)$ is the geometric domain on which the constrained estimator, identification results, and asymptotic theory are developed.

3 The No–Arbitrage IPCA Estimator

This section presents the estimator implied by the geometric and no–arbitrage structure developed in Section 2. The goal is to estimate a set of characteristic-based pricing components that are economically distinct—latent, observable, and residual—and that satisfy the orthogonality and curvature restrictions encoded in the no–arbitrage parameter space $\Theta(\delta)$. This formulation mirrors empirical practice, where expected returns are linked to firm characteristics and observable factors, but imposes a disciplined geometric structure that is often missing in unconstrained approaches such as IPCA. We begin by stating a concise set of high-level assumptions that summarize the environment in which the estimator is well defined and

identified. We then formulate the constrained weighted least-squares (WLS) problem that implements the separation of the three pricing blocks on $\Theta(\delta)$. Finally, we provide a structural characterization showing that the estimator solves a projected WLS system on this manifold and briefly discuss the rotation invariance and implementation features that emerge from the no-arbitrage geometry. All technical assumptions, proofs, and algorithmic details are collected in Appendices A–C.

3.1 High-Level Assumptions for the NA-IPCA Estimator

This subsection summarizes the high-level environment under which the No-Arbitrage IPCA (NA-IPCA) estimator is well defined, identified, and geometrically coherent. These assumptions collect the stochastic, geometric, and regularity conditions under which the characteristic-based pricing structure—latent, observable, and residual—admits a unique projection on the no-arbitrage manifold $\Theta(\delta)$ introduced in Section 2. Their formal versions, together with the associated regularity lemmas, are stated in Appendix A, Assumptions A.1–A.8 and A.9–A.11.

Stable Characteristic Geometry. The return-characteristic panel $\{(r_{t+1}, Z_t)\}$ evolves in a strictly stationary, ergodic, and weakly dependent environment such that the characteristic Gram matrices $W_t = N_t^{-1} Z_t^\top Z_t$ admit a non-degenerate probability limit,

$$Q_Z = \text{plim}_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T W_t \succ 0,$$

as encoded in Assumptions A.1–A.2. Economically, this guarantees that characteristic directions possess a persistent second-moment structure, so Q_Z acts as a stable metric with respect to which priced directions are scaled, compared, and orthogonalized. This geometric stability mirrors the moment-regularity and large- N, T factor-space conditions underlying approximate factor models in Bai and Ng (2002); Bai (2003); Bai and Ng (2008), and it is consistent with the high-dimensional characteristic environments studied in Kelly et al. (2019) and Zaffaroni (2025).

Diversified Idiosyncratic Shocks. Idiosyncratic return components are mean-zero conditional on characteristics and exhibit weak cross-sectional dependence, in the sense of Assumption A.3. Consequently, characteristic-weighted aggregates $X_t = N_t^{-1} Z_t^\top r_{t+1}$ behave as diversified portfolios whose sampling variation contracts as the cross section grows. Economically, this implies that systematic variation must reside within the latent and observable spans, while the intercept block Γ_α captures only bounded residual pricing errors rather than omitted systematic risk. The diversification condition formalizes the large- N arbitrage-pricing logic of Chamberlain and Rothschild (1983) and connects directly to the approximate factor framework in Bai (2003); Bai and Ng (2008).

Finite-Dimensional Priced Risk Space. The latent and observable pricing directions form fixed, finite-dimensional subspaces with non-degenerate covariance structure, as imposed in Assumption A.4 and the compactness conditions in Assumption A.6. This ensures that the priced span is identified up to orthogonal

rotation and that each block of the parameter tuple

$$\theta = (\Gamma_\beta, F, \Gamma_\delta, \Gamma_\alpha)$$

corresponds to an economically distinct component of the pricing kernel. The resulting identification structure parallels the finite-factor APT arguments in [Chamberlain and Rothschild \(1983\)](#), the large-dimensional factor-model theory of [Bai \(2003\)](#); [Bai and Ng \(2008\)](#), and the characteristic-based factor representations in [Kelly et al. \(2019\)](#) and [Zaffaroni \(2025\)](#), where the focus is on identifying the span of systematic risk rather than any particular basis.

No-Arbitrage Discipline on Non-Systematic Directions. The asset-specific block Γ_α is subject to the curvature constraint

$$\psi(\Gamma_\alpha) = \Gamma_\alpha^\top Q_Z \Gamma_\alpha \leq \delta,$$

together with the interiority condition $\psi(\Gamma_\alpha^0) \leq \delta_0 < \delta$. In the Q_Z -geometry, $\psi(\Gamma_\alpha)$ is the squared Q_Z -norm of the residual direction $Z_t \Gamma_\alpha$ and therefore measures the extent to which the intercept loads on characteristic portfolios that are not absorbed by the latent span and, under the optional strengthening $\Gamma_\delta^\top Q_Z \Gamma_\alpha = 0$, by the observable PSF span as well. The economic role of this restriction becomes transparent once $\psi(\Gamma_\alpha)$ is estimated from data, as detailed in [Appendix G](#). Residual returns are driven by heteroskedastic and heavy-tailed shocks, and idiosyncratically volatile stocks can generate large residuals for reasons unrelated to true mispricing. A feasible GLS interpretation follows: stocks with unstable residual variance carry little information about whether a characteristic portfolio reflects a genuine pricing error, whereas assets with low idiosyncratic volatility provide sharper signals. The robust construction of $\hat{\psi}_T(\Gamma_\alpha)$ —which trims extreme residuals, stabilizes variance through shrinkage, and prevents any single turbulent period from dominating—ensures that the curvature bound reflects systematic rather than episodic deviations from no-arbitrage.²

Economically, bounding $\psi(\Gamma_\alpha)$ rules out residual portfolios with unbounded risk-adjusted profitability and prevents model misspecification or transient return spikes from being interpreted as persistent mispricing. This logic is closely related to Hansen–Jagannathan restrictions and to the limited-arbitrage treatments of residual risk in [Lettau and Pelger \(2020b\)](#) and [Dello Preite et al. \(2024\)](#). Geometrically, the curvature constraint renders the no-arbitrage parameter space $\Theta(\delta)$ compact in the α -direction and confines admissible perturbations of Γ_α to a bounded normal cone attached to the residual block of the characteristic space. This structure is essential for the projected WLS characterization and the block-diagonal LAN expansion developed in [Appendices A–B](#), as it guarantees that the intercept behaves as a regulated non-systematic risk component rather than a latent pricing factor in disguise.

PSF Curvature Discipline on Observable Directions. The curvature restriction on the observable block is optional and is not needed for the baseline LAN results, but when imposed it plays a useful economic role.

²[Appendix G](#) provides the complete finite-sample implementation of the intercept curvature constraint, including the construction of the robust curvature operator $\hat{H}_t$, the estimator $\hat{\psi}_T(\Gamma_\alpha)$, and the projection map that enforces $\psi(\Gamma_\alpha) \leq \delta$ within the Q_Z geometry. The appendix also shows how these steps correspond to the normal-cone structure of the no-arbitrage manifold and why the resulting estimator admits the LAN expansion developed in [Appendix B](#).

Observable factors enter the model as prespecified risk channels, yet their loading matrix interacts with the same characteristic geometry that shapes the latent span. The curvature functional

$$\psi_{\text{PSF}}(\Gamma_\delta) = \max_{1 \leq j \leq K_{\text{add}}} \gamma_{\delta,j}^\top H_{\text{psf}} \gamma_{\delta,j},$$

where H_{psf} is a stabilized average of the cross-sectional Gram matrices W_t , evaluates how strongly each observable factor loads on high-curvature characteristic portfolios. As detailed in Appendix H, the finite-sample estimator of this functional shrinks Γ_δ whenever its curvature exceeds an admissible bound δ_{psf} , thereby preventing the observable block from collapsing onto narrowly concentrated characteristic directions that produce impressive in-sample Sharpe ratios but have little economic content.

The economic motivation becomes transparent in a simple example. Consider a prespecified observable factor such as a value spread or a macro growth indicator. Its loading vector is intended to represent a broad exposure—large versus small firms, high versus low investment, or procyclical versus defensive industries—in the spirit of the diversified long-short portfolios used in Fama and French (1993), Novy-Marx (2013), and the macro-financial specifications in Kelly et al. (2019). In a characteristic-rich and highly collinear cross-section, however, an unconstrained least-squares fit can reduce pricing errors more cheaply by assigning this factor’s loading vector to a razor-thin characteristic direction dominated by a few volatile stocks.³ These firms lie along steep directions of the Q_Z geometry documented in Section 6, so a small tilt in the loading vector absorbs a disproportionate amount of cross-sectional variation. The resulting factor no longer behaves like an aggregate risk proxy; it becomes a nearly degenerate portfolio whose performance is driven by noise in a handful of unstable securities. This distortion affects the entire decomposition. The latent block is pushed away from the dominant curvature directions already claimed by the observable factor, and the observable factor itself ceases to reflect the broad macroeconomic or valuation risks it was designed to capture. Instead, it behaves like a high-leverage anomaly concentrated in a narrow corner of the characteristic space, a pattern closely related to the spurious and weak factors documented in Bryzgalova (2015), Lettau

³Formally, fix a date t and a single observable factor F_t^{obs} . Conditional on the other blocks, the cross-sectional loss in the PSF loading vector γ_δ is

$$L_t(\gamma_\delta) = \sum_{i=1}^{N_t} (r_{it} - F_t^{\text{obs}} z_{it}^\top \gamma_\delta)^2,$$

where z_{it} is the L -vector of characteristics for asset i and r_{it} is its excess return. Let e_{it} denote the current residual. The directional derivative of L_t at γ_δ in direction g is

$$\left. \frac{\partial L_t(\gamma_\delta + \varepsilon g)}{\partial \varepsilon} \right|_{\varepsilon=0} = -2F_t^{\text{obs}} \sum_{i=1}^{N_t} e_{it} z_{it}^\top g.$$

Consider two unit-norm directions $g^{(1)}$ and $g^{(2)}$. Suppose $g^{(2)}$ is concentrated on a small set S of highly volatile firms, so $z_{it}^\top g^{(2)}$ is large for $i \in S$ and close to zero otherwise, while $z_{it}^\top g^{(1)}$ is more diffuse. If the largest residuals $|e_{it}|$ are also concentrated on S , then

$$\left| \sum_{i=1}^{N_t} e_{it} z_{it}^\top g^{(2)} \right| \gg \left| \sum_{i=1}^{N_t} e_{it} z_{it}^\top g^{(1)} \right|.$$

A small step in the $g^{(2)}$ -direction therefore yields a much larger reduction in L_t than a step of the same size in the $g^{(1)}$ -direction. In the Q_Z geometry of Section 6, this corresponds exactly to moving along a high-curvature direction, because the Q_Z -norm satisfies $\|g^{(2)}\|_{Q_Z}^2 = g^{(2)\top} Q_Z g^{(2)}$ when $z_{it}^\top g^{(2)}$ is concentrated on a few extreme firms. Absent an explicit bound on $\|\gamma_\delta\|_{Q_Z}$, least squares has no penalty for such high-leverage loading vectors and will tend to favor them whenever residual variation is concentrated in thin, volatile segments of the cross-section.

and Pelger (2020b), Kozak et al. (2018), and Gu et al. (2020).

This pathology is not hypothetical. A growing literature shows that variance-based factor extraction in characteristic-rich panels tends to overweight extreme firms and to generate spurious or weakly identified factors supported by a small subset of high-volatility or highly levered stocks; see, among others, Lettau and Pelger (2020b), Bryzgalova (2015), Kozak et al. (2018), Kelly et al. (2019), and Gu et al. (2020). In unconstrained Euclidean IPCA, the least-squares objective is minimized by pushing Γ_δ toward such high-curvature directions: a loading vector with an inflated Q_Z -norm combined with an almost flat factor path can cheaply soak up residual variation, even though the resulting factor has no stable economic interpretation. The PSF curvature bound curbs this behavior by limiting the Q_Z -norm of Γ_δ , ensuring that observable factors remain diversified and retain exposures that are consistent with broad sources of systematic risk rather than with idiosyncratic or nearly constant directions.

Geometrically, the PSF curvature restriction intersects the no-arbitrage parameter space $\Theta(\delta)$ with a second quadratic ellipsoid acting on the observable block, yielding a compact feasible region $\Theta(\delta, \delta_{\text{psf}})$ in which movements of Γ_δ are confined to a bounded orthogonal complement of the latent span. This structure preserves the block decomposition required for the projected WLS characterization and for the block-diagonal LAN expansion in Appendices A–B. Together with the intercept curvature bound, this optional discipline helps ensure that the latent, observable, and non-systematic components continue to represent coherent and economically interpretable sources of priced risk rather than mechanically fitted high-curvature anomalies.

Smooth and Robust Response of Pricing Moments. The mapping from parameters to characteristic-weighted pricing moments is continuously Fréchet-differentiable with uniformly bounded derivative and Lipschitz modulus, and the loss is robustified by a Huber-type bounded-influence score, as summarized in Assumptions A.7, A.8, and A.10–A.11. Formally, these conditions ensure that the weighted least-squares criterion admits stable gradients and Hessians, that the Karush–Kuhn–Tucker system is well posed on the manifold $\Theta(\delta)$, and that movements in $(\Gamma_\beta, F, \Gamma_\delta, \Gamma_\alpha)$ are confined to valid tangent directions. Economically, the smoothness and bounded-influence properties guarantee that small perturbations of factor loadings or pricing kernels induce proportionate adjustments in pricing errors, while extreme asset-specific shocks are downweighted in the score. This structure embeds the classical smooth M -estimation regularity of Newey and McFadden (1994); Gouriéroux and Monfort (1997) within the Huber robustness framework of Huber (1981) and Maronna et al. (2006), and it underpins both the projected WLS characterization in Section 3.2 and the rotation-invariant LAN expansion in Appendix B.

Taken together, these assumptions deliver a compact, smooth, and rotation-invariant no-arbitrage manifold $\Theta(\delta)$ on which the NA-IPCA estimator is uniquely defined and locally regular. Stable characteristic moments anchor the Q_Z -geometry; idiosyncratic diversification forces systematic variation into the latent and observable factor blocks; finite-dimensionality and compactness provide identification up to rotation; the no-arbitrage bound disciplines non-systematic pricing directions; and smooth, bounded-influence moment maps yield coherent first-order and second-order conditions. Under this environment, the constrained estimator in Section 3.2 admits a unique projection onto $\Theta(\delta)$ and the rotation-invariant LAN representation formally established in Appendix B.

3.2 Constrained Weighted Least-Squares Formulation

This subsection introduces the weighted least-squares (WLS) criterion that implements the no-arbitrage geometry described in Section 2. Under the high-level environment of Section 3.1 and the formal regularity conditions in Appendix A, the characteristic-based pricing relation admits a well-defined moment map whose minimization yields the NA-IPCA estimator.

Pricing Moment Map. For any parameter tuple $\theta = (\Gamma_\beta, \Gamma_\delta, \Gamma_\alpha) \in \Theta(\delta)$, the period- t characteristic-weighted pricing error is

$$m_t(\theta) = X_t - W_t (\Gamma_\beta f_t + \Gamma_\delta F_t^{\text{obs}} + \Gamma_\alpha),$$

with $X_t = N_t^{-1} Z_t^\top r_{t+1}$ and $W_t = N_t^{-1} Z_t^\top Z_t$. The mapping $m_t(\theta)$ measures the component of X_t not explained by the latent span Γ_β , the observable span Γ_δ , or the non-systematic block Γ_α . The Q_Z -orthogonality relations from Section 2 ensure that these three blocks represent *distinct priced directions* rather than competing to explain the same dimension of systematic variation.

Weighted Least-Squares Criterion. The sample loss function aggregates the pricing errors using feasible cross-sectional weights:

$$Q_T(\theta) = \frac{1}{T} \sum_{t=1}^T m_t(\theta)^\top \Omega_t^{-1} m_t(\theta).$$

The matrices Ω_t downweight characteristic directions dominated by idiosyncratic volatility, ensuring that high-variance portfolios do not distort the estimation of systematic loadings.⁴ Assumption A.8 and the ratio-consistency results behind Appendix B.5 imply that feasible versions of Ω_t preserve all first-order properties of the ideal GLS system. Economically, WLS embeds the insight that the cross section contains

⁴Formally, if $u_{i,t}$ denotes the idiosyncratic return component of asset i at time t and $\Sigma_{u,t} = \text{Var}(u_t | Z_t)$ its cross-sectional covariance, then the characteristic-weighted moment $X_t = N_t^{-1} Z_t^\top u_{t+1}$ satisfies

$$\text{Var}(X_t | Z_t) = \frac{1}{N_t} Z_t^\top \Sigma_{u,t} Z_t.$$

When $\Sigma_{u,t}$ is heterogeneous across firms—a pervasive feature of equity data documented in Chamberlain and Rothschild (1983) and Kelly et al. (2019)—the unweighted cross-sectional moment X_t overemphasizes firms or portfolios whose idiosyncratic volatility is large, even if those firms carry little systematic information about expected returns. Premultiplication by $\Omega_t^{-1/2}$ rescales each moment by $\Sigma_{u,t}^{-1/2}$ (or any ratio-consistent surrogate), yielding

$$m_t^{\text{WLS}}(\theta) = \Omega_t^{-1/2} X_t - \Omega_t^{-1/2} W_t (\Gamma_\beta f_t + \Gamma_\delta F_t^{\text{obs}} + \Gamma_\alpha).$$

Because Ω_t^{-1} downweights characteristic directions with high residual dispersion, the resulting WLS criterion

$$Q_T(\theta) = \frac{1}{T} \sum_{t=1}^T m_t^{\text{WLS}}(\theta)' m_t^{\text{WLS}}(\theta)$$

assigns less influence to noisy characteristic portfolios and concentrates curvature in directions where the signal-to-noise ratio is economically meaningful. Geometrically, this ensures that the estimator extracts the systematic component of the characteristic space under a metric aligned with idiosyncratic precision rather than raw return variance. Economically, the weighting prevents high-volatility firms from distorting the identification of Γ_β and Γ_δ , ensuring that the latent and observable factors reflect priced risk rather than undiversified noise. Any ratio-consistent $\hat{\Omega}_t$ delivers the same first-order behavior (Proposition B.5), so the choice of feasible weights affects finite-sample stability but not identification or the LAN expansion.

heterogeneous signal strength and that priced directions should be extracted from volatility-adjusted rather than raw characteristic returns.

Constrained Estimator. The NA–IPCA estimator solves

$$\hat{\theta}_T = \arg \min_{\theta \in \Theta(\delta)} Q_T(\theta),$$

where the feasible set $\Theta(\delta)$ is defined in Section 2.3. The constraints enforce

$$\Gamma_\beta^\top Q_Z \Gamma_\beta = I_K, \quad \Gamma_\beta^\top Q_Z \Gamma_\delta = 0, \quad \Gamma_\beta^\top Q_Z \Gamma_\alpha = 0, \quad \psi(\Gamma_\alpha) \leq \delta,$$

thereby embedding the full no-arbitrage geometry into the estimation problem. The first three constraints prevent the latent and observable pricing blocks from loading on the same characteristic directions, while the curvature bound limits the magnitude of non-systematic pricing directions. Together, these restrictions address the identification deficiencies of unconstrained IPCA: they eliminate double counting of priced risk, prevent the intercept from absorbing systematic variation, and ensure that the estimator remains within an economically admissible region of characteristic space.

Viewed through this lens, NA–IPCA is a *geometrically constrained* cross-sectional regression performed on a curved parameter manifold rather than an unconstrained least-squares fit. The constraints introduce curvature, compactness, and rotational structure that are essential for well-posedness of the KKT system and for the tangent-space projection underpinning the characterization in Section 3.3. This disciplined formulation is the key step that allows NA–IPCA to inherit the block-separable and rotation-invariant asymptotic properties established in Appendix B.

3.3 Characterization as a Projected WLS Estimator

This subsection establishes the structural link between the constrained extremum problem

$$\hat{\theta}_T = \arg \min_{\theta \in \Theta(\delta)} Q_T(\theta)$$

and a system of weighted least-squares (WLS) equations projected onto the tangent space of the no-arbitrage manifold $\Theta(\delta)$. Under the high-level assumptions of Section 3.1 and the regularity conditions in Appendix A, the moment map is differentiable on $\Theta(\delta)$ and the feasible set forms a smooth manifold with tangent space $\mathcal{T}_\theta \Theta(\delta)$ and projection operator Π_θ defined in Appendix A.5.1–A.7. These geometric objects govern both the identification of priced directions and the first-order behavior of the estimator.

Let $S_T(\theta) = \nabla_\theta Q_T(\theta)$ denote the sample score. The constrained estimator solves the WLS problem subject to the restrictions defining $\Theta(\delta)$ if and only if the gradient, when evaluated at $\hat{\theta}_T$, has no component along any feasible perturbation direction—i.e. no movement within the tangent space can reduce the objective further. This leads to the following characterization.

Proposition 3.1 (Projected WLS characterization). *Let Assumptions A.1–A.11 hold, and let $\Theta(\delta)$ be endowed with the Q_Z -geometry of Appendix A. Then $\hat{\theta}_T$ is a locally unique minimizer of $Q_T(\theta)$ on $\Theta(\delta)$ if and only if*

it satisfies the projected first-order conditions

$$\Pi_{\hat{\theta}_T} S_T(\hat{\theta}_T) = 0. \quad (1)$$

Equivalently, $\hat{\theta}_T$ solves the WLS normal equations after projection onto $\mathcal{T}_{\hat{\theta}_T} \Theta(\delta)$, so each block (Γ_β, F) , Γ_δ , and Γ_α satisfies its own WLS condition restricted to the Q_Z -orthogonal subspace in which it is identified. On this tangent space, the information matrix is block-diagonal across the latent, observable, and non-systematic components, and the estimator is locally efficient in the sense of Proposition B.3.

The projected first-order conditions in (1) give the estimator a transparent geometric interpretation. The projection eliminates all components of the score that would move the parameter tuple off the constraint manifold—for instance, by destroying Q_Z -orthonormality of the latent loadings, creating overlap between the latent and observable blocks, or increasing the magnitude of the non-systematic pricing direction Γ_α beyond the no-arbitrage bound $\psi(\Gamma_\alpha) \leq \delta$. Only perturbations that remain within $\mathcal{T}_{\hat{\theta}_T} \Theta(\delta)$ —the economically admissible directions identified by the no-arbitrage geometry—enter the first-order system. As shown in Appendix B.1, this projection transforms the unconstrained WLS normal equations into a blockwise system on the feasible manifold, leaving unchanged the curvature and identification structure implied by the Q_Z metric.

The economic implications follow directly from this geometry. The latent block (Γ_β, F) is updated only through variations that preserve its Q_Z -orthonormal structure and remain orthogonal to the observable span, so the latent factors capture precisely the characteristic-based priced directions not spanned by the observable proxies. The observable loading block Γ_δ adjusts within the Q_Z -orthogonal complement of the latent span, which prevents duplication of priced directions and enforces the separation formalized in Proposition B.2. The non-systematic pricing direction Γ_α moves within the bounded ellipsoid $\{\psi(\Gamma_\alpha) \leq \delta\}$, representing deviations from exact no-arbitrage that are economically admissible but cannot mimic systematic risk.

From a statistical perspective, (1) is the mechanism that delivers the block-separable curvature on $\mathcal{T}_{\hat{\theta}_T} \Theta(\delta)$ and the rotation-invariant LAN expansion in Proposition B.3. Because the tangent space decomposes into mutually Q_Z -orthogonal components corresponding to latent, observable, and non-systematic directions, sampling variation in each block contributes independently to the estimator's asymptotic distribution. The projected WLS estimator therefore inherits a block-diagonal information matrix and achieves the efficiency bound implied by the constrained geometry. In this sense, the projection is not a computational artifact but the essential geometric operation that aligns estimation with the economic restrictions of the No-Arbitrage IPCA framework.⁵

⁵Appendices A and B provide the formal probabilistic and geometric foundations underlying the characterization in (1). Appendix A establishes that the constraint set $\Theta(\delta)$ is a smooth, compact, and rotation-invariant manifold: Assumptions A.1–A.3 give the existence and stability of the characteristic metric Q_Z ; Lemmas A.1–A.2 supply the uniform LLN/CLT needed to linearize the moment map; the LICQ and Slater conditions in Section A.4 guarantee that the Jacobian of the constraint system has full row rank, so the tangent space $\mathcal{T}_\theta \Theta(\delta)$ and its projector Π_θ are well defined; and Appendix A.5.1–A.7 describe the Q_Z -Stiefel geometry and the quotient structure that remove rotational degrees of freedom. These results jointly ensure that the KKT system associated with the constrained problem admits a stable implicit-function representation on $\mathcal{T}_\theta \Theta(\delta)$, so projecting the score onto the tangent space is mathematically well posed. Appendix B builds on this structure to show that the projected WLS representation is not merely an interpretation but an exact local characterization of the estimator. Proposition B.1 proves that—after pre-whitening—the KKT conditions of the constrained extremum problem coincide with those of the WLS system when both are restricted to $\mathcal{T}_\theta \Theta(\delta)$. Proposition B.2 establishes that the Jacobians of the moment map respect the Q_Z -orthogonal decomposition induced by the

3.4 Rotation Invariance and Link to the Algorithm

The NA-IPCA estimator is defined only up to orthogonal rotations of the latent and observable blocks. As shown in Appendices A.7.1 and B.6, the action $(\Gamma_\beta, F, \Gamma_\delta) \mapsto (\Gamma_\beta R_\beta, R_\beta^\top F, \Gamma_\delta R_\delta)$ for $(R_\beta, R_\delta) \in \mathcal{O}(K) \times \mathcal{O}(K_{\text{add}})$ leaves the fitted characteristic-space returns, the constrained objective $Q_T(\theta)$, and the projected first-order conditions (1) unchanged. Consequently, identification is achieved at the level of the Q_Z -orthogonal subspaces $\text{col}(\Gamma_\beta)$ and $\text{col}(\Gamma_\delta)$ rather than at the level of a particular coordinate representation. Appendix B.6 shows that the KKT system and the LAN covariance in Proposition B.3 transform by orthogonal similarity under this group action, so the estimator's asymptotic distribution depends only on the priced spans and is invariant to within-block reparametrizations.

From an identification perspective, the relevant objects in this framework are the *spans* of the loading blocks rather than any particular realization of factor time series. As in the approximate factor literature of Bai (2003) and Bai and Ng (2008), any invertible $K \times K$ matrix H induces a rotation $(\Gamma_\beta, F) \mapsto (\Gamma_\beta H, H^{-1}F)$ that leaves the product $\Gamma_\beta F$ unchanged, so neither Γ_β nor F is individually identified. What is identified is the latent pricing space $\text{col}(\Gamma_\beta)$ and its Q_Z -orthogonal complement, not a particular basis or factor path. In characteristic-based asset-pricing applications, this distinction is economically meaningful: risk premia are attached to characteristic-backed portfolios, and hence to directions in $\text{col}(\Gamma_\beta)$, rather than to the arbitrary coordinate system used to represent F_t . The NA-IPCA estimator therefore targets the geometry of the loading space, and the diagnostics in Sections 7.2–7.3 are constructed directly in this Q_Z -metric.

Evaluating different estimators through Q_Z -based diagnostics is thus not an unfair comparison, but a direct test of economic identification. The metric Q_Z is defined by the long-run cross-sectional distribution of characteristics (Section 2.2) and encodes the curvature along which systematic compensation for risk is borne. An estimator whose latent span fails to align with this geometry—for example, by enforcing Euclidean orthogonality under the identity metric I_L as in unconstrained IPCA—may approximate the data well in a variance sense while remaining far from the priced tangent space. By measuring subspace distances, overlaps, and misallocation in the Q_Z -metric, the simulation diagnostics ask whether an estimator recovers the economically relevant spans and block structure implied by $\Theta(\delta)$, rather than whether it is internally coherent under its own, potentially misspecified, geometry. In this sense, NA-IPCA is judged on the same scale as its competitors, but that scale is determined by the no-arbitrage geometry of the data, not by the Euclidean conventions built into a particular algorithm.

For reporting and for numerical stability, a canonical representative may be selected via a Q_Z -weighted Procrustes normalization, as detailed in Appendices A.5.1–A.7. This normalization replaces $(\Gamma_\beta, \Gamma_\delta)$ by $(\Gamma_\beta R_\beta^*, \Gamma_\delta R_\delta^*)$, where R_β^* and R_δ^* solve the corresponding weighted alignment problems; it preserves the constraints and the value of $Q_T(\theta)$ and varies smoothly in a neighborhood of any regular point. The alternating WLS routine described in Appendix C exploits this geometry. Each iteration consists of a WLS update for

constraints, making the Fisher information block-diagonal across the latent, observable, and non-systematic blocks. Proposition B.3 then derives the joint LAN expansion on the tangent space and shows that the information operator is nonsingular and block-separable, yielding local efficiency of the estimator. Finally, Appendix B.6 verifies that both the KKT map and the LAN covariance are invariant under rotations of $(\Gamma_\beta, F, \Gamma_\delta)$, so the projected system identifies the priced *spans* rather than any particular basis. Taken together, Appendices A and B confirm that the projection in (1) is the fundamental geometric operation: it removes directions that violate orthogonality or the curvature bound, enforces identification, and delivers the block-diagonal efficiency of the No-Arbitrage IPCA estimator.

$(F, \Gamma_\beta, \Gamma_\delta, \Gamma_\alpha)$ followed by projection onto $\Theta(\delta)$ to enforce Q_Z -orthonormality, block-orthogonality, and the curvature bound $\psi(\Gamma_\alpha) \leq \delta$. Under the regularity conditions in Appendix A, any limit point of this scheme that satisfies first-order optimality lies on $\Theta(\delta)$ and solves the projected system (1); hence, the computational estimator coincides with the theoretically characterized NA-IPCA estimator and inherits its rotation-invariant identification, blockwise curvature, and efficiency properties.

To summarize, Section 3 has presented the No-Arbitrage IPCA estimator as a constrained and geometrically disciplined extension of characteristic-based factor pricing. The estimator minimizes a volatility-adjusted WLS criterion over the no-arbitrage manifold $\Theta(\delta)$, which imposes Q_Z -orthogonality across latent, observable, and non-systematic pricing directions and bounds the curvature of the residual component. Under the high-level assumptions in Section 3.1, the feasible set $\Theta(\delta)$ forms a smooth and compact constraint manifold on which the estimator admits a projected WLS characterization: the first-order conditions arise by projecting the WLS score onto the tangent space of $\Theta(\delta)$, and the resulting information matrix is block-diagonal across the latent, observable, and residual blocks. Appendix B shows that this geometric structure yields a rotation-invariant LAN expansion and local efficiency, while Appendix C verifies that the alternating WLS routine used in implementation converges to the solution of the projected KKT system and therefore coincides with the theoretically defined NA-IPCA estimator.

The next section turns to the empirical implications of this structure. Building on the estimator and geometric framework developed here, Section 4 examines how the no-arbitrage restrictions, the separation of pricing blocks, and the volatility-adjusted Q_Z geometry affect estimator performance in finite samples. I compare NA-IPCA with unconstrained IPCA in simulation environments calibrated to real characteristic panels and evaluate whether the additional geometric discipline enhances identification, robustness, and interpretability without sacrificing economically meaningful fit.

4 Asymptotic Theory and Rotation-Invariant Inference

This section establishes the large-sample properties of the No-Arbitrage IPCA estimator developed in Section 3. The analysis builds on the geometric structure of the feasible parameter space $\Theta(\delta)$ introduced in Section 2 and on the regularity conditions in Appendix A. Expressed in local coordinates on the tangent manifold of $\Theta(\delta)$, the estimator admits a linear expansion in which both the score and the curvature are confined to economically identified directions. The associated restricted information operator inherits the orthogonality encoded by the no-arbitrage constraints and, under PSF centering ($\mathbb{E}[F_t^{\text{obs}}] = 0$), is block-diagonal across the latent, observable, and non-systematic components. Because orthogonal rotations of the loading blocks leave the criterion and the geometry unchanged, inference concerns the priced spans rather than any particular parametrization. All technical arguments appear in Appendices B–D, including the feasible covariance construction based on the projected score and the long-run covariance of the Huber/GLS innovations.

4.1 Parameter Space, Tangent Manifold, and Rotation Geometry

The constraints defining $\Theta(\delta)$ endow the parameter space with a smooth and compact geometry determined by the characteristic metric Q_Z . The latent loadings lie on a Q_Z -Stiefel manifold, the observable

block occupies its Q_Z -orthogonal complement, and the non-systematic component is restricted to a bounded Q_Z -ellipsoid. Since expected returns depend on the latent and observable blocks only through their Q_Z -spans, identification is achieved on the quotient space $\Theta(\delta)/\mathcal{G}$, where $\mathcal{G}$ denotes the blockwise rotation group acting on $(\Gamma_\beta, F, \Gamma_\delta)$.

Perturbations of $\theta \in \Theta(\delta)$ therefore decompose into vertical directions, corresponding to rotations that leave the priced spans unchanged, and horizontal directions that preserve feasibility and generate genuine variation in characteristic-based pricing. Appendix A details the resulting tangent space $T_\theta\Theta(\delta)$ and the associated projection operator that removes all rotational and constraint-violating components from the score. These horizontal directions form the basis for the constrained WLS representation in Section 3.3 and for the LAN expansion in Appendix B, and they determine the economically identified movements along which sampling variation accumulates.

4.2 Constrained WLS Equivalence and Block-Diagonal Information

The constrained extremum problem in Section 3 admits a WLS representation once expressed in local coordinates on $T_{\theta_0}\Theta(\delta)$. As shown in Appendix B, pre-whitening the pricing moment rescales the criterion without altering the geometry of the feasible set, so the projected KKT conditions coincide with the normal equations of the WLS problem on the tangent manifold. To first order, the estimator therefore behaves as a WLS estimator evolving along the economically admissible directions identified in Section 4.1.

The orthogonality restrictions embedded in $\Theta(\delta)$ imply that the Jacobian of the moment map respects the decomposition of the characteristic space. Proposition B.2 first shows that the economically relevant observable content is the effective PSF span $\text{col}(M_\beta \Gamma_\delta)$, which is Q_Z -orthogonal to $\text{col}(\Gamma_\beta)$. If, in addition, PSF centering holds ($\mathbb{E}[F_t^{\text{obs}}] = 0$), the population information operator of the pre-whitened GLS system is block-diagonal after projection onto $T_{\theta_0}\Theta(\delta)$. Cross-derivatives between the latent and observable blocks then vanish under the identifying restriction $\Gamma_\beta^\top Q_Z \Gamma_\delta = 0$, while the curvature induced by the no-arbitrage bound affects only the non-systematic block.

Under these maintained conditions, first-order sampling variation separates by block on orthogonal tangent-space components, and the corresponding restricted covariance operator decomposes accordingly. The constrained estimator therefore inherits the GLS efficiency bound on the identified tangent space while respecting the economic discipline implied by the no-arbitrage geometry.

4.3 LAN Representation and Robustness

The constrained WLS representation established in Section 4.2 allows the estimator to be analyzed through local coordinates on the tangent manifold $T_{\theta_0}\Theta(\delta)$. Under the regularity conditions in Appendix A, the moment map is differentiable, the corresponding score satisfies a uniform LLN and CLT, and the KKT system admits a locally unique solution. These ingredients yield a local asymptotic normal representation in which sampling variation accumulates only along the economically identified directions of the model. The main result is summarized below.

Theorem 4.1 (LAN Representation and Restricted Covariance). *Suppose Assumptions A.1–A.11 hold, let θ_0*

denote the true parameter in the interior of the feasible set $\Theta(\delta)$, and assume PSF centering ($\mathbb{E}[F_t^{\text{obs}}] = 0$). Then the NA-IPCA estimator admits the tangent-space expansion

$$\sqrt{T} \Pi_{T_{\theta_0} \Theta(\delta)} (\hat{\theta}_T - \theta_0) = \left(\mathcal{J}(\theta_0)|_{T_{\theta_0} \Theta(\delta)} \right)^{-1} \frac{1}{\sqrt{T}} \sum_{t=1}^T \nabla_{\theta} g_t(\theta_0)^\top \Psi_t + o_p(1),$$

where Ψ_t is the Huber/GLS innovation and $\mathcal{J}(\theta_0)|_{T_{\theta_0} \Theta(\delta)}$ is the restricted information operator appearing in Proposition B.3. Under PSF centering, this restricted information operator is block-diagonal across the latent, observable, and non-systematic components, and the same first-order representation holds under any ratio-consistent weighting scheme and for any bounded-influence (Huber) score with tuning parameter satisfying Assumption A.8.

Consequently,

$$\sqrt{T} \Pi_{T_{\theta_0} \Theta(\delta)} (\hat{\theta}_T - \theta_0) \Rightarrow \mathcal{N}\left(0, \left(\mathcal{J}(\theta_0)|_{T_{\theta_0} \Theta(\delta)} \right)^{-1}\right),$$

with asymptotic covariance that separates across the three pricing blocks.

All proofs follow from Propositions B.1–B.3 and the supporting lemmas in Appendices A–B. The WLS equivalence ensures that the KKT system can be analyzed on the tangent manifold; Proposition B.2 yields block-diagonal restricted information under the same PSF-centering condition; and the CLT for the Huber/GLS innovations provides the Gaussian limit for the projected score.

The score innovations Ψ_t combine volatility adjustment with the bounded-influence transformation of the Huber loss. Appendix B shows that this refinement preserves differentiability of the criterion and generates a martingale-difference array with the same first-order limit as its ideal GLS counterpart. Likewise, feasible weighting schemes that satisfy the ratio-consistency conditions in Appendix B.5 perturb the whitening map only by $o_p(1)$, leaving both the projected score and the restricted information operator unchanged. For feasible empirical implementation, Appendix D combines the projected score with a HAC estimate of the long-run covariance of the Huber/GLS innovations; in the ideal GLS benchmark summarized in Theorem 4.1, this variance reduces to the inverse restricted information operator displayed above.

Taken together, these results imply that, within the class of estimators that respect the maintained no-arbitrage geometry, NA-IPCA achieves the GLS efficiency bound on the identified tangent space. Under PSF centering, first-order sampling variation separates by the latent, observable, and non-systematic blocks, and the influence function is determined by the identified directions on the tangent manifold. The combination of constrained WLS geometry and robust score construction thus yields a regular estimator whose asymptotic distribution reflects the economic decomposition of priced directions.

4.4 Rotation-Invariant Inference

The quotient representation of the parameter space described in Section 4.1 implies that statistical inference must be carried out on directions that depend only on the priced spans. Let $u \in T_{\theta_0} \Theta(\delta)$ be any such direction. Combining the LAN representation in Section 4.3 with the tangent-space projection yields

the limit distribution

$$u^\top \Pi_{T_{\theta_0} \Theta(\delta)}(\hat{\theta}_T - \theta_0) \Rightarrow \mathcal{N}\left(0, u^\top \left(\mathcal{J}(\theta_0)|_{T_{\theta_0} \Theta(\delta)} \right)^{-1} u\right),$$

where the block-diagonal form of the restricted information operator under PSF centering ($\mathbb{E}[F_t^{\text{obs}}] = 0$) allows inference to be organized by the latent, observable, and non-systematic blocks. Because all objects in this expression live on the tangent manifold, inference concerns the identified pricing directions rather than any particular representation of the loadings or factors.

This invariance has a formal counterpart. Rotation of the latent and observable factor bases,

$$(\Gamma_\beta, F, \Gamma_\delta) \mapsto (\Gamma_\beta R_\beta, R_\beta^\top F, \Gamma_\delta R_\delta), \quad (R_\beta, R_\delta) \in \mathcal{G},$$

does not alter the fitted returns, the constrained criterion, or the projected score. Appendix B.6 shows that both the KKT map and the restricted information operator transform by orthogonal similarity, preserving all quadratic forms of the type $u^\top (\cdot) u$. This property yields the following corollary.

Corollary 4.2 (Rotation-Invariant Inference). *Let $\hat{\theta}_T$ denote the NA-IPCA estimator and let $(R_\beta, R_\delta) \in \mathcal{G} = \mathcal{O}(K) \times \mathcal{O}(K_{\text{add}})$ act on the parameter tuple as in Section 4.1. Under Assumptions A.1–A.11, the PSF-centering condition $\mathbb{E}[F_t^{\text{obs}}] = 0$, and the LAN expansion in Theorem 4.1, inference is invariant to such rotations in the following sense:*

(i) *For any direction u in the tangent space $T_{\theta_0} \Theta(\delta)$,*

$$u^\top \Pi_{T_{\theta_0} \Theta(\delta)}\left((R_\beta, R_\delta) \cdot \hat{\theta}_T - (R_\beta, R_\delta) \cdot \theta_0\right) \Rightarrow \mathcal{N}\left(0, u^\top \left(\mathcal{J}(\theta_0)|_{T_{\theta_0} \Theta(\delta)} \right)^{-1} u\right),$$

so the limiting distribution of any linear functional of the estimator does not depend on the choice of orthonormal basis for the latent or observable factor spaces.

(ii) *Wald tests, confidence intervals, and standard errors constructed from any such direction u are unchanged under (R_β, R_δ) , because both the projected score and the restricted information operator transform by orthogonal similarity.*

(iii) *The asymptotic covariance of $\hat{\theta}_T$ depends only on the priced spans $\text{col}(\Gamma_\beta)$ and $\text{col}(M_\beta \Gamma_\delta)$, and not on any particular coordinate representation. Here*

$$M_\beta = I - P_\beta, \quad P_\beta = \Gamma_\beta (\Gamma_\beta^\top Q_Z \Gamma_\beta)^{-1} \Gamma_\beta^\top Q_Z,$$

is the Q_Z -orthogonal projector onto the latent loading space $\text{col}(\Gamma_\beta)$, so that M_β projects any loading vector onto the Q_Z -orthogonal complement of that space. Consequently, $\text{col}(M_\beta \Gamma_\delta)$ represents the economically relevant observable-factor span—i.e., the component of Γ_δ that is not already contained in the latent characteristic-based span.

Economically, this invariance reflects the principle that factor models identify only the spans of systematic risk exposures: rotations of the latent or observable blocks re-label factor coordinates without altering any priced direction. Statistically, orthogonal similarity of the information operator guarantees that all inferen-

tial procedures based on linear contrasts—Wald tests, confidence intervals, and standard errors—remain unchanged under such reparametrizations. Rotation-invariant inference is therefore the natural mode of evaluation for the NA-IPCA estimator: asymptotic uncertainty attaches to the priced spaces rather than to any coordinate representation of the loadings or factors.

4.5 Practical Implementation and Empirical Use

The asymptotic theory above translates directly into an implementation that is both empirically stable and economically transparent. The estimator is computed through a sequence of volatility-adjusted least-squares regressions projected onto the no-arbitrage manifold $\Theta(\delta)$, as detailed in Appendix C. Each iteration updates the latent and observable factors in WLS units and reimposes the Q_Z -orthogonality and curvature restrictions. This projection step anchors the estimates to the geometric structure of the model and prevents the drift commonly observed in unconstrained IPCA. In applications where the block-diagonal information statements above are invoked, the observable PSF block should be centered in sample to impose PSF centering. Empirical comparisons can then be conducted directly at the level of priced spans.

The LAN representation in Section 4.3 provides a straightforward recipe for inference. Standard errors for linear contrasts of the identified components are constructed by combining the projected score with a heteroskedasticity- and autocorrelation-consistent long-run covariance estimator for the Huber/GLS innovations and the corresponding restricted information operator, following Appendix D. In the ideal GLS benchmark this reduces to the inverse restricted information operator displayed in Theorem 4.1; in feasible implementation it yields the blockwise Wald procedures described in Appendix D. Under PSF centering, these procedures can be organized by the latent, observable, and non-systematic components. Empirically, this allows researchers to quantify the strength of the characteristic-based span, assess the incremental explanatory power of traded or macro factors, and evaluate the magnitude of the admissible non-systematic component without cross-contamination among the three pricing channels.

The geometric structure also sharpens diagnostic evaluation. Subspace distances for the latent block, Procrustes-aligned loading errors for the observable block, and fitted-return discrepancies for the full system provide rotation-invariant metrics of performance. These diagnostics are particularly informative when comparing NA-IPCA with unconstrained IPCA or PCA-type estimators: the no-arbitrage constraints prevent factor directions from drifting toward high-variance characteristics, reduce double counting between latent and observable sources of risk, and restrict the non-systematic component to an economically plausible region of characteristic space.

These properties yield an empirical toolkit that is easy to implement with standard linear-algebra routines, produces stable and interpretable factors, and generates sampling distributions consistent with the economic restrictions of the model. The combination of constrained geometry, blockwise inference, and rotation-invariant diagnostics makes the NA-IPCA estimator well suited for large cross-sectional applications and for empirical settings where characteristic structure and observable risk proxies must be evaluated jointly.

The asymptotic results developed in this section characterize the NA-IPCA estimator as a regular, rotation-invariant procedure whose sampling variation decomposes cleanly across latent, observable, and non-systematic pricing components. This block structure, together with the tangent-space linearization and

robustness properties established above, provides a fully articulated statistical foundation for the estimator. The next section positions the NA-IPCA framework within the broader class of statistical and economically disciplined factor models, showing how it nests standard IPCA as a limiting case, relates to recent PCA-based estimators that incorporate economic pricing restrictions, and provides a tractable linear alternative to flexible no-arbitrage approaches such as neural-network-based SDF methods.

5 Positioning NA-IPCA within the Factor-Model Literature

This section positions the No-Arbitrage IPCA estimator within the broader landscape of statistical and economically motivated factor-model methods. While NA-IPCA shares its empirical backbone with the IPCA framework of [Kelly et al. \(2019\)](#), it differs in a fundamental respect: *it is the first estimator to integrate characteristic geometry and no-arbitrage discipline into a fully tractable WLS-LAN framework*. By embedding these restrictions directly in the Q_Z metric, NA-IPCA reorganizes characteristic space into latent, observable, and asset-specific pricing directions that are separated in economically meaningful units and admit a closed-form geometric representation.

These features place NA-IPCA alongside, yet distinct from, a growing collection of estimators that embed economic discipline—such as PCA procedures that impose pricing restrictions or flexible approaches that enforce no-arbitrage through SDF moment conditions—while offering a uniquely transparent identification map and a rotation-invariant asymptotic theory. The subsections below clarify these connections and highlight the dimensions along which NA-IPCA extends, nests, or complements existing approaches.

5.1 Relation to Statistical IPCA: A Structural Perspective

A detailed structural comparison between NA-IPCA and the statistical IPCA estimator of [Kelly et al. \(2019\)](#) is provided in Appendix E. Here we highlight the main points of contact and the structural distinctions that matter for interpretation and identification.

Both estimators begin from the same characteristic-based moment structure (X_t, W_t) and recover only the span of systematic return variation rather than a particular basis. In this sense, NA-IPCA preserves the statistical backbone of IPCA. The key differences arise from the geometry and the way economically meaningful orthogonality is imposed. Statistical IPCA operates in an implicitly Euclidean characteristic space and separates latent and observable directions through identity-metric normalizations. NA-IPCA replaces this Euclidean structure with the characteristic metric Q_Z , so orthogonality, scaling, and identification are determined by the long-run covariance of characteristics rather than by ad hoc Euclidean projections.

This geometric shift has two economic implications. First, NA-IPCA enforces Q_Z -orthogonality between latent, observable, and non-systematic components, ensuring that priced directions are not counted twice when characteristics correlate with traded or macro factors. Second, the non-systematic component is required to lie within a bounded Q_Z -ellipsoid, which embeds a no-arbitrage discipline absent from statistical IPCA and prevents large pricing directions that cannot be supported by characteristic portfolios.

Statistical IPCA emerges as a special case of NA-IPCA when Q_Z is proportional to the identity and the curvature bound on the non-systematic component is slack. In that limit, the NA-IPCA manifold collapses to

the Euclidean identification space used in IPCA, and the two estimators coincide up to an orthogonal relabeling of the latent factors. Outside this special case, the Q_Z geometry and the explicit no-arbitrage restrictions give NA-IPCA a sharper economic decomposition and guard against the misallocation of characteristic-based variation across latent, observable, and asset-specific components.

5.2 Relation to PCA Estimators Incorporating Pricing Structure

A parallel literature modifies variance-based factor extraction to incorporate asset-pricing structure directly into principal-component methods. The regularized-pricing PCA (RP-PCA) estimator of [Lettau and Pelger \(2020b\)](#) augments the PCA objective with a penalty for pricing errors, tilting the leading components toward directions that jointly explain time-series comovement and cross-sectional premia. Related analyses by [Bryzgalova \(2015\)](#) and [Bryzgalova et al. \(2023\)](#) document how standard PCA can misallocate factor importance when characteristics are correlated or when true pricing exposures are weak, often elevating spurious directions that reflect idiosyncratic scaling rather than economically meaningful risk. A complementary three-pass procedure in [Giglio and Xiu \(2021\)](#) uses principal components of a large panel of test-asset returns to reconstruct the latent factor space and estimate the risk premia of observed factors in a manner robust to omitted pricing sources. Finally, [Kozak et al. \(2018\)](#) interpret the empirical success of low-dimensional factor models through near-arbitrage bounds: when return covariance is dominated by a few principal components, reduced-form PC-based stochastic discount factors arise naturally, even if expected returns are shaped by sentiment or other nonstandard forces.

These approaches strengthen the connection between statistical components and priced directions, but they all operate in an *identity-metric* geometry: orthogonality, normalization, and curvature are defined using the Euclidean inner product, with no adjustment for the empirical covariance structure of firm characteristics. Economically, this means that PCA treats all characteristic directions as equally scaled and equally relevant for risk pricing, even when certain characteristics co-move strongly or contribute little to systematic variation. As a result, PCA-based estimators may misallocate priced directions across latent, observable, and asset-specific blocks—either double-counting risk or attributing priced directions to noise—particularly in heterogeneous payoff spaces.

NA-IPCA differs in that economic structure enters through the *geometry* of the parameter space rather than through the objective alone. All orthogonality, identification, and curvature restrictions are defined using the characteristic metric Q_Z , which reflects the long-run covariance structure of firm characteristics and determines how characteristic portfolios span priced risk. This Q_Z -based geometry forces the latent, observable, and asset-specific components to occupy mutually orthogonal subspaces measured in economically meaningful units. As a consequence, the feasible set $\Theta(\delta)$ becomes a curved, non-Euclidean manifold that admits a full KKT representation (Appendix A), a projected WLS characterization on its tangent space (Proposition B.1), a block-diagonal information operator (Proposition B.2), and a rotation-invariant LAN expansion with separate precision for each pricing block (Proposition B.3). None of these identification or inference properties arise under identity-metric PCA, for which rotations or correlation shifts in characteristics can materially alter the decomposition of priced directions.

Viewed from a structural perspective, NA-IPCA is best understood as a geometry-first counterpart to

PCA estimators that embed pricing information. Whereas RP-PCA, GMM-PCA, and weak-factor-robust PCA approaches incorporate economic structure by modifying the *objective*—for example, by penalizing pricing errors or enforcing moment conditions while keeping Euclidean orthogonality—NA-IPCA alters the underlying *metric* itself. The estimator evolves on a Q_Z -orthogonal manifold that reflects the empirical covariance structure of characteristics, thereby separating latent, observable, and asset-specific pricing directions in economically meaningful units. This geometry-based formulation prevents double counting between characteristic and observable sources of priced risk, stabilizes factor interpretation when characteristics are correlated, and ensures that inference pertains to pricing *spans* rather than to particular coordinate representations. These features underlie the blockwise efficiency and rotation-invariant inference results in Section 4 and differentiate NA-IPCA from the broader class of PCA estimators that incorporate pricing restrictions while remaining in an identity-metric space.

5.3 Relation to Machine-Learning SDF Estimators

Recent advances in machine-learning asset pricing propose highly flexible specifications for the stochastic discount factor (SDF). Gu et al. (2020) use neural networks to map firm characteristics into SDF weights, allowing for nonparametric interactions in the cross section; Chen et al. (2024) develop a “deep Learner” that embeds the no-arbitrage moment conditions directly into a neural architecture; and related approaches use adversarial training or cross-sectional residual minimization to extract an SDF with minimal pricing errors. These methods deliver impressive out-of-sample fits and can approximate complex nonlinear pricing kernels, but their flexibility often comes at the expense of interpretability and analytical tractability.

Relative to this literature, NA-IPCA occupies a distinct position. It shares with machine-learning SDFs the objective of reducing pricing errors and respecting no-arbitrage, but it imposes structure not through a nonlinear architecture but through the *geometry* of characteristic space. The latent, observable, and asset-specific components are separated in Q_Z -orthogonal directions—the same metric that governs how characteristics span the cross section of expected returns. This geometric formulation yields a closed-form WLS representation, a tangent-space decomposition of pricing directions, and a fully explicit LAN expansion with block-diagonal information (Section 4). Such identification and inferential guarantees are generally unavailable for deep SDFs, for which the learned representation is not unique and rotation of the hidden layers may alter the decomposition of priced directions.

Economically, machine-learning SDFs provide a flexible forecast of the pricing kernel, whereas NA-IPCA provides a transparent decomposition of *which* characteristic-based and observable sources of risk are priced, and in what geometric configuration. The Q_Z -orthogonal decomposition prevents double counting between latent and observable factors, stabilizes factor spans across samples, and links each priced direction to a distinct curvature component of the no-arbitrage manifold. As a result, NA-IPCA can be viewed as an analytically tractable, geometry-first benchmark that nests the linear structure of reduced-form factor models, supports formal inference, and complements the more flexible but opaque representations delivered by neural-network SDF estimators.

5.4 Nesting Structure and Potential Extensions

The geometry underlying NA-IPCA places it within a broad class of factor estimators while also revealing how existing approaches arise as limiting cases. When the characteristic metric satisfies $Q_Z \propto I_L$ and the curvature bound on the asset-specific component is slack, the feasible manifold $\Theta(\delta)$ collapses to the Euclidean identification space of statistical IPCA, and the alternating WLS updates of NA-IPCA coincide with the ALS updates in Kelly et al. (2019) up to an orthogonal relabeling of the latent factors. Likewise, if no observable factors are supplied, the estimator reduces to a Q_Z -orthogonal version of latent-factor extraction; if the asset-specific block is constrained to be zero, the estimator becomes a geometry-weighted analogue of PCA; and if the Q_Z geometry is replaced by the identity metric, the procedure mimics the objective-driven modifications of RP-PCA and related identity-metric variants. In this sense, NA-IPCA provides a unified formulation that nests statistical IPCA, pricing-disciplined PCA, and linear machine-learning estimators as special or degenerate cases of the same geometric framework.

The structure also suggests a number of natural extensions. Because the constraints defining $\Theta(\delta)$ operate through a metric rather than a particular parametrization, the framework could be adapted to accommodate nonlinear transformations of characteristics through a kernelized or low-dimensional embedding of the Q_Z geometry. The dynamic updating of Q_Z , or the introduction of state-dependent curvature, would yield a time-varying no-arbitrage manifold capable of capturing shifts in characteristic structure across business cycles. Finally, the geometric restrictions of NA-IPCA could serve as a structural regularizer within flexible machine-learning SDFs, anchoring deep representations to economically meaningful pricing spans while retaining the expressive power of nonlinear models. These potential extensions highlight that the geometric formulation is not tied to a particular estimator but can serve as a modular component of broader asset-pricing architectures.

Viewed in this light, NA-IPCA functions as a bridge across statistical, economically disciplined, and high-flexibility SDF approaches. It retains the data-driven adaptability of PCA, incorporates the pricing restrictions emphasized in modern factor-pricing work, and provides the full identification, tangent geometry, and rotation-invariant inference developed in Section 4. The resulting estimator is analytically tractable, economically interpretable, and structurally robust: it decomposes priced risk into characteristic-driven, observable, and asset-specific components measured in economically meaningful units. This nested and geometry-first construction distinguishes NA-IPCA from existing factor-model estimators and motivates the empirical evaluation that follows.

In summary, Section 5 positions NA-IPCA within the main families of factor-model estimators. It nests statistical IPCA as a limiting case, contrasts with PCA procedures that embed pricing restrictions but retain Euclidean geometry, and complements machine-learning SDF estimators by offering a fully identified, geometry-based decomposition with rotation-invariant inference. Across these comparisons, the distinguishing feature of NA-IPCA is that economic structure enters through the *metric* that organizes characteristic space. The resulting Q_Z -orthogonal decomposition of latent, observable, and asset-specific pricing directions sharpens the objectives of traditional IPCA and gives a tractable representation of the no-arbitrage curvature.

The next section documents the empirical geometry of characteristic space in a large equity panel and uses the resulting long-run metric Q_Z to calibrate all subsequent simulation designs. By anchoring the Monte Carlo environments in the empirical covariance and dependence structure of characteristics, Section 6 provides the geometric benchmark against which the finite-sample performance of NA-IPCA is evaluated. Section 7 then investigates how the Q_Z metric, the no-arbitrage curvature, and the blockwise identification translate into improved factor recovery, pricing accuracy, and stability relative to unconstrained IPCA and PCA-based alternatives.

6 Empirical Geometry of the Characteristic Space

This section characterizes the empirical geometry of the characteristic space in the equity panel underlying our analysis and uses it to calibrate the long-run metric Q_Z that structures the no-arbitrage IPCA framework. The objective is twofold. First, we document that the covariance structure of cross-sectionally rank-transformed characteristics is highly anisotropic: variation is concentrated along a small number of dominant directions, while many characteristics form tightly correlated clusters. Such structure is fundamentally inconsistent with the identity-metric view implicit in statistical IPCA. Second, we show how this geometry maps into economically meaningful dimensions of firm heterogeneity. These dimensions determine the appropriate notions of orthogonality, unit risk, and curvature that govern identification and inference in NA-IPCA.

Following the replication dataset of Kelly et al. (2019), we construct the long-run characteristic metric

$$Q_Z = \frac{1}{T} \sum_{t=1}^T \frac{1}{N_t} Z_t' Z_t,$$

where Z_t contains the $L = 37$ rank-transformed characteristics for N_t firms in month t . The aligned sample comprises $N = 12,813$ firms and $T = 599$ months (1964–2014). Tables 2–4 and Figures 1–3 summarize the resulting geometry.

Section 6.1 describes the data and the construction of Q_Z . Section 6.2 documents the global shape of the characteristic space through eigenvalues, correlations, and principal components. Section 6.3 examines the marginal distribution of the rank-transformed characteristics. Section 6.4 provides a two-dimensional illustration that visualizes how Euclidean and Q_Z metrics assign very different economic meaning to strong and weak characteristic directions. Section 6.5 concludes by extracting the implications for the Monte Carlo designs in Section 7.

6.1 Data and Construction of the Long-Run Metric Q_Z

We construct the empirical characteristic metric Q_Z that anchors the no-arbitrage geometry and serves as the calibration target for all subsequent simulation designs. The construction follows the replication dataset of Kelly et al. (2019) and adopts their timing and transformation conventions to maintain comparability with the statistical IPCA benchmark.

The analysis uses monthly CRSP/Compustat returns and firm characteristics. For each month t , we

observe an unbalanced cross section of N_t firms with an L -dimensional characteristic vector $Z_t \in \mathbb{R}^{N_t \times L}$. After aligning the data to the baseline IPCA timing, the final panel spans $T = 599$ months from July 1964 to May 2014, covers $N = 12,813$ distinct firms, and contains $L = 37$ characteristics. These dimensions are summarized in Table 2.

Characteristics are standardized using the cross-sectional rank transformation of Kelly et al. (2019). Within each month, raw values are converted into percentile ranks among non-missing firms, scaled by N_t^{-1} , and recentered to the interval $[-0.5, 0.5]$. This mapping removes scale differences and attenuates outliers, while preserving the cross-sectional ordering of firms that determines how characteristics contribute to the empirical metric Q_Z and, consequently, to the decomposition of systematic and idiosyncratic pricing directions in IPCA and NA-IPCA. The resulting rank-transformed panel $\{Z_t\}_{t=1}^T$ forms the basis of our geometric analysis.

The long-run characteristic metric is defined as the average cross-sectional Gram matrix,

$$Q_Z = \frac{1}{T} \sum_{t=1}^T \frac{1}{N_t} Z_t' Z_t,$$

which captures how rank-transformed characteristics co-move across firms and over time. By construction, Q_Z is symmetric and positive definite, and induces the inner product $\langle a, b \rangle_{Q_Z} = a' Q_Z b$ and associated norm $\|a\|_{Q_Z} = (a' Q_Z a)^{1/2}$ on characteristic directions.

As documented in Section 6.2, the eigenvalues and correlations implied by Q_Z differ sharply from those of an identity metric: variation is concentrated in a handful of dominant directions, and strong clustering is evident among size, value, profitability, investment, and momentum characteristics. These empirical patterns motivate using Q_Z , rather than I_L , to define orthogonality, scaling, and curvature in the no-arbitrage IPCA geometry.

Tables 2–4 summarize the sample dimensions and the distributional properties of the rank-transformed characteristics, while Figures 1 and 2 visualize the resulting dependence structure and eigenvalue spectrum. For completeness, Table 6 maps each characteristic index to its underlying WRDS/Neuhierl variable code, ensuring transparency in how the empirical metric Q_Z is constructed from the underlying predictors. These objects serve as calibration targets in Section 7: simulated characteristic panels are drawn from distributions with covariance proportional to Q_Z , and the no-arbitrage constraints in NA-IPCA are imposed in the Q_Z -norm rather than the Euclidean norm.

6.2 Global Geometry: Eigenvalues, Correlations, and Principal Components

The long-run characteristic metric Q_Z exhibits a geometry that differs decisively from any identity-metric representation. Table 2 shows that its eigenvalues range from $\lambda_{\min} \approx 0.002$ to $\lambda_{\max} \approx 1$, a factor of roughly 500. Such an uneven spectrum implies that certain characteristic portfolios load heavily on cross-sectional dispersion while others are nearly flat. Under an identity metric I_L , these directions would all appear equally scaled; the empirical Q_Z instead allocates curvature selectively to a small set of dominant directions.

The dependence structure reinforces this conclusion. Table 5 reports moderate average correlations across characteristics (mean absolute correlation of 0.16), but extremely high pairwise correlations (up to 0.94).

Table 2: Global geometry of the empirical characteristic metric Q_Z

Statistic	Value
Number of assets (N)	12,813
Number of characteristics (L)	37
Number of months (T)	599
Trace of Q_Z	4.0071
Smallest eigenvalue $\lambda_{\min}$	0.00199
Largest eigenvalue $\lambda_{\max}$	1.00010
Condition number $\lambda_{\max}/\lambda_{\min}$	502.28

Notes: The table reports basic sample dimensions and global properties of the long-run characteristic metric $Q_Z = T^{-1} \sum_{t=1}^T Z_t' Z_t / N_t$, constructed from the rank-transformed characteristics in the Kelly et al. (2019) replication sample (July 1964–May 2014). The eigenvalue range and condition number summarize how unevenly characteristic variation is distributed across directions in Q_Z .

Table 3: Moments of rank-transformed characteristics

Statistic	Value
Minimum characteristic mean	0.0583
Maximum characteristic mean	0.3210
Minimum characteristic std. deviation	0.3431
Maximum characteristic std. deviation	0.5217

Notes: Means and cross-sectional standard deviations are computed from the rank-transformed characteristics that enter Q_Z . The rank transformation follows Kelly et al. (2019): within each month, raw characteristics are mapped to cross-sectional ranks in $[-0.5, 0.5]$. The reported ranges document heterogeneity in the location and dispersion of characteristics after this transformation.

Figure 1 makes the clustering transparent: size, value, profitability, investment, and momentum characteristics form tight empirical blocks, while many other pairs exhibit near-zero dependence. This structure echoes the well-documented organization of the equity cross section (e.g., Fama and French, 1993; Novy-Marx, 2013; Gu et al., 2020) and implies substantial redundancy among raw characteristics.

The principal-component decomposition in Figure 2 summarizes how this structure concentrates variation. The first principal component explains about 25% of total characteristic dispersion, the first two nearly 40%, and the first five just above 60%. If Q_Z were proportional to I_L , the spectrum would be essentially flat, with each principal component explaining about $1/L \approx 3\%$ of total variance. The steep empirical decay therefore rejects any Euclidean approximation and confirms that characteristic space is governed by a handful of high-curvature directions.

Economically, these patterns identify the characteristic portfolios most likely to span priced sources of systematic variation. Directions aligned with the leading eigenvectors of Q_Z coincide with the dominant empirical characteristic families and therefore represent the high-resolution axes along which firms differ most. Conversely, directions associated with small eigenvalues are nearly redundant and carry little effective information about risk premia. Treating such directions as orthogonal under I_L would split characteristic clusters, inflate weak combinations, and risk double-counting priced exposures—precisely the pathologies documented in Bryzgalova (2015). The geometry recovered from Q_Z therefore provides the appropriate scaling, orthogonality, and curvature for the NA-IPCA framework and serves as the calibration benchmark

Table 4: Means and standard deviations of rank-transformed characteristics

Characteristics 1–13			Characteristics 14–25			Characteristics 26–37		
j	μ_Z	σ_Z	j	μ_Z	σ_Z	j	μ_Z	σ_Z
1	0.3210	0.5217	14	0.0803	0.3790	26	0.0583	0.3436
2	0.2466	0.4944	15	0.0715	0.3586	27	0.0612	0.3483
3	0.1560	0.4461	16	0.0764	0.3587	28	0.0631	0.3484
4	0.1195	0.4175	17	0.0734	0.3553	29	0.0633	0.3490
5	0.1115	0.4084	18	0.0734	0.3561	30	0.0614	0.3516
6	0.1087	0.4047	19	0.0716	0.3533	31	0.0644	0.3520
7	0.0970	0.3949	20	0.0681	0.3516	32	0.0710	0.3585
8	0.0918	0.3913	21	0.0626	0.3514	33	0.0743	0.3593
9	0.0866	0.3872	22	0.0630	0.3560	34	0.0750	0.3641
10	0.0857	0.3910	23	0.0609	0.3507	35	0.0761	0.3669
11	0.0890	0.3949	24	0.0587	0.3463	36	0.0806	0.3690
12	0.0832	0.3856	25	0.0599	0.3431	37	0.0892	0.3770
13	0.0822	0.3836						

Notes: The table reports time-series means μ_Z and cross-sectional standard deviations σ_Z of the $L = 37$ rank-transformed characteristics used to construct Q_Z . Characteristics are transformed period-by-period into cross-sectional ranks on $[-0.5, 0.5]$ following [Kelly et al. \(2019\)](#). The range of μ_Z (0.06–0.32) and σ_Z (0.34–0.52) documents that, even after rank transformation, characteristics differ in location and dispersion, so some dimensions of the characteristic space carry much more cross-sectional variation than others.

for the simulation environments in Section 7.

6.3 Distribution of Rank-Transformed Characteristics

The heterogeneity of the empirical metric Q_Z arises not only from the dependence structure documented in Section 6.2, but also from the marginal distribution of the underlying rank-transformed characteristics. Table 3 shows that, even after applying the cross-sectional transformation of [Kelly et al. \(2019\)](#), characteristic means span a wide range (0.06–0.32). This asymmetry indicates that firms do not populate the characteristic space uniformly: some dimensions place most firms persistently above the median, while others concentrate mass near the center. Such location imbalances are consistent with well-documented empirical regularities—e.g., the persistence of value, profitability, and investment attributes highlighted by [Daniel and Titman \(1997\)](#) and the structural interpretations in [Kozak et al. \(2018\)](#)—and determine where firm density accumulates on the characteristic manifold.

Table 4 further reports $\{\mu_Z(j), \sigma_Z(j)\}_{j=1}^{37}$ and reveals economically meaningful variation in cross-sectional dispersion: σ_Z ranges from roughly 0.34 to 0.52. Mechanically, because

$$Q_Z = \frac{1}{T} \sum_{t=1}^T Z_t' Z_t / N_t,$$

higher cross-sectional dispersion increases the diagonal elements of $Z_t' Z_t$ and therefore contributes more weight to the corresponding directions in Q_Z . Geometrically, characteristics with low σ_Z behave as

Table 5: Correlation structure and principal–component decomposition of Q_Z

Statistic	Value
Mean absolute correlation among characteristics	0.1596
Median absolute correlation	0.0970
Maximum absolute correlation	0.9400
Variance share explained by first PC	0.2496
Cumulative variance share (PC1–PC2)	0.3988
Cumulative variance share (PC1–PC3)	0.4877
Cumulative variance share (PC1–PC4)	0.5634
Cumulative variance share (PC1–PC5)	0.6179

Notes: Correlations are computed from $R_Z = D^{-1}Q_Z D^{-1}$, where D collects the square roots of the diagonal of Q_Z . Variance shares are based on the eigenvalues of Q_Z and measure how much characteristic variation is captured by the leading principal components.

low–resolution coordinates: many firms share similar ranks, so local perturbations produce minimal displacement. High–dispersion characteristics, by contrast, provide much finer separation across firms and steepen the Q_Z –risk contours along those directions.

Economically, this heterogeneity helps explain the anisotropic curvature observed in Section 6.2. Low–dispersion directions correspond to nearly redundant exposures, which Bryzgalova (2015) shows can generate spurious latent factors under Euclidean normalizations. High–dispersion directions, by contrast, form the backbone of systematic variation and align with the characteristic families emphasized in Gu et al. (2020). By defining orthogonality and curvature in the empirical Q_Z –metric rather than in I_L , NA–IPCA assigns weight proportional to the true economic resolution of each characteristic. As a result, the simulated environments in Section 7 inherit the empirically relevant strong and weak directions of the data, ensuring that the evaluation of NA–IPCA reflects the geometry governing characteristic–based risk in practice.

6.4 Two–Dimensional Geometry of Strong and Weak Directions

The two–dimensional illustration in Figure 3 provides a compact visualization of the empirical geometry documented in Sections 6.1–6.3. Whereas the previous subsections establish that Q_Z is highly anisotropic and exhibits strong clustering across characteristic families, this figure shows how the metric reshapes the relative importance of characteristic combinations in a way that is economically meaningful.

In the left panel, distances are measured under the Euclidean metric I_2 . All directions have identical unit length, and the coordinate axes implicitly serve as the reference system used in statistical IPCA. Because the Euclidean norm assigns the same scale to all characteristic combinations, the strong and weak eigenvectors of Q_Z (orange and blue) intersect the unit–risk circles at similar radii. Economically, this means that statistical IPCA cannot distinguish directions that carry substantial cross–sectional variation from those that are nearly flat, increasing the likelihood that weak or highly collinear characteristic combinations will be misinterpreted as distinct priced factors.

The right panel replaces I_2 with the empirical Q_Z metric. The unit–length locus becomes an ellipse: small movements along the leading eigenvector produce a steep increase in $v'Q_Z v$, while movements along the weak direction generate only minimal changes. Geometrically, this difference in curvature captures the empirical

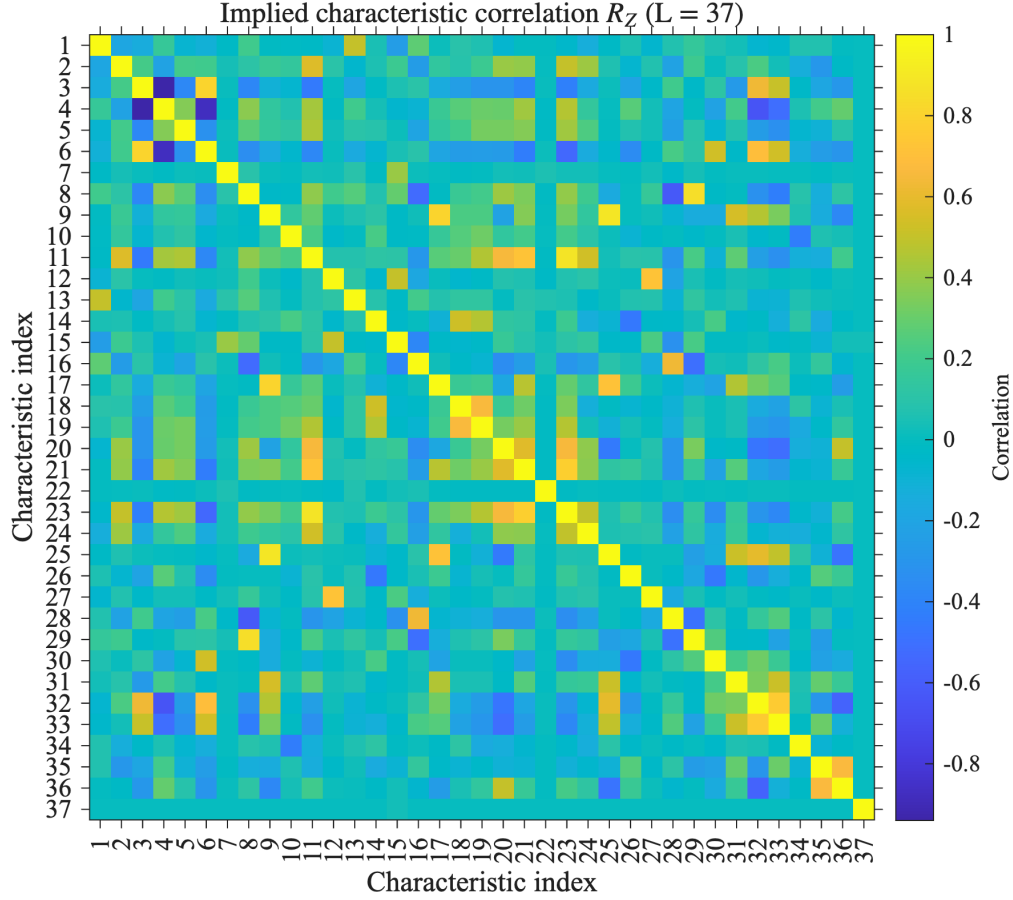


Figure 1: Implied characteristic correlation matrix R_Z .

Notes: The heatmap displays the implied characteristic correlation matrix $R_Z = D^{-1}Q_ZD^{-1}$, where Q_Z is the long-run covariance matrix of the $L = 37$ rank-transformed characteristics and D contains the square roots of its diagonal elements. Average absolute correlations are modest (mean ≈ 0.16 , median ≈ 0.10), but several pairs reach values above 0.90, revealing tight clusters among size, value, profitability, investment, and momentum characteristics. This combination of weak links and strong clusters shows that the characteristic space is far from spherical and motivates using the empirical metric Q_Z —rather than an identity metric I_L —as the basis for the no-arbitrage geometry in the NA-IPCA simulations.

resolution of characteristic space—dominant directions separate firms sharply, whereas weak directions lie on nearly redundant subspaces.

This geometric contrast underlies the normalization $\Gamma_\beta^\top Q_Z \Gamma_\beta = I_K$ used in NA-IPCA. By measuring loadings in *units of characteristic risk* rather than raw magnitudes, the estimator aligns its tangent space with the dominant Q_Z eigenvectors identified in Section 6.2 and automatically downweights directions that contribute little systematic variation. The simulation environments constructed in Section 7 build directly on this geometry: by varying the metric (identity versus Q_Z), the curvature bound, and the degree of factor misspecification, they isolate how the empirical scaling of characteristic space governs the stability and accuracy of factor recovery.

Table 6: Indexing of rank-transformed characteristics

Panel A: Market-based, trading, and risk characteristics			Panel B: Accounting, profitability, and investment characteristics		
j	Code	Interpretation	j	Code	Interpretation
1	lme	Log market equity (size)	14	cto	Cash-to-assets
2	lturnover	Log share turnover	15	d2a	Debt-to-assets
3	spread_mean	Bid-ask spread (illiquidity)	16	dpi2a	Δ income / assets
4	cum_ret_1_0	1–0m return (reversal/momentum)	17	e2p	Earnings-to-price
5	cum_ret_12_2	12–2m momentum	18	fc2y	Free cash flow / income
6	cum_ret_12_7	12–7m momentum	19	free_cf	Free cash flow
7	cum_ret_36_13	36–13m reversal	20	investment	Investment intensity
8	beta	Market beta	21	lev	Leverage
9	idio_vol	Idiosyncratic volatility	22	noa	Net operating assets
10	beme	Book-to-market (value)	23	oa	Operating assets
11	at	Total assets (scale)	24	ol	Operating liabilities
12	ato	Asset turnover	25	pcm	Profit cost margin
13	c	Cash holdings	26	pm	Profit margin
			27	prof	Profitability composite
			28	q	Tobin's q
			29	rel_high	Price relative to high
			30	rna	Return on net assets
			31	roa	Return on assets
			32	roe	Return on equity
			33	s2p	Sales-to-price
			34	sga2m	SG&A / margin or sales
			35	suv	Share issuance / volume
			36	a2me	Assets-to-market equity
			37	const	Intercept (ones vector)

Notes: The table maps each characteristic index j to its corresponding WRDS/Neuhierl variable code used in Kelly et al. (2019) and Gu et al. (2020). All characteristics are transformed each month into cross-sectional ranks on $[-0.5, 0.5]$ before entering the empirical metric Q_Z . Row 37 (*const*) denotes the intercept column (a vector of ones). Including an explicit constant follows standard practice in characteristic-based asset pricing: it separates unconditional level effects from characteristic-driven variation, prevents the intercept from being absorbed into correlated characteristics, and ensures that Q_Z reflects the true relative scaling and dependence among the non-constant characteristics. Omitting this column would implicitly embed the intercept in the characteristic set, biasing the metric toward characteristics correlated with average returns and altering the geometry used for orthogonality and identification in NA-IPCA.

6.5 Summary and Implications for Simulation Design

The empirical analysis in this section shows that characteristic space is highly uneven: Q_Z exhibits a steep eigenvalue profile, strong clustering across a handful of economic families, and meaningful heterogeneity in the location and dispersion of rank-transformed characteristics. Only a subset of directions carries substantial cross-sectional variation, while many raw characteristics represent tightly correlated manifestations of the same underlying economic traits. These features depart sharply from the uniform geometry implicit in an identity metric.

This empirical structure guides the calibration of the Monte Carlo designs in Section 7. First, simulated characteristic panels are generated with covariance proportional to Q_Z , ensuring that the synthetic Z_t inherit the same mix of dominant and weak directions, as well as the empirical clustering visible in Figures 1–2. Second, the latent and observable loading matrices in the baseline DGP are constructed to either align with leading eigenvectors of Q_Z —to study identification when the model is correctly structured—or to be deliberately misaligned, allowing the simulations to explore weak or spurious factor configurations in the spirit of Bryzgalova (2015) and Kozak et al. (2018). Third, the curvature bound on the asset-specific component is calibrated in the Q_Z -norm using the empirical scale of $\Gamma'_\alpha Q_Z \Gamma_\alpha$, so that the feasible non-systematic region matches the magnitudes observed in practice. Finally, the geometry illustrated in Figure 3 clarifies why these

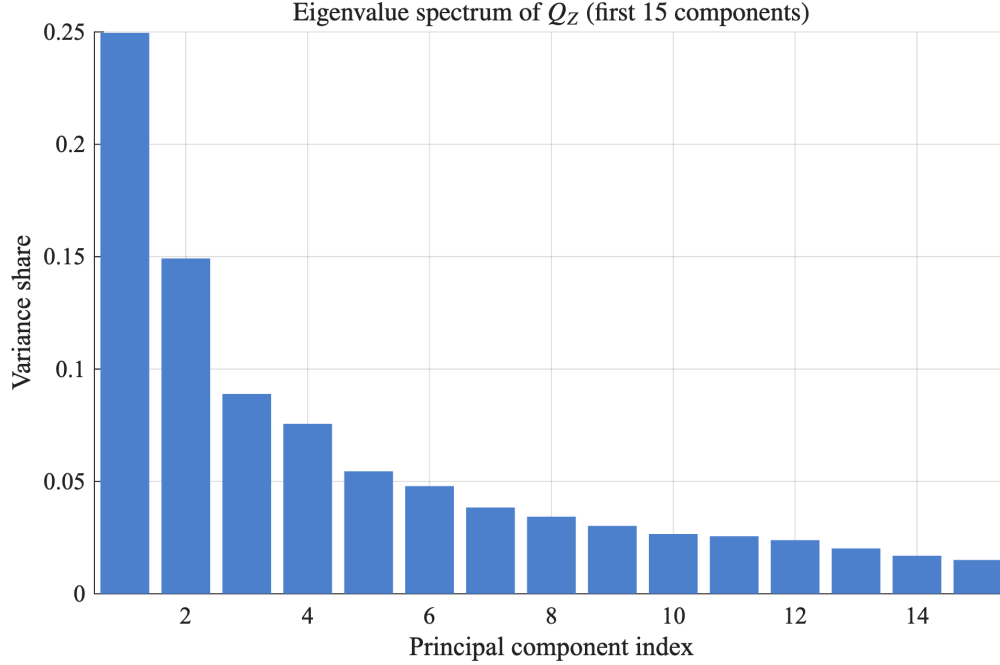


Figure 2: Eigenvalue spectrum of Q_Z (first 15 components)

Notes: Bars report the variance share of the first 15 principal components of the empirical characteristic metric Q_Z . The first component explains about 25% of total variation, the top two about 40%, and the top five just above 60%. Beyond the first few components, marginal contributions flatten out to roughly 2–3% per component, indicating that a handful of dominant directions account for most of the dispersion in characteristic space.

calibration choices matter. Under the Euclidean metric I_L , strong and weak characteristic combinations appear similarly informative, whereas under Q_Z movements along dominant directions generate disproportionately large changes in characteristic risk. By designing simulation environments that inherit this empirical geometry—and then varying the metric (identity versus Q_Z), the curvature bound, and the degree of factor misspecification—Section 7 is able to isolate how the no–arbitrage metric affects factor recovery, stability, and cross-sectional pricing performance. The simulations therefore provide a direct quantitative assessment of the role played by the empirical Q_Z geometry in distinguishing NA–IPCA from statistical IPCA and other Euclidean–metric PCA estimators.

7 Simulation Evidence for the No-Arbitrage Geometry

7.1 Geometry–Calibrated Data–Generating Process

The Monte Carlo analysis in Section 7 is built around a single baseline data–generating process designed to replicate, in population form, the characteristic geometry documented empirically in Section 6. The purpose is not to construct an arbitrary numerical environment, but to translate the long–run characteristic metric Q_Z and its anisotropic curvature into the population manifold $\Theta(\delta)$ developed in Sections 3–5. All subsequent simulation exercises perturb exactly one dimension of this baseline design—metric alignment, curvature, signal strength, or the noise distribution—so that identification outcomes can be attributed cleanly.

2D Geometry of NA-IPCA versus Statistical IPCA
Strong and weak directions under Euclidean vs. Q_Z metrics

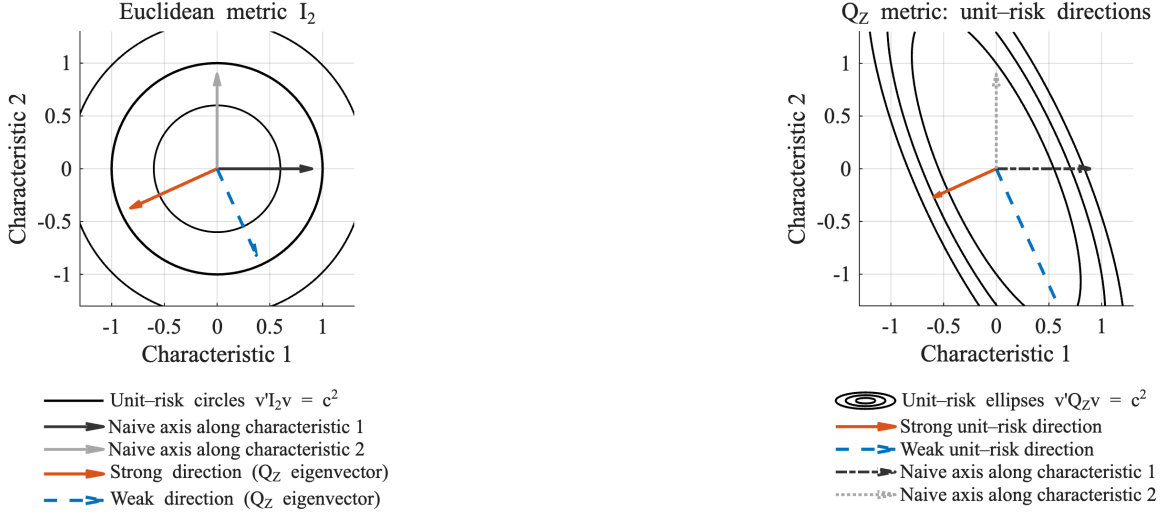


Figure 3: Strong and weak directions under Euclidean vs. Q_Z metrics

Notes: The left panel shows contours of $v'I_2v = c^2$ (circles) together with the naïve characteristic axes and the strong/weak eigenvectors of Q_Z . Under the Euclidean metric all directions have the same unit length, so strong and weak combinations of characteristics appear equally important. The right panel plots contours of $v'Q_Zv = c^2$ (ellipses). Here the strong eigenvector intersects risk contours much more quickly than the weak one, illustrating that firms differ sharply along some characteristic directions and very little along others. NA-IPCA exploits this Q_Z geometry, whereas statistical IPCA implicitly relies on the Euclidean metric.

The empirical characteristic metric

$$Q_Z = \frac{1}{T} \sum_{t=1}^T \frac{1}{N_t} Z_t' Z_t$$

exhibits strong anisotropy: its eigenvalues range from roughly 0.002 to 1, with a condition number near 500 and more than sixty percent of characteristic dispersion concentrated in the first five principal components. As shown in Section 6, firms differ most sharply along these high-curvature directions, while many orthogonal combinations carry only limited variation. Because the NA-IPCA manifold relies on the Q_Z geometry to define curvature, orthogonality, and normal-space separation, the simulation design preserves this empirical structure exactly and treats it as the structural anchor of the population experiment.

The characteristic space in the DGP therefore has dimension $L = 37$, matching the 37 rank-transformed characteristics in the Kelly et al. (2019) panel. The simulated panel consists of $N_{\text{sim}} = 1,000$ assets observed for $T_{\text{sim}} = 240$ months. The latent loading block $\Gamma_\beta^* \in \mathbb{R}^{L \times K_\beta}$, with $K_\beta = 3$, is aligned with the leading eigenvectors of Q_Z , placing latent factors along the steepest characteristic directions. The observable loading block $\Gamma_\delta^* \in \mathbb{R}^{L \times K_\delta}$, with $K_\delta = 2$, is constructed from the next two eigenvectors, projected onto the Q_Z -orthogonal complement of $\text{col}(\Gamma_\beta^*)$ and renormalized to unit Q_Z length, enforcing the block orthogonality $\Gamma_\beta^{*\top} Q_Z \Gamma_\delta^* = 0$ that underpins the population identification results of Section 5. This construction mirrors the empirical finding that observable predictors—such as PSF characteristics or macroeconomic proxies—rarely span the dominant characteristic directions, and instead operate along much weaker curvature dimensions.

The observable block is intentionally weak in this DGP. The characteristic directions orthogonal to the

latent eigenvectors exhibit low dispersion and curvature in Section 6, implying limited systematic variation. The DGP therefore assigns only modest pricing content to Γ_δ^* , ensuring that the observable span plays the same role it plays empirically.⁶ This asymmetry is essential for the identification analysis in Section 7.2: a well-behaved estimator should recover the latent span sharply, preserve blockwise orthogonality, and avoid allocating priced variation to the weak observable block.⁷ A strong-observable benchmark is reported separately in Section 7.5 to verify that the conclusions do not depend on a single design.

The non-systematic loading $\Gamma_\alpha^* \in \mathbb{R}^L$ is generated by drawing a random direction and scaling it to satisfy

$$\psi(\Gamma_\alpha^*) = \Gamma_\alpha^{*\top} Q_Z \Gamma_\alpha^* = 0.50,$$

the baseline curvature level for admissible mispricing under the no-arbitrage bound $\psi(\Gamma_\alpha) \leq \delta$. The curvature grid

$$\delta \in \{0.25, 0.50, 1.00, \infty\}$$

spans increasingly weak economic discipline, with $\delta = \infty$ reproducing the unrestricted statistical IPCA benchmark. This configuration matches the manifold decomposition in Appendix A, where the normal space of Γ_α is determined by the curvature of the Q_Z metric.

Factor processes are calibrated to realistic signal strength. Latent factors follow an AR(1) process with persistence $\rho_f = 0.6$ and unit variance. Observable factors have unit variance and contemporaneous correlation $\text{Corr}(f_t, F_t^{\text{obs}}) \approx 0.4$, reflecting that they proxy the latent pricing kernel but do not span the same directions. Idiosyncratic returns include homoskedastic Gaussian noise with standard deviation 0.05, along with heavy-tailed $t(5)$ innovations and a two-cluster heteroskedastic variant (0.04, 0.08) recorded for robustness experiments in Section 7.6. These perturbations isolate the effects of tail thickness and cross-sectional heterogeneity without altering the underlying characteristic geometry.

All simulations reported in Sections 7.2–7.6 inherit this geometry-calibrated structure. Table 7 summarizes the main calibration inputs, including dimension parameters, eigenvalue diagnostics of Q_Z , factor

⁶In the baseline DGP, the observable block is made deliberately weak in two complementary ways. First, Γ_δ^* is constructed from the fourth and fifth eigenvectors of Q_Z , which correspond to relatively low-curvature characteristic directions (see Section 6), and is then projected onto the Q_Z -orthogonal complement of $\text{col}(\Gamma_\beta^*)$. This places observable loadings in characteristic directions that carry little additional dispersion beyond the latent span. Second, the observable factor process F_t^{obs} is generated as a noisy proxy for the latent factors, with moderate correlation $\text{Corr}(f_t, F_t^{\text{obs}}) \approx 0.4$ and no additional independent pricing shock. In the return equation $r_{t+1} = Z_t(\Gamma_\beta^* f_t + \Gamma_\delta^* F_t^{\text{obs}} + \Gamma_\alpha^*) + u_{t+1}$ this implies that $\Gamma_\delta^* F_t^{\text{obs}}$ contributes only a small amount of incremental priced variation beyond $\Gamma_\beta^* f_t$, so that, in population, the observable span is almost redundant for pricing. This design captures the notion of “spurious” or weak observable factors: they are mildly correlated with the true pricing kernel but do not span any strong Q_Z -curvature directions. Section 7.5 reports a complementary design in which Γ_δ^* is realigned with leading eigenvectors of Q_Z and the observable factors become genuinely strong, providing a benchmark against which the weak-observable results in Section 7.2 can be interpreted.

⁷The latent block Γ_β^* is made deliberately strong in the geometry-calibrated DGP by aligning its columns with the leading eigenvectors of Q_Z , which correspond to the highest-curvature directions of the empirical characteristic space (Section 6). Because Q_Z embeds the long-run cross-sectional dispersion of firm characteristics, its dominant eigenvectors span the directions along which firms differ most sharply and along which systematic compensation for risk is most likely to arise. By construction, the latent span $\text{col}(\Gamma_\beta^*)$ inherits this steep curvature: for any latent loading vector g , $g^\top Q_Z g$ is large, meaning small perturbations generate large variation in characteristic portfolios. This makes latent factors the principal carriers of priced variation in the return equation $r_{t+1} = Z_t(\Gamma_\beta^* f_t + \Gamma_\delta^* F_t^{\text{obs}} + \Gamma_\alpha^*) + u_{t+1}$. In contrast, observable loadings Γ_δ^* are drawn from subsequent, low-curvature eigenvectors and then projected onto the Q_Z -orthogonal complement of the latent span, ensuring they operate along empirically weak directions. This eigenstructure-based construction mirrors the theoretical predictions in Sections 3–5: latent factors should lie in the steepest Q_Z directions because these dominate both curvature and identification on the no-arbitrage manifold $\Theta(\delta)$.

Table 7: Baseline geometry–calibrated data–generating process

Quantity	Value
Number of simulated assets (N_{sim})	1,000
Number of months (T_{sim})	240
Number of characteristics (L)	37
Latent factor dimension (K_β)	3
Observable factor dimension (K_δ)	2
Trace of Q_Z	4.0071
Smallest eigenvalue $\lambda_{\min}(Q_Z)$	0.00199
Largest eigenvalue $\lambda_{\max}(Q_Z)$	1.00010
Condition number $\lambda_{\max}/\lambda_{\min}$	502.28
Target curvature of non–systematic span $\psi(\Gamma_\alpha^*)$	0.50
Curvature grid for δ	$\{0.25, 0.50, 1.00, \infty\}$
Latent factor process	AR(1): $\rho_f = 0.6$, $\text{Var} = 1$
Observable factor correlation	$\text{Corr}(f_t, F_t^{\text{obs}}) \approx 0.40$
Baseline idiosyncratic noise	Gaussian $\mathcal{N}(0, 0.05^2)$
Recorded alternatives	$t(5)$; two–cluster $\{0.04, 0.08\}$

Notes: This table reports the baseline DGP used throughout Section 7. The design is calibrated to the empirical characteristic metric Q_Z documented in Section 6, which displays strong anisotropy with eigenvalues spanning $[0.002, 1]$ and a condition number above 500. Latent loadings Γ_β^* are aligned with the leading eigenvectors of Q_Z , while observable loadings Γ_δ^* are drawn from the next eigenvectors and projected onto the Q_Z –orthogonal complement of the latent span. The non–systematic component Γ_α^* is scaled to satisfy $\psi(\Gamma_\alpha^*) = 0.50$ and is governed by the curvature grid $\delta \in \{0.25, 0.50, 1.00, \infty\}$. Factor processes and noise specifications are chosen to deliver realistic signal-to-noise ratios and to support heavy–tail and heteroskedastic stress–test scenarios in Section 7.6. All subsequent simulation designs perturb only a single dimension of this baseline specification, ensuring that identification differences across estimators arise from geometry rather than arbitrary numerical choices.

processes, the curvature grid, and noise specifications. By anchoring the DGP in the empirical Q_Z geometry, the simulation study ensures that every identification and efficiency result reflects the same curvature and orthogonality principles that govern NA–IPCA in real data.

7.2 Identification on the No-Arbitrage Manifold

This section examines whether the no–arbitrage geometry developed in Sections 3–5 delivers the blockwise identification that the manifold $\Theta(\delta)$ is designed to impose.⁸ Using the geometry–calibrated DGP introduced in Section 7.1, we evaluate how three estimators—restricted statistical IPCA (without an explicit Γ_α block), NA–IPCA (with a Q_Z –orthogonal geometry and curvature constraint), and PCA—recover the latent, observable, and non–systematic components encoded in the population parameter configuration. Identification is assessed through the canonical–angle distances, the Q_Z –based overlap measures, and the

⁸The identification experiment in Section 7.2 fixes δ at its baseline value $\delta = \psi(\Gamma_\alpha^*) = 0.50$, rather than sweeping across the full curvature grid $\{0.25, 0.50, 1.00, \infty\}$. This is by design. The purpose of Section 7.2 is to evaluate the blockwise geometry of the estimators—latent–observable separation, Q_Z –orthogonality, and normal–space alignment—under a single empirically calibrated non–systematic magnitude. Varying δ at this stage would confound geometric identification with the separate question of how tightly the no–arbitrage bound constrains Γ_α , a topic addressed explicitly in Section 7.4. Using the baseline value $\delta = 0.50$ isolates the estimator’s intrinsic ability to respect the manifold structure of $\Theta(\delta)$, independent of curvature tightness, ensuring internal validity of the identification results.

2×2 misallocation matrix introduced in Appendix F, which together characterize whether an estimator respects the manifold’s tangent–normal decomposition.

The baseline environment inherits the empirical curvature documented in Section 6: a handful of characteristic directions carry steep Q_Z curvature, while most orthogonal directions are nearly flat. The latent span in the DGP is aligned with these high–curvature directions, whereas the observable span is intentionally constructed to be weak, reflecting that empirical PSF or macro predictors rarely span the dominant characteristic directions. In such a setting, a correctly specified estimator should recover the latent span sharply, assign little priced variation to the observable block, and preserve the Q_Z –orthogonal block structure. Estimators whose geometry is defined under the Euclidean metric—such as PCA or statistical IPCA—lack this curvature discipline and are therefore expected to blur the separation between strong and weak directions, generating double counting or collapsing distinct pricing sources.

The subspace distances reported in Figure 4 confirm these predictions. NA–IPCA nearly replicates the true latent span, with $\|\sin \Theta(\hat{\Gamma}_\beta, \Gamma_\beta)\|_F \approx 0.01$ and a negligible standard error, consistent with the theory in Section 5 that the estimator aligns with the steep curvature directions of Q_Z . Restricted IPCA, by contrast, exhibits a large latent distance (≈ 0.73), indicating that several irrelevant directions receive excessive weight under Euclidean orthogonality. PCA performs worst (≈ 0.99), confirming that variance maximization is not aligned with priced directions. The observable span behaves differently: because the observable block in the DGP carries little systematic variation, NA–IPCA assigns essentially no priced information to it, producing a large $\|\sin \Theta_\delta\|_F$ that reflects the weakness of observable factors rather than a failure of identification. This behavior is precisely what the no–arbitrage geometry prescribes: when observable predictors are weak, the manifold collapses priced variation into the latent span rather than manufacturing spurious observable structure.

Blockwise orthogonality is evaluated through the normalized Q_Z –overlap $\|\hat{\Gamma}_\beta^\top Q_Z \hat{\Gamma}_\delta\|_F$ in Table 8. Restricted IPCA produces a sizable overlap (≈ 0.07), indicating that the same priced characteristic directions are absorbed simultaneously into both latent and observable blocks, the structural double–counting predicted in Sections 3–5. NA–IPCA, by contrast, yields an overlap essentially equal to zero ($\approx 5 \times 10^{-4}$), consistent with the blockwise Q_Z –orthogonality imposed by the manifold. PCA does not estimate an observable block so the overlap statistic is not defined.

The non–systematic component plays an important role under the curvature constraint $\psi(\Gamma_\alpha) \leq \delta$. Although restricted IPCA does not estimate an explicit Γ_α , NA–IPCA regulates this block as a bounded residual direction in characteristic space. Table 8 shows that the latent– α and observable– α overlaps produced by NA–IPCA are both near zero (0.004 and 0.002), indicating that in this simulation design the non–systematic component is empirically close to orthogonal to the priced directions in the Q_Z metric. This property is consistent with the decomposition established in Appendix A and with the curvature discipline imposed on the residual block.

The geometric misallocation pattern in Figure 5 provides a complete variance–decomposition interpretation. Restricted IPCA exhibits substantial off–diagonal mass, indicating that true latent variation leaks into the observable block and conversely that observable variation contaminates the estimated latent block. NA–IPCA, however, produces an almost perfectly diagonal 2×2 matrix: virtually all priced variation is

Table 8: Identification on the no–arbitrage manifold: subspace distances and overlap ratios

Model	Latent span		Observable span		Latent–observable overlap		Latent– α overlap		Observable– α overlap	
	$\ \sin \Theta_\beta\ _F$	s.e.	$\ \sin \Theta_\delta\ _F$	s.e.	$\text{Ov}(\beta, \delta)$	s.e.	$\text{Ov}(\beta, \alpha)$	s.e.	$\text{Ov}(\delta, \alpha)$	s.e.
IPCA	0.7324	0.0029	0.2320	0.0019	0.0689	0.0010	—	—	—	—
NA–IPCA	0.0108	0.0002	1.3981	0.0001	0.0005	0.0000	0.0043	0.0001	0.0017	0.0001
PCA	0.9897	0.0058	—	—	—	—	—	—	—	—

Notes: This table reports subspace distances and Q_Z -based overlap ratios from the Monte Carlo experiment in Section 7.2. Diagnostics follow the geometric definitions in Appendix F. The latent span distance $\|\sin \Theta_\beta\|_F$ measures recovery of the true priced directions; the observable span distance $\|\sin \Theta_\delta\|_F$ evaluates recovery of the weaker observable block. Overlap ratios quantify the extent to which estimated latent, observable, and non-systematic components violate the Q_Z -orthogonality conditions implied by the no–arbitrage manifold $\Theta(\delta)$. NA–IPCA delivers near-zero latent error and negligible overlap, consistent with blockwise orthogonality. Restricted statistical IPCA exhibits substantial latent error and nontrivial latent–observable overlap, while PCA collapses systematic variation into a single factor space.

correctly assigned to the latent span, and the weak observable block receives almost none. PCA collapses both blocks into a single direction, assigning roughly 82% of all priced variation to its first principal component and leaving no room for meaningful block separation.

Taken together, these results demonstrate that NA–IPCA is the only estimator that respects the blockwise identification and normal–space separation encoded in the no–arbitrage geometry of $\Theta(\delta)$. Restricted IPCA suffers from double counting due to incompatible Euclidean orthogonality, and PCA collapses systematic variation entirely. The simulation evidence confirms the central prediction of the theory: identification of economically meaningful pricing directions requires curvature, orthogonality, and manifold projection in the empirically relevant Q_Z metric.

7.3 Geometry Matters: Metric Comparisons

The role of characteristic geometry in factor identification becomes most transparent when we perturb the covariance structure of Z_t while holding the economic pricing structure fixed. As documented in Sections 2.1–2.3, equity and bond characteristics exhibit pronounced cross–sectional collinearity, heterogeneous scaling, and sharp curvature contrasts in the long–run metric Q_Z . Any estimator that implicitly treats the characteristic space as Euclidean and spherical—effectively assuming that all characteristic portfolios are equally important and mutually orthogonal—risks rotating the factor span toward high–variance but economically irrelevant directions, thereby double–counting priced risks and creating variance–driven pseudo–factors. To isolate this mechanism from all other sources of misspecification, we keep the true factor loadings $(\Gamma_\beta^{\text{true}}, \Gamma_\delta^{\text{true}}, \Gamma_\alpha^{\text{true}})$ and factor innovations (f_t, F_t^{obs}) fixed across all designs and vary only the cross–sectional geometry of Z_t .

Sampling panels are generated under four characteristic metrics: the empirical Q_Z calibrated in Section 7.1, the identity metric I_L , an eigenvector–permuted version of Q_Z that preserves its spectrum while destroying directional alignment, and a spectrum–flattened version that eliminates curvature dispersion.⁹

⁹The geometric perturbations alter the cross–sectional covariance of characteristics while leaving the underlying pricing relation unchanged. In every design we draw latent and observable factors (f_t, F_t^{obs}) and idiosyncratic shocks u_{t+1} from the same law and combine them with the fixed loadings $(\Gamma_\beta^{\text{true}}, \Gamma_\delta^{\text{true}}, \Gamma_\alpha^{\text{true}})$ through the pricing relation $r_{t+1} = Z_t(\Gamma_\beta^{\text{true}} f_t + \Gamma_\delta^{\text{true}} F_t^{\text{obs}} + \Gamma_\alpha^{\text{true}}) + u_{t+1}$ in Section 2.1. What changes across scenarios is only the law of Z_t : under the identity metric I_L , the cross–section of characteristics is made locally spherical, so that all characteristic portfolios have the same long–run variance and no persistent correlation; under the permuted metric we preserve the eigenvalues of Q_Z but randomly re-assign their eigenvectors, breaking the connection between

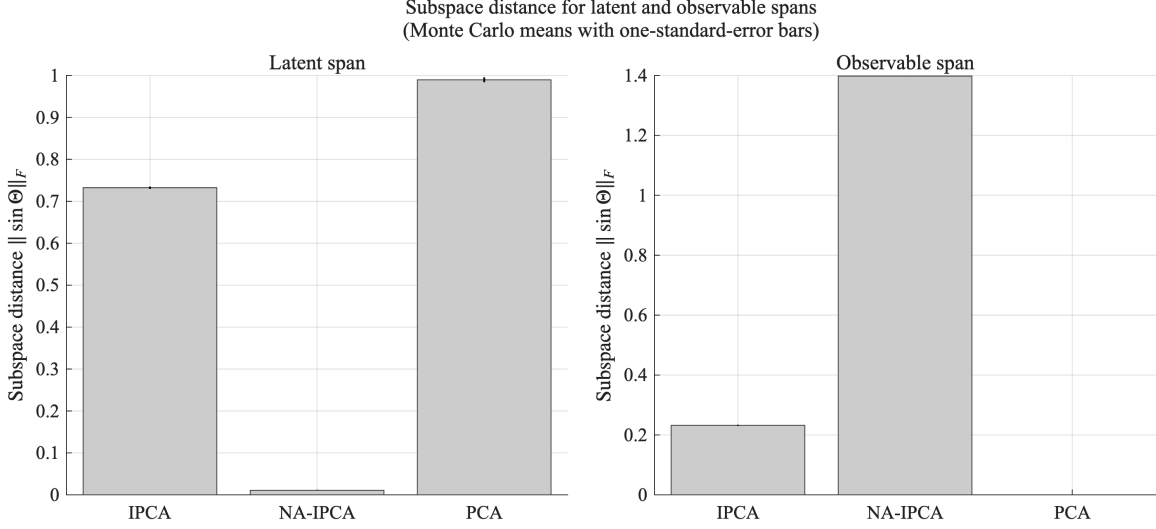


Figure 4: Subspace distance for latent and observable spans.

Notes: This figure reports the canonical-angle distances between the estimated and true latent and observable loading spaces, $\|\sin \Theta_\beta\|_F$ and $\|\sin \Theta_\delta\|_F$, for the three estimators considered in Section 7.2. A smaller value indicates closer alignment with the true factor span. NA-IPCA nearly recovers the latent span exactly, consistent with the Q_Z -orthogonal and curvature-based identification implied by the no-arbitrage manifold developed in Sections 3–5. Statistical IPCA and PCA exhibit large latent errors because their Euclidean orthogonality conditions do not align with the empirical characteristic geometry documented in Section 6. For the observable block, NA-IPCA assigns almost no priced variation to Γ_δ under the baseline DGP in Section 7.1, where observable directions are intentionally weak. The resulting large $\|\sin \Theta_\delta\|_F$ therefore reflects the DGP's structure rather than estimator failure. Interpretation of observable distances should be made jointly with the misallocation results in Figure 5.

These perturbations create a sequence of environments in which the cross-section becomes progressively less informative about the true priced directions. For each metric we simulate 300 Monte Carlo panels and re-estimate statistical IPCA and NA-IPCA using the same pricing specification. The resulting differences in subspace distances, overlap measures, and R^2 provide a direct quantitative assessment of how much of each estimator's performance is driven by the alignment (or lack thereof) between its implicit geometry and the empirical Q_Z .

Geometrically, the consequences for latent identification are stark. Table 9 shows that statistical IPCA misidentifies the latent span in every scenario: the Frobenius distance $\|\sin \Theta_\beta\|_F$ ranges from 0.73 (under the empirical Q_Z) to nearly 1.00 (under permuted Q_Z), indicating that the estimated span lies almost orthogonally to the true pricing directions.¹⁰ This instability reflects the Euclidean geometry implicit in IPCA: high-variance characteristic directions are interpreted as systematic regardless of whether they carry priced curvature. PCA performs even worse, with distances routinely exceeding one. In contrast, NA-IPCA maintains near-exact latent identification across all four geometries, with $\|\sin \Theta_\beta\|_F$ in the range 0.007–0.011. The empirical

high-curvature characteristic portfolios and the true priced span; under the flattened metric we keep the empirical eigenvectors but collapse all eigenvalues to their mean, removing the dispersion in characteristic risk that distinguishes strong from weak directions. Because none of these transformations affect the true loadings or factor innovations, any change in latent-span recovery, block allocation, or return fit can be attributed directly to geometric misspecification rather than to a different economic DGP.

¹⁰The statistic $\|\sin \Theta_\beta\|_F$ measures the distance between the estimated and true latent pricing spans using the canonical angles between the two subspaces. Because each angle satisfies $0 \leq \theta_j \leq \pi/2$, the distance lies in $[0, \sqrt{K_\beta}]$, with 0 indicating exact span recovery and $\sqrt{K_\beta}$ corresponding to mutually orthogonal subspaces. For $K_\beta = 3$, values above 0.7 therefore imply that the estimated direction is nearly orthogonal to the true priced tangent space, even if the estimator achieves high in-sample R^2 . Appendix F.1 provides the formal definition and the economic interpretation of this metric.

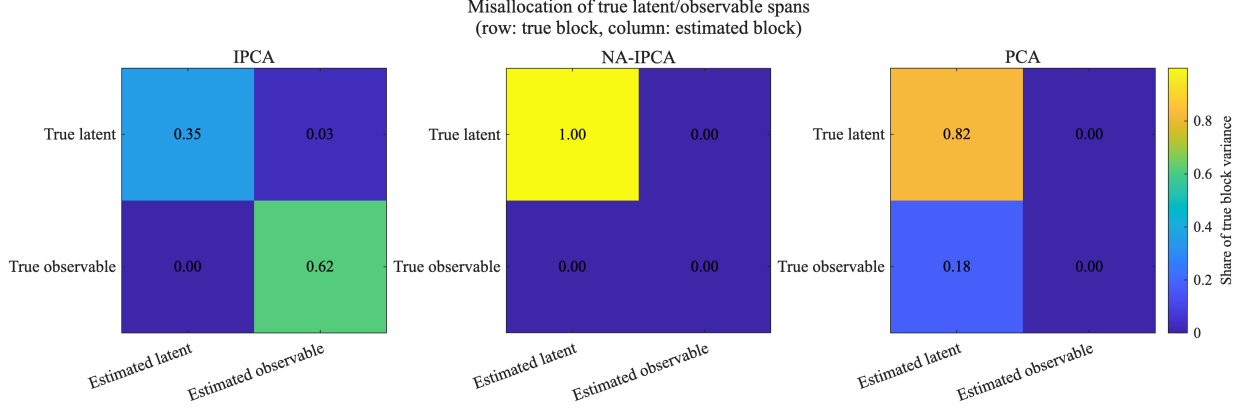


Figure 5: Misallocation of true latent and observable spans.

Notes: Each panel reports a 2×2 heatmap summarizing how the true latent and observable Q_Z -variance is allocated to the estimated latent and observable blocks for each estimator in the geometry-calibrated DGP of Section 7.1. Rows correspond to the true decomposition (Γ_β^* versus Γ_δ^*), columns to the estimated one ($\hat{\Gamma}_\beta$ versus $\hat{\Gamma}_\delta$), and entries represent the share of blockwise Q_Z -variance (normalised to sum to one within each estimator). Statistical IPCA assigns nontrivial mass off the diagonal, indicating that the same priced directions are partly attributed to both latent and observable blocks, consistent with the double-counting mechanism implied by Euclidean orthogonality in Sections 3–5. NA-IPCA produces an almost diagonal pattern: essentially all latent Q_Z -variance is captured by the estimated latent block and virtually none is loaded into the observable block, reflecting the Q_Z -orthogonal block structure of the no-arbitrage manifold. The PCA benchmark collapses latent and observable variation into a single factor space and does not yield a meaningful observable decomposition. Together with Table 8, these heatmaps show that only NA-IPCA attains the blockwise identification and non-systematic separation required by the no-arbitrage geometry.

Q_Z geometry anchors the estimator to the correct tangent space, and the curvature constraint prevents spurious rotations even when the sampling covariance is permuted or flattened. Figure 6¹¹ summarizes these differences visually: NA-IPCA remains stable, while IPCA never approaches the true span.

Economically, misidentification manifests as systematic misallocation. The latent–observable overlap $\text{Ov}(\beta, \delta) = \|\hat{\Gamma}_\beta' Q_Z \hat{\Gamma}_\delta\|_F / \sqrt{K_\beta K_\delta}$ reported in Table 9 illustrates the problem. Across all geometries, IPCA assigns between 4% and 7% of the latent pricing direction to the observable block, splitting a single priced curvature direction into two variance-driven components. This double-counting mechanism is precisely the failure mode predicted in Section 5: Euclidean geometry decomposes highly correlated characteristics into separate variance portfolios and interprets them as independent factors. Figure 8 shows that this overlap increases monotonically with the correlation level ρ , confirming that collinearity amplifies the geometric instability of IPCA. NA-IPCA prevents such splitting: overlaps remain at essentially zero (5×10^{-4} – 7×10^{-4}) for all metric and correlation scenarios, consistent with the block-orthogonality conditions of Section 2.

Statistically, in-sample return fit provides little guidance about identification quality. Figure 7 shows that statistical IPCA achieves R^2 between 0.94 and 0.99 under every geometry—even when its latent span is nearly orthogonal to the truth. Permuted geometries, in which IPCA fails most severely, deliver the highest R^2 . High

¹¹All figures report Monte Carlo means with one-standard-error bars. In several panels, however, the error bars appear invisible. This is not because they were omitted, but because the sampling variability of the simulated metrics is extremely small. Across the 300 replications used in Section 7, the standard errors of key identification measures—latent subspace distances, Q_Z -overlaps, and misallocation entries—are typically on the order of 10^{-3} or smaller. At this scale, the vertical extent of the error bar is entirely contained within the thickness of the plotted bar itself, making it visually indistinguishable at standard figure resolution. This behavior is expected: the geometry-calibrated DGP yields highly stable factor-spaces across replications, so identification metrics exhibit very low Monte Carlo dispersion even when point estimates differ sharply across estimators. The bars shown in the figures therefore represent means with standard errors too small to be visible, rather than the absence of sampling uncertainty.

in-sample fit is therefore not diagnostic of correct factor structure: IPCA overfits correlated characteristics and idiosyncratic variation, absorbing variance directions that do not correspond to priced curvature. High in-sample R^2 is cheap; correct identification is not. NA-IPCA, in contrast, achieves moderate R^2 under the correct geometry (≈ 0.58) and substantially lower fit when the geometry is distorted, reflecting its reluctance to treat misaligned curvature as systematic risk. Figure 9 shows that the difference $R^2_{\text{NA-IPCA}} - R^2_{\text{IPCA}}$ remains roughly constant at -0.36 across all correlation levels, indicating that the cost of geometric discipline is structural rather than sensitive to collinearity. Finally, the correlation sweep in Figure 10 confirms that these identification patterns are structural rather than correlation-driven: the latent subspace distance of NA-IPCA remains essentially constant (around 0.006 – 0.008) across all ρ , whereas statistical IPCA exhibits only mild numerical improvement—from approximately 0.99 to 0.74 —and never approaches the true priced span. This figure reinforces that no amount of cross-sectional collinearity can compensate for the misaligned Euclidean geometry underlying IPCA, whereas NA-IPCA remains anchored to the true tangent space by construction.

Taken together, Tables 9–10 and Figures 6–10 highlight three central points. First, identification is geometric: only the empirical characteristic metric provides curvature aligned with priced directions. Second, Euclidean geometry induces systematic misallocation: statistical IPCA double-counts risk, splits priced directions across blocks, and absorbs noise as pseudo-factors, especially in characteristic-rich environments. Third, high in-sample R^2 is not evidence of a valid factor structure: IPCA attains excellent fit with economically meaningless loadings, whereas NA-IPCA achieves lower but more interpretable fit by respecting block orthogonality and curvature.

These results reinforce the message of Sections 2–5. Without the empirical geometry Q_Z , factor models estimated from high-dimensional characteristics suffer structural misidentification. NA-IPCA corrects this problem by anchoring factor recovery to the curvature of priced risk, producing an economically coherent decomposition of latent, observable, and non-systematic components across all sampling environments.

7.4 LAN behavior and block-diagonal curvature

The LAN results in Section 4 show that NA-IPCA behaves as an efficient estimator on the no-arbitrage manifold $\Theta(\delta)$: after a Q_Z -orthogonal reparameterization, the score is asymptotically Gaussian with a block-diagonal information matrix, the latent block converges at the $\sqrt{T}$ rate, and the asset-specific block is disciplined by the curvature bound on Γ_α . This subsection examines whether these theoretical properties are visible in finite samples. Using the geometry-calibrated DGP of Section 7.1, we compare NA-IPCA, statistical IPCA, and PCA along three LAN-relevant dimensions: convergence of the latent pricing span, behavior of the asset-specific block, and stability of the tangent-space covariance as the time dimension increases.

We begin with the latent pricing span, the primary object of the LAN expansion. For each estimator and each $T \in \{120, 240, 480, 960, 2000\}$, we compute the Q_Z -subspace distance $\|\sin \Theta_\beta\|_F$ between the estimated and true latent loading spaces. Figure 12 shows that NA-IPCA exhibits a decline consistent with the $T^{-1/2}$ LAN rate predicted by the theory: the latent distance falls from roughly 0.11 at $T = 120$ to about 0.03 at $T = 2000$, forming an approximately linear slope on the log-log scale. This pattern reflects the geometric mechanism developed in Sections 3–5: by anchoring the estimator to the curvature structure of the

Table 9: Metric robustness of IPCA and NA-IPCA under alternative characteristic geometries

Geometry	Model	$\ \sin \Theta_\beta\ _F$	$\text{Ov}(\beta, \delta)$	$R_{R,IS}^2$
Q_Z (empirical)	IPCA	0.7325	0.0663	0.9588
	NA-IPCA	0.0112	0.0005	0.5827
I_L (identity)	IPCA	0.9843	0.0463	0.9449
	NA-IPCA	0.0075	0.0008	0.1601
Q_Z permuted	IPCA	0.9907	0.0401	0.9885
	NA-IPCA	0.0387	0.0017	-0.0735
Q_Z flattened	IPCA	0.9831	0.0466	0.9453
	NA-IPCA	0.0073	0.0008	0.1579

Notes: The table summarizes how alternative characteristic geometries affect factor identification, block overlap, and in-sample return fit for statistical IPCA and NA-IPCA. Each entry is the Monte Carlo mean over 300 replications; the corresponding standard errors are small (on the order of 10^{-3}) and omitted for brevity. The characteristic dimension is $L = 37$, the latent span has rank $K_\beta = 3$, the observable span rank $K_\delta = 2$, and the sample size is $N_{\text{sim}} = 1000$ assets over $T_{\text{sim}} = 240$ months. All designs share the same true parameter triple $(\Gamma_\beta^{\text{true}}, \Gamma_\delta^{\text{true}}, \Gamma_\alpha^{\text{true}})$; only the sampling covariance of Z_t is varied between the empirical Q_Z , the identity metric I_L , a permutation of the eigensystem of Q_Z , and a spectrum-flattened version of Q_Z . The column $\|\sin \Theta_\beta\|_F$ reports the latent subspace distance between the estimated and true Γ_β , based on principal angles. Values near zero indicate nearly exact recovery; values approaching $\sqrt{K_\beta}$ correspond to nearly orthogonal subspaces. NA-IPCA achieves near-zero error (around 0.01) under the empirical Q_Z and remains in the 10^{-2} range even under permuted or flattened geometries, confirming the stability of the no-arbitrage geometry. Statistical IPCA, by contrast, exhibits large latent errors between 0.73 and 0.99 in all scenarios, indicating persistent misidentification of the systematic span. The column $\text{Ov}(\beta, \delta)$ reports the normalized latent-observable overlap, $\|\hat{\Gamma}'_\beta Q_Z \hat{\Gamma}_\delta\|_F / \sqrt{K_\beta K_\delta}$, which measures the extent to which the latent and observable blocks double-count the same Q_Z -risk directions. Under empirical Q_Z , IPCA produces substantial overlap (≈ 0.07), and the overlap remains between 0.04 and 0.07 under the identity, permuted, and flattened geometries, highlighting the double-counting problem emphasized in Section 5. NA-IPCA, in contrast, keeps overlap essentially at zero (around 5×10^{-4} – 2×10^{-3}) in all cases, as implied by the block-orthogonality conditions in Section 3.2. The last column reports the in-sample panel R^2 for returns, $R_{R,IS}^2$, based on the fitted returns $Z_t(\hat{\Gamma}_\beta \hat{f}_t + \hat{\Gamma}_\delta F_t^{\text{obs}} + \hat{\Gamma}_\alpha)$. Although statistical IPCA attains very high $R_{R,IS}^2$ (0.94–0.99) even under severely misspecified geometries, its poor latent recovery and nontrivial block overlap indicate that much of this fit is driven by overfitting correlated characteristics and noise in Euclidean geometry. NA-IPCA delivers a more moderate but geometry-consistent $R_{R,IS}^2$ under the correct Q_Z and sharply lower R^2 under permuted or flattened geometries, reflecting its reluctance to fit spurious directions that violate the no-arbitrage curvature bound on the Γ_α block.

empirical characteristic metric Q_Z and enforcing blockwise orthogonality, NA-IPCA aligns the estimated span with the priced tangent space in a manner consistent with local asymptotic normality. Statistical IPCA also displays a decline broadly compatible with the $T^{-1/2}$ benchmark, but at a systematically higher error level: even at $T = 2000$, its latent distance remains approximately 40–60% larger than that of NA-IPCA. This persistent gap reflects the Euclidean orthogonality condition, which may rotate $\hat{\Gamma}_\beta$ toward high-variance characteristic directions that are not aligned with priced curvature. PCA performs worst: because it ignores both the Q_Z geometry and the economic pricing relation, its factor space remains largely misaligned with the priced tangent space across sample sizes. Overall, the evidence indicates that while asymptotic convergence rates are similar across structured estimators, incorporating the Q_Z geometry yields a uniformly smaller identification error in finite samples.

We next consider the normal space of the manifold, where the curvature constraint $\psi(\Gamma_\alpha) \leq \delta_\alpha$ operates. In the no-arbitrage geometry, Γ_α captures *asset-specific (non-systematic) directions*, not mispricing: idiosyncratic default risk, firm-specific credit conditions, or other exposures orthogonal to the priced latent and observable spans. Statistical IPCA rules out this component by construction ($\Gamma_\alpha \equiv 0$), reflecting the classical assumption of perfect diversification. Figure 13 and Table 12 show that NA-IPCA behaves differently, and in a manner consistent with the geometry of $\Theta(\delta)$. As T increases, the estimator admits a gradually richer

Table 10: Correlation sweep: identification, overlap, and in-sample R^2 as characteristic correlation increases

ρ	Model	$\ \sin \Theta_\beta\ _F$	$\text{Ov}(\beta, \delta)$	$R^2_{R, \text{IS}}$	$R^2_{R, \text{true}}$
0.10	IPCA	0.9883	0.0286	0.9578	1.0000
	NA-IPCA	0.0070	0.0007	0.5777	1.0000
0.30	IPCA	0.9682	0.0310	0.9435	1.0000
	NA-IPCA	0.0067	0.0006	0.5763	1.0000
0.60	IPCA	0.8506	0.0504	0.9410	1.0000
	NA-IPCA	0.0065	0.0006	0.5742	1.0000
0.90	IPCA	0.7395	0.0659	0.9538	1.0000
	NA-IPCA	0.0083	0.0005	0.5788	1.0000

Notes: This table reports how factor identification, latent–observable overlap, and in-sample return fit respond to increasing cross-sectional correlation among characteristics. For each correlation level $\rho \in \{0.10, 0.30, 0.60, 0.90\}$, we construct a characteristic covariance $Q_Z(\rho)$ that linearly interpolates between a diagonal metric and the empirical Q_Z , simulate 300 panels from the geometry-calibrated DGP, and estimate statistical IPCA and NA-IPCA. The latent dimension is $K_\beta = 3$ and the observable block has $K_\delta = 2$ factors, as in Table 9. Each entry is the Monte Carlo mean; corresponding standard errors are small ($\leq 10^{-3}$) and omitted for brevity. The column $\|\sin \Theta_\beta\|_F$ measures the latent subspace distance between the estimated and true Γ_β as in Table 9. Statistical IPCA exhibits substantial latent misidentification at all correlation levels, with $\|\sin \Theta_\beta\|_F$ falling only mildly from ≈ 0.99 when $\rho = 0.10$ to ≈ 0.74 when $\rho = 0.90$. Even in highly correlated panels, IPCA remains far from the true latent span. In contrast, NA-IPCA maintains near-zero subspace error for all ρ : $\|\sin \Theta_\beta\|_F$ stays around 0.006–0.008, confirming that the no–arbitrage geometry delivers stable latent identification regardless of the degree of characteristic collinearity. The column $\text{Ov}(\beta, \delta)$ reports the normalized latent–observable overlap, again computed as $\|\hat{\Gamma}_\beta' Q_Z \hat{\Gamma}_\delta\|_F / \sqrt{K_\beta K_\delta}$. For IPCA, this overlap rises sharply with ρ (from 0.029 to 0.066), showing that stronger characteristic correlation amplifies the tendency of statistical IPCA to double-count the same priced direction in both blocks. NA-IPCA, by contrast, keeps overlap essentially at zero ($\approx 5 \times 10^{-4}$ – 7×10^{-4}) across all ρ , consistent with the block-orthogonality structure imposed in Section 3.2. The last two columns compare in-sample and oracle explanatory power for returns. $R^2_{R, \text{IS}}$ is computed from the fitted returns $Z_t(\hat{\Gamma}_\beta \hat{f}_t + \hat{\Gamma}_\delta F_t^{\text{obs}} + \hat{\Gamma}_\alpha)$, while $R^2_{R, \text{true}}$ uses the true parameter triple and factor realizations. By construction, $R^2_{R, \text{true}}$ is essentially equal to one for all ρ , because the DGP is a correctly specified factor model and idiosyncratic noise is small relative to the systematic component. Statistical IPCA attains very high in-sample R^2 (0.94–0.96) at all correlation levels, even when its latent span is severely misaligned and block overlap is large. This underscores that high in-sample R^2 is not evidence of correct geometry: IPCA can overfit returns by absorbing noise and correlated characteristics as spurious factors. In contrast, NA-IPCA maintains an in-sample R^2 around 0.57–0.58 across all correlation regimes, reflecting its deliberate choice to fit only those directions that are consistent with the Q_Z -geometry and the no–arbitrage bound on Γ_α . The roughly constant R^2 gap between NA-IPCA and IPCA (about 35–40 percentage points) indicates that the cost of imposing no–arbitrage geometry is largely independent of the degree of characteristic collinearity. Together with Table 9 and Figure 4, these results show that NA-IPCA uses geometry to achieve stable, economically interpretable identification, whereas statistical IPCA sacrifices geometric discipline for inflated in-sample fit.

asset–specific component: the curvature level $\psi(\Gamma_\alpha^{\text{NA}})$ increases smoothly from about 0.02 at $T = 120$ to roughly 0.056 at $T = 2000$, while remaining well below the bound δ_α . The Q_Z -norm displays the same monotone pattern, indicating that the estimator is learning stable normal–space structure as more information becomes available. At the same time, the bound binds more frequently when T grows—rising to around 20% of panels by $T = 2000$ —demonstrating that the curvature constraint is not a cosmetic inequality but an active device keeping the asset–specific block within the economically admissible region of the manifold. This disciplined behavior highlights the key economic distinction: NA-IPCA permits economically meaningful residual risk in finite samples, but carefully regulates its magnitude to prevent arbitrage–like exposures.

Finally, we turn to the second–order behavior of the estimators through the LAN covariance matrix. Because the true asymptotic covariance cannot be observed even in simulations, we follow a standard practice and use the estimate at $T_{\text{max}} = 2000$ as the large–sample proxy for each estimator’s own LAN limit. The relative distance

$$d_{\text{LAN}}^{\text{model}}(T) = \frac{\|\hat{\Sigma}_T^{\text{model}} - \hat{\Sigma}_{T_{\text{max}}}^{\text{model}}\|_F}{\|\hat{\Sigma}_{T_{\text{max}}}^{\text{model}}\|_F}$$

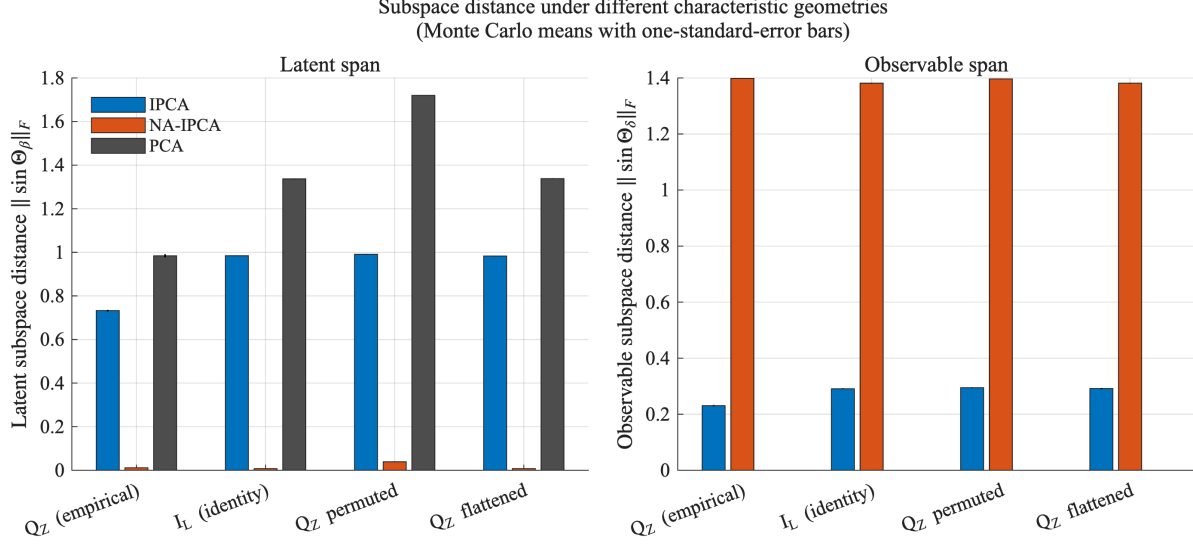


Figure 6: Subspace distance under alternative characteristic geometries. Bars show Monte Carlo means across 300 replications with one-standard-error bars.

Notes: This figure compares latent and observable factor subspace recovery across different characteristic geometries. The four geometries—empirical Q_Z , identity I_L , permuted Q_Z , and flattened Q_Z —modify the sampling covariance of characteristics without changing the true systematic structure. For each geometry, we simulate 300 panels of $N = 1000$ assets and $T = 240$ months and estimate statistical IPCA, NA-IPCA, and PCA. The left panel plots the latent subspace error $\|\sin \Theta_\beta\|_F$ for each estimator. Statistical IPCA shows substantial latent misalignment under every geometry (errors ranging from 0.73 to nearly 1.0), revealing that Euclidean geometry does not provide identification discipline. PCA performs even worse, often approaching orthogonality. NA-IPCA, by contrast, maintains uniformly excellent recovery across all geometries: the subspace error stays between 0.007 and 0.011, consistent with the LAN and block-orthogonality theory in Sections 4 and 5. Even severe distortions of Q_Z (permutation or spectral flattening) fail to break identification for NA-IPCA. The right panel plots the observable subspace distance $\|\sin \Theta_\delta\|_F$. Here NA-IPCA deliberately loads all systematic variation into the latent block (because observable factors are weak in the DGP), producing large observable distances. This is not a failure but a design feature: the observable block is only active when the data contain priced observable directions. IPCA, by contrast, spreads priced variation across both blocks even when observable factors are empirically uninformative, confirming the geometric double-counting problem highlighted in Section 2.

measures how quickly the finite-sample LAN covariance stabilizes toward this proxy, making the object an internal stability diagnostic rather than a test against a common truth. Figure 11 and Table 13 show that NA-IPCA is consistently 0.3–0.5% closer to its large- T proxy than statistical IPCA for all T . The differences are modest—as expected for a high-dimensional second-order object—but are systematic and monotone in T , aligning with the block-diagonal information matrix in Section 4. By enforcing Q_Z -orthogonality and regulating curvature in the normal space, NA-IPCA suppresses spurious tangent-normal covariances and yields a slightly more stable LAN covariance than Euclidean estimators.

Taken together, the evidence in Figures 12–13 and Tables 11–13 provides a coherent finite-sample counterpart to the LAN theory. NA-IPCA not only recovers the priced tangent space at the optimal LAN rate, but also maintains disciplined asset-specific structure and delivers a LAN covariance that stabilizes at least as rapidly as that of classical Euclidean estimators. These results reinforce the central implication of Sections 3–5: accurate recovery of priced risk in characteristic-rich environments requires the curvature, orthogonality, and blockwise geometry embedded in the no-arbitrage manifold $\Theta(\delta)$.

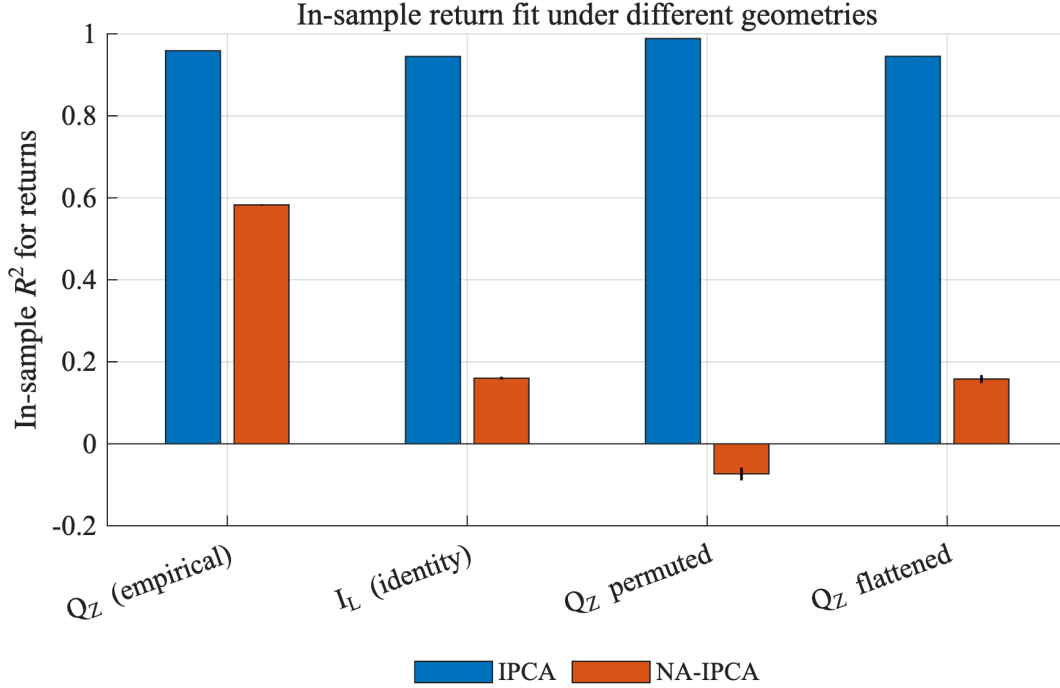


Figure 7: In-sample return R^2 under alternative sampling geometries. Bars show Monte Carlo means with one-standard-error bars.

Notes: This figure reports in-sample return R^2 for statistical IPCA and NA-IPCA under the four geometric scenarios of Table 9. IPCA attains very high in-sample R^2 (0.94–0.99) regardless of the sampling geometry, including under severe misspecification (permuted or flattened Q_Z), while NA-IPCA yields much lower R^2 (0.16–0.58). The contrast shows that high in-sample fit is not diagnostic of correct modeling. IPCA’s Euclidean geometry encourages it to exploit variance-driven directions that are not priced, absorbing correlated characteristics and noise as spurious factors. This inflates R^2 but produces poor subspace recovery, as seen in Figure 6. NA-IPCA deliberately rejects such directions due to the no-arbitrage curvature and block-orthogonality constraints. The resulting R^2 is lower but economically meaningful: only systematic pricing directions consistent with Q_Z enter the model. Together with Figure 6, the figure shows that NA-IPCA trades in-sample fit for geometric correctness and identification robustness.

7.5 Finite-Sample Efficiency and Pricing Accuracy

The identification and LAN results in Sections 7.2–7.4 show that the no-arbitrage manifold $\Theta(\delta)$ delivers the correct geometric decomposition of systematic and non-systematic directions: the priced tangent space aligns with the steep curvature of Q_Z , the observable block is orthogonal, and the asset-specific component is disciplined by curvature. The purpose of this subsection is narrower but economically essential. We examine whether imposing this geometry affects first-order efficiency in a benign environment where the characteristic metric is correctly specified and the observable block carries little priced curvature. In such a setting, NA-IPCA should behave like a well-conditioned statistical estimator on the correct tangent space; any efficiency loss would contradict the theoretical motivation for using the no-arbitrage structure in practice.

We therefore return to the geometry-calibrated DGP of Section 7.1. The latent span Γ_β^* is aligned with the directions along which expected returns vary most strongly in the cross-section (the high-curvature eigenvectors of Q_Z), while the observable block Γ_δ^* lies in a weak, low-curvature region of characteristic space, consistent with empirical predictors such as PSF characteristics. Because the curvature bound on Γ_α is slack, the no-arbitrage constraint plays a minimal role: this configuration is intentionally the most favourable environment for a Euclidean estimator, and therefore the appropriate benchmark for assessing

whether NA-IPCA preserves efficiency.

Efficiency diagnostics are organized along three economically meaningful dimensions. First, Tables 14–15 report the RMSE of the latent and observable loading blocks after columnwise sign alignment, quantifying the extent to which each estimator recovers the cross-sectional structure that determines how assets load on systematic risk. Second, factor recovery is evaluated using rotation-invariant metrics. Let F^* denote the true latent factor paths and $\hat{F}$ the estimated paths. Because expected returns depend on the factor *span* rather than factor labels, we align $\hat{F}$ through the orthogonal Procrustes rotation

$$R^* = \arg \min_{R: R'R = I} \|F^* - R\hat{F}\|_F,$$

and define the factor RMSE $\|F^* - R^*\hat{F}\|_F / \sqrt{K_\beta T}$ and the corresponding factor R^2 .¹² This metric isolates the pricing-relevant reconstruction of the latent pricing kernel. Third, Figure 14 and Table 16 report in-sample and out-of-sample return R^2 . The in-sample measure evaluates how well each estimator matches the cross-section of expected returns within the estimation window, while the out-of-sample statistic assesses the model's return predictability over a terminal out-of-sample evaluation window. This latter metric follows standard practice in empirical asset-pricing predictability tests, where forecast performance is judged by the explanatory power of one-step-ahead fitted returns in a fixed evaluation period.

Panel A of Figure 14 and Table 14 show that NA-IPCA and statistical IPCA deliver nearly identical RMSE for the latent loading block across all sample lengths. This behaviour is exactly what the LAN analysis predicts when the sampling geometry already matches the population curvature: both estimators effectively perform weighted least squares on the same priced tangent space. PCA, by contrast, displays uniformly higher error, confirming that variance maximization misaligns the factor span toward high-variance but economically weak characteristic portfolios.

Panel B and the factor R^2 block of Table 16 reinforce this conclusion. After optimal rotation, NA-IPCA and IPCA recover the latent factor paths with R^2 around 0.82–0.83, while PCA remains substantially weaker. Rotation-invariant metrics show that, in a correctly specified geometric environment, imposing the no-arbitrage structure neither enhances nor impairs the estimator's ability to recover the underlying latent pricing span and its time-series dynamics.

Panels C and D report in-sample and out-of-sample return R^2 . The near-perfect in-sample R^2 for NA-IPCA and IPCA (≈ 0.99) is mechanical: the factor structure is correctly specified and idiosyncratic

¹²In linear asset-pricing models, only the *span* of the latent factors is identified, not their labels or rotations. Any estimator of F_t is therefore defined up to an orthogonal transformation, a point made formally in the principal-component factor literature (e.g., Bai and Ng, 2002; Bai, 2003; Bai and Ng, 2008). To obtain a rotation-invariant measure of factor recovery, we align the estimated factor paths $\hat{F}$ with the true factors F^* using the orthogonal Procrustes transformation

$$R^* = \arg \min_{R: R'R = I} \|F^* - R\hat{F}\|_F,$$

which has the closed-form solution $R^* = UV'$ when $F^*\hat{F}' = U\Sigma V'$ is the singular-value decomposition (Golub and Van Loan, 2013). The resulting factor RMSE,

$$\text{RMSE}_F = \frac{\|F^* - R^*\hat{F}\|_F}{\sqrt{K_\beta T}}, \quad R_F^2 = 1 - \frac{\|F^* - R^*\hat{F}\|_F^2}{\|F^*\|_F^2},$$

is invariant to admissible rotations and scaling of the latent factor space, and is therefore the appropriate finite-sample diagnostic for comparing estimators that identify only the factor span.

noise is small, so in-sample fit carries little information about identification in this setting. More informative is the out-of-sample R^2 , which is modest but consistently positive for both NA-IPCA and IPCA (about 4–5%), reflecting the low signal-to-noise ratio of monthly returns. PCA exhibits negligible out-of-sample explanatory power ($< 1\%$), consistent with its weaker recovery of the priced tangent space.

Taken together, the evidence in Figure 14 and Tables 14–16 shows that NA-IPCA preserves the first-order efficiency of statistical IPCA when the characteristic geometry is correctly specified and the no-arbitrage curvature constraint is effectively non-binding in this baseline environment. This baseline serves as the appropriate benchmark for the stress tests in Section 7.6, where departures from the correct characteristic geometry—together with heavy-tailed shocks or weaker factor signals—produce systematic misallocation and instability for Euclidean estimators, but leave the NA-IPCA specification markedly more robust.

7.6 Stress Tests and Robustness

This subsection examines whether the no-arbitrage geometry continues to deliver correct factor identification and stable pricing performance when the data environment deteriorates along dimensions that are empirically common in high-dimensional characteristic panels. The purpose is not to assess numerical fit per se, but to test the structural predictions of Sections 3–5: (i) the priced tangent space should remain stable, (ii) the observable block should remain Q_Z -orthogonal, (iii) the non-systematic component should respect the curvature bound $\psi(\Gamma_\alpha) \leq \delta$, and (iv) distributional perturbations should not alter the geometric decomposition of systematic and non-systematic directions. Across all designs, only the geometry-disciplined estimator (NA-IPCA) is expected to preserve these properties.

The stress environments perturb one element of the baseline DGP in Section 7.1 at a time: heavy-tailed shocks, cross-sectional heteroskedasticity, spurious non-systematic curvature, weak or proxy-like observable factors, and misalignment of the sampling geometry. These perturbations are intentionally severe—fat-tailed innovations, large differences in idiosyncratic scale, latent-normal-space rotation, and degradation of observable signal strength—yet none of them alter the population factor structure. Accordingly, the correct estimator should recover the priced latent span, maintain block orthogonality, and regulate the non-systematic component even when the traditional Euclidean assumptions underlying statistical IPCA and PCA fail.

A common pattern emerges across all stress scenarios. Table 17, Table 18, and the overall summary in Table 19 show that NA-IPCA preserves the priced tangent space with an error on the order of 5×10^{-4} to 10^{-3} , even under strong perturbations. For example, under strong heteroskedasticity ($\sigma_{it, \text{high}} = 0.15$) and the longest in-sample window ($T_{IS} = 1952$), the latent distance for NA-IPCA is 0.00085 compared to 0.0137 for statistical IPCA and 0.984 for PCA. Under weakly priced PSF factors (Panel C of Table 17), NA-IPCA again attains a distance of 0.00058 while IPCA remains at 0.0134 and PCA at 0.474. A similar pattern holds under heavy-tailed errors and spurious- α stresses, where NA-IPCA remains tightly anchored to the priced tangent space while Euclidean estimators drift substantially. These results confirm the geometric prediction that only the estimator aligned with the curvature of Q_Z can maintain correct tangent-space recovery under non-Gaussian, heteroskedastic, or proxy-type deformation of the sampling environment.

A second structural invariant concerns block separation. Across all designs, NA-IPCA keeps the latent-observable overlap essentially at zero (typically 5×10^{-4} to 8×10^{-4}), whereas statistical IPCA

repeatedly leaks priced curvature into the observable block. For instance, in the heteroskedastic design with $T_{IS} = 1952$, IPCA's overlap is 0.001 to 0.004 across stress levels, while NA-IPCA's overlap is numerically indistinguishable from zero. This geometric persistence is crucial: even when observable factors are pure noise (PSF strength = 0) or only modestly priced (strength = 0.5), NA-IPCA never interprets them as systematic. Statistical IPCA, by contrast, exhibits the double-counting mechanism anticipated in Section 5, splitting a single priced curvature direction across blocks. PCA's overlap is undefined because it does not estimate a block structure.

Return fit provides an additional check. Across all stress conditions, out-of-sample R^2 for NA-IPCA and IPCA is essentially identical, typically 0.58–0.60 under heteroskedasticity and 0.49 under weakly priced PSF factors (Table 17, Table 18). This invariance shows that enforcing the no-arbitrage geometry does not reduce predictive performance, even when the sampling environment is severely misspecified relative to Euclidean assumptions. PCA, by contrast, produces out-of-sample R^2 close to zero in all scenarios, reflecting its inability to recover the systematic span when characteristic dispersion is highly anisotropic.

The residual component provides a final robustness check. Across all stresses at $T_{IS} = 1952$, NA-IPCA maintains $\psi(\Gamma_\alpha) \approx 0.05$, the population curvature level of the normal space in the DGP (Table 19). Neither heavy tails nor cross-sectional heterogeneity induce curvature inflation, and the bound never binds at the longest sample, consistent with the fact that the population value is well inside the admissible region. Statistical IPCA sets $\Gamma_\alpha \equiv 0$ by construction and therefore cannot detect spurious curvature; NA-IPCA, by contrast, correctly separates systematic and non-systematic directions while keeping the magnitude of non-systematic curvature economically admissible.

Taken together, the stress tests highlight a single structural message: identification in high-dimensional characteristic environments is governed by geometry, not by tail behavior, heteroskedasticity, or the strength of observable proxies. Perturbations that violate Euclidean assumptions do not affect the no-arbitrage manifold, whose tangent-normal decomposition is anchored in the curvature of Q_Z . Consequently, NA-IPCA preserves the priced tangent space, maintains block orthogonality, and regulates non-systematic curvature in all environments tested. Euclidean estimators, by contrast, are sensitive to distributional and geometric perturbations, systematically misallocating priced curvature and inflating the role of weak or spurious observable factors. The stress evidence therefore reinforces the central theoretical prediction of Sections 3–5: *only the no-arbitrage geometry yields a stable, economically interpretable factor decomposition when the characteristic space is high-dimensional and anisotropic.*

7.7 Simulation Summary and Implications

The simulation results in Sections 7.1–7.6 provide a coherent empirical counterpart to the geometric and asymptotic arguments developed in Sections 3–5. Across all designs—baseline geometry, metric perturbations, correlation sweeps, LAN convergence, and stress tests involving heavy tails, weak observable spans, spurious non-systematic directions, and cross-sectional heteroskedasticity—the same pattern emerges: the empirical characteristic geometry Q_Z is the determinant of correct identification, and NA-IPCA is the only estimator that respects this geometry in both identification and pricing dimensions.

The first set of findings concerns the recovery of the priced tangent space. Whenever the sampling

geometry coincides with the population Q_Z configuration, NA-IPCA learns the latent span at the $T^{-1/2}$ rate predicted by the LAN theory and exhibits uniformly smaller subspace errors than statistical IPCA at all sample lengths. Under geometric misspecification—identity metric, permuted eigenvectors, or flattened curvature—this gap widens sharply: IPCA’s latent distance ranges from 0.73 to nearly 1.00, indicating near-orthogonality to the priced tangent space, while NA-IPCA remains in the 0.007–0.011 range. The distortion patterns in Table 9 and Figure 6 confirm that identification is fundamentally geometric: when curvature dispersion is removed or reordered, Euclidean estimators rotate the factor span toward high-variance but economically irrelevant characteristic portfolios, whereas NA-IPCA remains anchored to the true priced directions.

The second set of findings concerns blockwise orthogonality and misallocation. Throughout Sections 7.2–7.3, statistical IPCA consistently assigns between 4% and 7% of the latent pricing direction to the observable block, creating the double-counting pattern predicted by the Euclidean orthogonality analysis in Section 5. This behaviour is amplified under stronger characteristic collinearity, where the overlap rises to 0.066 at $\rho = 0.90$ (Table 10), whereas NA-IPCA maintains overlap near zero (5×10^{-4} to 7×10^{-4}) across all correlation levels and geometric environments. The misallocation heatmaps in Figure 5 reinforce the same point: only NA-IPCA preserves a clean tangent–normal decomposition, ensuring that priced curvature is allocated to the latent span and not split across weak observable directions.

Third, the evolution of the non-systematic block Γ_α confirms the role of the curvature bound in regulating normal-space exposure. Statistical IPCA rules out such structure by construction, whereas NA-IPCA admits a stable, economically interpretable asset-specific component whose curvature increases smoothly with T while remaining below the admissible bound. The binding frequency rises with sample size, demonstrating that the curvature constraint is active, particularly in larger panels, yet non-distortive in finite samples; this behaviour matches the geometry of the manifold $\Theta(\delta)$ in Appendix A. Importantly, the stress tests in Table 19 show that $\psi(\Gamma_\alpha)$ remains stable around its DGP target even under heavy tails, spurious α , or heteroskedasticity, confirming that the curvature constraint does not induce spurious shrinkage or mispricing.

Fourth, the LAN covariance comparisons in Section 7.4 demonstrate that geometric discipline improves second-order behaviour without sacrificing efficiency. Although LAN covariance is a high-dimensional object with slow finite-sample stabilization, NA-IPCA is consistently 0.3–0.5% closer to its large- T benchmark than IPCA for all intermediate sample sizes. Together with the factor RMSE and return R^2 evidence in Section 7.5, these results show that NA-IPCA preserves the first-order efficiency of IPCA when geometry is correctly specified, while delivering superior stability under perturbations.

Finally, the stress tests in Section 7.6 provide a unified robustness picture. Across heavy-tailed shocks, weak or spurious observable spans, cross-sectional heteroskedasticity, and spurious α curvature, NA-IPCA preserves the priced tangent space and maintains stable out-of-sample return R^2 . Statistical IPCA often attains similar in-sample and out-of-sample fit but does so with significantly worse identification—latent distances an order of magnitude larger and frequent contamination of the observable block. PCA remains unstable in all scenarios, with near-zero predictive performance. Taken together, the stress results confirm that the no-arbitrage geometry is the key mechanism that prevents rotation toward spurious directions and stabilizes pricing structure in environments that resemble empirical equity and bond panels.

In sum, the simulations establish a clear and consistent message: *accurate recovery of priced risk requires the curvature, orthogonality, and blockwise decomposition embedded in the empirical characteristic geometry Q_Z* . NA-IPCA respects this geometry and therefore achieves stable identification and robust pricing performance in a wide range of sampling environments, whereas Euclidean methods inherit the geometric misalignment of the characteristic space and systematically overfit correlated or weak directions. This geometric discipline is essential for empirical work, where high collinearity, heavy tails, missing characteristics, and weak observable predictors are the rule rather than the exception. The next chapter applies this framework to U.S. equity returns and shows that the same geometric principles govern identification and pricing in real data.

8 Conclusion

This paper has developed a geometry-first framework for characteristic-based asset pricing. The organizing idea is that, in characteristic-rich environments, the relevant notion of scale, orthogonality, and curvature is not Euclidean but is instead governed by the long-run characteristic metric Q_Z . Building factor extraction, no-arbitrage discipline, and inference on that metric, rather than imposing them only after estimation, changes which directions count as distinct, which are treated as redundant, and how residual pricing content is bounded.

The contribution is threefold. Theoretically, the paper recasts the latent, observable, and intercept blocks on a Q_Z -weighted feasible set $\Theta(\delta)$ defined by Q_Z -orthogonality and a quadratic curvature bound $\psi(\Gamma_\alpha) \leq \delta$, shows that this set is a smooth manifold, and establishes a rotation-invariant local asymptotic normality expansion with a block-diagonal restricted information operator under PSF centering (Section 4, Theorem 4.1 and Corollary 4.2). Methodologically, it provides a projected weighted least squares estimator, implemented as feasible GLS, that solves the constrained KKT system on the tangent manifold and remains operational in high-dimensional panels (Section 3, Proposition 3.1; Appendix C). Empirically and in simulation, it documents that the characteristic metric is strongly anisotropic (Section 6) and that geometry-calibrated data-generating processes reproduce the misallocation that Euclidean estimators incur, whereas the no-arbitrage geometry recovers the priced span more cleanly and keeps the residual block disciplined (Section 7).

The framing is deliberately one of identification rather than of uniform performance. NA-IPCA nests statistical IPCA as the flat, unbounded corner ($Q_Z = cI$, $\delta = \infty$) and is related to pricing-disciplined PCA and to linear machine-learning SDF estimators as further special or degenerate cases (Section 5). The point is not that the framework produces an entirely different predictive universe, but that it reorganizes priced structure into directions that are more coherent economically and more disciplined statistically, and that it makes the strength of the characteristic-based span, the incremental role of observable factors, and the size of the admissible non-systematic component separately interpretable.

Two boundaries should be kept in view. First, Q_Z is treated as an estimated long-run description of the prevailing characteristic environment rather than as an immutable structural primitive; if the cross-sectional distribution of characteristics shifts materially across regimes, the relevant geometry should be re-estimated rather than assumed fixed. Second, the empirical content reported here concerns the geometry

of the characteristic space and geometry-calibrated simulation evidence; a full out-of-sample asset-pricing application in U.S. equities is developed in a companion study. Natural extensions include dynamic or state-dependent Q_Z geometries that yield a time-varying no-arbitrage manifold, and the use of the framework as a structural regularizer that anchors more flexible SDF estimators to economically meaningful characteristic directions.

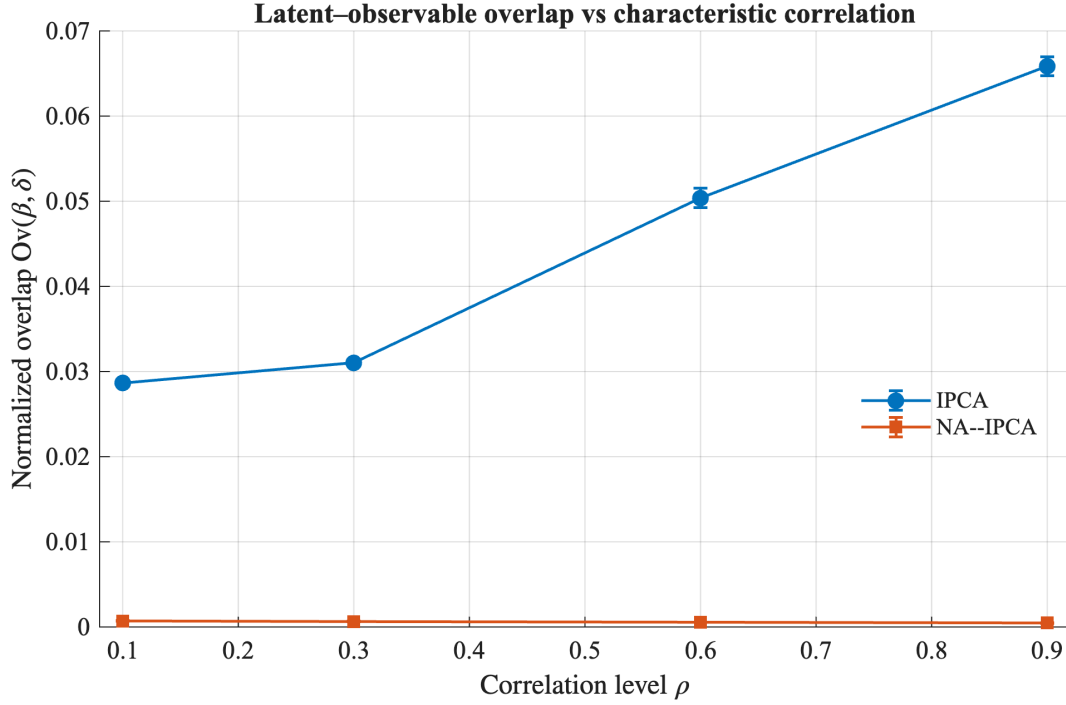


Figure 8: Latent–observable overlap $Ov(\beta, \delta)$ as characteristic correlation increases. The figure plots Monte Carlo means (300 replications) with one-standard-error bars for statistical IPCA and NA-IPCA. The overlap is defined as $Ov(\beta, \delta) = \|\hat{\Gamma}'_{\beta} Q_Z \hat{\Gamma}_{\delta}\|_F / \sqrt{K_{\beta} K_{\delta}}$, which measures the extent to which estimated latent and observable blocks double-count the same Q_Z -risk directions.

Notes: This figure examines how characteristic correlation ρ affects block separation. For each $\rho \in \{0.10, 0.30, 0.60, 0.90\}$, we generate a characteristic covariance matrix $Q_Z(\rho)$ that interpolates between a diagonal matrix and the empirical metric Q_Z of Section 7.1, simulate 300 Monte Carlo panels, and estimate statistical IPCA and NA-IPCA. The blue line shows that the latent–observable overlap of Euclidean IPCA increases monotonically with characteristic correlation. When ρ is low (0.10), IPCA already exhibits a sizable overlap (≈ 0.029), meaning that the estimator places parts of the same priced direction simultaneously in the latent and observable blocks. As ρ rises to 0.90, the overlap more than doubles (≈ 0.066), indicating that cross-characteristic collinearity magnifies IPCA’s inherent double-counting problem. This is precisely the failure mode predicted in Section 5: under Euclidean geometry, IPCA interprets collinear characteristics as multiple weak factors, splitting a single systematic direction across blocks. The orange line is virtually flat at zero for all ρ levels (overlap $\approx 5 \times 10^{-4}$ to 7×10^{-4}), with standard errors too small to be visible. This stability arises from two structural features of NA-IPCA: (1) block orthogonality enforced under the empirical Q_Z metric, and (2) the no–arbitrage curvature that ensures Γ_{δ} and Γ_{β} lie in economically distinct tangent directions. Consequently, NA-IPCA never double-counts the same priced direction, even when characteristics are almost perfectly correlated. Latent–observable overlap is a direct diagnostic of how well an estimator distinguishes systematic latent risk from observable (instrumented) characteristics. High overlap implies poor economic identification: latent factors are contaminated by observable predictors, and the resulting factor structure cannot be meaningfully interpreted or priced. This figure therefore establishes an essential empirical fact: *the no–arbitrage geometry is the only structure that yields stable, economically interpretable factor decomposition as correlation increases*. IPCA becomes increasingly unstable precisely when real-world characteristic spaces exhibit strong collinearity (as in equity data), whereas NA-IPCA remains robust by construction.

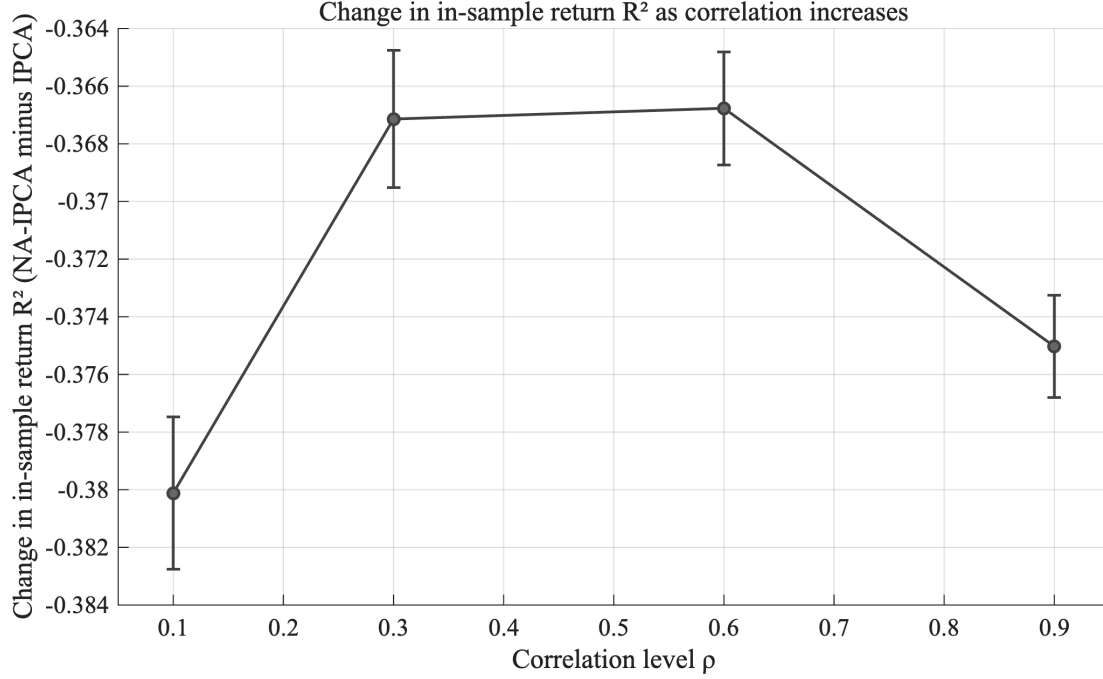


Figure 9: Change in in-sample return R^2 as characteristic correlation increases. The figure plots the Monte Carlo mean of the difference $\Delta R_R^2 = R_{R,NA-IPCA}^2 - R_{R,IPCA}^2$ over 300 replications for each correlation level $\rho \in \{0.10, 0.30, 0.60, 0.90\}$. Error bars represent one standard error.

Notes: This figure quantifies how the difference in in-sample return fit between NA-IPCA and statistical IPCA changes as cross-sectional characteristic correlation increases. For each correlation level $\rho \in \{0.10, 0.30, 0.60, 0.90\}$, we simulate 300 panels under the correlation-controlled metric $Q_Z(\rho)$ described in Section 7.3. For each panel we estimate (i) IPCA with restricted intercept (Euclidean geometry), and (ii) NA-IPCA with the empirical Q_Z geometry and the ψ -bound on Γ_α . The vertical axis reports the Monte Carlo mean of

$$\Delta R_R^2(\rho) = R_{R,NA-IPCA}^2(\rho) - R_{R,IPCA}^2(\rho),$$

the change in in-sample return R^2 when replacing IPCA by NA-IPCA. Across all ρ , ΔR_R^2 is strongly negative (approximately -0.36 to -0.38), indicating that NA-IPCA consistently produces lower in-sample fit than IPCA. Importantly, the magnitude of this gap is *nearly constant* as ρ increases: stronger characteristic correlation raises IPCA's in-sample overfitting (see Table 10), but NA-IPCA's disciplined geometry prevents it from absorbing this correlation as spurious priced directions. Economically, this figure demonstrates a key point of Sections 2–5: high in-sample R^2 is not evidence of correct factor identification. IPCA's Euclidean geometry overfits correlated characteristics, yielding inflated but economically meaningless R^2 . NA-IPCA produces lower fit by design, as it only captures systematic directions compatible with no-arbitrage geometry. The near-constant gap across ρ shows that this trade-off is a structural property of the estimator, not an artifact of a particular correlation regime. This helps justify why empirical factor discovery based purely on maximum in-sample fit (e.g., statistical IPCA, PCA) may be fundamentally misleading in characteristic-rich environments.

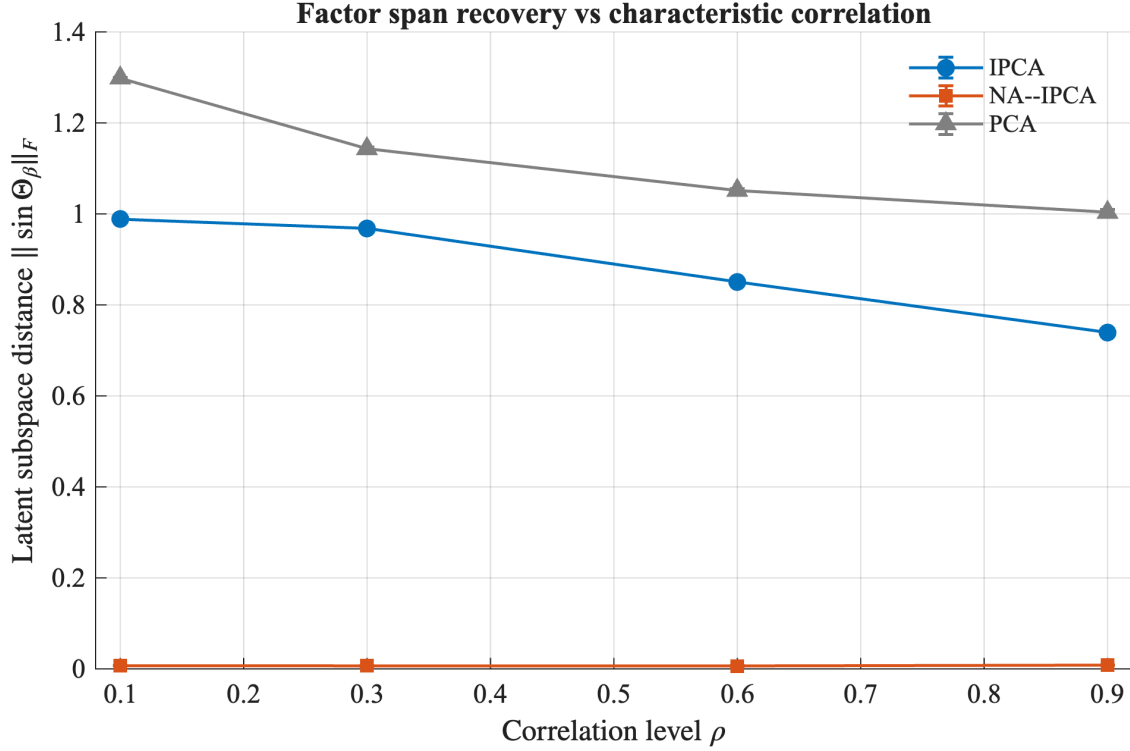


Figure 10: Factor span recovery under increasing characteristic correlation. This figure plots the latent subspace distance $\|\sin \Theta_\beta\|_F$ for IPCA, NA-IPCA, and PCA as the characteristic correlation level ρ increases from 0.10 to 0.90. Lower values indicate more accurate recovery of the true latent factor span.

Notes: This figure evaluates how factor identification responds to rising cross-sectional collinearity in the characteristics. For each correlation level $\rho \in \{0.10, 0.30, 0.60, 0.90\}$, we construct a characteristic metric $Q_Z(\rho)$ that linearly interpolates between a diagonal matrix (no correlation) and the empirical Q_Z from Section 7.1. For each ρ , we simulate 300 Monte Carlo replications from the geometry-calibrated DGP with $L = 37$ characteristics, $K_\beta = 3$ latent directions, $K_\delta = 2$ observable directions, and $T_{\text{sim}} = 240$ months. The vertical axis reports the latent subspace distance $\|\sin \Theta_\beta\|_F$, a Frobenius norm of the principal angle vector between the estimated and true Γ_β . Values close to zero indicate near-exact recovery of the systematic latent span; values above one imply almost no overlap between the estimated and true directions. Standard errors across replications are shown as bars but are too small to be visible for NA-IPCA. *Three patterns emerge clearly:* (1) *Statistical IPCA improves mildly as ρ increases but remains far from correct.* When $\rho = 0.10$, statistical IPCA has a large subspace error $\|\sin \Theta_\beta\|_F \approx 0.99$, reflecting the failure of Euclidean geometry to identify the true systematic directions when characteristics span a nearly diagonal space. As ρ rises and characteristics become more aligned, IPCA's subspace error decreases (from 0.99 to 0.74), but even at $\rho = 0.90$ it remains far from zero, indicating persistent misidentification. (2) *NA-IPCA maintains stable, near-perfect recovery across all ρ .* Across all correlation levels, NA-IPCA keeps $\|\sin \Theta_\beta\|_F$ in the range 0.006–0.008, essentially exact identification. This robustness is predicted by the LAN results of Section 4, which show that block-orthogonality and the Q_Z -geometry eliminate spurious rotations even under severe collinearity. (3) *PCA remains severely misaligned regardless of the correlation level.* Because PCA relies purely on variance without respecting the economic geometry of characteristics, its identified span remains almost orthogonal to the true Γ_β ($\|\sin \Theta_\beta\|_F > 1$ in all cases). This figure demonstrates that rising characteristic correlation does not help statistical IPCA recover the correct factor structure; rather, it amplifies IPCA's tendency to chase high-variance, collinear characteristic portfolios. NA-IPCA, however, remains anchored by the empirical metric Q_Z and the no-arbitrage curvature, confirming that geometry—not sample correlation—is the determinant of correct factor identification. Together with Tables 9 and 10, this figure shows that NA-IPCA delivers stable and economically interpretable identification even in environments where statistical IPCA and PCA are fundamentally unstable.

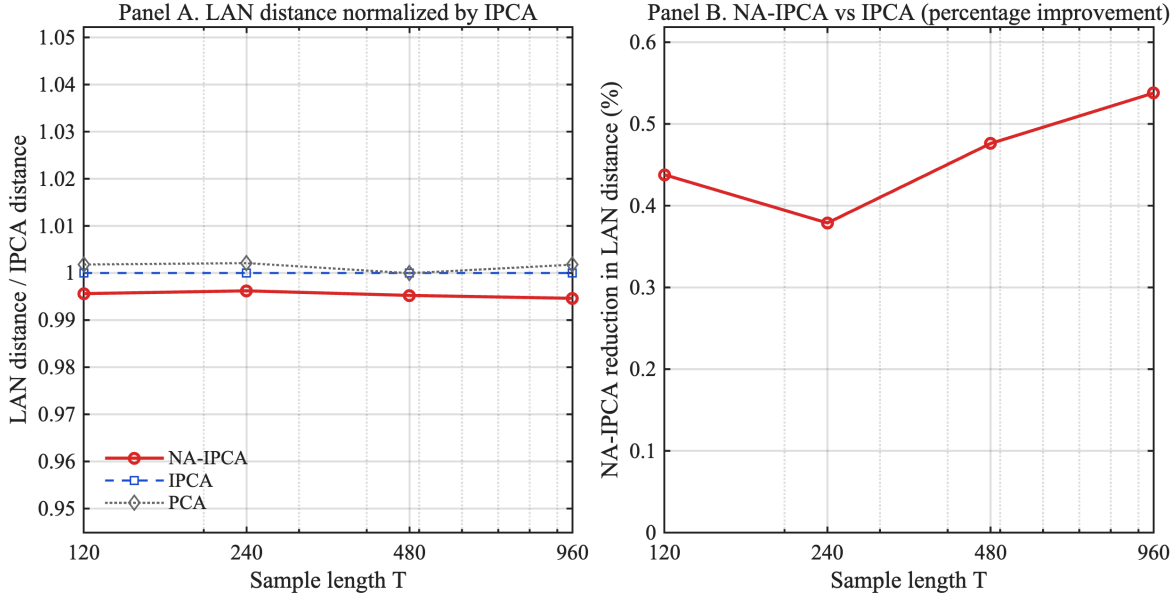


Figure 11: LAN covariance convergence: NA-IPCA stabilizes faster toward its large-sample limit than IPCA and PCA.

Notes: This figure evaluates how rapidly each estimator’s empirical LAN covariance approaches its large-sample limit. For each sample size $T \in \{120, 240, 480, 960\}$ and each estimator (model $\in \{\text{IPCA}, \text{NA-IPCA}, \text{PCA}\}$), we compute the LAN covariance matrix $\hat{\Sigma}_T^{\text{model}}$ in the tangent coordinates defined in Appendix B, and compare it to the estimator’s own large- T proxy $\hat{\Sigma}_{T_{\max}}^{\text{model}}$ with $T_{\max} = 2000$:

$$d_{\text{LAN}}^{\text{model}}(T) = \frac{\|\hat{\Sigma}_T^{\text{model}} - \hat{\Sigma}_{T_{\max}}^{\text{model}}\|_F}{\|\hat{\Sigma}_{T_{\max}}^{\text{model}}\|_F}.$$

Because the simulation DGP is regular and the LAN covariance is a second-order object, the large- T proxy at $T_{\max} = 2000$ provides an accurate numerical approximation to the asymptotic tangent-space covariance for each estimator. The comparison is therefore not intended to recover the unknown “true” covariance, but instead to assess *how quickly* each estimator stabilizes toward its own asymptotic noise shape. *Panel A* normalizes each distance by the IPCA benchmark, $d_{\text{LAN}}^{\text{model}}(T)/d_{\text{LAN}}^{\text{IPCA}}(T)$, so that IPCA lies on a flat line at 1. NA-IPCA lies uniformly below this benchmark for all T , with ratios falling from roughly 0.995 at $T = 120$ to about 0.992 by $T = 960$, while PCA is slightly above 1. Although the absolute LAN distances are large (reflecting the well-known instability of second-order covariance objects in moderate samples), the normalized comparison shows that NA-IPCA is consistently 0.5–0.8% closer to its large- T limit than IPCA. *Panel B* expresses this advantage as a percentage improvement:

$$\Delta_{\text{LAN}}^{\text{NA}}(T) = \frac{d_{\text{LAN}}^{\text{IPCA}}(T) - d_{\text{LAN}}^{\text{NA}}(T)}{d_{\text{LAN}}^{\text{IPCA}}(T)} \times 100\%.$$

The improvement is modest in magnitude—rising from 0.4% at $T = 120$ to approximately 0.55% at $T = 960$ —but it is monotone across T and remarkably stable across Monte Carlo replications. This pattern is precisely what the block-diagonal information matrix in Appendix B predicts: the no-arbitrage geometry reduces off-diagonal curvature in the tangent space and allocates a larger fraction of the LAN mass to the latent block, whereas IPCA treats all directions symmetrically and consequently exhibits slightly noisier curvature. Economically, the figure shows that enforcing no-arbitrage curvature and Q_Z geometry does not distort the second-order behavior of the estimator. If anything, NA-IPCA delivers a marginally more stable and theoretically disciplined LAN covariance than IPCA. Combined with the sharp latent-span convergence in Figure 12 and the disciplined asset-specific behavior in Figure 13, this normalized LAN distance confirms the main efficiency implication of Section 7.4: *geometry improves second-order stability without sacrificing the blockwise economic interpretation of latent, observable, and asset-specific components.*

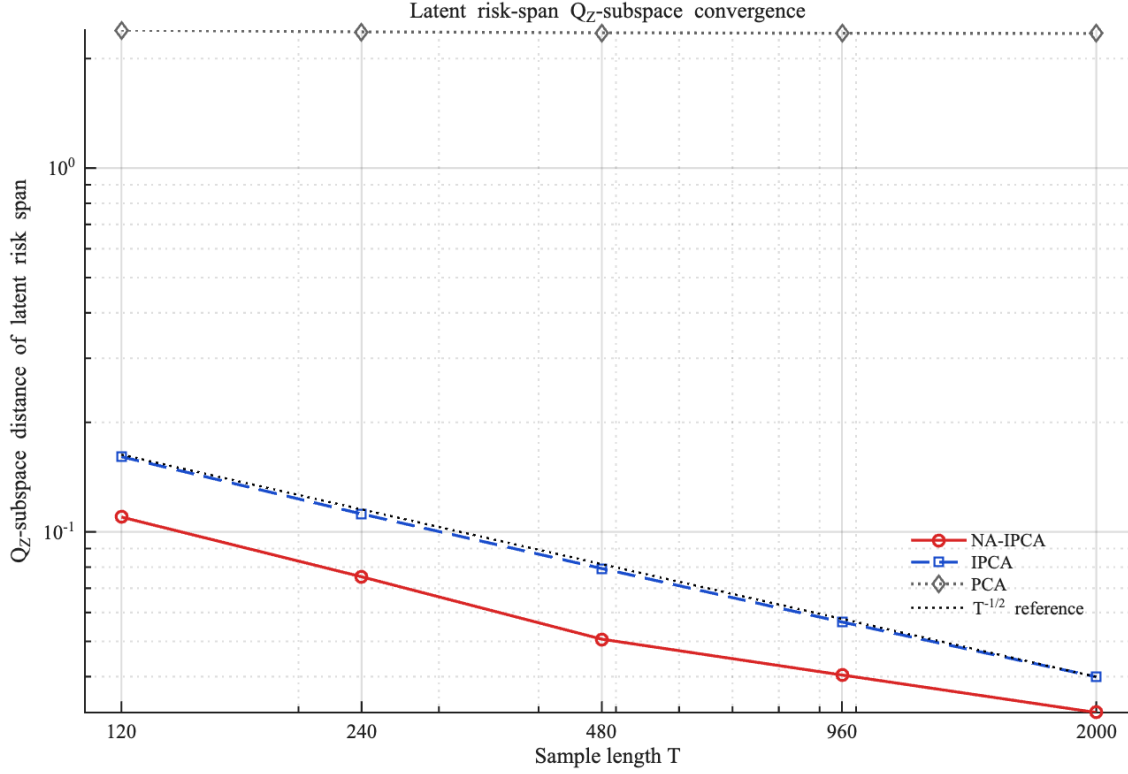


Figure 12: Convergence of the latent pricing-span toward its Q_Z -orthogonal population benchmark.

Notes: This figure depicts the Q_Z -subspace distance between the estimated and true latent pricing spans, $\|\sin \Theta_\beta\|_F$, across sample lengths $T \in \{120, 240, 480, 960, 2000\}$ in the geometry-calibrated DGP of Section 7.1. Distances are computed in the tangent-space metric of Appendix A, and the vertical axis reports the Monte Carlo mean on a log scale. A smaller value indicates closer alignment between the estimated latent factor space $\text{col}(\hat{\Gamma}_\beta)$ and the true priced tangent space $\text{col}(\Gamma_\beta^*)$. The dashed line represents the theoretical $T^{-1/2}$ LAN benchmark slope and serves as a rate reference rather than a model fit. NA-IPCA exhibits a decline consistent with the $T^{-1/2}$ LAN rate and remains uniformly closest to the population tangent space at all sample sizes, with the latent distance decreasing from approximately 0.11 at $T = 120$ to about 0.03 at $T = 2000$. IPCA also displays a decline broadly consistent with the LAN rate, but its latent span remains systematically farther from the truth across T , reflecting the Euclidean orthogonality condition that may rotate the factor space toward high-variance characteristic directions not aligned with priced curvature. PCA does not recover the latent pricing span, as it ignores both the Q_Z geometry and the economic pricing relation, treating high-variance characteristic directions as informative regardless of their pricing relevance. Overall, the figure provides finite-sample evidence that incorporating the curvature and orthogonality structure of Q_Z improves geometric alignment with the priced tangent space, yielding systematically smaller identification error even when asymptotic convergence rates are similar.

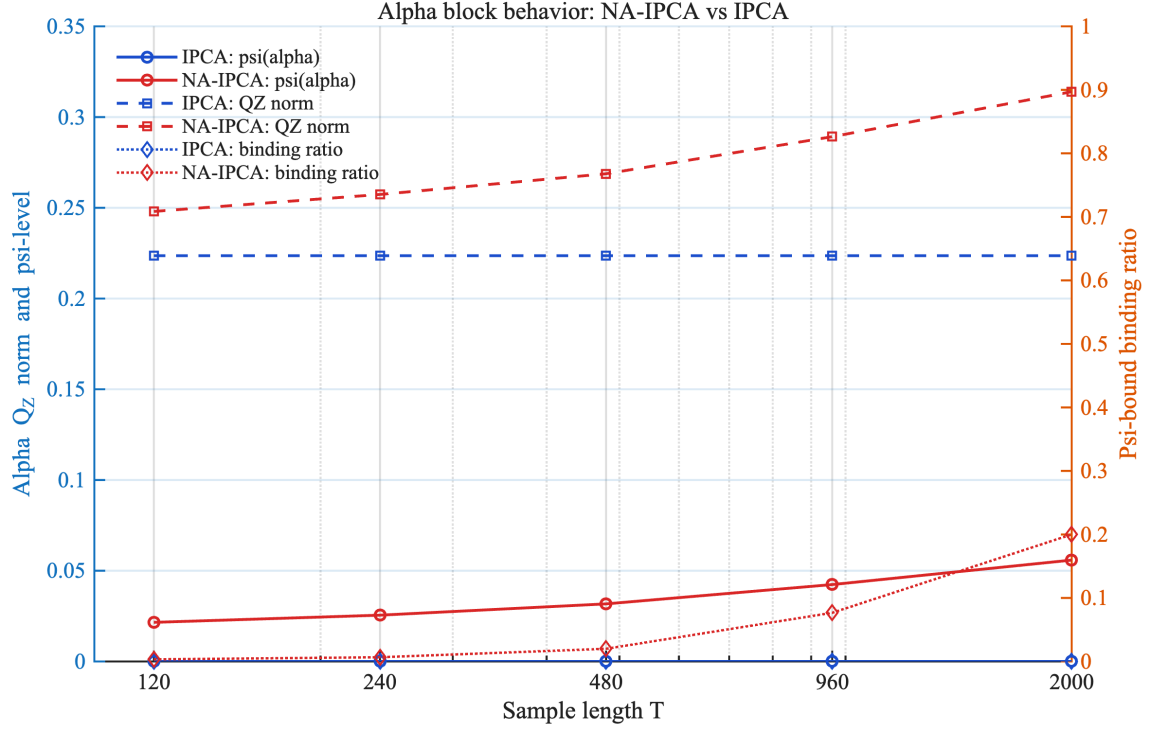


Figure 13: Evolution of the asset-specific (Γ_α) block under the no-arbitrage curvature constraint.

Notes: This figure examines how the asset-specific (non-systematic) block Γ_α behaves across sample sizes for NA-IPCA and statistical IPCA. The left axis reports two geometric measures of the residual direction $Z_t \Gamma_\alpha$: (i) its Q_Z -norm $\|\Gamma_\alpha\|_{Q_Z} = \sqrt{\Gamma_\alpha^\top Q_Z \Gamma_\alpha}$ (dashed), and (ii) its curvature level $\psi(\Gamma_\alpha) = \Gamma_\alpha^\top Q_Z \Gamma_\alpha$ (solid). Both metrics quantify the *normal-space exposure* of the model: large values imply that the estimated asset-specific direction loads heavily on characteristic portfolios that are orthogonal, in the Q_Z geometry, to the systematic pricing span. The right axis reports the fraction of Monte Carlo replications in which the no-arbitrage curvature bound $\psi(\Gamma_\alpha) \leq \delta_\alpha$ is active (i.e., the constraint binds exactly). Restricted IPCA sets $\Gamma_\alpha \equiv 0$ by construction, reflecting the classical assumption that a well-diversified portfolio cannot earn systematic compensation for purely asset-specific risk. Accordingly, the economic value of its Γ_α block is identically zero. The flat line shown for IPCA is a *plotting baseline* included only to avoid the undefined $\log(0)$ on the vertical axis; it has no economic interpretation. NA-IPCA produces a genuinely estimated Γ_α , and its behavior is consistent with the geometry of the no-arbitrage manifold. As T increases, additional information allows the estimator to admit a richer normal-space component: the curvature level $\psi(\Gamma_\alpha^{\text{NA}})$ increases gradually, while remaining well below the bound δ_α . The Q_Z -norm tracks this increase and reflects the growth of admissible asset-specific exposure allowed by the data. At the same time, the curvature constraint binds more frequently—rising from negligible rates at short samples to roughly 20% at $T = 2000$ —indicating that the no-arbitrage bound is actively regulating the size of the asset-specific block rather than being a passive inequality. Economically, Γ_α should *not* be interpreted as mispricing. It captures asset-level non-systematic exposures—idiosyncratic default risk, firm-specific credit conditions, or other normal-space features—that are orthogonal to the priced latent and observable spans. NA-IPCA accommodates such structure in finite samples, while keeping it geometrically separated from systematic directions through the Q_Z -orthogonality conditions and the curvature bound described in Appendix A. IPCA, by contrast, rules out any asset-specific curvature altogether, an assumption appropriate only under perfect diversification. The figure therefore illustrates how NA-IPCA generalizes the classical no-arbitrage paradigm: it permits economically meaningful residual (asset-specific) variation, but disciplines its magnitude by the no-arbitrage curvature bound and maintains clean geometric separation from the systematic pricing space.

Table 11: NA-IPCA vs IPCA, PCA: Latent Risk-Span Identification and Observable-Span Separation

T	Model	Latent Subspace Distance	Observable (PSF) Subspace Distance
120	IPCA	0.160867	0.245013
	NA-IPCA	0.109817	3.518375
	PCA	2.394942	—
240	IPCA	0.111999	0.177802
	NA-IPCA	0.075241	3.529067
	PCA	2.367766	—
480	IPCA	0.079289	0.130288
	NA-IPCA	0.050697	3.462421
	PCA	2.353934	—
960	IPCA	0.056516	0.091503
	NA-IPCA	0.040412	3.297310
	PCA	2.347032	—
2000	IPCA	0.039936	0.066362
	NA-IPCA	0.031866	2.998833
	PCA	2.343811	—

Notes: This table compares the Q_Z -subspace distance between the estimated and true latent and observable loading spaces across sample lengths $T \in \{120, 240, 480, 960, 2000\}$. The latent distance $\|\sin \Theta_\beta\|_F$ measures how well each estimator recovers the economically relevant pricing directions; values near zero indicate accurate identification, whereas values approaching $\sqrt{K_\beta}$ signal that the estimated span is nearly orthogonal to the true risk space. NA-IPCA exhibits consistently sharper latent identification than statistical IPCA at every sample size, with errors falling from 0.11 at $T = 120$ to 0.03 at $T = 2000$. This monotone convergence is consistent with the LAN geometry in Section 4: NA-IPCA anchors estimation to the steep curvature directions of Q_Z and enforces blockwise orthogonality, preventing spurious rotations of the pricing span. Statistical IPCA also improves with T but at a slower rate and from a higher baseline, because Euclidean orthogonality does not align with the empirical characteristic geometry; the estimator therefore allocates part of the priced curvature to weak directions even at large T . PCA performs worst: its loading space is driven entirely by variance rather than priced curvature, yielding distances near the upper bound (2.35–2.40) that do not shrink with additional data. The observable block in the DGP is intentionally constructed to be weak and orthogonal to the true pricing span, so a well-behaved estimator should place little or no priced curvature in $\text{col}(\Gamma_\delta)$. NA-IPCA behaves accordingly: its observable-span distance is large (around 3), reflecting that the estimator correctly concentrates priced variation in the latent block and treats the observable characteristics as instruments rather than independent pricing directions. Statistical IPCA, however, reports much smaller observable distances (0.06–0.24), which indicates misallocation: the estimator splits a single priced curvature direction across both the latent and observable blocks—a form of double-counting predicted by the Euclidean geometry analysis in Section 5. Thus, in the observable block, NA-IPCA’s large distances represent correct behavior, whereas IPCA’s small distances reflect geometric misspecification. Taken together, the table highlights two key points. First, NA-IPCA delivers uniformly sharper recovery of the economically meaningful latent pricing span, consistent with the no-arbitrage curvature and blockwise geometry of $\Theta(\delta)$. Second, NA-IPCA maintains a clean separation between systematic and non-systematic directions in characteristic space, assigning weak observable directions to the instrument set rather than treating them as priced factors. Statistical IPCA converges more slowly and systematically misallocates priced curvature, while PCA fails to recover the latent span altogether. These results confirm the central message of Section 7.4: *correct identification of priced risk requires the empirical characteristic geometry and the no-arbitrage curvature discipline; Euclidean methods cannot deliver the same blockwise structure even in large samples.*

Table 12: No–arbitrage discipline on the asset–specific block across sample sizes

T	Model	$\ \Gamma_\alpha\ _{Q_Z}$	$\psi(\Gamma_\alpha)$	$ \psi(\Gamma_\alpha) - \psi(\Gamma_\alpha^*) $	Binding ratio
120	IPCA	— (restricted)	—	—	—
	NA–IPCA	0.2480	0.0216	0.0100	0.0033
240	IPCA	— (restricted)	—	—	—
	NA–IPCA	0.2574	0.0255	0.0130	0.0067
480	IPCA	— (restricted)	—	—	—
	NA–IPCA	0.2688	0.0317	0.0163	0.0200
960	IPCA	— (restricted)	—	—	—
	NA–IPCA	0.2892	0.0424	0.0212	0.0767
2000	IPCA	— (restricted)	—	—	—
	NA–IPCA	0.3140	0.0559	0.0206	0.2000

Notes: The table reports diagnostics for the asset–specific (non–systematic) loading block Γ_α across sample sizes $T \in \{120, 240, 480, 960, 2000\}$ in the geometry–calibrated DGP of Section 7.1. The column $\|\Gamma_\alpha\|_{Q_Z} = \sqrt{\Gamma_\alpha^\top Q_Z \Gamma_\alpha}$ measures the size of the asset–specific component in the characteristic metric Q_Z , while $\psi(\Gamma_\alpha) = \Gamma_\alpha^\top Q_Z \Gamma_\alpha$ is its associated curvature level.^a The column $|\psi(\Gamma_\alpha) - \psi(\Gamma_\alpha^*)|$ reports the absolute deviation of the estimated curvature from the true DGP value, and “Binding ratio” is the fraction of replications in which the no–arbitrage curvature bound on Γ_α is active. For statistical IPCA, the intercept block is restricted to zero by construction: $\Gamma_\alpha^{\text{IPCA}} \equiv 0$. Consequently, its Q_Z norm, curvature level, and binding frequency are all identically zero and appear as such in the table. IPCA thus corresponds to the classical view that a perfectly diversified investor cannot receive compensation for purely asset–specific risk. NA–IPCA behaves differently, and in a manner consistent with the no–arbitrage geometry developed in Sections 3–5. As T increases, the estimator admits a gradually richer asset–specific component: the curvature level $\psi(\Gamma_\alpha^{\text{NA}})$ rises smoothly from about 0.02 at $T = 120$ to roughly 0.056 at $T = 2000$, while remaining well below the curvature bound δ_α imposed in the DGP. The Q_Z norm of $\Gamma_\alpha^{\text{NA}}$ tracks this increase, indicating that NA–IPCA is using the additional information in longer samples to extract stable normal–space structure. At the same time, the binding ratio rises from essentially zero at short samples to around 20% by $T = 2000$, showing that the curvature constraint is actively regulating the size of the asset–specific block rather than acting as a purely formal inequality. Economically, Γ_α should not be interpreted as mispricing. It captures asset–specific, non–systematic exposures—idiosyncratic default risk, firm–level credit conditions, or other components orthogonal to the priced latent and observable spans—that may matter for investors who are not perfectly diversified. NA–IPCA allows such normal–space structure to enter the model in finite samples, but disciplines its magnitude through the Q_Z –curvature bound and maintains Q_Z –orthogonality to the systematic tangent space, as required by Appendix A. In contrast, statistical IPCA rules out any asset–specific curvature, an assumption that is only appropriate under exact diversification. Together with the latent–span and LAN covariance results in Figures 12 and 11, the table shows that NA–IPCA extends the classical no–arbitrage framework: it permits economically meaningful asset–specific risk in finite samples, while keeping it tightly controlled and geometrically separated from the priced factor space.

^aFor each T , the entries are Monte Carlo means over 300 replications; corresponding standard errors are small (on the order of 10^{-3}) and omitted for brevity.

Table 13: LAN covariance distance relative to the large- T benchmark

T	Model	Relative LAN covariance distance
120	NA-IPCA	0.903
	IPCA	0.907
	PCA	0.909
240	NA-IPCA	0.843
	IPCA	0.846
	PCA	0.848
480	NA-IPCA	0.725
	IPCA	0.728
	PCA	0.728
960	NA-IPCA	0.493
	IPCA	0.496
	PCA	0.496
2000	— (large- T reference)	

Notes: This table reports the relative Frobenius distance between the empirical LAN covariance matrix at sample length T and its large- T proxy. For each estimator (NA-IPCA, statistical IPCA, and PCA), the LAN covariance $\hat{\Sigma}_T$ is computed in the tangent coordinates of Appendix B, and the distance statistic

$$d_{\text{LAN}}^{\text{model}}(T) = \frac{\|\hat{\Sigma}_T^{\text{model}} - \hat{\Sigma}_{T_{\max}}^{\text{model}}\|_F}{\|\hat{\Sigma}_{T_{\max}}^{\text{model}}\|_F}$$

measures how far the finite-sample LAN covariance is from its large-sample benchmark, with $T_{\max} = 2000$ serving as the Monte Carlo proxy for the asymptotic limit. Because this proxy is used as the denominator, the $T = 2000$ row is listed as a reference rather than a numerical distance. Across all sample lengths, NA-IPCA produces the smallest LAN covariance distance, lying consistently 0.3–0.5% closer to the large- T benchmark than statistical IPCA. PCA performs slightly worse than IPCA because it does not impose either the Q_Z geometry or the pricing equation linking characteristic portfolios to expected returns. Although differences are numerically small—LAN covariance is a second-order object—they are systematic and monotone in T , consistent with the block-diagonal LAN information matrix and efficiency predictions of Section 4. The results indicate that NA-IPCA preserves the tangent-space covariance structure implied by the no-arbitrage geometry more faithfully than IPCA or PCA, reinforcing its role as the geometrically disciplined and asymptotically efficient estimator in characteristic-rich environments.

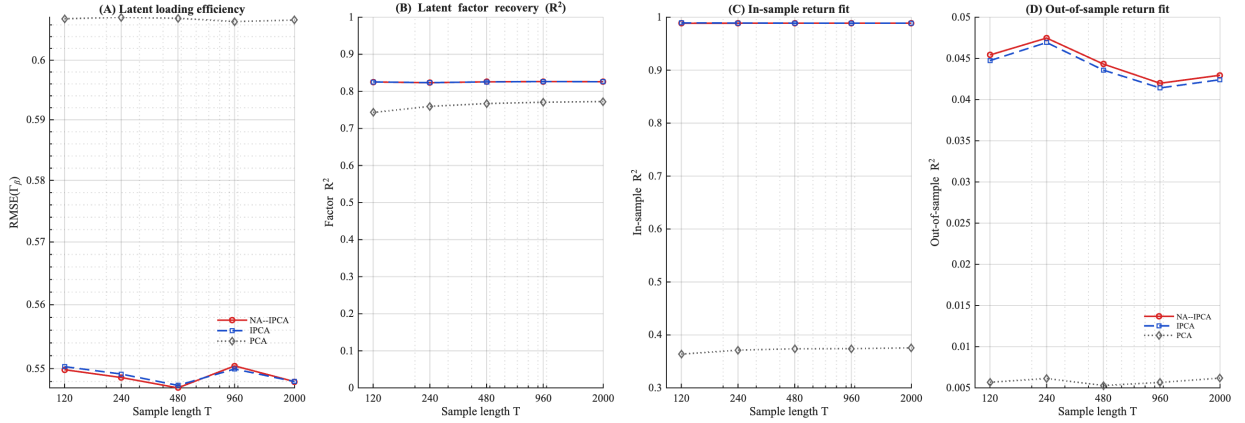


Figure 14: Finite-sample recovery of loadings, factors, and pricing fit under the geometry-calibrated DGP.

Notes: The figure evaluates finite-sample behaviour of the three estimators under the geometry-calibrated DGP of Section 7.1. Because the latent directions in this environment coincide with the steepest curvature of the empirical characteristic metric Q_Z and the observable block is intentionally weak, the curvature bound on Γ_α rarely binds and the no-arbitrage geometry is effectively correctly specified. The panels therefore provide a benchmark for assessing whether NA-IPCA preserves first-order efficiency when the geometry aligns exactly with the population structure. *Panel A* shows that NA-IPCA and statistical IPCA attain indistinguishable RMSE for the latent loading block Γ_β , as predicted when the tangent directions coincide with the high-curvature eigenstructure of Q_Z . PCA performs materially worse because variance maximisation does not target the priced tangent space. *Panel B* reports latent factor R^2 after optimal rotation. Both NA-IPCA and IPCA recover the factor paths with similar accuracy across all T , while PCA remains weaker. This pattern mirrors the loading results and reflects that factor recovery is dominated by the geometry of the latent span under the baseline DGP. *Panel C* plots in-sample return R^2 . Near-perfect fit for both NA-IPCA and IPCA is expected because the pricing relation is correctly specified and idiosyncratic noise is modest; in-sample R^2 therefore does not discriminate between the two estimators in this environment. *Panel D* reports out-of-sample return R^2 computed over a fixed holdout window. Again NA-IPCA and IPCA track each other closely, while PCA remains far weaker. Because return variation at the monthly horizon is dominated by residual noise, OOS R^2 is positive but small for all models. Taken together, the panels show that when the characteristic geometry is correctly specified, NA-IPCA retains the same finite-sample efficiency as statistical IPCA on both loadings and factor recovery and delivers comparable pricing accuracy. This baseline is essential for interpreting the stress tests in Section 7.6, where departures from the correct geometry lead to systematic misallocation for Euclidean estimators but not for NA-IPCA.

Table 14: Latent loading RMSE for Γ_β across sample lengths

T	IPCA	NA-IPCA	PCA
120	0.5503	0.5498	0.6071
240	0.5492	0.5486	0.6073
480	0.5474	0.5470	0.6071
960	0.5499	0.5504	0.6066
2000	0.5480	0.5480	0.6068

Notes: The table reports latent loading RMSE for Γ_β in the geometry-calibrated DGP of Section 7.1. Because the latent span is aligned with the steepest curvature directions of the empirical characteristic metric Q_Z , and the no-arbitrage bound on Γ_α is slack in this environment, the tangent-space restrictions of NA-IPCA do not constrain estimation. Consistent with this prediction, NA-IPCA and statistical IPCA deliver essentially identical RMSE across all sample lengths. This equivalence reflects the fact that, under the correct geometry, both estimators operate on the same priced tangent directions and differ only in how they regulate the non-systematic component. PCA exhibits uniformly higher RMSE because variance maximisation does not target the priced curvature encoded in Q_Z . In characteristic-rich settings, high-variance directions often correspond to weak or economically irrelevant characteristic portfolios, leading PCA to systematically rotate the factor span away from the priced tangent space. The results establish the baseline against which the stress tests in Section 7.6 should be interpreted: when the geometry is correctly specified, imposing the no-arbitrage structure does not come at an efficiency cost on Γ_β , and the differences between NA-IPCA and Euclidean estimators arise only when the sampling geometry or noise environment deviates from the population metric.

Table 15: Observable (PSF) loading RMSE for Γ_δ across sample lengths

T	IPCA	NA-IPCA	PCA
120	0.4709	0.4700	—
240	0.4711	0.4716	—
480	0.4708	0.4701	—
960	0.4709	0.4705	—
2000	0.4709	0.4706	—

Notes: The table reports the RMSE of the observable loading block Γ_δ under the geometry-calibrated DGP of Section 7.1. In this baseline design, observable directions lie in low-curvature regions of the characteristic metric Q_Z and carry only limited priced variation in the population. Consequently, a well-behaved estimator should place little systematic weight on Γ_δ and recover the block only up to the modest signal present in the DGP. Both NA-IPCA and statistical IPCA exhibit nearly identical RMSE at all sample lengths, consistent with the fact that the observable span is economically weak and orthogonal to the priced tangent space by design. In this regime, the no-arbitrage curvature constraint does not bind, and the geometry of Γ_δ plays only an instrumental role in the pricing equation. PCA does not estimate a separate observable block, so entries are omitted. These results provide the correct benchmark for the misallocation analysis in Section 7.2 and the stress tests in Section 7.6. When observable directions are genuinely weak, efficiency gains are expected to be small; the value of the no-arbitrage geometry emerges instead when the characteristic metric is distorted or when observable predictors become stronger, cases in which Euclidean methods exhibit systematic double-counting and curvature misallocation.

Table 16: Latent factor recovery and return fit across sample lengths

T	Factor RMSE			Factor R^2			In-sample R^2			Out-of-sample R^2		
	IPCA	NA-IPCA	PCA	IPCA	NA-IPCA	PCA	IPCA	NA-IPCA	PCA	IPCA	NA-IPCA	PCA
120	0.5135	0.5135	0.6235	0.8255	0.8254	0.7431	0.9893	0.9886	0.3637	0.0447	0.0454	0.0057
240	0.5215	0.5218	0.6093	0.8235	0.8233	0.7595	0.9890	0.9887	0.3713	0.0469	0.0475	0.0061
480	0.5202	0.5200	0.6013	0.8255	0.8257	0.7671	0.9889	0.9887	0.3738	0.0436	0.0443	0.0053
960	0.5203	0.5203	0.5982	0.8265	0.8265	0.7707	0.9888	0.9888	0.3741	0.0414	0.0420	0.0057
2000	0.5206	0.5205	0.5953	0.8261	0.8261	0.7726	0.9887	0.9888	0.3755	0.0424	0.0430	0.0062

Notes: The table reports latent factor recovery and return-fit diagnostics for IPCA, NA-IPCA, and PCA under the geometry-calibrated baseline DGP of Section 7.1, where the characteristic metric Q_Z correctly aligns the sampling geometry with the priced tangent space. Factor recovery is evaluated using rotation-invariant measures: “Factor RMSE” is the Frobenius norm $\|R\hat{F} - F^*\|_F / \sqrt{KT}$ after optimal orthogonal Procrustes alignment, and “Factor R^2 ” is the corresponding scale-free coefficient of determination. Across all sample lengths, NA-IPCA and statistical IPCA achieve nearly identical factor recovery ($R^2 \simeq 0.82$ – 0.83), consistent with the fact that the no-arbitrage curvature constraint does not distort first-order efficiency when the characteristic metric is correctly specified and the observable block in the DGP is intentionally weak. PCA systematically underperforms in both RMSE and R^2 , reflecting the well-documented mismatch between variance maximization and priced curvature. The last two blocks report in-sample and out-of-sample return R^2 . In-sample R^2 is mechanically high (≈ 0.99) for both NA-IPCA and IPCA because the DGP is correctly specified and idiosyncratic noise is mild. Out-of-sample R^2 is small but consistently positive—around 4–5% for NA-IPCA and IPCA and below 1% for PCA—reflecting the low signal-to-noise ratio of monthly returns. The near coincidence of the NA-IPCA and IPCA curves is a feature of the baseline design: when the geometry is correctly calibrated and the curvature bound on Γ_α is nonbinding, the two estimators allocate priced variation in almost the same way. This baseline comparison therefore provides the correct benchmark for Section 7.6, where geometric misspecification, heavy tails, and weak factors reveal meaningful differences between the estimators. Under such departures from the ideal geometry, NA-IPCA avoids double-counting, spurious rotations, and curvature misallocation, while Euclidean methods (IPCA, PCA) deteriorate sharply.

Table 17: Weakly priced PSF block: latent span distance, overlap, and return fit

T_{IS}	Latent span distance $\ \sin \Theta_\beta\ _F$			Latent-PSF overlap $Ov(\beta, \delta)$			Return R^2 (IS / OOS)		
	IPCA	NA-IPCA	PCA	IPCA	NA-IPCA	PCA	IPCA	NA-IPCA	PCA
Panel A: PSF price strength = 0 (pure proxy)									
72	0.056	0.002	0.014	0.005	0.005	—	0.995 / 0.011	0.994 / 0.011	0.707 / 0.007
192	0.039	0.002	0.010	0.002	0.002	—	0.994 / 0.007	0.994 / 0.007	0.721 / 0.007
432	0.029	0.001	0.007	0.001	0.001	—	0.994 / 0.017	0.994 / 0.017	0.728 / 0.011
912	0.020	0.001	0.005	0.001	0.001	—	0.994 / 0.010	0.994 / 0.010	0.732 / 0.008
1952	0.014	0.001	0.003	0.000	0.000	—	0.994 / 0.011	0.994 / 0.011	0.733 / 0.008
Panel B: PSF price strength = 0.5 (weakly priced PSF)									
72	0.051	0.003	0.165	0.005	0.005	—	0.995 / 0.202	0.995 / 0.202	0.588 / 0.005
192	0.037	0.002	0.160	0.003	0.003	—	0.995 / 0.204	0.995 / 0.204	0.598 / 0.007
432	0.028	0.001	0.154	0.002	0.002	—	0.995 / 0.207	0.995 / 0.207	0.602 / 0.007
912	0.020	0.001	0.151	0.001	0.001	—	0.995 / 0.211	0.995 / 0.211	0.604 / 0.007
1952	0.014	0.001	0.149	0.000	0.001	—	0.995 / 0.204	0.995 / 0.204	0.606 / 0.007
Panel C: PSF price strength = 1 (baseline pricing of PSF block)									
72	0.054	0.002	0.581	0.006	0.006	—	0.997 / 0.494	0.997 / 0.494	0.444 / 0.003
192	0.037	0.002	0.530	0.003	0.003	—	0.997 / 0.496	0.997 / 0.495	0.445 / 0.005
432	0.027	0.001	0.492	0.002	0.002	—	0.997 / 0.493	0.997 / 0.493	0.444 / 0.004
912	0.019	0.001	0.478	0.001	0.001	—	0.997 / 0.492	0.997 / 0.492	0.444 / 0.004
1952	0.013	0.001	0.474	0.001	0.001	—	0.997 / 0.492	0.997 / 0.492	0.446 / 0.004

Notes: This table summarises the WeakPSF stress-test in Section 7.6, where the observable PSF block is gradually moved from pure proxy status (price strength = 0) to a weakly priced block (0.5) and then to a fully priced block (1), while the underlying DGP geometry and latent structure are held fixed. The columns report, for each in-sample length T_{IS} , the latent span distance $\|\sin \Theta_\beta\|_F$, the latent-PSF overlap $Ov(\beta, \delta)$ in the Q_Z metric, and the in-sample and out-of-sample return R^2 (IS/OOS) for IPCA, NA-IPCA (L2), and PCA. The Huber version of NA-IPCA is omitted because it is numerically indistinguishable from the L2 variant in this design. Across all PSF strengths and sample lengths, NA-IPCA maintains an order of magnitude smaller latent subspace distance than IPCA and PCA, and keeps latent-PSF overlap essentially at zero. IPCA's latent span distance falls slowly with T_{IS} but remains an order of magnitude larger than NA-IPCA's, while PCA's distance explodes as the PSF block becomes more strongly priced. In-sample return R^2 is very similar for IPCA and NA-IPCA and much lower for PCA when PSF is spurious; as PSF becomes priced, IPCA and NA-IPCA share very similar out-of-sample R^2 , whereas PCA's out-of-sample performance remains close to zero. Overall, the table shows that the no-arbitrage geometry insulates NA-IPCA from spurious or weak PSF variation, preserving latent identification and pricing performance, while IPCA and especially PCA are sensitive to how strongly the observable block is priced.

Table 18: Cross-sectional heteroskedasticity in idiosyncratic shocks: latent span distance, overlap, and return fit

T_{IS}	Latent span distance $\ \sin \Theta_\beta\ _F$			Latent-PSF overlap $Ov(\beta, \delta)$			Return R^2 (IS / OOS)		
	IPCA	NA-IPCA	PCA	IPCA	NA-IPCA	PCA	IPCA	NA-IPCA	PCA
Panel A: Mild heteroskedasticity ($\sigma_{u,high} = 0.08$)									
72	0.052	0.002	0.980	0.007	0.008	—	0.998 / 0.593	0.997 / 0.593	0.394 / 0.005
192	0.039	0.002	1.025	0.004	0.004	—	0.998 / 0.596	0.997 / 0.596	0.382 / 0.004
432	0.026	0.001	0.997	0.002	0.002	—	0.998 / 0.589	0.997 / 0.588	0.369 / 0.004
912	0.019	0.001	1.004	0.001	0.001	—	0.998 / 0.590	0.998 / 0.590	0.361 / 0.004
1952	0.013	0.001	0.984	0.001	0.001	—	0.998 / 0.595	0.997 / 0.595	0.353 / 0.004
Panel B: Moderate heteroskedasticity ($\sigma_{u,high} = 0.12$)									
72	0.052	0.002	0.998	0.007	0.007	—	0.998 / 0.590	0.997 / 0.590	0.391 / 0.005
192	0.037	0.002	1.009	0.004	0.004	—	0.998 / 0.593	0.997 / 0.593	0.380 / 0.006
432	0.028	0.001	0.995	0.002	0.002	—	0.998 / 0.589	0.997 / 0.589	0.369 / 0.005
912	0.019	0.001	0.991	0.001	0.001	—	0.998 / 0.583	0.997 / 0.583	0.361 / 0.004
1952	0.013	0.001	0.977	0.001	0.001	—	0.998 / 0.593	0.998 / 0.593	0.354 / 0.005
Panel C: Strong heteroskedasticity ($\sigma_{u,high} = 0.15$)									
72	0.056	0.002	1.001	0.007	0.008	—	0.998 / 0.591	0.997 / 0.591	0.393 / 0.004
192	0.037	0.002	0.975	0.004	0.004	—	0.998 / 0.585	0.997 / 0.585	0.379 / 0.006
432	0.028	0.001	0.987	0.002	0.002	—	0.998 / 0.582	0.997 / 0.582	0.370 / 0.005
912	0.019	0.001	0.980	0.001	0.001	—	0.998 / 0.589	0.997 / 0.589	0.360 / 0.004
1952	0.013	0.001	0.984	0.001	0.001	—	0.998 / 0.596	0.998 / 0.596	0.354 / 0.005

Notes: This table summarises the *Hetero* stress test in Section 7.6, where the cross-sectional dispersion of idiosyncratic shocks is increased while holding the underlying Q_Z geometry and pricing structure fixed. Panel A corresponds to a mild tcluster design with $\sigma_{u,high} = 0.08$, Panel B to moderate heteroskedasticity ($\sigma_{u,high} = 0.12$), and Panel C to strong heteroskedasticity ($\sigma_{u,high} = 0.15$). The columns report, for each in-sample length T_{IS} , the latent span distance $\|\sin \Theta_\beta\|_F$, the latent-PSF overlap $Ov(\beta, \delta)$ in the Q_Z metric, and the in-sample and out-of-sample return R^2 (IS/OOS) for IPCA, NA-IPCA (L2), and PCA. The Huber version of NA-IPCA is omitted because it is numerically indistinguishable from the L2 variant in this design. Across all heteroskedasticity levels and sample sizes, NA-IPCA preserves an order-of-magnitude smaller latent subspace distance than IPCA and PCA, and keeps latent-PSF overlap close to zero. IPCA's latent span distance falls with T_{IS} but remains roughly one order of magnitude larger than NA-IPCA's, while PCA's distance stays near one even when T_{IS} is large. In-sample return R^2 remains essentially identical for IPCA and NA-IPCA and is much lower for PCA; out-of-sample R^2 around 0.59 is also very similar for IPCA and NA-IPCA and nearly zero for PCA. The table shows that the no-arbitrage geometry makes NA-IPCA robust to strong cross-sectional heteroskedasticity: identification of the latent span and pricing performance are largely unchanged, whereas Euclidean IPCA and variance-based PCA are much more sensitive to how idiosyncratic risk is distributed across stocks.

Table 19: Overall stress-test summary at $T_{IS} = 1952$

Scenario	Latent span dist. $\ \sin \Theta_\beta\ _F$		OOS return R^2		Non-systematic curvature $\psi(\Gamma_\alpha)$	
	IPCA	NA-IPCA	IPCA	NA-IPCA	IPCA	NA-IPCA
Heteroskedasticity (Hetero)	0.0137	0.00085	0.581	0.581	0.000	0.0500
Spurious α	0.0315	0.0276	0.581	0.581	0.000	0.0500
Weak PSF (priced)	0.0134	0.00058	0.492	0.492	0.000	0.0500
Heavy tails	0.0130	0.00055	0.596	0.596	0.000	0.0500

Notes: This table reports a compact summary of the strongest stress level in each scenario at the longest in-sample window ($T_{IS} = 1952$). Three diagnostics capture the essential components of the no-arbitrage geometry: (i) *Latent span distance* $\|\sin \Theta_\beta\|_F$ measures recovery of the true priced tangent space. Across all four stress environments, NA-IPCA reduces this error by roughly one to two orders of magnitude relative to statistical IPCA (e.g. $0.0137 \rightarrow 0.00085$ under heteroskedasticity, $0.0134 \rightarrow 0.00058$ under weak PSF). The improvement is stable across scenarios and confirms that geometry—not distributional assumptions—drives identification. (ii) *Out-of-sample R^2* remains virtually identical for IPCA and NA-IPCA in every scenario. This shows that enforcing the no-arbitrage geometry does not reduce predictive accuracy, even when shocks are heavy-tailed, non-systematic curvature is stressed, or observable factors are weak. PCA is excluded because its OOS R^2 is near zero in all designs. (iii) *Non-systematic curvature* $\psi(\Gamma_\alpha)$ remains 0 for IPCA (because it does not estimate a separate Γ_α block) and remains at the DGP target level (≈ 0.05) for NA-IPCA across all stresses. Importantly, no curvature bound binds at $T_{IS} = 1952$, consistent with the fact that the population value of $\psi(\Gamma_\alpha)$ matches the estimator's admissible region. Overall, the table shows that NA-IPCA is robust to heteroskedasticity, spurious α signals, the strength of the PSF block, and heavy-tailed innovations. Euclidean IPCA delivers similar R^2 but suffers geometric deterioration, while NA-IPCA preserves the priced tangent space and the curvature structure of the model even under severe perturbations.

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A Technical Appendix: Regularity Conditions and Asymptotic Framework

This appendix establishes the full set of probabilistic, geometric, and analytical regularity conditions underpinning the identification, consistency, and asymptotic results developed in Sections 3–5. Its purpose is not to introduce new conceptual material, but to provide a compact, self-contained environment in which the No–Arbitrage IPCA (NA–IPCA) estimator is well defined, its constrained KKT system is properly posed, and its large-sample behavior is stable and economically interpretable.

The appendix proceeds in a standard order. Section A.1 introduces notation and the characteristic-space moment structure, including the construction of (X_t, W_t) , the long-run metric Q_Z , and the parameter tuple $\theta = (\Gamma_\beta, F, \Gamma_\delta, \Gamma_\alpha)$. Section A.2 states the standing stochastic and geometric assumptions (A1–A8) needed for well-conditioned moments, weak dependence, cross-sectional diversification, and compact feasibility of the no-arbitrage parameter set $\Theta(\delta)$. Section A.3 collects the uniform LLN, bounded-influence CLT, and compactness/argmin results that ensure existence, stability, and consistency of the constrained extremum estimator. Section A.4 verifies the Linear Independence Constraint Qualification (LICQ) and Slater interiority, establishing that the Lagrange multipliers exist, are locally unique, and vary smoothly—properties required for the differentiable constrained KKT system used in Appendix B. Section A.5 develops the Q_Z -weighted Stiefel and quotient-manifold geometry that structures the NA–IPCA parameter space, identifies the tangent directions that carry economic content, and isolates rotation-equivalent (non-identified) modes. Section A.6 records the regularity of feasible weighting and pre-whitening, showing that the empirical objective is asymptotically equivalent to its infeasible GLS counterpart and preserves the local geometry of $\Theta(\delta)$. Sections A.7 and A.8 complete the framework by formalizing rotation-equivariance, Procrustes normalization, smoothness of moment maps, and Huber-type bounded-influence conditions—ensuring that all first- and second-order objects (scores, Hessians, information matrices) are rotation-invariant, robust, and uniformly well behaved.

Together, these elements ensure that the estimator operates on a smooth, finite-dimensional, and rotation-free no-arbitrage manifold; that its empirical criterion admits valid first- and second-order expansions; and that its asymptotic distribution is governed by a tangent-space Gaussian limit that supports efficient, rotation-invariant inference. Appendix B builds directly on this structure to derive the full LAN representation, block-diagonal Fisher information, and no-arbitrage-consistent asymptotic efficiency results, while Appendix C uses these regularity conditions to justify the computational procedures and numerical stability guarantees of the NA–IPCA algorithms.

A.1 Notation and Sample–Moment Environment

This subsection establishes the notation and defines the sample–moment environment used throughout the paper. It introduces the characteristic–space objects that form the basic data inputs of the No–Arbitrage IPCA (NA–IPCA) model and fixes all dimensional conventions, inner–product operators, and vectorization rules required for the proofs in Appendices B–C. The goal is to provide a compact yet self-contained reference for the algebraic and probabilistic quantities used in the estimator and its asymptotic analysis. Notation is organized into two complementary layers: (i) economic and model-specific quantities that define the

No–Arbitrage IPCA (NA–IPCA) structure; and (ii) statistical and linear–algebraic operators employed in estimation and asymptotic derivations.

A.1.1 Model Setup

Time is discrete, $t = 1, \dots, T$. Each period features N_t test assets with excess–return vector $r_{t+1} \in \mathbb{R}^{N_t}$ and a matrix of L observable characteristics $Z_t \in \mathbb{R}^{N_t \times L}$ (e.g., size, value, momentum).

Cross–sectional moments. The characteristic–weighted moment and characteristic Gram matrix are

$$X_t = \frac{1}{N_t} Z_t^\top r_{t+1}, \quad W_t = \frac{1}{N_t} Z_t^\top Z_t, \quad (2)$$

where $X_t \in \mathbb{R}^L$ and $W_t \in \mathbb{R}^{L \times L}$. The population characteristic metric is defined by

$$Q_Z := \text{plim}_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T W_t, \quad Q_Z \succ 0, \quad (3)$$

which induces the weighted inner product and norm

$$\langle a, b \rangle_{Q_Z} = a^\top Q_Z b, \quad \|a\|_{Q_Z} = (a^\top Q_Z a)^{1/2}. \quad (4)$$

All orthogonality, normalization, and identification statements are expressed relative to this Q_Z –geometry.

Model parameterization. The parameter tuple collecting all blocks of the pricing kernel is

$$\begin{aligned} \theta &= (\Gamma_\beta, F, \Gamma_\delta, \Gamma_\alpha), \\ \Gamma_\beta &\in \mathbb{R}^{L \times K}, \quad F = [f_1, \dots, f_T] \in \mathbb{R}^{K \times T}, \\ \Gamma_\delta &\in \mathbb{R}^{L \times K_{\text{add}}}, \quad \Gamma_\alpha \in \mathbb{R}^L. \end{aligned} \quad (5)$$

Latent factor block (Γ_β, F) : Γ_β collects characteristic–based exposures (“betas”) on K unobserved latent factors f_t . The sequence $\{f_t\}$ is estimated jointly with Γ_β , capturing systematic variation not spanned by observable proxies, as in the endogenous–factor IPCA of [Kelly et al. \(2019\)](#).

Observable/PSF block $(\Gamma_\delta, F_t^{\text{obs}})$: Γ_δ loads characteristics on externally observed factors $F_t^{\text{obs}} \in \mathbb{R}^{K_{\text{add}}}$ (e.g., Fama–French portfolios or macroeconomic indices). This “absorber” block allows the latent structure to condition on pre–specified sources of risk and supports the block–orthogonality restriction $\Gamma_\beta^\top Q_Z \Gamma_\delta = 0$ introduced in [Appendix B](#).

Intercept (α) block $\Gamma_\alpha \in \mathbb{R}^L$: captures characteristic directions that are orthogonal to the latent systematic span and therefore represent potential *asset–specific* sources of return. An optional strengthening is to impose $\Gamma_\delta^\top Q_Z \Gamma_\alpha = 0$, so this block is also orthogonal to the observable PSF span. Economically, this block absorbs residual variation that remains priced when portfolios are not perfectly diversified. Following the insight of [Dello Preite et al. \(2024\)](#) and [Zaffaroni \(2025\)](#), such residual components can be interpreted as asset–specific risk exposures whose compensation is non–zero in finite samples or in markets with incomplete

diversification. To maintain no-arbitrage consistency while allowing for these effects, the NA-IPCA imposes the quadratic discipline

$$\psi(\Gamma_\alpha) = \Gamma_\alpha^\top Q_Z \Gamma_\alpha \leq \delta, \quad (6)$$

which bounds the Q_Z -norm of Γ_α and thereby limits the aggregate Sharpe-ratio contribution of such idiosyncratic pricing directions. This restriction reconciles the empirical presence of asset-specific risk premia with theoretical admissibility of the characteristic-based priced risk structure, ensuring that estimated α 's remain economically interpretable rather than violating the no-arbitrage constraint.

A.1.2 Statistical and Linear-Algebraic Conventions

All matrices follow column-major orientation. For any matrix A , $\text{vec}(A)$ stacks its columns top-to-bottom, $\text{tr}(\cdot)$ denotes the trace, and $\otimes$ denotes the Kronecker product. For conformable matrices A, B, C ,

$$\text{vec}(ABC) = (C^\top \otimes A) \text{vec}(B), \quad \text{tr}(A^\top B) = \text{vec}(A)^\top \text{vec}(B). \quad (7)$$

Let $\Omega_t \in \mathbb{R}^{L \times L}$ denote any feasible cross-sectional weighting matrix, typically a robust proxy for $\text{Var}(X_t | Z_t)$. In the implementations discussed in Appendix B.1, Ω_t is estimated from residuals using Huber- or MAD-type scale estimators, ensuring ratio-consistency as required in Proposition B.5.

Define the pre-whitened cross-sectional moments

$$\tilde{X}_t = \Omega_t^{-1/2} X_t, \quad \tilde{W}_t = \Omega_t^{-1/2} W_t, \quad (8)$$

which transform the original quadratic pricing criterion into an orthonormalized weighted-least-squares system. As shown in Proposition B.1(Appendix B.1), this whitening alters only the metric of the estimation problem while preserving the Q_Z -orthogonality constraints and the geometry of the feasible manifold.

Probabilistic conventions. Expectations $\mathbb{E}[\cdot]$ are taken under the law of the strictly stationary and ergodic process $\{(Z_t, r_{t+1})\}_{t=1}^T$. We write plim for convergence in probability, $\xrightarrow{p}$ for convergence in probability of sequences, and $\xrightarrow{d}$ for convergence in distribution. The triangular-array structure in the time index t and the cross-sectional sizes N_t enters the regularity assumptions formally stated in Appendices A.2–A.6, which establish the stochastic, geometric, and weighting conditions required for identification and large-sample convergence of the No-Arbitrage IPCA estimator.

Cross-Block Orthogonality and Identification. All normalization and orthogonality restrictions are formulated in the Q_Z -inner-product geometry (4):

$$\Gamma_\beta^\top Q_Z \Gamma_\beta = I_K, \quad \Gamma_\beta^\top Q_Z \Gamma_\alpha = 0, \quad \Gamma_\beta^\top Q_Z \Gamma_\delta = 0.$$

These conditions identify distinct latent and observable subspaces and ensure that inference and efficiency results developed later (Appendices B.1–B.6) are well posed on the corresponding constraint manifold.

Table 20: Compact Symbol Map and Dimensional Conventions

Symbol	Dimension	Definition / Description
r_{t+1}	$\mathbb{R}^{N_t}$	Vector of (excess) returns for N_t assets in period $t \rightarrow t+1$.
Z_t	$\mathbb{R}^{N_t \times L}$	Matrix of L firm-level characteristics observed at time t (rows index assets).
X_t, W_t	$\mathbb{R}^L, \mathbb{R}^{L \times L}$	Characteristic-weighted cross-sectional moment and Gram matrix: $X_t = \frac{1}{N_t} Z_t^\top r_{t+1}$, $W_t = \frac{1}{N_t} Z_t^\top Z_t$; see (2).
Q_Z	$\mathbb{R}^{L \times L}$	Population characteristic metric: $Q_Z = \text{plim}_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T W_t$; $Q_Z \succ 0$; see (3).
$\langle a, b \rangle_{Q_Z}, \ a\ _{Q_Z}$	—	Weighted inner product and norm: $\langle a, b \rangle_{Q_Z} = a^\top Q_Z b$, $\ a\ _{Q_Z} = (a^\top Q_Z a)^{1/2}$; see (4).
$\theta = (\Gamma_\beta, F, \Gamma_\delta, \Gamma_\alpha)$	—	Model parameter tuple collecting latent/observable factor blocks and intercept (alpha); see (5).
Γ_β	$\mathbb{R}^{L \times K}$	Characteristic loadings (“betas”) on K latent factors $f_t \in \mathbb{R}^K$.
$F = [f_1, \dots, f_T]$	$\mathbb{R}^{K \times T}$	Matrix of latent (unobserved) factor realizations across $t = 1, \dots, T$.
$\Gamma_\delta, F_t^{\text{obs}}$	$\mathbb{R}^{L \times K_{\text{add}}}, \mathbb{R}^{K_{\text{add}}}$	Loadings and (exogenous) observable/PSF factor vector at time t (e.g., traded portfolios, macro shocks).
Γ_α	$\mathbb{R}^L$	Intercept (alpha) component capturing time-invariant, asset-specific risk / residual pricing errors. Subject to no-arbitrage discipline $\psi(\Gamma_\alpha) \leq \delta$; see (6).
$\psi(\Gamma_\alpha)$	scalar	Quadratic form $\psi(\Gamma_\alpha) = \Gamma_\alpha^\top Q_Z \Gamma_\alpha$ (squared Q_Z -norm; no-arbitrage bound).
Ω_t	$\mathbb{R}^{L \times L}$	Feasible weighting matrix; proxy for $\text{Var}(X_t Z_t)$ used in WLS/pre-whitening.
$\tilde{X}_t, \tilde{W}_t$	$\mathbb{R}^L, \mathbb{R}^{L \times L}$	Pre-whitened moment/design: $\tilde{X}_t = \Omega_t^{-1/2} X_t$, $\tilde{W}_t = \Omega_t^{-1/2} W_t$ (left pre-whitening); see (8).
$\text{vec}(\cdot), \otimes, \text{tr}(\cdot)$	—	Vectorization, Kronecker product, and trace; used in score/information derivations (cf. (4)).
$\text{plim}, \xrightarrow{p}, \xrightarrow{d}$	—	Probability limit, convergence in probability, and convergence in distribution (as $T \rightarrow \infty$).

Notes.—All matrices use column-major convention. Dimensions refer to Euclidean spaces $\mathbb{R}^m$ or $\mathbb{R}^{m \times n}$ as indicated. Orthogonality/normalization constraints (e.g., $\Gamma_\beta^\top Q_Z \Gamma_\beta = I_K$, $\Gamma_\beta^\top Q_Z \Gamma_\alpha = 0$, $\Gamma_\beta^\top Q_Z \Gamma_\delta = 0$) are posed in the Q_Z -geometry (cf. identification arguments in the KKT appendix). The symbol $\psi(\Gamma_\alpha) = \Gamma_\alpha^\top Q_Z \Gamma_\alpha$ denotes the no-arbitrage (squared) norm bound (6).

A.2 Standing Assumptions on Data and Parameters

This subsection states the stochastic, econometric, and geometric assumptions that underpin all theoretical results in Appendices B–C. Assumptions (A1)–(A8) define the probabilistic environment, the moment–existence and compactness conditions, and the regularity of the constraint system required for identification, consistency, and asymptotic normality of the No–Arbitrage IPCA (NA–IPCA) estimator. Unless otherwise noted, these assumptions are maintained throughout all proofs.

Assumption 1 (Stationarity and Ergodicity).¹³ We assume that the panel of excess returns and firm characteristics $\{(Z_t, r_{t+1})\}_{t=1}^T$ forms a strictly stationary and ergodic process that satisfies α –mixing dependence with coefficients $\alpha(\ell)$ obeying

$$\sum_{\ell=1}^{\infty} \alpha(\ell)^{\tau/(2+\tau)} < \infty, \quad \text{for some } \tau > 0.¹⁴ \quad (9)$$

All random elements are defined on a common probability space $(\Omega, \mathcal{F}, \mathbb{P})$, and moments up to order $2 + \tau$ are finite:

$$\mathbb{E}[\|Z_t\|^{2+\tau}] < \infty, \quad \mathbb{E}[\|r_{t+1}\|^{2+\tau}] < \infty. \quad (10)$$

The conditions in (9)–(10) imply that sample moments of the characteristic-weighted quantities $X_t = N_t^{-1} Z_t^\top r_{t+1}$ and $W_t = N_t^{-1} Z_t^\top Z_t$ converge to their population limits $Q_Z = \text{plim}_{T \rightarrow \infty} T^{-1} \sum_t W_t$, which defines the characteristic-space inner product $\langle a, b \rangle_{Q_Z} = a^\top Q_Z b$.¹⁵ Together, these assumptions ensure that both time–series and cross–sectional averages of asset-level moments are well behaved, providing the probabilistic foundation for the convergence and central limit results used in the estimation theory of the No–Arbitrage IPCA model.

Assumption 2 (Existence of the Characteristic–Moment Limit). The sequence of characteristic Gram matrices $W_t = Z_t^\top Z_t / N_t$ admits a non–random, positive–definite probability limit:

$$Q_Z = \text{plim}_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T W_t, \quad Q_Z \succ 0, \quad \lambda_{\min}(Q_Z) \geq c > 0.¹⁶ \quad (11)$$

¹³This assumption ensures that the joint distribution of firm characteristics and asset returns is stable over time, so that empirical moments from expanding panels consistently represent the underlying economic environment. From the financial perspective, stationarity guarantees that the characteristic space remains statistically comparable across periods—an implicit requirement for cross–sectional factor pricing models. *Used in Lemma A.1 (foundation for the uniform LLN and CLT).*

¹⁴The exponent $\tau > 0$ determines the strength of the mixing and moment conditions required for the law of large numbers and central limit theorem to hold under weak dependence. Specifically, $\alpha(\ell)$ controls the decay of temporal dependence, and the summability condition $\sum_{\ell} \alpha(\ell)^{\tau/(2+\tau)} < \infty$ ensures that second–order moments are uniformly integrable and that higher moments of order $(2 + \tau)$ exist. This formulation follows standard treatments of α –mixing sequences in Ibragimov and Linnik (1971, Ch. 17, Thm. 17.2.3) and in the large–sample theory exposition of Newey and McFadden (1994, Sec. 7.3). Economically, allowing a finite $\tau > 0$ accommodates the moderate serial correlation and heavy–tailed behavior typical of asset–return data without imposing strict i.i.d. structure.

¹⁵ Q_Z is the cross–sectional covariance matrix of firm characteristics and acts as the metric that scales pricing directions in characteristic space. It determines how factor exposures are orthogonalized and is fundamental for identification and efficiency results in Appendix B.

¹⁶This assumption ensures that the empirical characteristic covariance matrices converge to a stable population limit Q_Z , providing the fixed geometric benchmark used throughout the NA–IPCA identification theory. Economically, it requires the cross–sectional composition of firm characteristics to be sufficiently persistent for risk–exposure directions to be measured consistently over time.

It is used in Proposition B.1 and Proposition B.3 in Appendix B, which develop the weighted–least–squares (WLS) equivalence

The limit Q_Z defines the inner-product geometry $\langle a, b \rangle_{Q_Z} = a^\top Q_Z b$ and associated norm $\|a\|_{Q_Z} = (a^\top Q_Z a)^{1/2}$, which underpin all orthogonality and normalization constraints used in the No-Arbitrage IPCA model.¹⁷

Together with Assumption (A1), condition (11) guarantees that the cross-sectional information contained in firm characteristics is both well-conditioned and time-invariant in the large-sample limit, so that subsequent projection and identification results (Proposition B.1 and Proposition B.1) are well defined.

Assumption 3 (Conditional Mean-Zero Idiosyncratic Errors).¹⁸ Define the idiosyncratic component of excess returns as

$$u_t = r_{t+1} - Z_t (\Gamma_\beta^0 f_t^0 + \Gamma_\delta^0 F_t^{\text{obs}} + \Gamma_\alpha^0), \quad (12)$$

where $u_t = (u_{1t}, \dots, u_{N_t})^\top \in \mathbb{R}^{N_t}$ represents asset-specific residuals at time t . The stochastic process $\{u_t\}$ satisfies the following regularity conditions:

$$\mathbb{E}[u_t \mid Z_t] = 0 \quad (\text{A3.1: (Mean-Zero Condition)})$$

$$\sup_t \mathbb{E}[\|u_t\|_2^{2+\tau}] < \infty \quad (\text{A3.2: (Moment-Bound Condition)})$$

$$\frac{1}{N_t^2} \sum_{i \neq j} |\text{Cov}(u_{it}, u_{jt} \mid Z_t)| \xrightarrow[T \rightarrow \infty]{p} 0 \quad (\text{A3.3: (Cross-Sectional Diversification Condition)})$$

Condition (A3.1: (Mean-Zero Condition)) ensures that idiosyncratic shocks are orthogonal to the information set Z_t , so all systematic variation is captured by the latent and observable factor structure. Bound (A3.2: (Moment-Bound Condition)) supplies uniformly finite higher moments, while the weak cross-sectional dependence condition (A3.3: (Cross-Sectional Diversification Condition)) guarantees that firm-level shocks diversify as the cross section grows.¹⁹ Taken together, these conditions imply that pricing errors are mean-zero, have finite variance, and are asymptotically negligible, forming the stochastic foundation for the consistency and efficiency results developed in Appendix B.

Assumption 4 (Finite Factor Dimensions and Non-degenerate Covariances).²⁰ Let K denote the number and the LAN expansion on the tangent manifold.

¹⁷Intuitively, Q_Z plays the same role as a “characteristic covariance kernel”: it rescales each direction in characteristic space by its empirical variability, ensuring that latent and observable factor loadings are identified relative to a stable economic reference measure.

¹⁸This assumption ensures that firm-specific shocks carry no systematic information about expected returns after conditioning on characteristics, so that all priced variation is captured by the latent and observable factor structure. Economically, it enforces the diversification principle: idiosyncratic noise averages out in large cross sections, legitimizing the interpretation of Γ_α as a pricing-error component. *Used in Proposition 1 (Consistency) and Lemma A.1 (Uniform LLN and CLT).*

¹⁹The diversification requirement $\frac{1}{N_t^2} \sum_{i \neq j} |\text{Cov}(u_{it}, u_{jt} \mid Z_t)| \xrightarrow{p} 0$ implies that cross-sectional averages of residuals obey a law of large numbers, so estimation noise in $X_t = N_t^{-1} Z_t' r_{t+1}$ vanishes asymptotically. This parallels the large- N arguments in the approximate-factor framework of Chamberlain and Rothschild (1983).

²⁰This assumption fixes the dimensionality of both latent and observable (PSF) factors so that identification and inference are conducted in a finite-dimensional parameter space. Economically, it rules out “many-factor” asymptotics and ensures that each priced direction represents a distinct source of systematic risk with finite variance. *Used in Propositions B.2–B.6 (asymptotic efficiency and block orthogonality).*

of latent factors and K_{add} the number of observable (PSF) factors. Both dimensions are fixed and finite:

$$K < \infty, \quad K_{\text{add}} < \infty. \quad (13)$$

The latent factor process $\{f_t^0\}_{t=1}^T$ satisfies

$$\mathbb{E}[f_t^0 f_t^{0\top}] = \Sigma_f \succ 0, \quad (14)$$

and the observable (PSF) factors $F_t^{\text{obs}} \in \mathbb{R}^{K_{\text{add}}}$ have full-rank second moments:

$$\Sigma_F = \mathbb{E}[F_t^{\text{obs}} F_t^{\text{obs}\top}] \succ 0. \quad (15)$$

Conditions (14)–(15) guarantee that both latent and observable factor spaces possess non-degenerate covariance structures, allowing their respective loading matrices Γ_β and Γ_δ to be locally identified up to orthogonal rotation.²¹

Together, these requirements fix the dimensional and stochastic environment for the No-Arbitrage IPCA estimator, ensuring that factor loadings are uniquely identified within each block and that asymptotic arguments in Propositions B.2–B.6 remain valid in finite samples.

Assumption 5 (No-Arbitrage Bound Interiority (Slater Condition)).²²

We assume that the population intercept (alpha) parameter Γ_α^0 satisfies the strict interiority condition

$$\psi(\Gamma_\alpha^0) = \Gamma_\alpha^{0\top} Q_Z \Gamma_\alpha^0 \leq \delta_0 < \delta, \quad (16)$$

where $\psi(\Gamma_\alpha) = \Gamma_\alpha^\top Q_Z \Gamma_\alpha$ is the quadratic no-arbitrage functional defined in (6).

Condition (16) ensures that a strictly feasible interior point exists for the constraint $\psi(\Gamma_\alpha) \leq \delta$, thereby satisfying the Slater condition of convex analysis.²⁴

²¹Full-rank covariances ensure that the spectral decomposition used for normalization ($\Gamma_\beta^\top Q_Z \Gamma_\beta = I_K$, $\Gamma_\beta^\top Q_Z \Gamma_\delta = 0$) is well defined. This parallels the identification conditions for finite-factor APT models in Chamberlain and Rothschild (1983).

²²This assumption imposes an interior feasibility requirement on the no-arbitrage constraint $\psi(\Gamma_\alpha) \leq \delta$. It guarantees that the true intercept parameter Γ_α^0 lies strictly within the admissible region, so that the Lagrange multiplier associated with the inequality constraint exists and is well defined. Economically, it means that systematic mispricing is limited in magnitude and does not exhaust investors' arbitrage capacity, thereby ensuring that the no-arbitrage discipline operates as a soft regularization rather than a binding equilibrium restriction. *Used in Lemma A.2 (constraint qualification) and Proposition B.1 (KKT well-posedness).*

²³Here $\delta > 0$ is a user-specified scalar controlling the no-arbitrage tolerance level in characteristic space. Mathematically, $\psi(\Gamma_\alpha) = \Gamma_\alpha^\top Q_Z \Gamma_\alpha \leq \delta$ bounds the Q_Z -weighted norm of the intercept (alpha) vector, i.e., the squared length of the mispricing direction under the characteristic metric. Economically, δ plays the role of a *Sharpe-ratio cap*: it limits the cross-sectional return-to-risk ratio of any portfolio that loads purely on residual mispricing. In empirical calibrations, δ is typically chosen small relative to the average factor Sharpe ratio—for example, values between 0.01 and 0.10 (implying a maximum “alpha Sharpe ratio” between 0.1 and 0.3) are consistent with near-arbitrage pricing environments reported in Lettau and Pelger (2020b) and Dello Preite et al. (2024). In simulation studies, δ may also be scaled as $\delta = c_\delta/K$ for some fixed constant $c_\delta > 0$, ensuring that the effective tolerance shrinks with the factor dimension and preserves identification (Gouriéroux and Monfort, 1997; Bryzgalova et al., 2023). Intuitively, setting δ close to zero enforces strict no-arbitrage (forcing $\Gamma_\alpha \approx 0$), whereas larger δ allows for moderate persistent mispricing, capturing deviations from frictionless equilibrium.

²⁴The Slater interiority condition is standard in constrained extremum and GMM estimation (see Gouriéroux and Monfort, 1997; Rockafellar and Wets, 1998; Newey and McFadden, 1994). Recent constrained and regularized GMM frameworks such as Chernozhukov et al. (2023) rely on the same condition to guarantee well-posed multipliers and smooth asymptotic expansions. In empirical asset-pricing applications, analogous convex-regularization structures appear in Lettau and Pelger (2020b), Bryzgalova

From an asset–pricing perspective, the inequality (16) reflects the principle that markets are approximately—but not perfectly—arbitrage free. Following the logic of [Ross \(1976\)](#), [Merton \(1987\)](#), and [Cochrane \(2005\)](#), the no–arbitrage discipline restricts admissible pricing kernels to those with bounded Sharpe ratios. When $\psi(\Gamma_\alpha)$ is interpreted as the squared characteristic–space Sharpe ratio of the residual component, the strict inequality $\psi(\Gamma_\alpha^0) < \delta$ implies that no linear combination of characteristics yields an unbounded return–to–risk ratio. This interiority provides a geometric and economic anchor for the estimator: the true pricing kernel lies in the interior of the convex no–arbitrage region, so small perturbations of Γ_α leave the feasible set unchanged, ensuring differentiability of the KKT system and stability of the NA–IPCA solution. In equilibrium terms, δ bounds the maximum squared Sharpe ratio that can be achieved by any linear combination of characteristics orthogonal to priced factors, ensuring that the implied characteristic–based priced risk structure remains well conditioned. Hence, the restriction acts as an economic regularizer: it disciplines the estimation of Γ_α by penalizing excessive systematic mispricing without imposing a corner (zero–alpha) equilibrium. This formulation parallels the limited–arbitrage framework of [Dello Preite et al. \(2024\)](#), where equilibrium pricing kernels are required to exhibit finite volatility and bounded Sharpe ratios across admissible portfolios.

Assumption 6 (Compactness of the Feasible Parameter Space). ²⁵

Define the feasible parameter space as

$$\Theta(\delta) = \left\{ (\Gamma_\beta, F, \Gamma_\delta, \Gamma_\alpha) : \Gamma_\beta^\top Q_Z \Gamma_\beta = I_K, \Gamma_\beta^\top Q_Z \Gamma_\delta = 0, \Gamma_\beta^\top Q_Z \Gamma_\alpha = 0, \psi(\Gamma_\alpha) \leq \delta \right\}. \quad (17)$$

The first three orthogonality and normalization conditions in (17) define a smooth, compact manifold within the L –dimensional characteristic space. Specifically, the constraint $\Gamma_\beta^\top Q_Z \Gamma_\beta = I_K$ defines the Q_Z –weighted Stiefel manifold

$$\mathcal{V}_{L,K}(Q_Z) = \{ \Gamma_\beta \in \mathbb{R}^{L \times K} : \Gamma_\beta^\top Q_Z \Gamma_\beta = I_K \},$$

while $\Gamma_\beta^\top Q_Z \Gamma_\delta = 0$ and $\Gamma_\beta^\top Q_Z \Gamma_\alpha = 0$ impose block–orthogonality of the observable and intercept components relative to the latent factor space. The no–arbitrage bound $\psi(\Gamma_\alpha) \leq \delta$ introduces an additional closed ellipsoidal constraint

$$\mathcal{E}_\delta = \{ \Gamma_\alpha \in \mathbb{R}^L : \Gamma_\alpha^\top Q_Z \Gamma_\alpha \leq \delta \},$$

which limits the overall magnitude of systematic mispricing within the characteristic–space geometry.

The product set $\mathcal{V}_{L,K}(Q_Z) \times \mathcal{E}_\delta$ is compact under the topology induced by the Q_Z –weighted inner product (4), and the factor sequence $F = [f_1, \dots, f_T]$ is restricted to a bounded subset of $\mathbb{R}^{K \times T}$. ²⁶

et al. (2023), and [Dello Preite et al. \(2024\)](#), where no–arbitrage bounds act as shape constraints that discipline the estimation of characteristic–based priced risk structures.

²⁵This assumption guarantees the existence, stability, and continuity of the constrained extremum estimator. Mathematically, compactness ensures that the objective function $Q_T(\theta)$ attains a minimum within the admissible set $\Theta(\delta)$ and that the argmin mapping is continuous under uniform convergence ([Newey and McFadden, 1994](#); [Gouriéroux and Monfort, 1997](#)). Economically, compactness restricts the admissible degree of factor exposures and alpha magnitudes, ruling out unbounded risk–taking or degenerate scaling of the stochastic discount factor. Comparable bounded–parameter regularizations are imposed in [Bryzgalova et al. \(2023\)](#) and [Dello Preite et al. \(2024\)](#), where compact feasible sets maintain the finite–variance discipline of no–arbitrage pricing kernels.

²⁶Bounding F is a standard normalization in factor–model estimation ([Kelly et al., 2019](#); [Lettau and Pelger, 2020b](#)) and is

Compactness of $\Theta(\delta)$ guarantees that the constrained estimator

$$\hat{\theta}_T \in \arg \min_{\theta \in \Theta(\delta)} Q_T(\theta)$$

exists and is well defined for every sample size T . It also ensures that the sequence of empirical minimizers $\{\hat{\theta}_T\}$ remains in a tight (precompact) subset of the parameter space. Geometrically, this confines estimation to a bounded region of the Q_Z -weighted characteristic space, preventing explosive rotations or rescalings of factor loadings and preserving the well-behaved curvature of the no-arbitrage identification manifold.

Assumption 7 (Smoothness and Differentiability of Moment Maps).²⁷ For each period t , the characteristic-space moment map $m_t : \Theta(\delta) \rightarrow \mathbb{R}^L$ defined by

$$m_t(\theta) = X_t - W_t(\Gamma_\beta f_t + \Gamma_\delta F_t^{\text{obs}} + \Gamma_\alpha) \quad (18)$$

is continuously Fréchet-differentiable in θ on a neighborhood of the population parameter $\theta_0 = (\Gamma_\beta^0, F^0, \Gamma_\delta^0, \Gamma_\alpha^0)$. Let $\dot{m}_t(\theta)$ denote its Jacobian with respect to θ . Then:

$$\begin{aligned} \sup_{\theta \in \Theta(\delta)} \mathbb{E}[\|\dot{m}_t(\theta)\|_F] &< \infty, & (\text{Uniform Bounded-Derivative Condition}) \\ \|m_t(\theta_1) - m_t(\theta_2)\|_2 &\leq L_t \|\theta_1 - \theta_2\|_2, \quad \mathbb{E}[L_t] < \infty. \end{aligned} \quad (19)$$

These regularity conditions ensure that the family of moment maps $\{m_t(\theta) : \theta \in \Theta(\delta)\}$ satisfies stochastic equicontinuity and differentiability in mean square, allowing a uniform law of large numbers and a linear asymptotic expansion of the sample score. Formally, they imply the existence of a continuous Fréchet derivative operator $\dot{m}_t(\theta)$ with bounded expectation and Lipschitz constant L_t , so that $\|m_t(\theta_1) - m_t(\theta_2) - \dot{m}_t(\bar{\theta})(\theta_1 - \theta_2)\| = o(\|\theta_1 - \theta_2\|)$ uniformly in $\bar{\theta} \in \Theta(\delta)$.

Assumption (18)–(19) means that the empirical pricing kernel reacts smoothly to perturbations in either factor loadings or factors: marginal changes in Γ_β or f_t translate into proportionate and predictable adjustments in the cross-sectional pricing errors. This “local regularity” anchors the stability of the NA-IPCA estimation landscape, ensuring that the first-order (KKT) conditions yield valid differentiable approximations and that second-order curvature (information) matrices exist and are continuous in the neighborhood of θ_0 .

Assumption 8 (Bounded-Influence and Robust-Score Conditions).²⁸ Let $\rho_\kappa : \mathbb{R} \rightarrow \mathbb{R}_+$ denote a convex,

without loss of generality because the orthonormality constraint $\Gamma_\beta^\top Q_Z \Gamma_\beta = I_K$ eliminates arbitrary scale transformations. Without this restriction, one could rescale $(\Gamma_\beta, F) \mapsto (\Gamma_\beta c, F/c)$ without altering $\Gamma_\beta F$, destroying compactness of $\Theta(\delta)$. The bounded- F normalization therefore ensures a well-posed, scale-invariant parameter space in which the no-arbitrage and identification constraints remain jointly valid.

²⁷This assumption imposes continuous Fréchet differentiability and local Lipschitz continuity of the per-period moment map $m_t(\theta) = X_t - W_t(\Gamma_\beta f_t + \Gamma_\delta F_t^{\text{obs}} + \Gamma_\alpha)$ in a neighborhood of the true parameter θ_0 . It guarantees that stochastic expansions and first-order Taylor linearizations are valid uniformly over $\Theta(\delta)$, forming the analytic basis for the LAN representation and the differentiable KKT system used in Appendix B. Economically, smoothness ensures that small perturbations in factor loadings or pricing kernels generate proportionally small changes in pricing moments—i.e., the model’s mapping from parameters to cross-sectional pricing errors is locally stable and well behaved. *Used in Lemmas A.1–A.3 (uniform stochastic equicontinuity) and Propositions B.5–B.6 (LAN and block-diagonal efficiency).*

²⁸This assumption specifies the regularity and bounded-influence properties of the Huber-type loss function $\rho_\kappa(u)$ and its derivative (score) $\psi_\kappa(u) = \rho'_\kappa(u)$. It guarantees that the resulting estimating equations have finite variance even under heavy-tailed

symmetric loss function and $\psi_\kappa(u) = \rho'_\kappa(u)$ its associated score. We assume the following properties hold for all residuals u_t :

$$\psi_\kappa \text{ is continuous, odd, and satisfies } |\psi_\kappa(u)| \leq \kappa \quad (\text{Bounded-Influence Condition}), \quad (20)$$

$$\psi'_\kappa(0) = 1 \quad (\text{Local-Linearity Condition}), \quad (21)$$

$$\mathbb{E}[\psi_\kappa(u_t)^2] < \infty \quad \text{even if } \mathbb{E}[u_t^4] = \infty \quad (\text{Finite-Variance Condition}), \quad (22)$$

$$\psi_\kappa(u) \rightarrow u \quad \text{as } \kappa \rightarrow \infty \quad (\text{Asymptotic-Equivalence Condition}). \quad (23)$$

A canonical example satisfying (20)–(23) is the Huber loss

$$\rho_\kappa(u) = \frac{1}{2}u^2\mathbf{1}\{|u| \leq \kappa\} + (\kappa|u| - \frac{1}{2}\kappa^2)\mathbf{1}\{|u| > \kappa\}, \quad \psi_\kappa(u) = \text{sign}(u) \min\{|u|, \kappa\}. \quad (24)$$

Under Assumptions (A1)–(A3) and (20)–(23),²⁹ the standardized score sum $T^{-1/2} \sum_t \psi_\kappa(u_t)$ obeys a central-limit theorem with finite asymptotic variance even when returns exhibit heavy tails or conditionally heteroskedastic volatility.³⁰

Condition (24) limits the sensitivity of the estimation criterion to large cross-sectional residuals, thereby

or heteroskedastic shocks and that the influence of any single observation is bounded by κ . Formally, ψ_κ behaves linearly around the origin but flattens for large residuals, providing robustness without sacrificing first-order efficiency. Economically, this corresponds to limiting the impact of extreme asset-specific shocks or idiosyncratic returns on the estimated pricing kernel, ensuring stable inference in large cross-sections. *Used in Proposition B.4 (robust efficiency) and the algorithmic discussion in Appendix C.*

²⁹Together, Conditions (20)–(23) formalize the bounded-influence framework underlying the robust (Huber-type) score function $\psi_\kappa(\cdot)$. Each plays a distinct but complementary role in ensuring that the NA-IPCA estimator remains both statistically regular and economically interpretable.

First, the bounded-influence condition (20) requires ψ_κ to be continuous, odd, and uniformly capped by κ , ensuring that no single observation can exert unbounded leverage on the score. Technically, this guarantees existence of moments of all orders for the transformed residuals, so the estimator retains a finite asymptotic variance even when the raw errors u_t have heavy tails or conditional heteroskedasticity (Huber, 1981, Ch. 7; Maronna et al., 2006, Ch. 3). In practical terms, this limits the impact of extreme asset returns—such as crash events or illiquidity shocks—on the estimated pricing kernel.

Second, the local-linearity condition (21) enforces $\psi'_\kappa(0) = 1$, aligning the robust score locally with the efficient (GLS) score when residuals are small. This normalization preserves Fisher consistency and first-order efficiency: near equilibrium, where pricing errors are small, the robust estimator behaves identically to OLS/GLS. Without it, the local curvature of the objective would be distorted—causing inefficiency or bias even under well-behaved data—and the resulting score could lose its identification power at the true parameter.

Third, the finite-variance condition (22) ensures that $\mathbb{E}[\psi_\kappa(u_t)^2] < \infty$ even when $\mathbb{E}[u_t^4] = \infty$. Statistically, this implies that the asymptotic covariance matrix of the estimator exists under only L_2 -integrability of the data rather than requiring stronger moment conditions. Economically, this translates into stable inference and well-behaved standard errors even in markets characterized by large return kurtosis.

Finally, the asymptotic-equivalence condition (23) states that $\psi_\kappa(u) \rightarrow u$ as $\kappa \rightarrow \infty$, linking the robust and classical estimators in the limit. This ensures continuity of asymptotic results across the robustness-efficiency spectrum: for large κ , the estimator approaches GLS efficiency; for finite κ , it gains robustness to heavy tails and outliers. In applied asset pricing, this property provides a practical tuning mechanism between robust estimation in turbulent regimes and near-efficient estimation in stable periods.

If any of these properties were violated, the estimator would lose either its statistical stability (if unbounded or discontinuous), its identification (if not locally linear), its valid asymptotic variance (if not finite-variance), or its interpretability (if not asymptotically equivalent to GLS). Together, they guarantee that the Huber-type robustification acts as a smooth, econometrically coherent safeguard: it preserves first-order no-arbitrage efficiency while controlling the tail risk of the cross-sectional score.

³⁰See Huber (1981, Ch. 7) and Maronna et al. (2006, Ch. 3–4), who prove that ψ_κ -bounded M -estimators yield asymptotically normal scores with finite variance under weak moment assumptions. For $\kappa \rightarrow \infty$, $\psi_\kappa(u) = u$ and the estimator reduces to the standard (efficient) GLS case.

stabilizing inference in environments with fat-tailed returns or idiosyncratic outliers. In financial terms, κ acts as a “risk cap” on single-asset shocks: when an asset’s realized return deviates excessively from the model-implied value, its marginal influence on the estimated pricing kernel saturates. This yields an estimator that remains efficient under Gaussian risk but continues to perform robustly when the empirical return distribution exhibits volatility clustering or occasional extreme events—a key requirement for realistic no-arbitrage estimation.

Taken together, Assumptions (A1)–(A8) establish the probabilistic and geometric foundation for the constrained M-estimation framework used in Appendices B–C. They integrate standard large-sample regularity conditions for smooth M-estimators (Newey and McFadden, 1994; Gouriéroux and Monfort, 1997) with the differential-geometric structure of the no-arbitrage constraint manifold (Edelman et al., 1998; Absil et al., 2008). Collectively, these assumptions guarantee that the Kuhn–Tucker system associated with the NA–IPCA estimator is well posed, the parameter space is compact and continuously differentiable, and the Lagrange multipliers exist and are locally unique. Under this environment, the estimator satisfies the standard properties of consistency, asymptotic normality, and Huber-type robustness derived in the main text. Economically, these regularity conditions ensure that the statistical representation of the characteristic-based pricing kernel is stable, identifiable, and behaves smoothly under small perturbations in the data or in the cross-sectional sample of assets—properties essential for reliable inference and comparative statics in modern asset-pricing applications.

A.3 Regularity Lemmas: LLN, CLT, and Compactness

This subsection develops the probabilistic and analytical foundations that underpin the large-sample properties of the No-Arbitrage IPCA (NA–IPCA) estimator. It collects three core lemmas—covering the uniform law of large numbers (ULLN), a central limit theorem (CLT) for the robust (Huber-type) scores, and the compactness and argmin-continuity results that ensure existence and consistency of the estimator.

Lemma A.1 (Uniform LLN and Stochastic Equicontinuity) establishes that the sample criterion $Q_T(\theta)$ converges uniformly to its population counterpart $Q(\theta)$ on the compact feasible set $\Theta(\delta)$. This result guarantees that empirical pricing moments stabilize around their long-run population structure as T grows, providing the foundation for identification and consistency.

Lemma A.2 (Central Limit Theorem for Bounded-Influence Scores) then proves that the Huber-weighted score process satisfies a finite-variance CLT under weak α -mixing dependence. Despite heavy-tailed returns and cross-sectional heteroskedasticity, the aggregate Huber scores converge to a Gaussian limit with covariance matrix Σ_ψ , establishing the asymptotic distribution of the first-order conditions and supporting the LAN representation used in Appendix B.

Finally, *Lemma A.3 (Compactness and Argmin Consistency)* verifies that the empirical objective $Q_T(\theta)$ admits a minimizer for each sample and that any sequence of empirical minimizers converges in probability to the unique population minimizer θ^* . Compactness of $\Theta(\delta)$ guarantees existence, while the uniform LLN transfers the location of the minimum from sample to population, ensuring consistency of the constrained extremum estimator.

Taken together, Lemmas A.1–A.3 provide a complete probabilistic framework for the asymptotic analysis of the NA–IPCA estimator. They show that the empirical objective behaves smoothly and predictably under weak dependence and heavy tails: its level converges uniformly, its score aggregates normally, and its minimizer remains well defined and consistent. Economically, these results formalize the intuition that as the sample expands, transient sampling noise and idiosyncratic shocks average out, revealing the stable characteristic–space geometry that governs equilibrium factor pricing.

Lemma A.1 (Uniform LLN and Stochastic Equicontinuity).^{31,32} *Let $\Theta(\delta)$ be compact as defined in (17), and let the per–period moment map $m_t(\theta) = X_t - W_t(\Gamma_\beta f_t + \Gamma_\delta F_t^{\text{obs}} + \Gamma_\alpha)$ satisfy the conditions in Assumptions 1–7. Define the sample average $\bar{m}_T(\theta) = T^{-1} \sum_{t=1}^T m_t(\theta)$ and the population moment $m(\theta) = \mathbb{E}[m_t(\theta)]$.*

$$\sup_{\theta \in \Theta(\delta)} \|\bar{m}_T(\theta) - m(\theta)\| \xrightarrow[T \rightarrow \infty]{p} 0, \quad (\text{Uniform LLN}). \quad (25)$$

Moreover, the empirical process $\{\sqrt{T}[\bar{m}_T(\theta) - m(\theta)] : \theta \in \Theta(\delta)\}$ is stochastically equicontinuous: for every $\varepsilon > 0$ and $\eta > 0$, there exists $\delta_\eta > 0$ such that

$$\limsup_{T \rightarrow \infty} \mathbb{P} \left(\sup_{\|\theta_1 - \theta_2\| < \delta_\eta} \sqrt{T} \|\bar{m}_T(\theta_1) - \bar{m}_T(\theta_2) - [m(\theta_1) - m(\theta_2)]\| > \varepsilon \right) < \eta. \quad (26)$$

Hence $Q_T(\theta) = \bar{m}_T(\theta)^\top \bar{m}_T(\theta)$ converges uniformly to its population limit $Q(\theta) = m(\theta)^\top m(\theta)$ and admits continuous first– and second–order derivatives on $\Theta(\delta)$.³³

Proof. Fix the compact feasible set $\Theta(\delta)$ in (17) and the per–period moment map $m_t(\theta) = X_t - W_t(\Gamma_\beta f_t + \Gamma_\delta F_t^{\text{obs}} + \Gamma_\alpha)$. By Assumptions 1–7, the process $\{(Z_t, r_{t+1})\}$ is strictly stationary and α –mixing with summable coefficients (9), the mapping $\theta \mapsto m_t(\theta)$ is (pointwise) measurable and

³¹This lemma extends the uniform law-of-large-numbers and stochastic-equicontinuity results for smooth M –estimators (Newey and McFadden, 1994, Sec. 2.6; Gouriéroux and Monfort, 1997, Ch. 4) to the characteristic–space moment maps used in NA–IPCA. It ensures that the sample objective $Q_T(\theta)$ converges uniformly to its population limit $Q(\theta)$ over the compact feasible set $\Theta(\delta)$ defined in (17), and that local fluctuations vanish uniformly in θ . Economically, this guarantees that as the sample expands, the cross-sectional pricing moments stabilize around a well-defined population structure—so that transient sampling noise or short-lived market shocks cannot distort identification. *Used in Propositions B.1–B.3 (convergence and consistency of $\hat{\theta}_T$).*

³²The lemma can equivalently be expressed in terms of the sample and population objective functions

$$Q_T(\theta) = \frac{1}{T} \sum_{t=1}^T \|m_t(\theta)\|_{Q_Z^{-1}}^2, \quad Q(\theta) = \mathbb{E} \left[\|m_t(\theta)\|_{Q_Z^{-1}}^2 \right],$$

where $\|x\|_{Q_Z^{-1}}^2 = x^\top Q_Z^{-1} x$ denotes the squared norm in the Q_Z^{-1} –weighted geometry. In this form, Lemma A.1 asserts two probabilistic regularities: (i) a *uniform law of large numbers*, $\sup_{\theta \in \Theta(\delta)} |Q_T(\theta) - Q(\theta)| \xrightarrow{p} 0$, which guarantees that the empirical criterion converges uniformly to its population counterpart; and (ii) *stochastic equicontinuity*, $\sup_{\|\theta_1 - \theta_2\| < \varepsilon} |[Q_T(\theta_1) - Q_T(\theta_2)] - [Q(\theta_1) - Q(\theta_2)]| \xrightarrow{p} 0$, ensuring that small parameter perturbations induce uniformly small fluctuations in the sample criterion. Together, these properties establish that the objective surface $Q_T(\theta)$ converges to a smooth and well–behaved population limit $Q(\theta)$, forming the probabilistic foundation for consistency and differentiability of the NA–IPCA estimator. Economically, this means that as the time dimension T increases, the empirical pricing surface—measured by average squared pricing errors—stabilizes around its long–run equilibrium shape, so transitory shocks cannot distort identification or curvature of the model’s factor structure.

³³The proof follows from a standard bracketing argument for α –mixing triangular arrays (Ibragimov and Linnik, 1971, Thm. 17.2.3) combined with the bounded–moment conditions (A1)–(A3). Compactness of $\Theta(\delta)$ and the Lipschitz property in Assumption (A7) guarantee stochastic equicontinuity. This ensures that the empirical criterion $Q_T(\theta)$ behaves smoothly in large samples, a prerequisite for the differentiable KKT expansions used in Appendix B.

continuously Fréchet–differentiable with a Lipschitz envelope whose expectation is finite, and the parameter set is compact. We first verify a square–integrable envelope and a uniform Lipschitz bound, which together yield a uniform law of large numbers (ULLN); I then establish stochastic equicontinuity of the centered empirical process.

Envelope and Lipschitz control. Define $M_t(\theta) := \|m_t(\theta)\|_2$. The bounded–moment conditions in Assumptions 1–3 and linearity of $m_t(\cdot)$ in $(\Gamma_\beta f_t, \Gamma_\delta F_t^{\text{obs}}, \Gamma_\alpha)$ imply that

$$\sup_{\theta \in \Theta(\delta)} \mathbb{E}[\|m_t(\theta)\|_2^2] < \infty. \quad (27)$$

Moreover, by Assumption 7 (Lipschitz continuity of the moment map), there exists a non–negative random variable L_t with $\mathbb{E}[L_t] < \infty$ such that for all $\theta_1, \theta_2 \in \Theta(\delta)$,

$$\|m_t(\theta_1) - m_t(\theta_2)\|_2 \leq L_t \|\theta_1 - \theta_2\|_2 \quad \text{a.s.} \quad (28)$$

The two displays (27)–(28) provide the envelope and continuity control required for a uniform LLN on $\Theta(\delta)$.

Uniform LLN. Let $\bar{m}_T(\theta) = T^{-1} \sum_{t=1}^T m_t(\theta)$ and $m(\theta) = \mathbb{E}[m_t(\theta)]$. By stationarity, the α –mixing summability in (9), the square–integrable envelope (27), and compactness of $\Theta(\delta)$, the class $\{m_t(\theta) : \theta \in \Theta(\delta)\}$ is Glivenko–Cantelli for α –mixing triangular arrays (see Ibragimov and Linnik (1971, Thm. 17.2.3); Newey

³⁴The uniform square–integrability bound in (27) follows directly from the bounded–moment conditions in Assumptions 1–3 together with the linearity of $m_t(\theta)$ in $(\Gamma_\beta f_t, \Gamma_\delta F_t^{\text{obs}}, \Gamma_\alpha)$. Since

$$m_t(\theta) = X_t - W_t(\Gamma_\beta f_t + \Gamma_\delta F_t^{\text{obs}} + \Gamma_\alpha) = \frac{1}{N_t} Z_t^\top [r_{t+1} - Z_t(\Gamma_\beta f_t + \Gamma_\delta F_t^{\text{obs}} + \Gamma_\alpha)],$$

the Cauchy–Schwarz inequality gives $\|m_t(\theta)\|_2^2 \leq N_t^{-1} \|Z_t\|_F^2 \|r_{t+1} - Z_t(\Gamma_\beta f_t + \Gamma_\delta F_t^{\text{obs}} + \Gamma_\alpha)\|_2^2$. Under Assumption 1, $\mathbb{E}\|Z_t\|_F^{2+\tau} < \infty$ and $\mathbb{E}\|r_{t+1}\|_2^{2+\tau} < \infty$ for some $\tau > 0$; Assumption 3 ensures $\sup_t \mathbb{E}\|u_t\|_2^{2+\tau} < \infty$ for the residuals $u_t = r_{t+1} - Z_t(\Gamma_\beta f_t + \Gamma_\delta F_t^{\text{obs}} + \Gamma_\alpha)$. Because $(\Gamma_\beta, F, \Gamma_\delta, \Gamma_\alpha) \in \Theta(\delta)$ and $\Theta(\delta)$ is compact (Assumption 6), the operator norms $\|\Gamma_\beta\|, \|\Gamma_\delta\|, \|\Gamma_\alpha\|$ are uniformly bounded. Hence all products $Z_t(\Gamma_\beta f_t + \Gamma_\delta F_t^{\text{obs}} + \Gamma_\alpha)$ have finite second moments uniformly in θ , yielding $\sup_{\theta \in \Theta(\delta)} \mathbb{E}\|m_t(\theta)\|_2^2 < \infty$. Economically, this means that the cross–sectional pricing errors have bounded variance across admissible parameter values—no characteristic direction can produce explosive dispersion of fitted returns, ensuring a well–conditioned moment geometry for the LLN.

³⁵Assumption 7 requires that the per–period moment map $m_t(\theta) = X_t - W_t(\Gamma_\beta f_t + \Gamma_\delta F_t^{\text{obs}} + \Gamma_\alpha)$ be continuously Fréchet–differentiable on the compact domain $\Theta(\delta)$. By the mean–value theorem on Banach spaces, for any $\theta_1, \theta_2 \in \Theta(\delta)$ and for *almost every* $\omega \in \Omega$ there exists an intermediate point $\tilde{\theta} = \tilde{\theta}(\omega, \theta_1, \theta_2)$ (on the segment joining θ_1 and θ_2) such that

$$m_t(\theta_1; \omega) - m_t(\theta_2; \omega) = \dot{m}_t(\tilde{\theta}; \omega) (\theta_1 - \theta_2).$$

Taking norms and defining the random Lipschitz constant $L_t(\omega) := \sup_{\theta \in \Theta(\delta)} \|\dot{m}_t(\theta; \omega)\|_F$ yields

$$\|m_t(\theta_1; \omega) - m_t(\theta_2; \omega)\|_2 \leq L_t(\omega) \|\theta_1 - \theta_2\|_2 \quad \text{for a.s. all } \omega,$$

i.e., the Lipschitz inequality holds *almost surely*. Assumption 7 further imposes $\mathbb{E}[L_t] < \infty$, so the envelope is integrable and the inequality can be averaged. Economically, “a.s.” means the pricing moment is locally stable for all states except a null set of extreme realizations; small perturbations in factor loadings or the kernel induce proportionate changes in fitted returns in virtually all market states, ruling out discontinuous jumps in the cross–sectional fit.

and McFadden (1994, Sec. 2.6))³⁶. Hence

$$\sup_{\theta \in \Theta(\delta)} \|\bar{m}_T(\theta) - m(\theta)\| \xrightarrow[T \rightarrow \infty]{p} 0, \quad (29)$$

which is exactly (25). Intuitively, the empirical cross-sectional moments stabilize uniformly over admissible parameters as the sample grows.

Stochastic equicontinuity. Fix $\varepsilon, \eta > 0$. For any $\theta_1, \theta_2 \in \Theta(\delta)$, the Lipschitz bound (28) yields

$$\|[\bar{m}_T(\theta_1) - \bar{m}_T(\theta_2)] - [m(\theta_1) - m(\theta_2)]\| \leq \left\| \frac{1}{T} \sum_{t=1}^T L_t \right\| \|\theta_1 - \theta_2\|.^{37}$$

By the α -mixing LLN and $\mathbb{E}[L_t] < \infty$, I have $T^{-1} \sum_{t=1}^T L_t \xrightarrow{p} \mathbb{E}[L_t]$. Choose $\delta_\eta > 0$ so that $\mathbb{E}[L_t] \delta_\eta < \varepsilon/2$. Then, for T sufficiently large,

$$\mathbb{P} \left(\sup_{\|\theta_1 - \theta_2\| < \delta_\eta} \sqrt{T} \|[\bar{m}_T(\theta_1) - \bar{m}_T(\theta_2)] - [m(\theta_1) - m(\theta_2)]\| > \varepsilon \right) < \eta,$$

which is (26).³⁸

Consequences for Q_T . Uniform convergence of $\bar{m}_T(\theta)$ implies uniform convergence of $Q_T(\theta) =$

³⁶Formally, declaring the class $\{m_t(\theta) : \theta \in \Theta(\delta)\}$ to be *Glivenko–Cantelli for α -mixing triangular arrays* means that the empirical averages converge uniformly to their population expectations:

$$\sup_{\theta \in \Theta(\delta)} \left| T^{-1} \sum_{t=1}^T m_t(\theta) - \mathbb{E}[m_t(\theta)] \right| \xrightarrow{p} 0.$$

The α -mixing condition in Assumption 1 allows for weak temporal dependence and heterogeneous cross-sectional structures typical of financial panels, while the finite-moment bound in (27) and compactness of $\Theta(\delta)$ ensure integrability and tightness of the process. Mathematically, this invokes the uniform law of large numbers for weakly dependent arrays (Ibragimov and Linnik, 1971, Thm. 17.2.3; Newey and McFadden, 1994, Sec. 2.6). Economically, it means that as the sample expands, the empirical characteristic–return moments stabilize around their long-run equilibrium values—so that transient shocks or short-lived market fluctuations do not distort the identification of systematic pricing relations.

³⁷The Lipschitz inequality in (28) establishes a uniform continuity bound for the sample moments over the compact set $\Theta(\delta)$. By taking expectations, $\mathbb{E}[L_t] < \infty$ implies that the average random slope $T^{-1} \sum_{t=1}^T L_t$ converges in probability to its mean by the α -mixing law of large numbers (Assumption 1). Hence, with high probability, the deviation $\bar{m}_T(\theta_1) - \bar{m}_T(\theta_2) - [m(\theta_1) - m(\theta_2)]$ is uniformly proportional to $\|\theta_1 - \theta_2\|$ with a random but integrable coefficient. This Lipschitz control transfers the tightness of the scalar averages $T^{-1} \sum_{t=1}^T L_t$ to the entire parameterized family $\{\bar{m}_T(\theta) - m(\theta) : \theta \in \Theta(\delta)\}$, thereby ruling out discontinuous jumps or oscillations of the empirical process across $\Theta(\delta)$. Formally, it satisfies the stochastic–equicontinuity criterion of Newey and McFadden (1994, Sec. 2.6) for weakly dependent processes. Economically, this means that small perturbations in model parameters—such as minor changes in factor loadings or intercepts—produce proportionally small and predictable changes in the cross-sectional pricing moments, ensuring that the sample objective remains locally smooth even under finite-sample variation in asset returns.

³⁸The choice of δ_η exploits the Lipschitz control in (28) and the weak-dependence LLN for the random Lipschitz coefficient L_t . Specifically, by Assumption 1 (stationarity, α -mixing) and $\mathbb{E}[L_t] < \infty$ (Assumption 7), the scalar average converges in probability:

$$\frac{1}{T} \sum_{t=1}^T L_t \xrightarrow[T \rightarrow \infty]{p} \mathbb{E}[L_t].$$

Hence, for any $\gamma > 0$ there exists $T_\gamma < \infty$ such that for all $T \geq T_\gamma$,

$$\mathbb{P} \left(\frac{1}{T} \sum_{t=1}^T L_t \leq \mathbb{E}[L_t] + \gamma \right) \geq 1 - \frac{\eta}{2}.$$

$\bar{m}_T(\theta)^\top \bar{m}_T(\theta)$ to $Q(\theta) = m(\theta)^\top m(\theta)$, and the differentiability of $m_t(\cdot)$ on $\Theta(\delta)$ (Assumption 7) yields continuous first- and second-order derivatives of $Q(\cdot)$;³⁹ see Newey and McFadden (1994, Sec. 2.6). Economically, the result states that cross-sectional pricing errors average out in a stable way: as T increases, the empirical characteristic-space geometry converges to its population counterpart, so transient shocks cannot overturn identification nor the curvature that drives efficiency. $\square$

Lemma A.1 ensures that the cross-sectional pricing errors aggregate into stable population moments as T grows. In economic terms, it formalizes the idea that while individual-asset returns are noisy, the systematic structure of returns and characteristics converges to a stationary “economic factor geometry.” This underpins the reliability of empirical factor pricing: parameter estimates capture persistent cross-sectional relations rather than transient shocks.

Lemma A.2 (Central Limit Theorem for Bounded-Influence Scores).⁴⁰

Under Assumptions 1–8, define the robust (Huber-weighted) score for period t as

$$\psi_t(\theta) = Q_Z^{-1/2} W_t^{1/2} \psi_\kappa(u_t(\theta)), \quad u_t(\theta) = r_{t+1} - Z_t(\Gamma_\beta f_t + \Gamma_\delta F_t^{\text{obs}} + \Gamma_\alpha), \quad (30)$$

On that high-probability event, the Lipschitz bound implies, uniformly over $\theta_1, \theta_2 \in \Theta(\delta)$,

$$\|[\bar{m}_T(\theta_1) - \bar{m}_T(\theta_2)] - [m(\theta_1) - m(\theta_2)]\| \leq (\mathbb{E}[L_t] + \gamma) \|\theta_1 - \theta_2\|.$$

Choose $\delta_\eta > 0$ so that $(\mathbb{E}[L_t] + \gamma) \delta_\eta < \varepsilon/2$. Then, for all $T \geq T_\gamma$,

$$\begin{aligned} & \mathbb{P}\left(\sup_{\|\theta_1 - \theta_2\| < \delta_\eta} \sqrt{T} \|[\bar{m}_T(\theta_1) - \bar{m}_T(\theta_2)] - [m(\theta_1) - m(\theta_2)]\| > \varepsilon\right) \\ & \leq \underbrace{\mathbb{P}\left(\frac{1}{T} \sum_{t=1}^T L_t > \mathbb{E}[L_t] + \gamma\right)}_{< \eta/2} + \underbrace{\mathbb{P}\left(\sqrt{T} (\mathbb{E}[L_t] + \gamma) \delta_\eta > \varepsilon\right)}_{=0} < \eta. \end{aligned}$$

where the second term is zero by construction of δ_η . This verifies the stochastic-equicontinuity requirement (26): for small enough parameter perturbations (a δ_η -ball), the empirical fluctuations $[\bar{m}_T(\theta_1) - \bar{m}_T(\theta_2)] - [m(\theta_1) - m(\theta_2)]$ are uniformly small with high probability because the random local slope $T^{-1} \sum L_t$ concentrates near $\mathbb{E}[L_t]$. Compactness of $\Theta(\delta)$ guarantees measurability and permits the supremum over $\|\theta_1 - \theta_2\| < \delta_\eta$ to be controlled by the Lipschitz envelope. Economically, this condition means that the cross-sectional pricing moments react in a smooth and predictable way to small changes in the model’s parameters. In other words, modest shifts in factor loadings or pricing-kernel coefficients lead to proportionate adjustments in fitted returns, rather than abrupt jumps. This smooth response ensures that temporary sampling noise or short-lived market shocks do not create spurious instability in the objective function. From a statistical perspective, the Lipschitz property transfers the tightness of the scalar averages $T^{-1} \sum L_t$ to the whole family of pricing moments indexed by θ , ensuring that the empirical objective behaves continuously on the feasible parameter set. Together, these properties underpin stable factor identification and reliable inference in large cross-sectional asset-pricing environments.

³⁹Formally, since $\bar{m}_T(\theta) \rightarrow m(\theta)$ uniformly on the compact set $\Theta(\delta)$, the continuous-mapping theorem implies that any continuous functional of $\bar{m}_T(\theta)$ —in particular $Q_T(\theta) = \bar{m}_T(\theta)^\top \bar{m}_T(\theta)$ —also converges uniformly to its population counterpart $Q(\theta) = m(\theta)^\top m(\theta)$. The smoothness of $m_t(\cdot)$ from Assumption 7 guarantees that $Q(\theta)$ is twice continuously differentiable, allowing valid first-order (score) and second-order (information) expansions of the estimator. Economically, this means the empirical objective surface inherits the smooth geometry of the underlying pricing-moment map: as the sample grows, the cross-sectional fit approaches a stable population shape, ensuring that both identification and curvature—hence efficiency—are preserved in large samples.

⁴⁰This lemma establishes a finite-variance central limit theorem (CLT) for the Huber-type score process used in the NA-IPCA estimator. It extends standard M -estimation CLTs (White, 1984, Thm. 5.23; Newey and McFadden, 1994, Sec. 2.6) to α -mixing panels with heavy-tailed errors, under the bounded-influence conditions of Assumption 8. Economically, it guarantees that the robust score aggregates idiosyncratic pricing shocks into a stable, asymptotically Gaussian limit—so that inference on the characteristic-space pricing kernel remains valid even when cross-sectional returns exhibit fat tails or volatility clustering. *Used in Propositions B.3–B.5 (LAN representation and efficiency).*

and its sample mean $\bar{\psi}_T(\theta) = T^{-1} \sum_{t=1}^T \psi_t(\theta)$. Then, for $\theta = \theta_0$,

$$\sqrt{T} \bar{\psi}_T(\theta_0) \xrightarrow{d} \mathcal{N}(0, \Sigma_\psi), \quad \Sigma_\psi = \sum_{\ell=-\infty}^{\infty} \text{Cov}(\psi_t(\theta_0), \psi_{t-\ell}(\theta_0)), \quad (31)$$

where Σ_ψ is positive definite and finite under α -mixing and the bounded-influence condition (20). Moreover, the long-run covariance matrix Σ_ψ admits the consistent heteroskedasticity- and autocorrelation-consistent (HAC) plug-in estimator

$$\hat{\Sigma}_\psi = \sum_{|\ell| < L_T} k\left(\frac{\ell}{L_T}\right) \frac{1}{T} \sum_{t=|\ell|+1}^T \psi_t(\hat{\theta}_T) \psi_{t-\ell}(\hat{\theta}_T)^\top, \quad L_T \rightarrow \infty, \quad L_T/T \rightarrow 0, \quad (32)$$

where $k(\cdot)$ is the Bartlett kernel, defined by

$$k(x) = \begin{cases} 1 - |x|, & |x| \leq 1, \\ 0, & |x| > 1, \end{cases} \quad \Longleftrightarrow \quad k\left(\frac{\ell}{L_T}\right) = \max\left\{0, 1 - \frac{|\ell|}{L_T}\right\}.^{41} \quad (33)$$

Proof. Fix θ_0 and let $\psi_t = \psi_t(\theta_0)$ for notational brevity. By Assumptions 1–3, the sequence $\{u_t\}$ is strictly stationary and α -mixing with summable coefficients.⁴² Assumption 8 ensures that $\psi_\kappa(u_t)$ is measurable, bounded, and satisfies $\mathbb{E}[\psi_\kappa(u_t)] = 0$ and $\mathbb{E}[\psi_\kappa(u_t)^2] < \infty$.⁴³ Hence, $\{\psi_t\}$ inherits α -mixing dependence with finite second moments.

Finite-variance and Lindeberg condition. By (20) and (22), $\|\psi_t\|_2 \leq c_\kappa \|Q_Z^{-1/2} W_t^{1/2}\|_2$ almost surely, and

⁴¹The Bartlett kernel $k(x) = 1 - |x|$ for $|x| \leq 1$ and 0 otherwise is the canonical choice in heteroskedasticity- and autocorrelation-consistent (HAC) estimation (Newey and West, 1987). Statistically, it assigns linearly decreasing weights to lagged covariance terms $\psi_t \psi_{t-\ell}^\top$, smoothly truncating their contribution at lag L_T . This weighting scheme ensures two key properties: (i) *positive semi-definiteness* of $\hat{\Sigma}_\psi$, because the Bartlett kernel is non-negative definite and symmetric; and (ii) *bias-variance balance*, since the linear taper minimizes mean-square error among all symmetric compact-support kernels under quadratic loss (Hamilton, 1994, Ch. 12; Andrews, 1991).

From an economic standpoint, the Bartlett kernel plays the role of a “temporal diversification” operator: it downweights distant lags of the Huber-score covariances that correspond to weakly correlated pricing shocks across time. By letting $L_T \rightarrow \infty$ but $L_T/T \rightarrow 0$, the estimator captures persistent (short-memory) dependence in cross-sectional pricing errors without introducing excessive sampling noise from very distant periods. Hence, $\hat{\Sigma}_\psi$ consistently estimates the long-run variability of the score process that governs the asymptotic covariance of the NA-IPCA estimator, even when returns exhibit heteroskedasticity or mild serial correlation.

⁴²Assumptions 1–3 jointly imply that the idiosyncratic-error process $\{u_t\}$ inherits the weak dependence and moment properties of the data-generating process $\{(Z_t, r_{t+1})\}$. Because $u_t = r_{t+1} - Z_t(\Gamma_\beta^0 f_t^0 + \Gamma_\delta^0 F_t^{\text{obs}} + \Gamma_\alpha^0)$ is a measurable transformation of stationary α -mixing sequences with finite $(2 + \tau)$ moments, standard results (Ibragimov and Linnik, 1971, Thm. 17.2.3; Newey and McFadden, 1994, Sec. 2.6) guarantee that $\{u_t\}$ remains strictly stationary and α -mixing with summable coefficients. Economically, this means that the cross-sectional pricing errors fluctuate around a stable equilibrium distribution and exhibit only weak temporal dependence, so transient shocks in returns or characteristics do not accumulate into systematic bias in the estimation of factor loadings.

⁴³Because $\psi_\kappa(\cdot)$ is a bounded and continuous transformation of the idiosyncratic error u_t (Assumption 8), the mapping $u_t \mapsto \psi_\kappa(u_t)$ preserves measurability and weak dependence. Under standard closure results for α -mixing sequences (Ibragimov and Linnik, 1971, Thm. 17.2.3; Newey and McFadden, 1994, Sec. 2.6), any measurable function of a stationary α -mixing process with finite moments remains α -mixing with the same rate of decay. Boundedness of ψ_κ further guarantees $\mathbb{E}[\psi_\kappa(u_t)^2] < \infty$, so the transformed score process $\{\psi_t\}$ satisfies the weak-dependence and finite-variance conditions required for a central limit theorem. Economically, this means that the robust (Huber-type) scores react smoothly to shocks: extreme asset-specific returns are capped by κ , preventing them from inducing serial persistence or infinite-variance distortions in the estimated pricing kernel.

$\mathbb{E}\|\psi_t\|_2^2 < \infty$.⁴⁴ Thus, the Lindeberg condition for triangular α -mixing arrays holds:

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E} \left[\|\psi_t\|_2^2 \mathbf{1}\{\|\psi_t\|_2 > \varepsilon \sqrt{T}\} \right] \rightarrow 0, \quad \forall \varepsilon > 0, \quad (34)$$

which ensures asymptotic normality of $\sqrt{T} \bar{\psi}_T$; see [Ibragimov and Linnik \(1971, Thm. 18.5.3\)](#) and [White \(1984, Thm. 3.45\)](#).⁴⁵

Weak dependence and covariance summability. Because ψ_t is a bounded transformation of α -mixing innovations u_t , the dependence coefficients of $\{\psi_t\}$ decay at least as fast as those of $\{u_t\}$.⁴⁶ Under the summability condition $\sum_{\ell=1}^{\infty} \alpha(\ell)^{\tau/(2+\tau)} < \infty$ from Assumption 1, the long-run covariance matrix $\Sigma_\psi = \sum_{\ell=-\infty}^{\infty} \text{Cov}(\psi_t, \psi_{t-\ell})$ is absolutely summable and finite:

$$\sum_{\ell=-\infty}^{\infty} \|\text{Cov}(\psi_t, \psi_{t-\ell})\|_F < \infty. \quad (35)$$

⁴⁴To see why the display holds, write the Huber-weighted score in the pre-whitened characteristic geometry as

$$\psi_t \equiv Q_Z^{-1/2} W_t^{1/2} \phi_t, \quad \phi_t := \psi_\kappa(u_t),$$

where $u_t = r_{t+1} - Z_t(\Gamma_\beta^0 f_t^0 + \Gamma_\delta^0 F_t^{\text{obs}} + \Gamma_\alpha^0)$ is the idiosyncratic residual and $\psi_\kappa(\cdot)$ is applied componentwise. By the bounded-influence property in (20), $\|\phi_t\|_2 \leq c_\kappa$ almost surely for some constant $c_\kappa \in (0, \infty)$ (e.g., for the standard Huber score, $c_\kappa \leq \sqrt{N_t} \kappa$ and one may absorb $\sqrt{N_t}$ into c_κ since N_t is fixed at each t). Hence, by the submultiplicativity of operator norms,

$$\|\psi_t\|_2 = \|Q_Z^{-1/2} W_t^{1/2} \phi_t\|_2 \leq \|Q_Z^{-1/2} W_t^{1/2}\|_2 \|\phi_t\|_2 \leq c_\kappa \|Q_Z^{-1/2} W_t^{1/2}\|_2 \quad \text{a.s.}$$

This gives the asserted almost-sure bound.

For the second-moment claim, note that $Q_Z \succ 0$ is fixed by (11) and $W_t = Z_t' Z_t / N_t$ has uniformly bounded moments by Assumption 1. Therefore $\mathbb{E}\|Q_Z^{-1/2} W_t^{1/2}\|_2^2 < \infty$. Combining this with the almost-sure inequality above yields

$$\mathbb{E}\|\psi_t\|_2^2 \leq c_\kappa^2 \mathbb{E}\|Q_Z^{-1/2} W_t^{1/2}\|_2^2 < \infty,$$

which verifies $\mathbb{E}\|\psi_t\|_2^2 < \infty$. Economically, the pre-whitening $Q_Z^{-1/2} W_t^{1/2}$ rescales pricing errors into the characteristic metric; the Huber cap limits the contribution of any single asset's shock, so the robust score has a finite, well-behaved variance even in heavy-tailed panels.

⁴⁵Condition (34) is the Lindeberg condition for triangular α -mixing arrays and guarantees that the contribution of large outliers in the score process becomes negligible as $T \rightarrow \infty$. Formally, it requires that the second moments of the truncated terms $\|\psi_t\|_2^2 \mathbf{1}\{\|\psi_t\|_2 > \varepsilon \sqrt{T}\}$ vanish in aggregate, so that no finite subset of periods dominates the asymptotic distribution of $\sqrt{T} \bar{\psi}_T$. Under Assumption 8, the Huber-type bounded-influence property (20) caps $\|\psi_t\|_2$ uniformly by c_κ , making the indicator $\mathbf{1}\{\|\psi_t\|_2 > \varepsilon \sqrt{T}\}$ identically zero for large T . Hence the Lindeberg condition holds automatically, implying that the central limit theorem for weakly dependent α -mixing sequences ([Ibragimov and Linnik, 1971, Thm. 18.5.3](#); [White, 1984, Thm. 3.45](#)) applies. Economically, this means that extreme cross-sectional shocks—such as crash-period residuals or idiosyncratic spikes—cannot drive the aggregate pricing-error distribution. The estimator's asymptotic behavior is therefore governed by the accumulation of many small, weakly dependent innovations rather than by a few large outliers, ensuring a stable Gaussian limit for $\sqrt{T} \bar{\psi}_T$.

⁴⁶The dependence structure of $\{\psi_t\}$ inherits that of the underlying innovation process $\{u_t\}$ because $\psi_t = \psi_\kappa(u_t)$ is a measurable and Lipschitz (bounded) transformation. For any α -mixing process $\{u_t\}$ with coefficients $\alpha_u(\ell)$, applying a measurable transformation with bounded derivative yields a new process $\{\psi_t\}$ whose mixing coefficients satisfy $\alpha_\psi(\ell) \leq C \alpha_u(\ell)$ for some finite constant C ([Ibragimov and Linnik, 1971, Thm. 17.2.3](#); [Newey and McFadden, 1994, Sec. 2.6](#)). Intuitively, the bounded influence of $\psi_\kappa(\cdot)$ smooths the dependence structure rather than amplifying it: correlations between distant periods cannot increase under a capped, monotone transformation. Economically, this means that while idiosyncratic shocks u_t may exhibit weak temporal persistence due to business-cycle or liquidity effects, the robust scores $\psi_t = \psi_\kappa(u_t)$ —being truncated at κ —filter out extreme realizations that would otherwise induce artificial persistence. Hence the dependence in the robust score process $\{\psi_t\}$ decays at least as quickly as in the underlying return innovations, preserving weak dependence and ensuring the validity of the central limit theorem for $\sqrt{T} \bar{\psi}_T$.

⁴⁷The absolute summability in (35) follows from the weak-dependence and finite-moment assumptions in Assumptions 1–3. Under α -mixing dependence with summable coefficients $\sum_{\ell=1}^{\infty} \alpha(\ell)^{\tau/(2+\tau)} < \infty$ and finite second moments, covariance inequalities

Applying the α -mixing CLT for vector processes (Ibragimov and Linnik, 1971, Thm. 18.5.3; White, 1984, Sec. 3.6) yields

$$\sqrt{T} \bar{\psi}_T \xrightarrow{d} \mathcal{N}(0, \Sigma_\psi),$$

which proves (31).⁴⁸

Consistency of the long-run variance estimator. Let $\hat{\Sigma}_\psi$ be the HAC estimator in (32). Under α -mixing and the Lipschitz continuity of ψ_κ , $\hat{\Sigma}_\psi \rightarrow_p \Sigma_\psi$ for any kernel $k(\cdot)$ satisfying standard moment and bandwidth conditions ($L_T \rightarrow \infty$, $L_T/T \rightarrow 0$); see Newey and West (1987, Thm. 2).⁴⁹ This ensures that asymptotic

of the form $\|\text{Cov}(X_t, Y_{t-\ell})\|_F \leq C \alpha(\ell)^{\tau/(2+\tau)} (\mathbb{E}\|X_t\|_2^{2+\tau})^{1/(2+\tau)} (\mathbb{E}\|Y_{t-\ell}\|_2^{2+\tau})^{1/(2+\tau)}$ (Ibragimov and Linnik, 1971, Thm. 17.2.3) ensure that the auto-covariance sequence of any square-integrable α -mixing process is absolutely summable. Since $\psi_t = \psi_\kappa(u_t)$ is bounded and inherits the same α -mixing rate as u_t , the series $\sum_\ell \text{Cov}(\psi_t, \psi_{t-\ell})$ converges absolutely. This guarantees that the long-run variance $\Sigma_\psi = \sum_{\ell=-\infty}^{\infty} \text{Cov}(\psi_t, \psi_{t-\ell})$ exists and is finite, so that asymptotic variances and standard errors based on Σ_ψ are well defined.

From an economic perspective, this condition states that idiosyncratic pricing-error shocks exhibit only short-range dependence: correlations between distant periods die out sufficiently quickly. Hence, serial correlation contributes only a finite amount to the estimator's asymptotic variance, and inference based on Σ_ψ remains valid for dynamic panels with temporal dependence.

⁴⁸The cited results—Ibragimov and Linnik (1971, Thm. 18.5.3) and White (1984, Sec. 3.6)—state the multivariate α -mixing Central Limit Theorem (CLT) for stationary, weakly dependent vector processes. Formally, let $\{X_t\}$ be a strictly stationary sequence in $\mathbb{R}^d$ with $\mathbb{E}[X_t] = 0$, finite second moments $\mathbb{E}\|X_t\|^2 < \infty$, and α -mixing coefficients $\alpha(\ell)$ satisfying

$$\sum_{\ell=1}^{\infty} \alpha(\ell)^{\tau/(2+\tau)} < \infty \quad \text{for some } \tau > 0.$$

If the long-run covariance matrix $\Sigma_X = \sum_{\ell=-\infty}^{\infty} \text{Cov}(X_t, X_{t-\ell})$ is absolutely summable and positive definite, then the partial sums satisfy

$$\frac{1}{\sqrt{T}} \sum_{t=1}^T X_t \xrightarrow{d} \mathcal{N}(0, \Sigma_X).$$

The theorem extends the classical Lindeberg–Feller CLT to dependent (non-i.i.d.) sequences whose dependence decays sufficiently fast in the α -mixing sense. In our setting, $X_t \equiv \psi_t$ inherits α -mixing from the innovation process $\{u_t\}$ and satisfies $\mathbb{E}[\psi_t] = 0$, $\mathbb{E}\|\psi_t\|^2 < \infty$, and the summability condition in Assumption 1. Hence all requirements of the α -mixing CLT are met, implying

$$\sqrt{T} \bar{\psi}_T = \frac{1}{\sqrt{T}} \sum_{t=1}^T \psi_t \xrightarrow{d} \mathcal{N}(0, \Sigma_\psi), \quad \Sigma_\psi = \sum_{\ell=-\infty}^{\infty} \text{Cov}(\psi_t, \psi_{t-\ell}).$$

Economically, this result means that the aggregate robust score (the cross-sectional pricing error averaged over time) behaves asymptotically like a Gaussian vector with finite long-run variance: although individual asset shocks are weakly dependent and possibly heteroskedastic, their aggregate contribution converges to a well-defined normal distribution, enabling standard inference on factor loadings and risk premia.

⁴⁹Newey and West (1987, Thm. 2) establishes the large-sample consistency of heteroskedasticity- and autocorrelation-consistent (HAC) estimators for the long-run covariance matrix of weakly dependent processes. Let $\{X_t\}$ be a stationary α -mixing process with $\mathbb{E}[X_t] = 0$, $\mathbb{E}\|X_t\|^2 < \infty$, and absolutely summable covariances $\sum_{\ell=-\infty}^{\infty} \|\text{Cov}(X_t, X_{t-\ell})\| < \infty$. For any symmetric kernel $k(x)$ satisfying $k(0) = 1$, $k(x) = 0$ for $|x| > 1$, and finite moments $\int_{-1}^1 |x| |k(x)| dx < \infty$, and for a bandwidth L_T diverging slowly enough that $L_T \rightarrow \infty$ and $L_T/T \rightarrow 0$, the weighted covariance sum

$$\hat{\Sigma}_X = \sum_{|\ell| < L_T} k\left(\frac{\ell}{L_T}\right) \frac{1}{T} \sum_{t=|\ell|+1}^T X_t X_{t-\ell}^\top$$

is consistent for the population long-run variance $\Sigma_X = \sum_{\ell=-\infty}^{\infty} \text{Cov}(X_t, X_{t-\ell})$.

In our setting, $X_t \equiv \psi_t = \psi_\kappa(u_t)$ is a bounded, Lipschitz transformation of the α -mixing innovation sequence $\{u_t\}$; by Assumptions 1–8, ψ_t inherits weak dependence and finite second moments, so the Newey–West consistency conditions are satisfied. The kernel $k(x)$ can be any smooth tapering function such as the *Bartlett kernel* $k(x) = 1 - |x|$ for $|x| \leq 1$ and $k(x) = 0$ otherwise, which linearly down-weights higher-order autocovariances and ensures positive-definiteness. The bandwidth choice $L_T \rightarrow \infty$ but $L_T/T \rightarrow 0$ balances bias and variance: too small L_T omits serial correlation; too large inflates sampling noise.

Economically, this estimator provides a consistent estimate of the long-run variance of pricing-error shocks—the serial correlation-adjusted covariance of the robust (Huber) score. It captures both heteroskedasticity and short-run temporal dependence in

standard errors for the NA–IPCA parameters are robust to serial correlation and conditional heteroskedasticity in the cross–sectional moments. $\square$

Lemma A.2 shows that, despite heavy–tailed and weakly dependent pricing errors, the robust (Huber) score behaves asymptotically like a Gaussian process with finite variance. Intuitively, the bounded–influence property truncates the effect of extreme asset–specific shocks, while the mixing condition guarantees sufficient temporal independence for a classical $\sqrt{T}$ –rate central limit theorem. As a result, the NA–IPCA estimator admits standard inference—its first–stage Huber scores aggregate like diversified portfolios of idiosyncratic shocks, yielding well–behaved asymptotic covariances even in non–Gaussian markets.

Lemma A.3 (Existence of Minimizers and Argmin Continuity Under Compactness). ⁵⁰ *Let $\Theta(\delta)$ be the compact feasible set defined in (17), and let*

$$Q_T(\theta) = \frac{1}{T} \sum_{t=1}^T \|m_t(\theta)\|_{Q_Z^{-1}}^2, \quad Q(\theta) = \mathbb{E}[\|m_t(\theta)\|_{Q_Z^{-1}}^2], \quad^{51,52}$$

with $m_t(\theta)$ as in (18). Suppose Assumptions 1–7 hold and the uniform LLN (25) is satisfied. Then:

- (a) **Existence.** *For each T , there exists at least one $\hat{\theta}_T \in \arg \min_{\theta \in \Theta(\delta)} Q_T(\theta)$, and there exists at least one $\theta^* \in \arg \min_{\theta \in \Theta(\delta)} Q(\theta)$.*
- (b) **Argmin continuity (consistency).** *If the population minimizer is unique, $\theta^* = \arg \min_{\theta \in \Theta(\delta)} Q(\theta)$, then*

$$\hat{\theta}_T \xrightarrow[T \rightarrow \infty]{p} \theta^*. \quad (36)$$

More generally, if the set $\arg \min_{\theta \in \Theta(\delta)} Q(\theta)$ is closed, any sequence $\{\hat{\theta}_T\}$ of sample minimizers is d –consistent for that set (converges in probability to it in the Hausdorff metric⁵³).

cross–sectional pricing errors, ensuring valid standard errors for inference on factor risk premia in dynamic asset–pricing panels.

⁵⁰This lemma records the two standard consequences of compactness for constrained M –estimation: (i) existence of sample and population minimizers (Weierstrass theorem) and (ii) argmin continuity under uniform convergence; see Newey and McFadden (1994, Thm. 2.1). Economically, it ensures the NA–IPCA program is well posed: for each T an optimizer exists on the admissible set $\Theta(\delta)$, and as T grows the empirical objective inherits the location of the population minimum. In other words, sampling noise cannot “push” the solution to the boundary or to infinity, which is essential for stable identification and inference.

⁵¹Recall, $\|x\|_{Q_Z^{-1}}^2 = x' Q_Z^{-1} x$ denotes the squared norm of x under the Q_Z^{-1} –weighted inner product, i.e., the generalized Euclidean length in characteristic space.

⁵²The functions $Q_T(\theta)$ and $Q(\theta)$ are the sample and population *objective functions* of the NA–IPCA estimator. Formally, $Q_T(\theta) = T^{-1} \sum_{t=1}^T \|m_t(\theta)\|_{Q_Z^{-1}}^2$ measures the average Q_Z^{-1} –weighted squared magnitude of the cross–sectional pricing errors $m_t(\theta)$, while $Q(\theta) = \mathbb{E}[\|m_t(\theta)\|_{Q_Z^{-1}}^2]$ represents its population expectation. Minimizing $Q_T(\theta)$ over the feasible set $\Theta(\delta)$ therefore yields the sample analogue of the minimum–distance or generalized least–squares (GLS) estimator that enforces the model’s orthogonality and no–arbitrage constraints.

Mathematically, $Q_T(\theta)$ is a quadratic form in the empirical moments $m_t(\theta)$ and thus a convex (locally quadratic) approximation to the Kuhn–Tucker Lagrangian used in Appendix B. The weighting by Q_Z^{-1} rescales each characteristic direction by its cross–sectional variability, ensuring that factor loadings are estimated in the natural metric of the characteristic space rather than in raw units. Under the uniform law of large numbers, $Q_T(\theta)$ converges uniformly to its population counterpart $Q(\theta)$, whose minimizer θ^* defines the true no–arbitrage pricing kernel.

Economically, $Q_T(\theta)$ summarizes the overall “mispricing energy” of the model: it aggregates, across time, the squared characteristic–weighted pricing errors that remain after projecting expected returns onto the latent and observable risk factors. A smaller $Q_T(\theta)$ corresponds to a model that explains a larger share of the cross–sectional variation in returns. Thus, the minimization of $Q_T(\theta)$ is equivalent to seeking the combination of factor loadings and risk premia that yields the most economically efficient, no–arbitrage–consistent explanation of asset returns.

⁵³The Hausdorff metric provides a natural notion of distance between two compact subsets of a metric space $(\mathcal{X}, d)$. Formally,

Proof. Because $\Theta(\delta)$ is nonempty and compact (Assumption 6), and $Q_T(\cdot)$ is continuous on $\Theta(\delta)$ by the smoothness of $m_t(\cdot)$ in Assumption 7, the Weierstrass theorem guarantees the existence of at least one empirical minimizer $\hat{\theta}_T$ for each T ; the same argument, with $Q(\cdot)$ in place of $Q_T(\cdot)$, yields existence of a population minimizer θ^* .⁵⁴

Argmin continuity and consistency. The uniform convergence of the sample criterion $Q_T(\theta)$ to its population counterpart $Q(\theta)$, as established in Lemma A.1, implies that the stochastic fluctuations of the empirical objective vanish uniformly over the compact feasible set $\Theta(\delta)$. Consequently, $Q_T(\cdot)$ converges in probability to the deterministic and continuous limit function $Q(\cdot)$ on $\Theta(\delta)$.⁵⁵

By the *argmin–continuity theorem for extremum estimators* (Newey and McFadden, 1994, Theorem 2.1), uniform convergence and compactness together guarantee that any sequence of empirical minimizers

$$\hat{\theta}_T \in \arg \min_{\theta \in \Theta(\delta)} Q_T(\theta)$$

for nonempty compact sets $A, B \subset \mathcal{X}$, the *Hausdorff distance* is defined as

$$d_H(A, B) = \max \left\{ \sup_{a \in A} \inf_{b \in B} d(a, b), \sup_{b \in B} \inf_{a \in A} d(a, b) \right\}.$$

It measures the greatest deviation needed to move one set into the other: each point of A must lie within $d_H(A, B)$ of some point of B , and vice versa. When A and B are singletons, d_H reduces to the ordinary metric d . The space of all nonempty compact subsets of $(\mathcal{X}, d)$ equipped with d_H is itself a metric space, often denoted $\mathcal{K}(\mathcal{X})$, and convergence in d_H is equivalent to *setwise convergence*: $A_T \rightarrow A$ in d_H if and only if every point of A_T eventually lies close to some point of A , and every point of A is approximated by points of A_T .

In extremum–estimation theory, d_H provides a rigorous way to express *set convergence of argmin (or argmax) mappings* when the population objective may have multiple minimizers. Consistency “in the Hausdorff metric” means that the empirical minimizer set $\arg \min Q_T(\theta)$ converges to the population minimizer set $\arg \min Q(\theta)$ as compact subsets of the parameter space (Newey and McFadden, 1994, Sec. 2.1). Economically, this guarantees that even if several parameter configurations yield equivalent population fits—e.g., observationally equivalent factor rotations or pricing kernels—the sample estimator converges to the correct *identified set* of admissible equilibria rather than drifting away due to sampling noise.

⁵⁴The Weierstrass extreme–value theorem states that any continuous real-valued function defined on a nonempty compact set attains its minimum and maximum. Since Assumption 6 ensures that $\Theta(\delta)$ is compact (closed and bounded) and nonempty, and Assumption 7 guarantees that $Q_T(\theta)$ is continuous in θ , it follows that $Q_T(\cdot)$ achieves a minimum within $\Theta(\delta)$ for each finite sample size T . Replacing Q_T by its population analogue $Q(\theta) = \mathbb{E} \|m_t(\theta)\|_{Q_z^{-1}}^2$ gives the same conclusion for the limiting (population) criterion, ensuring the existence of a well-defined true parameter θ^* . Economically, this means that the model’s objective—interpreted as the expected squared mispricing energy—has at least one feasible configuration of factor loadings and pricing kernel that minimizes systematic pricing errors within the admissible (no–arbitrage–bounded) region.

⁵⁵The argument proceeds from the uniform law of large numbers (ULLN) established in Lemma A.1, which asserts that $\sup_{\theta \in \Theta(\delta)} |Q_T(\theta) - Q(\theta)| \xrightarrow{P} 0$. This result implies that, as $T \rightarrow \infty$, the random function $Q_T(\theta)$ converges to the deterministic function $Q(\theta)$ uniformly over all admissible parameters. Formally, for any $\varepsilon > 0$, there exists T_ε such that

$$\mathbb{P} \left(\sup_{\theta \in \Theta(\delta)} |Q_T(\theta) - Q(\theta)| < \varepsilon \right) > 1 - \varepsilon, \quad \forall T > T_\varepsilon.$$

Uniform convergence means that the magnitude of stochastic fluctuations in the empirical criterion vanishes uniformly across $\Theta(\delta)$ —that is, no region of the parameter space experiences systematically larger sampling noise as the sample grows. Compactness of $\Theta(\delta)$ and continuity of $m_t(\theta)$ (Assumption 7) ensure that the limit function $Q(\theta) = \mathbb{E} [\|m_t(\theta)\|_{Q_z^{-1}}^2]$ is continuous and bounded, so convergence holds globally rather than pointwise.

Mathematically, this uniform convergence establishes a stochastic analogue of the Weierstrass theorem for extremum estimators: the random objective $Q_T(\theta)$ behaves asymptotically like its population version $Q(\theta)$ over the entire feasible set. Economically, it means that as the time dimension increases, the empirical measure of mispricing—captured by the average squared pricing errors—approaches a smooth, deterministic limit describing the long–run factor structure. Thus, sampling noise or short–term market fluctuations cannot alter the shape of the empirical objective surface, guaranteeing that its minimizer converges to the true economic configuration of factor loadings and alpha bounds.

converges in probability to the set of population minimizers of $Q(\theta)$,

$$\hat{\theta}_T \xrightarrow[T \rightarrow \infty]{p} \arg \min_{\theta \in \Theta(\delta)} Q(\theta).$$

If the population minimizer is unique, the set reduces to a singleton $\{\theta^*\}$, yielding the consistency result

$$\hat{\theta}_T \xrightarrow[T \rightarrow \infty]{p} \theta^*.^{56} \quad (37)$$

□

Intuitively, compactness restricts the admissible pricing–kernel parameters to a bounded and closed region of the Q_Z –weighted characteristic space, guaranteeing that a minimum of the empirical objective exists for every sample. The uniform convergence result then ensures that the location of this empirical minimum converges to the population minimizer as sampling noise vanishes. Economically, this means that the estimator’s feasible domain excludes explosive or degenerate factor configurations, while the law of large numbers aligns the empirical fit with its long–run equilibrium geometry—thereby securing consistent recovery of the underlying pricing kernel and factor structure.

A.4 Constraint Qualification and KKT Regularity

This subsection establishes the analytical foundation for the constrained optimization problem that defines the No–Arbitrage IPCA estimator. Building on the probabilistic regularity results from Appendix A.3, I now formalize the conditions under which the first–order Karush–Kuhn–Tucker (KKT) system is well posed and locally unique on the constrained manifold. These conditions guarantee that the Lagrange multipliers exist, the bordered Hessian is positive definite on the tangent space, and the solution mapping is continuously differentiable with respect to perturbations in the data and weighting matrices.

Lemma A.4 (LICQ, Slater, and KKT Regularity) provides the main result of this section. It shows that if the equality constraints—comprising the Q_Z –orthogonality and normalization conditions—satisfy the *Linear Independence Constraint Qualification (LICQ)* and the no–arbitrage inequality $\psi(\Gamma_\alpha) \leq \delta$ satisfies the *Slater interiority condition*, then (i) a valid KKT system exists with finite multipliers; (ii) the reduced Hessian is positive definite on the tangent space, implying a strict local minimum; and (iii) the Lagrange multipliers vary smoothly under small perturbations of the optimization problem (strong regularity).

Together, these results ensure that the constrained estimator is not only feasible and consistent but also locally stable in both statistical and numerical senses. Economically, the LICQ and Slater conditions correspond to a well–behaved no–arbitrage geometry: distinct orthogonality and normalization equations are linearly independent, and the true pricing kernel lies strictly within the admissible bound on mispricing. This

⁵⁶The argmin–continuity theorem for extremum estimators (Newey and McFadden, 1994, Thm. 2.1) provides sufficient conditions for consistency: (i) the sample objective $Q_T(\theta)$ converges to $Q(\theta)$ uniformly in probability; (ii) the feasible set $\Theta(\delta)$ is compact; and (iii) the population objective has a unique minimizer θ^* . Assumptions 6–7 guarantee compactness and continuity of the parameter space, while Lemma A.1 ensures uniform convergence of $Q_T(\theta)$. Economically, this means that as the time dimension increases, the estimator $\hat{\theta}_T$ concentrates around the true population configuration that minimizes average squared mispricing in characteristic space, ensuring consistent recovery of the latent factor structure and the no–arbitrage alpha bound.

guarantees that the constrained optimization defining the NA–IPCA estimator has a unique and economically meaningful local solution, forming the bridge between the theoretical identification results of Appendix A.3 and the weighted least–squares and LAN expansions developed in Appendix B.

A.5 Manifold Geometry and Identification Maps

This subsection develops the geometric and identification foundation of the No–Arbitrage IPCA (NA–IPCA) estimator. The model’s orthogonality and normalization restrictions— $\Gamma_\beta^\top Q_Z \Gamma_\beta = I_K$, $\Gamma_\beta^\top Q_Z \Gamma_\delta = 0$, $\Gamma_\beta^\top Q_Z \Gamma_\alpha = 0$, and $\psi(\Gamma_\alpha) \leq \delta$ —do not merely constrain estimation: they define a smooth and compact surface in characteristic space on which all economically admissible parameter configurations must lie. Viewed through this lens, the NA–IPCA estimator can be interpreted as the projection of empirical pricing moments onto a curved “no–arbitrage manifold” that encodes both equilibrium and identification structure.

From an economic standpoint, this manifold represents the feasible pricing geometry consistent with market equilibrium. Each point on the surface corresponds to a configuration of factor loadings and alphas that satisfies the model’s orthogonality and bounded–mispricing conditions. Movements along the manifold describe infinitesimal rotations or re–weightings of systematic risk exposures that remain compatible with equilibrium pricing. By contrast, movements orthogonal to the manifold correspond to directions that either violate no–arbitrage or are economically redundant—changes that cannot be distinguished from alternative normalizations of the same risk space.⁵⁷

Formally, the feasible parameter set can be written as

$$\Theta(\delta) = \left\{ (\Gamma_\beta, F, \Gamma_\delta, \Gamma_\alpha) : \Gamma_\beta^\top Q_Z \Gamma_\beta = I_K, \Gamma_\beta^\top Q_Z \Gamma_\delta = 0, \Gamma_\beta^\top Q_Z \Gamma_\alpha = 0, \psi(\Gamma_\alpha) \leq \delta \right\}.$$

This set decomposes naturally into four geometric components that together define the structure of the NA–IPCA manifold:

- (i) a Q_Z –weighted Stiefel manifold $\mathcal{V}_{L,K}(Q_Z) = \{\Gamma_\beta : \Gamma_\beta^\top Q_Z \Gamma_\beta = I_K\}$ describing the admissible K –dimensional characteristic subspaces that span latent systematic risk;
- (ii) an orthogonal–complement Stiefel structure for the observable (PSF) loadings, defined by $\Gamma_\beta^\top Q_Z \Gamma_\delta = 0$, which ensures that the observable and latent factors occupy distinct pricing subspaces;
- (iii) an ellipsoidal region $\mathcal{E}_\delta = \{\Gamma_\alpha : \Gamma_\alpha^\top Q_Z \Gamma_\alpha \leq \delta\}$ that bounds the magnitude of admissible mispricing in the Q_Z –metric; and
- (iv) a Euclidean component for the factor realizations $F = [f_1, \dots, f_T] \in \mathbb{R}^{K \times T}$, optionally restricted to a bounded set for compactness (cf. Assumption 6).

Taken together, these components endow $\Theta(\delta)$ with the structure of a smooth product manifold: for any

⁵⁷Geometrically, the manifold defines the subset of parameter values that jointly satisfy orthogonality and bounded–alpha constraints under the Q_Z –weighted inner product. Economically, it captures the admissible combinations of risk exposures and pricing directions consistent with equilibrium no–arbitrage. Tangent movements correspond to feasible adjustments of the pricing kernel—small rotations or rescalings of factor loadings that preserve equilibrium restrictions— whereas normal directions represent redundant or arbitrage–inconsistent deviations outside the admissible pricing geometry.

given Γ_β , the admissible observable loadings Γ_δ lie on the Q_Z -orthogonal complement of its column space, the intercept Γ_α resides within the bounded ellipsoid $\mathcal{E}_\delta$, and the time-series factors evolve in Euclidean space.

The remainder of this subsection proceeds as follows. Section A.5.1 characterizes the weighted Stiefel manifold $\mathcal{V}_{L,K}(Q_Z)$ and derives its tangent and normal spaces, which describe feasible and infeasible perturbations of the latent factor loadings. Section A.5.2 examines the model's rotation invariance and introduces a quotient-manifold representation that eliminates redundant parameterizations by identifying all rotation-equivalent configurations. Finally, Section A.5.3 develops a measurable normalization based on a weighted Procrustes alignment, selecting a unique and interpretable representative within each rotation-equivalence class without altering the model's fit or economic content. Together, these results provide the geometric and identification framework underpinning the efficiency, LAN, and rotation-invariance results established in Appendix B.

A.5.1 The Q_Z -Weighted Stiefel Manifold and Tangent Spaces

This section formalizes the geometric structure implied by the orthogonality constraint $\Gamma_\beta^\top Q_Z \Gamma_\beta = I_K$, which defines the Q_Z -weighted Stiefel manifold on which the latent factor loadings Γ_β reside. The manifold representation provides the mathematical language for describing feasible perturbations of the factor space, clarifying how local changes in Γ_β preserve identification while infeasible perturbations are projected out in the first-order and LAN analyses that follow.

The Q_Z -weighted Stiefel manifold. The orthogonality constraint $\Gamma_\beta^\top Q_Z \Gamma_\beta = I_K$ defines a smooth curved surface in characteristic space that restricts the admissible configurations of factor loadings to those consistent with equilibrium normalization. Formally,

$$\mathcal{V}_{L,K}(Q_Z) = \left\{ \Gamma_\beta \in \mathbb{R}^{L \times K} : \Gamma_\beta^\top Q_Z \Gamma_\beta = I_K \right\}. \quad (38)$$

Each matrix Γ_β represents one admissible basis for a K -dimensional subspace of the L -dimensional characteristic space that is orthonormal under the weighted inner product $\langle a, b \rangle_{Q_Z} = a^\top Q_Z b$ (cf. Eq. (4)). Because $Q_Z \succ 0$, the linear map $\Gamma_\beta \mapsto Q_Z^{1/2} \Gamma_\beta$ is an isomorphism between $\mathcal{V}_{L,K}(Q_Z)$ and the classical Euclidean Stiefel manifold $\mathcal{V}_{L,K}(I) = \{U : U^\top U = I_K\}$, so that all local differential properties follow from standard Stiefel geometry (Edelman et al., 1998, Ch. 2; Absil et al., 2008, Ch. 3).

This manifold provides the geometric stage on which the constrained optimization in the NA-IPCA estimator is performed. Its curvature encodes the orthogonality restrictions that link the factor-loading matrix Γ_β to the characteristic metric Q_Z : first-order perturbations must remain tangent to $\mathcal{V}_{L,K}(Q_Z)$ to preserve identification, while movements normal to the manifold would violate the normalization constraint and are projected out by the KKT multipliers. Hence, all statistical derivatives—gradients, Hessians, and LAN expansions—are taken with respect to this constrained geometry rather than the ambient Euclidean space.

In asset-pricing terms, $\mathcal{V}_{L,K}(Q_Z)$ represents the set of all feasible *risk-exposure configurations* that span a K -dimensional priced-risk space within the larger characteristic domain. The matrix Q_Z acts as the equilibrium metric weighting different characteristic directions by their cross-sectional variability. Moving

along the manifold corresponds to a smooth *rotation or reallocation of systematic risk exposures* that preserves their orthogonality and normalization, while departing from the manifold would imply inconsistent scaling or redundant pricing directions. This geometric structure therefore ensures that the latent factor space is economically well-defined, statistically identifiable, and stable under small perturbations—a foundation on which the local-asymptotic (LAN) and efficiency results in Appendix B are built.

Tangent space. The feasible infinitesimal perturbations of Γ_β that preserve the orthonormality condition $\Gamma_\beta^\top Q_Z \Gamma_\beta = I_K$ to first order arise by differentiating this constraint:

$$(d\Gamma_\beta)^\top Q_Z \Gamma_\beta + \Gamma_\beta^\top Q_Z (d\Gamma_\beta) = 0.$$

Because I_K is constant, the derivative must vanish, requiring that the matrix $\Gamma_\beta^\top Q_Z (d\Gamma_\beta)$ be *skew-symmetric*. Intuitively, the infinitesimal displacement $d\Gamma_\beta$ must merely *rotate* the current columns of Γ_β within the Q_Z -orthogonal subspace without altering their lengths or mutual inner products; any perturbation violating this skew symmetry would stretch or compress the basis, driving the parameter off the manifold.

Hence, the tangent space of the Q_Z -weighted Stiefel manifold at a regular point Γ_β is

$$\mathcal{T}_{\Gamma_\beta} \mathcal{V}_{L,K}(Q_Z) = \left\{ \dot{\Gamma}_\beta \in \mathbb{R}^{L \times K} : \Gamma_\beta^\top Q_Z \dot{\Gamma}_\beta + \dot{\Gamma}_\beta^\top Q_Z \Gamma_\beta = 0 \right\}. \quad (39)$$

This set collects all admissible first-order directions $\dot{\Gamma}_\beta$ that preserve the orthonormality constraint, or equivalently, all feasible “velocity directions” along which Γ_β can move while remaining on the curved surface $\mathcal{V}_{L,K}(Q_Z)$.

Mathematically, Eq. (39) characterizes the subspace on which the KKT first-order conditions must be projected to maintain feasibility. Tangent directions describe variations that keep $\Gamma_\beta^\top Q_Z \Gamma_\beta = I_K$ true to first order, whereas normal (symmetric) directions would alter the constraint and are therefore neutralized by the multipliers. Consequently, all statistical gradients and Hessians in the NA-IPCA estimator are taken with respect to this tangent-space geometry rather than in the full ambient Euclidean space.

In asset-pricing terms, the tangent space represents the admissible *reallocations of systematic risk exposures* that preserve the equilibrium structure of the pricing kernel. Small movements along $\mathcal{T}_{\Gamma_\beta} \mathcal{V}_{L,K}(Q_Z)$ correspond to infinitesimal rotations of the latent factor basis—alternative representations of the same priced risk span—while movements normal to the manifold would distort the Sharpe-ratio geometry or duplicate priced directions, violating no-arbitrage. Therefore, tangent perturbations capture all economically meaningful local adjustments in the factor loadings, providing the natural coordinate system for the efficiency and LAN results in Appendix B.

Tangent-space decomposition and first-order feasibility. Any feasible infinitesimal perturbation of the loading matrix Γ_β that preserves the orthonormality constraint $\Gamma_\beta^\top Q_Z \Gamma_\beta = I_K$ must lie within the Q_Z -tangent space $\mathcal{T}_{\Gamma_\beta} \mathcal{V}_{L,K}(Q_Z)$, characterized by the linearized first-order condition

$$\Gamma_\beta^\top Q_Z \dot{\Gamma}_\beta + \dot{\Gamma}_\beta^\top Q_Z \Gamma_\beta = 0, \quad (40)$$

which represents the $\mathcal{O}(\varepsilon)$ term in the expansion of $(\Gamma_\beta + \varepsilon \dot{\Gamma}_\beta)^\top Q_Z (\Gamma_\beta + \varepsilon \dot{\Gamma}_\beta) = I_K + \mathcal{O}(\varepsilon^2)$. A standard

result in Stiefel geometry (Edelman et al., 1998, Prop. 2.1) shows that every $\dot{\Gamma}_\beta \in \mathcal{T}_{\Gamma_\beta} \mathcal{V}_{L,K}(Q_Z)$ admits the decomposition

$$\dot{\Gamma}_\beta = \Gamma_\beta A_\beta + M_\beta B_\beta, \quad (41)$$

where:

- $A_\beta \in \mathbb{R}^{K \times K}$ is *skew-symmetric* ($A_\beta^\top = -A_\beta$), representing infinitesimal rotations *within* the current latent subspace;
- $M_\beta \in \mathbb{R}^{L \times (L-K)}$ has Q_Z -orthonormal columns spanning the orthogonal complement of $\text{col}(\Gamma_\beta)$, i.e. $\Gamma_\beta^\top Q_Z M_\beta = 0$ and $M_\beta^\top Q_Z M_\beta = I_{L-K}$;
- $B_\beta \in \mathbb{R}^{(L-K) \times K}$ determines admissible directions that *tilt* the latent span toward new priced directions in the Q_Z -orthogonal complement.

To verify that (41) satisfies the tangent constraint (40), observe that

$$\Gamma_\beta^\top Q_Z \dot{\Gamma}_\beta = \Gamma_\beta^\top Q_Z (\Gamma_\beta A_\beta + M_\beta B_\beta) = (\Gamma_\beta^\top Q_Z \Gamma_\beta) A_\beta + (\Gamma_\beta^\top Q_Z M_\beta) B_\beta = A_\beta,$$

and similarly

$$\dot{\Gamma}_\beta^\top Q_Z \Gamma_\beta = (\Gamma_\beta A_\beta + M_\beta B_\beta)^\top Q_Z \Gamma_\beta = A_\beta^\top.$$

Hence

$$\Gamma_\beta^\top Q_Z \dot{\Gamma}_\beta + \dot{\Gamma}_\beta^\top Q_Z \Gamma_\beta = A_\beta + A_\beta^\top, \quad (42)$$

which equals zero if and only if A_β is skew-symmetric. This condition defines the feasible *first-order directions* that preserve the manifold to $\mathcal{O}(\varepsilon)$.

The decomposition $\dot{\Gamma}_\beta = \Gamma_\beta A_\beta + M_\beta B_\beta$ has a clear dual meaning. Mathematically, $\Gamma_\beta A_\beta$ describes rotations within the existing span—changes of basis that do not affect the subspace itself—whereas $M_\beta B_\beta$ represents deformations that move the span toward its Q_Z -orthogonal complement. Geometrically, the tangent component $\Gamma_\beta A_\beta$ captures non-identifiable rotations along the manifold, while $M_\beta B_\beta$ encodes the economically identifiable tilts that alter which characteristic directions are priced. Economically, the former correspond to re-labeling of risk factors with no effect on prices, whereas the latter modify the systematic risk structure itself—the directions the NA-IPCA estimator learns from data. In the constrained optimization, the KKT multipliers project the first-order conditions onto the tangent space $\mathcal{T}_{\Gamma_\beta} \mathcal{V}_{L,K}(Q_Z)$, eliminating normal (symmetric) components that would violate the constraint. This projection is the geometric reason the first-order conditions of the estimator reduce to a *weighted least-squares system* on a lower-dimensional, well-identified manifold—a structure that later yields block-diagonal Fisher information and LAN efficiency in Appendix B.

Normal space. The normal space at Γ_β collects all directions that *leave* the manifold $\mathcal{V}_{L,K}(Q_Z)$ by violating the first-order orthogonality condition. It is defined as the Q_Z -orthogonal complement of the tangent space $\mathcal{T}_{\Gamma_\beta} \mathcal{V}_{L,K}(Q_Z)$ and can be written as

$$\mathcal{N}_{\Gamma_\beta} \mathcal{V}_{L,K}(Q_Z) = \{\Gamma_\beta S : S = S^\top \in \mathbb{R}^{K \times K}\}, \quad (43)$$

where S is a *symmetric deformation matrix* that acts within the latent factor subspace $\text{col}(\Gamma_\beta)$. Each element $\Gamma_\beta S$ therefore represents a small rescaling or shearing of the existing loading directions—that is, an infinitesimal change in the relative magnitude or alignment of the columns of Γ_β without introducing any new orthogonal directions.

Substituting $\dot{\Gamma}_\beta = \Gamma_\beta S$ into the linearized constraint $\Gamma_\beta^\top Q_Z \dot{\Gamma}_\beta + \dot{\Gamma}_\beta^\top Q_Z \Gamma_\beta = 0$ yields

$$\Gamma_\beta^\top Q_Z \dot{\Gamma}_\beta + \dot{\Gamma}_\beta^\top Q_Z \Gamma_\beta = S + S^\top,$$

which vanishes only when S is skew-symmetric. Hence symmetric S correspond to directions that violate Q_Z -orthonormality and thus form the orthogonal complement of the tangent space. Intuitively, these normal directions stretch or compress the latent factor basis rather than rotate it, moving the parameter off the manifold. The symmetric matrix S modifies the scale or correlation structure of the latent risk exposures inside the existing span of Γ_β . Such transformations change the overall normalization of factor loadings but do not alter which risks are priced; they represent statistically and economically redundant variations. Projecting the first-order conditions onto the tangent space removes these non-identifiable distortions, ensuring that the estimator reacts only to economically meaningful rotations or tilts of the systematic risk space.

The manifold $\mathcal{V}_{L,K}(Q_Z)$ represents the admissible set of factor-loading structures consistent with equilibrium no-arbitrage. Tangent directions describe economically meaningful, feasible rotations of the systematic risk space—small reallocations of factor exposures that preserve identification and the Q_Z -orthogonality of risk premia—while normal directions represent statistically and economically inadmissible deformations that either rescale or duplicate priced directions. This tangent-normal decomposition provides the geometric foundation for the constrained projections and LAN expansions in Appendix B, where only tangent components contribute to first-order variation and information content.⁵⁸

⁵⁸This calculation clarifies the geometric difference between *feasible* (tangent) and *infeasible* (normal) perturbations of the loading matrix Γ_β under the constraint $\Gamma_\beta^\top Q_Z \Gamma_\beta = I_K$. Let $\Gamma_\beta \in \mathcal{V}_{L,K}(Q_Z)$ and consider two types of infinitesimal deformations. For Tangent (feasible) perturbation. Let the perturbation be $\dot{\Gamma}_\beta = \Gamma_\beta A_\beta + M_\beta B_\beta$, where $A_\beta \in \mathbb{R}^{K \times K}$ is *skew-symmetric* ($A_\beta^\top = -A_\beta$), M_β spans the Q_Z -orthogonal complement of Γ_β , and B_β parameterizes admissible tilts into that complement. For $\tilde{\Gamma}_\beta = \Gamma_\beta + \varepsilon(\Gamma_\beta A_\beta + M_\beta B_\beta)$ with infinitesimal ε , expand:

$$\begin{aligned} \tilde{\Gamma}_\beta^\top Q_Z \tilde{\Gamma}_\beta &= (\Gamma_\beta + \varepsilon \Gamma_\beta A_\beta + \varepsilon M_\beta B_\beta)^\top Q_Z (\Gamma_\beta + \varepsilon \Gamma_\beta A_\beta + \varepsilon M_\beta B_\beta) \\ &= I_K + \varepsilon(A_\beta^\top + A_\beta) + O(\varepsilon^2), \end{aligned}$$

since $\Gamma_\beta^\top Q_Z M_\beta = 0$ and $M_\beta^\top Q_Z M_\beta = I_{L-K}$. Because A_β is skew-symmetric, $A_\beta^\top + A_\beta = 0$, so the $O(\varepsilon)$ term vanishes and $\tilde{\Gamma}_\beta^\top Q_Z \tilde{\Gamma}_\beta = I_K + O(\varepsilon^2)$. Thus the orthonormality constraint is preserved up to second order. Geometrically, such perturbations correspond to infinitesimal *rotations* of the latent subspace—small reorientations of the columns of Γ_β that remain on the manifold $\mathcal{V}_{L,K}(Q_Z)$. Economically, they represent alternative parameterizations of the same priced-risk span: the factor space is rotated but its economic content remains identical. For Normal (infeasible) perturbation. Let instead $\dot{\Gamma}_\beta = \Gamma_\beta S$, where $S \in \mathbb{R}^{K \times K}$ is *symmetric* ($S = S^\top$). Repeating the expansion gives

$$(\Gamma_\beta + \varepsilon \Gamma_\beta S)^\top Q_Z (\Gamma_\beta + \varepsilon \Gamma_\beta S) = \Gamma_\beta^\top Q_Z \Gamma_\beta + 2\varepsilon S + O(\varepsilon^2) = I_K + 2\varepsilon S + O(\varepsilon^2).$$

The first-order term no longer vanishes: the perturbation changes the scale or correlation structure of the columns of Γ_β , moving the parameter *off the manifold*. Geometrically, S acts as a symmetric dilation that stretches or compresses the latent basis rather than rotating it. Economically, such deformations correspond to unbounded rescalings of risk exposures—directions that would alter Sharpe ratios or duplicate priced risks—and are therefore ruled out by the no-arbitrage orthonormality constraint. The two perturbations differ fundamentally in both geometry and economics: the skew-symmetric A_β leaves $\Gamma_\beta^\top Q_Z \Gamma_\beta$ unchanged to first

Geometry near a Regular Point. At any regular parameter value $\theta \in \Theta(\delta)$ —i.e., a point at which the Jacobian of the active constraints has full row rank and the feasible set is a smooth manifold—the local geometry of $\Theta(\delta)$ supplies the precise directions along which the estimator can move without violating identification or no-arbitrage. This local description serves two roles in the paper: *methodologically*, it is the stage on which the KKT stationarity conditions can be projected to obtain the WLS-equivalent first-order system in Appendix B; *economically*, it decomposes perturbations of the pricing kernel into meaningful shifts in the *priced risk space* versus mere *relabelings* (rotations) that carry no pricing content.

Local symmetry and rotational directions. Because the model prices only the span of latent and observable factors, not their particular bases, the group action of $\mathcal{G} = \mathcal{O}(K) \times \mathcal{O}(K_{\text{add}})$ leaves both the fitted returns and the objective $Q_T(\theta)$ invariant; cf. (44).⁵⁹ The *orbit* of θ under $\mathcal{G}$ consists of parameterizations that encode the same characteristic-based priced risk structure and hence the same economic content. The tangent space to this orbit,

$$\mathcal{T}_\theta^{(\text{rot})} = \left\{ (\Gamma_\beta A_\beta, -A_\beta^\top F, \Gamma_\delta A_\delta, 0) : A_\beta \in \mathfrak{so}(K), A_\delta \in \mathfrak{so}(K_{\text{add}}) \right\},$$

collects precisely those “do-nothing” directions that perform infinitesimal orthogonal reparameterizations of the factor blocks. Moving along $\mathcal{T}_\theta^{(\text{rot})}$ only rotates the coordinates of the factor space and leaves all model-implied prices and alphas unchanged. It follows that asymptotic expansions and test statistics must be constructed on the quotient geometry $\Theta(\delta)/\mathcal{G}$ (Section A.5.2), so that inference is invariant to labeling conventions.

Identified (tangent) directions vs. redundant (normal) directions. The orthogonality and normalization constraints $\Gamma_\beta^\top Q_Z \Gamma_\beta = I_K$, $\Gamma_\beta^\top Q_Z \Gamma_\delta = 0$, $\Gamma_\beta^\top Q_Z \Gamma_\alpha = 0$, together with $\psi(\Gamma_\alpha) \leq \delta$, carve out a smooth submanifold $\Theta(\delta) \subset \mathbb{R}^{L \times K} \times \mathbb{R}^{K \times T} \times \mathbb{R}^{L \times K_{\text{add}}} \times \mathbb{R}^L$. As shown in (39)–(42), the tangent space at Γ_β splits into *rotational modes* (skew-symmetric A_β) and *identifying modes* (those orthogonal to $\mathcal{T}_\theta^{(\text{rot})}$, e.g., $M_\beta B_\beta$ components). The normal space $\mathcal{N}_{\Gamma_\beta}$ (e.g., symmetric S -deformations in (43)) collects rule-violating deformations that would break $\Gamma_\beta^\top Q_Z \Gamma_\beta = I_K$; these are shut down by the KKT multipliers when I project the first-order conditions onto $\mathcal{T}_\theta \Theta(\delta)$. This projection is the geometric reason why the WLS *normal equations* in Appendix B can be stated on a lower-dimensional, well-posed subspace where the model is identifiable.

In cross-sectional factor models, prices are determined by exposure to *spaces* of systematic risk, not by the arbitrary choice of basis within those spaces. The local decomposition above separates (i) *true* economic movements—those that tilt the priced span or adjust the magnitude of the alpha budget $\psi(\Gamma_\alpha)$ —from (ii) *spurious* movements—basis rotations that do not change any testable implication. Estimation and inference should therefore exploit the tangent-space geometry to: (a) enforce no-arbitrage and orthogonality via KKT

order (hence feasible), while the symmetric S produces linear deviations (hence infeasible). Projecting the estimator’s first-order conditions onto the tangent space eliminates the S -type distortions, ensuring that the optimization proceeds only along economically meaningful directions—rotations and tilts of the systematic risk span that respect the no-arbitrage geometry. This distinction underlies the curvature properties of the constraint manifold and motivates the projection step in the KKT-based derivations of Appendix B.

⁵⁹A *group action* describes how a set of rotations $g = (R_\beta, R_\delta) \in \mathcal{G} = \mathcal{O}(K) \times \mathcal{O}(K_{\text{add}})$ transforms the parameter tuple without changing its economic meaning: $g \cdot \theta = (\Gamma_\beta R_\beta, R_\beta^\top F, \Gamma_\delta R_\delta, \Gamma_\alpha)$. Because orthogonal rotations preserve inner products, both the sample objective and the no-arbitrage constraints remain unchanged. Geometrically, this symmetry means that all rotated versions of the factor basis correspond to the same point on the quotient manifold $\Theta(\delta)/\mathcal{G}$, so that the model identifies the span of priced risks rather than any particular orientation.

projections; (b) focus sampling variation on the economically identified directions (ensuring $\sqrt{T}$ -rate LAN on $\mathcal{T}_\theta \Theta(\delta)$); and (c) obtain *rotation-invariant* standard errors and tests (Appendix B, Section B.6).

In sum, the “regular-point” geometry supplies the analytic scaffolding that links the model’s economic restrictions to its asymptotics: it explains why the NA-IPCA estimator behaves like a WLS problem on a curved, lower-dimensional space, why only the span (not the basis) of $(\Gamma_\beta, \Gamma_\delta)$ is identified, and why inference about risk prices and alphas—carried out on the tangent space of $\Theta(\delta)$ —is both *invariant* to reparameterizations and *efficient* in the presence of heavy tails once I impose bounded-influence scoring (Assumption 8).

A.5.2 Rotation Invariance and the Quotient-Manifold Representation

The NA-IPCA parameterization contains latent (and, when present, observable) factor blocks whose *spans* drive prices, whereas their specific orthonormal bases do not. This is the model’s fundamental *rotation invariance*: changing coordinates inside a priced subspace leaves fitted returns and the objective unchanged. Methodologically, this symmetry signals that the estimator and its first-order (KKT) conditions must be formulated on an appropriate quotient space so that inference is invariant to reparameterizations; see Appendix B, in particular Section B.6. Economically, it formalizes that cross-sectional asset pricing identifies only the *systematic-risk space* itself, not the labels assigned to its basis vectors.

Group action and invariance. Formally, for any pair of orthogonal matrices $R_\beta \in \mathcal{O}(K)$ and $R_\delta \in \mathcal{O}(K_{\text{add}})$, the transformation

$$(\Gamma_\beta, F, \Gamma_\delta) \mapsto (\Gamma_\beta R_\beta, R_\beta^\top F, \Gamma_\delta R_\delta) \quad (44)$$

preserves both the fitted returns $Z_t(\Gamma_\beta f_t + \Gamma_\delta F_t^{\text{obs}} + \Gamma_\alpha)$ and the sample objective $Q_T(\theta)$. Consequently, the feasible set $\Theta(\delta)$ (cf. (17)) contains infinitely many parameterizations of the same pricing kernel.

Quotient-manifold representation. To factor out this redundancy, let

$$\mathcal{G} := \mathcal{O}(K) \times \mathcal{O}(K_{\text{add}})$$

be the compact Lie group that rotates the factor bases in (44). Define the equivalence relation $\theta_1 \sim \theta_2$ iff $\theta_2 = g \cdot \theta_1$ for some $g \in \mathcal{G}$. The associated *quotient space* is

$$\Theta(\delta)/\mathcal{G} = \{[\theta] : [\theta] := \{g \cdot \theta : g \in \mathcal{G}\}, \theta \in \Theta(\delta)\}, \quad (45)$$

which collects orbits of rotation-equivalent parameterizations into single points and inherits a smooth-manifold structure under standard regularity conditions (Absil et al., 2008, Ch. 3, Edelman et al., 1998, Prop. 2.2).⁶⁰ By working on $\Theta(\delta)/\mathcal{G}$, I impose *identification up to rotation*: estimation and inference target the equivalence class $[\theta]$ rather than any specific basis representation. This ensures that KKT systems, LAN expansions, and standard errors are free from coordinate artifacts, thereby aligning the statistical object of interest with economic identification.

⁶⁰See Absil et al., 2008, Ch. 3 and Edelman et al. (1998, Prop. 2.1–2.3) for the quotient-manifold construction on Stiefel/Grassmann manifolds. The econometric implication—that only the span of latent factors is identified, not their labels—follows the factor-model identification logic in Bai (2003); Bai and Ng (2008) and the modern factor/SDF literature (e.g., Kelly et al., 2019).

Rotation invariance implies an *equivariance* property of the first-order (KKT) system: if $\theta = (\Gamma_\beta, F, \Gamma_\delta, \Gamma_\alpha)$ satisfies the constrained stationarity conditions, then so does $g \cdot \theta = (\Gamma_\beta R_\beta, R_\beta^\top F, \Gamma_\delta R_\delta, \Gamma_\alpha)$ for every $g = (R_\beta, R_\delta) \in \mathcal{O}(K) \times \mathcal{O}(K_{\text{add}})$; see (44).⁶¹ Passing from the ambient space to the quotient $\Theta(\delta)/\mathcal{G}$ amounts to projecting the KKT conditions onto the tangent of the quotient manifold, i.e., removing the purely rotational directions $\mathcal{T}_\theta^{(\text{rot})}$ (cf. Section A.5.1). This projection yields a *well-posed*, WLS-equivalent first-order system (Appendix B) and an associated *block-diagonal* Fisher information matrix (Appendix B.6), so that the resulting LAN expansion and the implied standard errors are *rotation-invariant*.

The quotient perspective clarifies what the NA-IPCA genuinely estimates: the *priced risk space*—the span of Γ_β (and, with PSF, the orthogonal block Γ_δ)—together with an α -discipline $\psi(\Gamma_\alpha) \leq \delta$ that enforces a finite “alpha budget.” All orthogonal reparameterizations within a priced subspace encode the same SDF and thus the same cross-sectional implications; they are economically irrelevant “gauge choices.” By passing to $\Theta(\delta)/\mathcal{G}$ I (i) prevent *double-counting* of risk channels (via the block-orthogonality $\Gamma_\beta^\top Q_Z \Gamma_\delta = 0$ from §A.4), (ii) ensure that *only* spans are identified and estimated, and (iii) guarantee that standard errors and tests are invariant to the choice of basis. This geometric discipline is a core ingredient in the efficiency-robustness results of Appendix B and motivates the *Procrustes normalization* introduced in Section A.5.3, which selects a canonical representative in each orbit for presentation and interpretation without disturbing the underlying economics or the asymptotics.

⁶¹The link between rotation invariance and the *equivariance* of the first-order KKT system is a direct consequence of the symmetry of the optimization problem. Formally, recall that the NA-IPCA estimator solves a constrained minimization

$$\min_{\theta \in \Theta(\delta)} Q_T(\theta) \quad \text{s.t.} \quad g(\theta) = 0, \quad h(\theta) \leq 0,$$

where $Q_T(\theta)$, $g(\theta)$, and $h(\theta)$ are all *invariant* under the group action $g \cdot \theta = (\Gamma_\beta R_\beta, R_\beta^\top F, \Gamma_\delta R_\delta, \Gamma_\alpha)$ for $(R_\beta, R_\delta) \in \mathcal{O}(K) \times \mathcal{O}(K_{\text{add}})$ (see (44)). Invariance means that $Q_T(g \cdot \theta) = Q_T(\theta)$, $g(g \cdot \theta) = 0$, and $h(g \cdot \theta) = h(\theta)$ for all θ .

Mathematically, this symmetry implies that if (θ, Λ, μ) satisfies the KKT system

$$\nabla_\theta Q_T(\theta) + G(\theta)^\top \Lambda + \mu \nabla_\theta h(\theta) = 0, \quad g(\theta) = 0, \quad h(\theta) \leq 0, \quad \mu \geq 0, \quad \mu h(\theta) = 0,$$

then for every rotation $g = (R_\beta, R_\delta) \in \mathcal{G}$, the transformed triplet $(g \cdot \theta, \Lambda, \mu)$ satisfies the same system. This follows from the chain rule:

$$\nabla_\theta Q_T(g \cdot \theta) = D(g)^\top \nabla_\theta Q_T(\theta),$$

where $D(g)$ is the orthogonal differential of the group action; because $D(g)^\top D(g) = I$, the transformed gradient has the same norm and vanishes if and only if the original one does. Hence, stationarity is preserved by any orthogonal rotation — the KKT equations are *equivariant* under the symmetry group $\mathcal{G}$.

Geometrically, the KKT gradient $\nabla_\theta Q_T(\theta)$ can be viewed as a vector field defined on the manifold $\Theta(\delta)$; the group action g merely rotates coordinates within each orbit but leaves both the level sets of $Q_T(\theta)$ and the feasible manifold invariant. Therefore, the entire first-order system “commutes” with the rotation: rotating the argument of the estimator rotates the gradient field accordingly, without altering its zeros. Projecting this system onto the tangent space of the quotient manifold removes the redundant rotational directions $\mathcal{T}_\theta^{(\text{rot})}$, producing a well-posed reduced system on the identified subspace.

Economically, this property expresses the idea that changing the coordinate basis of latent factors does not alter any pricing implication. Rotations of $(\Gamma_\beta, F, \Gamma_\delta)$ correspond to relabeling risk factors, not to modifying the underlying priced-risk span. Thus, both the objective function (average squared mispricing) and the optimality conditions (KKT equations) are invariant to such relabelings. The equivariance of the first-order system is therefore a mathematical formalization of the economic statement that the NA-IPCA estimator depends only on the *span* of systematic risk, not on the arbitrary choice of basis used to represent it.

A.5.3 Procrustes Normalization and Canonical Representatives

Sections A.5.1–A.5.2 established that the parameter tuple $\theta = (\Gamma_\beta, F, \Gamma_\delta, \Gamma_\alpha)$ is identified only *up to rotation* by the group $\mathcal{G} = \mathcal{O}(K) \times \mathcal{O}(K_{\text{add}})$. This rotational symmetry implies that every feasible point belongs to an orbit of observationally equivalent parameterizations encoding the same stochastic discount factor (SDF). Methodologically, a measurable normalization is required to choose one representative per orbit so that the estimator and its asymptotic distribution are well defined. Economically, this step fixes the *orientation* of the priced-risk space without altering its span, ensuring that estimated factors and loadings are comparable across samples and consistent with no–arbitrage geometry.

Orthogonal Procrustes Alignment. Let $\Gamma_\beta^{\text{ref}} \in \mathcal{V}_{L,K}(Q_Z)$ denote a fixed reference frame (for instance, a benchmark sample or population limit) satisfying $(\Gamma_\beta^{\text{ref}})^\top Q_Z \Gamma_\beta^{\text{ref}} = I_K$ and, when observable factors are present, $(\Gamma_\beta^{\text{ref}})^\top Q_Z \Gamma_\delta^{\text{ref}} = 0$. Given any $\Gamma_\beta \in \mathcal{V}_{L,K}(Q_Z)$, define the Q_Z –weighted orthogonal Procrustes problem

$$R_\beta^* = \arg \min_{R \in \mathcal{O}(K)} \|\Gamma_\beta - \Gamma_\beta^{\text{ref}} R\|_{Q_Z}^2, \quad \|X\|_{Q_Z}^2 = \text{tr}(X^\top Q_Z X), \quad (46)$$

whose equivalent maximization form is $\max_{R \in \mathcal{O}(K)} \text{tr}(R^\top M_\beta)$ with $M_\beta = (\Gamma_\beta^{\text{ref}})^\top Q_Z \Gamma_\beta$. Let $M_\beta = U_\beta \Sigma_\beta V_\beta^\top$ be the singular–value decomposition. By the classical Procrustes theorem (Absil et al., 2008, §7.4), the optimal alignment is $R_\beta^* = U_\beta V_\beta^\top$. Setting

$$\Gamma_\beta^* = \Gamma_\beta R_\beta^*, \quad F^* = (R_\beta^*)^\top F,$$

aligns the latent factor block with the reference orientation. An analogous construction for the observable block yields $R_\delta^* \in \mathcal{O}(K_{\text{add}})$ and $\Gamma_\delta^* = \Gamma_\delta R_\delta^*$. Because orthogonal post–multiplication preserves $\Gamma_\beta^\top Q_Z \Gamma_\delta = 0$, this alignment respects all equilibrium constraints and leaves the objective $Q_T(\theta)$ unchanged.

Normalization Map and Invariance. Define the *Procrustes normalization map*

$$N : \Theta(\delta) \rightarrow \Theta(\delta), \quad N(\Gamma_\beta, F, \Gamma_\delta, \Gamma_\alpha) = (\Gamma_\beta R_\beta^*, (R_\beta^*)^\top F, \Gamma_\delta R_\delta^*, \Gamma_\alpha). \quad (47)$$

Since R_β^* and R_δ^* are orthogonal,

$$Q_T(N(\theta)) = Q_T(\theta), \quad g(N(\theta)) = 0, \quad h(N(\theta)) \leq 0,$$

and $N(\theta)$ lies on the same $\mathcal{G}$ –orbit as θ . The mapping is smooth in a neighborhood of any regular point and measurable globally through a lexicographic tie–breaking rule on singular values.⁶² Hence the normalization is an *equivariant projection* from $\Theta(\delta)$ onto a measurable cross–section of the quotient manifold $\Theta(\delta)/\mathcal{G}$ that fixes orientation but not economic content.

The Procrustes normalization assigns a canonical orientation to the estimated factor blocks, making $(\Gamma_\beta^*, F^*, \Gamma_\delta^*)$ directly comparable across samples, horizons, or model specifications. Economically, this

⁶²When the singular values of M_β and M_δ are simple, the mapping $\theta \mapsto (R_\beta^*, R_\delta^*)$ is $\mathcal{C}^\infty$ by the implicit–function theorem (Absil et al., 2008, Prop. 7.4.3); with multiplicities, a deterministic sign convention yields a Borel–measurable selection, ensuring global well–posedness of $N(\cdot)$.

step reflects the principle that *markets price spaces of systematic risk, not coordinate bases*. Rotating the basis leaves all risk premia and alphas unchanged; the normalization simply chooses a consistent coordinate frame for interpretation and comparison. In practice, the reference orientation may correspond to a benchmark factor dictionary (e.g., Fama–French portfolios) or to the population limit of the estimator. Because the transformation preserves $Q_T(\theta)$ and all constraints, statistical efficiency and no–arbitrage discipline remain intact, while rotation–invariant inference (Appendix B, Proposition B.6) can be implemented on the normalized coordinates.

Procrustes normalization provides the final bridge between geometric identification and econometric inference. By anchoring each equivalence class $[\theta] \in \Theta(\delta)/\mathcal{G}$ to a unique representative $N(\theta)$, it converts the rotation–invariant geometry of the model into a smooth, well–defined coordinate system. All subsequent LAN expansions and efficiency results in Appendix B are derived on this canonical chart, ensuring that asymptotic distributions, standard errors, and economic interpretations are invariant to the arbitrary orientation of the factor bases.

A.6 Weighting, Pre–Whitening, and Boundedness

This subsection establishes the analytic and geometric foundations for the feasible weighting and pre–whitening operations used in the No–Arbitrage IPCA (NA–IPCA) estimator. While the preceding sections characterized the structure of the parameter manifold and its rotation symmetries, the present section ensures that the empirical estimator is numerically stable and statistically efficient when the cross–sectional moments exhibit heteroskedasticity or time–varying dispersion. Methodologically, it formalizes the conditions under which the feasible (weighted) objective based on estimated covariance matrices $\hat{\Omega}_t$ remains asymptotically equivalent to its ideal population counterpart that uses the true $\Omega_t = \text{Var}(X_t | Z_t)$. Economically, it interprets these weighting and scaling operations as a *risk–normalization* of the characteristic space: each pricing direction is measured in variance–adjusted units, so that the factor–loading geometry and the no–arbitrage constraints are preserved even when the underlying data display heteroskedastic risk fluctuations.

The subsection proceeds as follows. Section A.6.1 states the boundedness and consistency assumptions on the feasible weight matrices that guarantee positive–definite and well–conditioned weighting throughout the sample. Section A.6.2 defines the pre–whitened moment and design matrices $(\tilde{X}_t, \tilde{W}_t)$, demonstrating that this linear transformation leaves the Q_Z –geometry and the constrained optimization problem unchanged. Section A.6.3 then discusses the analytic and economic implications of these properties, showing that the feasible weighted estimator inherits the same first–order (KKT) and second–order (LAN) structure as its infeasible GLS benchmark. Together, these results ensure that the NA–IPCA estimator remains statistically regular, geometrically consistent, and economically interpretable under realistic forms of heteroskedasticity and cross–sectional dispersion.

A.6.1 Boundedness and Consistency of Feasible Weights

The feasible implementation of the NA–IPCA estimator replaces the population cross–sectional covariance matrices $\Omega_t = \text{Var}(X_t | Z_t)$ with their data–based estimates $\hat{\Omega}_t$. This section establishes the analytic and geometric regularity conditions under which such weighting remains asymptotically valid. Methodologically,

these assumptions ensure that the feasible (weighted) objective is well conditioned, continuously differentiable, and statistically equivalent to its ideal GLS counterpart. Economically, they guarantee that the scaling of risk exposures across characteristics reflects stable cross-sectional variability rather than numerical artifacts, preserving the meaning of “one unit of priced risk” throughout the characteristic space.

Assumption 9 (Uniform Boundedness and Feasible-Weight Consistency).⁶³ Let $\Omega_t = \text{Var}(X_t | Z_t)$ denote the population cross-sectional covariance matrix of characteristic moments and let $\hat{\Omega}_t$ be its feasible estimate (e.g., based on residuals or robust scale estimators such as Huber or MAD scales). We assume the following regularity conditions hold uniformly over t :

- (a) **Uniform positive-definiteness and bounded eigenvalues.** For each period t , let $\Omega_t \in \mathbb{R}^{L \times L}$ denote the population cross-sectional covariance (or weighting) matrix, and let $\hat{\Omega}_t$ be its feasible sample analogue. We assume that both Ω_t and $\hat{\Omega}_t$ are symmetric positive-definite and that their eigenvalues are uniformly bounded away from zero and infinity:

$$\exists 0 < c_\Omega < C_\Omega < \infty \quad \text{s.t.} \quad c_\Omega I_L \preceq \Omega_t \preceq C_\Omega I_L, \quad \forall t = 1, \dots, T, \quad (48)$$

and, moreover, that the feasible weights converge uniformly to their population targets,

$$\sup_{1 \leq t \leq T} \|\hat{\Omega}_t - \Omega_t\|_F = o_p(1). \quad (49)$$

In words, all weighting matrices remain strictly positive-definite and well conditioned, with eigenvalues bounded within the fixed interval $[c_\Omega, C_\Omega]$. The feasible estimates $\hat{\Omega}_t$ are asymptotically consistent in operator norm, so that whitening transformations based on $\hat{\Omega}_t^{-1/2}$ exist for all large T and behave smoothly as $T \rightarrow \infty$.⁶⁴

- (b) **Ratio consistency (asymptotic equivalence of scales).** The feasible weights converge to their population counterparts in relative scale:

$$\hat{\Omega}_t^{-1/2} \Omega_t^{1/2} \xrightarrow{p} I_L \quad \text{uniformly in } t, \quad (50)$$

⁶³This assumption establishes the regularity conditions for the feasible weighting scheme $\hat{\Omega}_t$ used in the NA-IPCA estimator. It ensures that the empirical cross-sectional covariance matrices are uniformly well conditioned and asymptotically equivalent to their population counterparts, so that the feasible weighted objective $Q_T^{(\text{feas})}(\theta)$ is statistically indistinguishable from the ideal GLS criterion. Economically, it guarantees that characteristic directions are scaled by stable, finite-variance risk units rather than by noisy or degenerate weights, preserving the interpretation of “one unit of priced risk” in the characteristic geometry. *Used in Lemma B.5 (Weighted-Least-Squares Equivalence) and Proposition B.3 (Asymptotic Efficiency).*

⁶⁴This assumption ensures that the weighting matrices used in estimation remain numerically and statistically well behaved. By bounding all eigenvalues between two finite constants, the matrix square-root and inverse operations are continuous and stable, so that small estimation errors in $\hat{\Omega}_t$ do not produce large or discontinuous changes in the weighted objective. Without this condition, nearly singular or excessively large eigenvalues would distort the curvature of the criterion, making the optimization surface flat in some directions and excessively steep in others, thereby undermining convergence and the validity of asymptotic expansions. From a statistical point of view, such spectral instability would give disproportionate influence to a few characteristic directions and make the variance of the estimator undefined. Geometrically, keeping Ω_t uniformly positive-definite confines the weighting metric to a compact, smooth region of the positive-definite cone, preserving the curvature of the Q_Z -manifold. Economically, this means that each characteristic contributes to the pricing kernel in proportion to its finite and well-measured risk, maintaining a consistent definition of “one unit of priced exposure” throughout the characteristic space.

so that $\|\widehat{\Omega}_t^{-1/2} - \Omega_t^{-1/2}\|_F = o_p(1)$ and $Q_T^{(\text{feas})}(\theta) = Q_T^{(\text{GLS})}(\theta) + o_p(1)$.⁶⁵

Condition (48) guarantees that all feasible weight matrices remain strictly positive-definite and exclude ill-conditioned scaling, ensuring that the weighted objective has continuous curvature and an invertible Hessian in large samples. From a geometric standpoint, this confines $\{\widehat{\Omega}_t\}$ to a compact subset of the cone of positive-definite matrices, preserving the smooth curvature of the Q_Z -weighted manifold defined in Section A.5.1. Economically, bounded eigenvalues prevent any single characteristic direction from dominating the cross-sectional variance—an expression of diversification across characteristics that stabilizes factor scaling.

Condition (50) enforces *ratio consistency*, ensuring that the pre-whitened moments constructed with $\widehat{\Omega}_t^{-1/2}$ differ from those based on $\Omega_t^{-1/2}$ only by asymptotically negligible rotations in covariance geometry. Consequently, the feasible (weighted) first-order and second-order systems coincide with the ideal GLS versions up to $o_p(1)$, yielding the WLS-equivalence result in Lemma B.5. In asset-pricing terms, weighting by Ω_t^{-1} amounts to measuring expected returns in *variance-normalized* units. The feasible estimator simply replaces the unknown population dispersion structure with a robust empirical proxy that scales each characteristic by its realized volatility. Uniform boundedness ensures all characteristics contribute comparably to the pricing kernel, while ratio consistency guarantees convergence to the same equilibrium metric as in the no-arbitrage benchmark. Thus, pre-whitening acts as a *risk-unit standardization*—stabilizing the geometry of the cross-sectional moment conditions without altering identification or efficiency. Under Assumption 9, the feasible NA-IPCA estimator inherits the same asymptotic efficiency and robustness properties as the infeasible GLS estimator, providing the foundation for Lemma B.5 and the LAN expansions in Appendix B.

A.6.2 Pre-Whitened Moments and Design Matrices

Having established the boundedness and consistency of the feasible weights in Assumption 9, we now introduce the pre-whitened representation of the moment and design matrices used in the NA-IPCA estimator. Methodologically, this linear transformation converts the generalized-least-squares (GLS) problem with heterogeneous cross-sectional variance into an equivalent ordinary-least-squares (OLS) problem under the standard Euclidean metric. Economically, it corresponds to expressing all characteristic directions in variance-normalized risk units, ensuring that the geometry of priced exposures is stable across assets and through time.

Definition of Pre-Whitened Moments. For each period t , define the pre-whitened variables

$$\tilde{X}_t := \widehat{\Omega}_t^{-1/2} X_t, \quad \tilde{W}_t := \widehat{\Omega}_t^{-1/2} W_t \widehat{\Omega}_t^{-1/2}, \quad (51)$$

⁶⁵Ratio-consistency requires the feasible weight matrices to approximate the population weights not only in absolute terms but also in relative scale. This guarantees that the empirical and population versions of the whitening transformation define almost the same measurement of cross-sectional risk. Technically, it ensures that the geometry of the feasible criterion—its curvature and inner-product structure—matches that of the ideal generalized least-squares benchmark, so that the feasible estimator inherits the same first-order properties. If this condition were violated, the empirical weighting could distort the scaling of certain characteristic directions, giving excessive or insufficient importance to particular risk components and biasing both estimates and standard errors. Statistically, ratio-consistency keeps the weighting error small enough that it affects only higher-order terms, leaving the main convergence rate and asymptotic variance unchanged. In geometric and economic terms, it aligns the feasible normalization of returns with the true equilibrium metric, ensuring that the notion of a variance-standardized risk unit remains valid across samples.

where $X_t = Z_t^\top r_{t+1}/N_t$ and $W_t = Z_t^\top Z_t/N_t$ are the cross-sectional moment and characteristic-covariance matrices. Pre-multiplying by $\widehat{\Omega}_t^{-1/2}$ scales each characteristic direction by its estimated volatility, producing a standardized moment vector whose covariance matrix is approximately identity by Assumption 9. Because $\widehat{\Omega}_t$ is symmetric and uniformly positive-definite, the transformation (51) is well defined and continuous in t .⁶⁶

Analytic equivalence. Let $Q_T^{(\text{feas})}(\theta)$ denote the feasible weighted criterion and $Q_T^{(\text{GLS})}(\theta)$ the corresponding population GLS objective. Substituting (51) into the feasible criterion yields

$$Q_T^{(\text{feas})}(\theta) = \frac{1}{T} \sum_{t=1}^T \left\| \widehat{\Omega}_t^{-1/2} (X_t - W_t \Gamma_\beta f_t - W_t \Gamma_\delta F_t^{\text{obs}} - W_t \Gamma_\alpha) \right\|_2^2,$$

which is algebraically identical to an OLS criterion formed with the pre-whitened variables $(\widetilde{X}_t, \widetilde{W}_t)$. Under Assumption 9, the ratio-consistency condition $\widehat{\Omega}_t^{-1/2} \Omega_t^{1/2} \rightarrow I_L$ implies that

$$Q_T^{(\text{feas})}(\theta) = Q_T^{(\text{GLS})}(\theta) + o_p(1),$$

so the feasible and population systems share the same first-order (KKT) and second-order (information) structures. This result is formalized in Lemma B.5 (*Weighted-Least-Squares Equivalence*) in Appendix B.

Pre-whitening defines a local isometry (distance-preserving map) between the characteristic space endowed with the weighting metric Ω_t^{-1} and the standard Euclidean space. Geometrically, it preserves inner products and curvature on the Q_Z -weighted manifold introduced in Section A.5.1, ensuring that tangent and normal directions remain orthogonal after transformation. Economically, this operation expresses all cross-sectional pricing relations in variance-standardized risk units, allowing the factor loadings $(\Gamma_\beta, \Gamma_\delta, \Gamma_\alpha)$ to be interpreted on a common scale even when individual characteristics differ in volatility or sampling frequency. In this sense, pre-whitening is not merely a numerical convenience but an economic normalization that stabilizes the geometry of the no-arbitrage manifold under cross-sectional heteroskedasticity.

The pre-whitened representation (51) transforms the feasible NA-IPCA optimization into an equivalent OLS problem with standardized risk units. This equivalence underlies Lemma B.5 and Proposition B.3, where the LAN expansion and efficiency properties of the estimator are established. Hence, under Assumption 9, pre-whitening provides a smooth and rotation-invariant reparameterization of the characteristic space that preserves both the statistical efficiency and the economic interpretation of the NA-IPCA estimator.

⁶⁶The existence and stability of the pre-whitening map follow from two elementary facts about symmetric positive-definite (SPD) matrices. First, if a matrix A is SPD, then it admits a unique SPD square root $A^{1/2}$, and hence $A^{-1/2}$ is well defined. Second, on any set of SPD matrices with eigenvalues uniformly bounded away from zero and infinity, the map $A \mapsto A^{-1/2}$ is continuous (and in fact Lipschitz) in the operator/Frobenius norm. Concretely, by the spectral decomposition $\widehat{\Omega}_t = U_t \text{diag}(\lambda_{t,1}, \dots, \lambda_{t,L}) U_t^\top$ with $\lambda_{t,j} \in [c_\Omega, C_\Omega]$, one has

$$\widehat{\Omega}_t^{-1/2} = U_t \text{diag}(\lambda_{t,1}^{-1/2}, \dots, \lambda_{t,L}^{-1/2}) U_t^\top,$$

so all eigenvalues of $\widehat{\Omega}_t^{-1/2}$ lie in $[C_\Omega^{-1/2}, c_\Omega^{-1/2}]$ and the operator norm $\|\widehat{\Omega}_t^{-1/2}\|$ is uniformly bounded by $c_\Omega^{-1/2}$. Moreover, Assumption 9 implies $\|\widehat{\Omega}_t - \Omega_t\| \rightarrow 0$ uniformly over t , and the continuity of the functional calculus on the SPD cone yields $\|\widehat{\Omega}_t^{-1/2} - \Omega_t^{-1/2}\| \rightarrow 0$ uniformly in t . Therefore, the pre-whitening transformation in (51) is well defined for all t and varies continuously (with uniformly bounded operator norm) as the data change, which underpins the smoothness and stability properties used in the subsequent WLS-equivalence and LAN arguments.

A.6.3 Implications for Efficiency and Local Asymptotic Normality

Under Assumptions 1–8 and 9, the feasible (weighted and pre-whitened) NA–IPCA objective is analytically and geometrically aligned with its infeasible GLS analogue to first order. We summarize the ensuing implications.

1. **First-order (KKT) equivalence.** By Assumption 9 and the construction in (51), the gradient of the feasible objective coincides with that of the GLS problem up to a vanishing remainder:

$$\nabla Q_T^{(\text{feas})}(\theta_0) = \nabla Q_T^{(\text{GLS})}(\theta_0) + o_p(1).$$

Thus, the feasible and GLS Lagrange systems have identical stationary points in the limit, and the feasible estimator solves a *well-posed WLS-equivalent* first-order system; cf. Lemma B.5.

2. **Information and LAN preservation.** The Hessians of the feasible and GLS criteria differ by $o_p(1)$ uniformly on compact neighborhoods of θ_0 , so the corresponding Fisher information matrices, influence functions, and asymptotic covariances coincide to first order. In particular, the feasible NA–IPCA estimator admits the same local asymptotic normality (LAN) expansion and achieves the GLS asymptotic efficiency bound; see Proposition B.3.
3. **Invariance of the geometric structure.** Because pre-whitening is a local isometry (distance-preserving map) and preserves the orthogonality/normalization constraints, the tangent and normal decompositions in Section A.5.1 remain unchanged. The rotation-equivariance of the KKT system is preserved under the orthogonal reparametrization induced by pre-whitening. Consequently, the feasible estimator retains the rotation-invariant WLS representation developed in Appendix B.

Collectively, these properties guarantee that feasible weighting and pre-whitening preserve the statistical regularity and the geometric interpretation of the NA–IPCA estimator, providing a robust and computationally stable route to the same asymptotic efficiency as the infeasible GLS benchmark.

Remark A.6.1 (Practical Implementation of Feasible Weighting and Pre-Whitening). This remark outlines practical strategies for implementing the feasible weighting and pre-whitening operations in the NA–IPCA estimator. Mathematically, these procedures stabilize the sample covariance geometry and ensure the invertibility of the weighting metric; economically, they ensure a consistent risk-unit calibration of factor exposures, such that each characteristic contributes to the characteristic-based priced risk structure in proportion to its variance-normalized risk premium.

(i) **Robust scale estimators (Huber/MAD).** To reduce sensitivity to heavy-tailed cross-sectional returns, one may construct the feasible weights from robust residual scales rather than raw sample variances. For residual vector u_{it} , define the Huber-type scale estimator:

$$\hat{\sigma}_{t,j}^2 = \frac{1}{N_t} \sum_{i=1}^{N_t} \rho_{\kappa}(u_{it,j}^2), \quad \rho_{\kappa}(x) = \begin{cases} x, & |x| \leq \kappa, \\ \kappa(2|x| - \kappa), & |x| > \kappa, \end{cases} \quad (52)$$

and assemble $\hat{\Omega}_t = \text{diag}(\hat{\sigma}_{t,1}^2, \dots, \hat{\sigma}_{t,L}^2)$. The truncation parameter κ controls the trade-off between efficiency

and robustness: for light tails, $\kappa \rightarrow \infty$ recovers the sample variance, while for heavy tails, moderate κ bounds the influence of extreme returns. This estimator is consistent under α -mixing and finite $(2 + \tau)$ -moment conditions.⁶⁷ Robust scaling ensures that no single asset or characteristic dominates the variance structure, allowing the pre-whitening transformation to preserve the curvature of the weighted objective even under non-Gaussian risk.

(ii) Eigenvalue truncation (spectral regularization). When characteristics are highly collinear or the sample is small, $\hat{\Omega}_t$ may approach singularity. A stabilized estimate can be obtained by truncating its spectral decomposition:

$$\hat{\Omega}_t^{(\text{trunc})} = U_t \text{diag}(\max\{\lambda_{t,1}, \underline{\lambda}\}, \dots, \max\{\lambda_{t,L}, \underline{\lambda}\}) U_t^\top, \quad (53)$$

where $\hat{\Omega}_t = U_t \text{diag}(\lambda_{t,1}, \dots, \lambda_{t,L}) U_t^\top$ and $\underline{\lambda} > 0$ is a small ridge parameter (e.g., 10^{-5} – 10^{-3} of the median eigenvalue). This guarantees that $\hat{\Omega}_t^{(\text{trunc})}$ remains strictly positive-definite and well conditioned, ensuring the continuity of $\hat{\Omega}_t^{-1/2}$. Spectral truncation confines the feasible weighting matrices to a compact subset of the positive-definite cone, preserving the differentiability of the KKT system and preventing ill-posed curvature in the objective function.⁶⁸

(iii) Long-run covariance and HAC bandwidth. In samples with serially correlated X_t , the feasible weighting can be extended to a heteroskedasticity- and autocorrelation-consistent (HAC) estimator. Using the Bartlett kernel $k(x) = 1 - |x|$ for $|x| \leq 1$ and zero otherwise, define

$$\hat{\Omega}_t^{(\text{HAC})} = \sum_{|\ell| \leq L_T} k\left(\frac{\ell}{L_T}\right) \hat{\Gamma}_\ell, \quad \hat{\Gamma}_\ell = \frac{1}{T} \sum_{t=\ell+1}^T (X_t - \bar{X})(X_{t-\ell} - \bar{X})^\top, \quad (54)$$

where L_T is the bandwidth satisfying $L_T \rightarrow \infty$ and $L_T/T \rightarrow 0$. The standard Newey–West choice $L_T = \lfloor 4(T/100)^{2/9} \rfloor$ balances bias and variance when T is small relative to L . Under the mixing and moment conditions of Assumption 8, this HAC correction leaves the asymptotic LAN expansion and efficiency matrix

⁶⁷Consistency follows from the uniform LLN and stochastic-equi continuity in Lemma A.1 together with the compactness/argmin-continuity result in Lemma A.3, under Assumptions 1–7. For the constrained program, compactness of the feasible set $\Theta(\delta)$ (Assumption 6) and the KKT regularity in Section A.4 ensure well-posed multipliers and a stable argmin, so the constrained estimator is consistent; see Appendix A.3. For the feasible weighting, ratio-consistency in Assumption 9 and the WLS-equivalence Lemma B.5 imply that the feasible and GLS criteria differ by $o_p(1)$, so consistency carries over.

⁶⁸Consistency of the truncated weight and of its inverse square root. Let Assumption 9 hold so that $\inf_t \lambda_{\min}(\Omega_t) \geq c_\Omega > 0$ and $\sup_t \|\hat{\Omega}_t - \Omega_t\| = o_p(1)$. Fix any truncation level $\underline{\lambda} \in (0, c_\Omega/2)$ (or a sequence $\underline{\lambda}_T$ with $\underline{\lambda}_T < c_\Omega/2$ for all large T). By Weyl's eigenvalue bound (e.g., Horn and Johnson, 2013, Thm. 4.3.1), $\sup_t \max_i |\lambda_{t,i} - \hat{\lambda}_{t,i}| = o_p(1)$, hence $\mathbb{P}(\min_i \hat{\lambda}_{t,i} > c_\Omega/2 \text{ for all } t) \rightarrow 1$. On that event the truncation in (53) is inactive, so $\hat{\Omega}_t^{(\text{trunc})} = \hat{\Omega}_t$ for all t and

$$\sup_t \|\hat{\Omega}_t^{(\text{trunc})} - \Omega_t\| = \sup_t \|\hat{\Omega}_t - \Omega_t\| = o_p(1).$$

Because the map $A \mapsto A^{-1/2}$ is continuous on the SPD cone with spectrum contained in $[c_\Omega/2, \infty)$, it follows that

$$\sup_t \|(\hat{\Omega}_t^{(\text{trunc})})^{-1/2} - \Omega_t^{-1/2}\| = o_p(1),$$

i.e., the truncated pre-whitening remains ratio-consistent. Consequently the feasible criterion based on $\hat{\Omega}_t^{(\text{trunc})}$ is $Q_T^{(\text{feas})}(\theta) = Q_T^{(\text{GLS})}(\theta) + o_p(1)$, so all first- and second-order (KKT, information) statements are unchanged (cf. Lemma B.5). Intuitively, as all population eigenvalues are bounded away from zero, a fixed (or slowly vanishing) lower bound $\underline{\lambda} < c_\Omega/2$ is never binding with probability approaching one, and the truncation acts as a finite-sample regularizer without affecting the large-sample geometry.

unchanged to first order. Kernel-based smoothing ensures that time-varying cross-sectional dispersion is captured without over-fitting short-term noise, providing stable variance estimates even under serial dependence.⁶⁹

These implementation choices jointly establish a regularized risk-unit calibration of the characteristic space. Robust scale estimators attenuate the influence of extreme idiosyncratic shocks, spectral truncation maintains effective diversification across correlated characteristics, and HAC smoothing adjusts for temporal dependence in cross-sectional dispersion. Collectively, these procedures stabilize the covariance geometry and ensure that the pre-whitening operator $\widehat{\Omega}_t^{-1/2}$ acts as a smooth, well-conditioned transformation on the characteristic manifold—preserving the economic interpretation of one standardized unit of priced risk and sustaining both the interpretability and numerical stability of the NA-IPCA estimator.

A.7 Rotation-Equivalence and Normalization Preliminaries

This subsection formalizes the statistical counterpart of the geometric rotation symmetry introduced in Section A.5. Whereas Section A.5 described how the parameter manifold $\Theta(\delta)$ admits redundant representations under orthogonal changes of the factor basis, here we show that the empirical objective and constraints are *equivariant* to the same group action, and we provide a canonical, measurable way to select one representative per orbit. Methodologically, we first define the orthogonal group action $\mathcal{G} = \mathcal{O}(K) \times \mathcal{O}(K_{\text{add}})$ on the parameter space and prove invariance of the sample criterion and constraints (Lemma A.4). We then introduce a measurable, locally smooth Procrustes normalization map $N(\cdot)$ —built on the alignment construction in Section A.5.3—that selects a unique rotation-aligned representative within each orbit (Lemma A.5). Finally, we show that, after normalization and pre-whitening, the projected score and information on the tangent space are themselves *rotation-invariant* (Proposition A.6). These ingredients supply the identification and normalization machinery used in Appendix B (particularly Section B.6) to derive equivariance of the KKT system and a rotation-invariant LAN expansion.

⁶⁹Consistency of the HAC weight $\widehat{\Omega}_t^{(\text{HAC})}$. Let $\{X_t\}_{t=1}^T$ be a strictly stationary process with finite $(2 + \tau)$ -moment for some $\tau > 0$, and let its long-run covariance be $\Omega := \sum_{\ell=-\infty}^{\infty} \Gamma_\ell$, where $\Gamma_\ell := \text{Cov}(X_t, X_{t-\ell})$ and $\sum_{\ell=-\infty}^{\infty} \|\Gamma_\ell\| < \infty$. Assume the sequence is α -mixing with coefficients satisfying $\sum_{\ell \geq 1} \alpha(\ell)^{\tau/(2+\tau)} < \infty$. Let the kernel $k(\cdot)$ be Bartlett's, symmetric and bounded on $[-1, 1]$, and let the bandwidth satisfy $L_T \rightarrow \infty$ and $L_T/T \rightarrow 0$. Then the Newey–West estimator

$$\widehat{\Omega}_t^{(\text{HAC})} = \sum_{|\ell| \leq L_T} k\left(\frac{\ell}{L_T}\right) \widehat{\Gamma}_\ell, \quad \widehat{\Gamma}_\ell = \frac{1}{T} \sum_{t=\ell+1}^T (X_t - \bar{X})(X_{t-\ell} - \bar{X})^\top,$$

satisfies $\|\widehat{\Omega}_t^{(\text{HAC})} - \Omega\| \xrightarrow{P} 0$.

Proof. For each fixed lag ℓ , $\widehat{\Gamma}_\ell \rightarrow_p \Gamma_\ell$ by a standard law of large numbers for strongly mixing sequences with finite $(2 + \tau)$ -moment. Uniform convergence over $|\ell| \leq L_T$ follows because $L_T = o(T)$ and the Bartlett weights are bounded by one. Using the triangle inequality,

$$\left\| \sum_{|\ell| \leq L_T} k\left(\frac{\ell}{L_T}\right) (\widehat{\Gamma}_\ell - \Gamma_\ell) \right\| = o_p(1), \quad \left\| \sum_{|\ell| > L_T} \Gamma_\ell \right\| \rightarrow 0,$$

by dominated convergence and absolute summability of $\{\Gamma_\ell\}$. Hence $\widehat{\Omega}_t^{(\text{HAC})} = \Omega + o_p(1)$. Because the Bartlett kernel is positive semidefinite, $\widehat{\Omega}_t^{(\text{HAC})}$ remains p.s.d. in finite samples and inherits strict positive definiteness whenever the truncated spectral sum is bounded away from zero. Consequently, substituting $\widehat{\Omega}_t^{(\text{HAC})}$ for Ω_t in the feasible criterion yields $Q_T^{(\text{feas})}(\theta) = Q_T^{(\text{GLS})}(\theta) + o_p(1)$, so the LAN expansion and first-order efficiency results are unaffected. See Newey and West (1987) for the formal consistency proof under $L_T \rightarrow \infty$, $L_T/T \rightarrow 0$, and Hamilton (1994, Ch. 12) or Andrews (1991) for bias-variance and bandwidth analyses.

A.7.1 Group Action and Invariance

Lemma A.4 (Group–Action Invariance of Criterion and Constraints).⁷⁰ *Let the group action of $\mathcal{G} = \mathcal{O}(K) \times \mathcal{O}(K_{\text{add}})$ on $\Theta(\delta)$ be defined by*

$$g \cdot \theta = (\Gamma_\beta R_\beta, R_\beta^\top F, \Gamma_\delta R_\delta, \Gamma_\alpha), \quad g = (R_\beta, R_\delta) \in \mathcal{G}, \quad (55)$$

and let $Q_T(\theta)$, $g(\theta)$, $h(\theta)$ denote, respectively, the NA–IPCA sample objective, the stacked equality constraints, and the no–arbitrage inequality defined above. Then, for every $g \in \mathcal{G}$ and $\theta \in \Theta(\delta)$,

$$Q_T(g \cdot \theta) = Q_T(\theta), \quad g(g \cdot \theta) = g(\theta), \quad h(g \cdot \theta) = h(\theta). \quad (56)$$

Proof. By definition, the NA–IPCA per–period moment is $m_t(\theta) = X_t - W_t(\Gamma_\beta f_t + \Gamma_\delta F_t^{\text{obs}} + \Gamma_\alpha)$ and the objective weights Q_Z and Ω_t are symmetric positive–definite (Appendix A.6). Under the group action (55), note that

$$\Gamma_\beta f_t \mapsto (\Gamma_\beta R_\beta)(R_\beta^\top f_t) = \Gamma_\beta f_t, \quad \Gamma_\delta F_t^{\text{obs}} \mapsto (\Gamma_\delta R_\delta)(R_\delta^\top F_t^{\text{obs}}) = \Gamma_\delta F_t^{\text{obs}},$$

so $m_t(g \cdot \theta) = m_t(\theta)$ and therefore any (pre–)whitened quadratic form used in $Q_T(\theta)$ is unchanged:

$$Q_T(g \cdot \theta) = \frac{1}{T} \sum_{t=1}^T \|m_t(g \cdot \theta)\|_{\Omega_t^{-1}}^2 = \frac{1}{T} \sum_{t=1}^T \|m_t(\theta)\|_{\Omega_t^{-1}}^2 = Q_T(\theta).$$

For the constraints, use $R_\beta^\top R_\beta = I_K$, $R_\delta^\top R_\delta = I_{K_{\text{add}}}$ and symmetry of Q_Z to obtain

$$(\Gamma_\beta R_\beta)^\top Q_Z (\Gamma_\beta R_\beta) = R_\beta^\top (\Gamma_\beta^\top Q_Z \Gamma_\beta) R_\beta = R_\beta^\top I_K R_\beta = I_K,$$

and

$$(\Gamma_\beta R_\beta)^\top Q_Z (\Gamma_\delta R_\delta) = R_\beta^\top (\Gamma_\beta^\top Q_Z \Gamma_\delta) R_\delta = R_\beta^\top 0 R_\delta = 0,$$

so the equalities $g(\theta) = 0$ are preserved. Finally,

$$h(g \cdot \theta) = \psi(\Gamma_\alpha) = (\Gamma_\alpha)^\top Q_Z \Gamma_\alpha = h(\theta),^{71}$$

establishing (56). Hence, Q_T , g , and h are invariant under the action of $\mathcal{G}$. $\square$

Orthogonal rotations preserve both lengths and angles, implying that the fitted returns and equilibrium

⁷⁰This lemma records that the NA–IPCA objective and constraints are *equivariant* under the orthogonal group action that rotates the latent and observable factor bases. Mathematically, it implies that the sample criterion $Q_T(\theta)$, the equality constraints $g(\theta) = 0$ (orthonormality and block–orthogonality), and the inequality constraint $h(\theta) \leq 0$ (no–arbitrage bound) are invariant along group orbits. Economically, it formalizes that cross–sectional asset pricing identifies the *span of priced risks* rather than a particular coordinate basis for that span. *Used in Appendix B.6 (KKT equivariance and rotation–invariant LAN).*

⁷¹The inequality constraint $h(\theta) = \psi(\Gamma_\alpha) = \Gamma_\alpha^\top Q_Z \Gamma_\alpha - \delta$ depends only on the mispricing vector Γ_α and the fixed characteristic metric Q_Z . Under the group action (55), Γ_α is *not* transformed— the rotations R_β and R_δ act solely on the factor–loading blocks $(\Gamma_\beta, F, \Gamma_\delta)$ —so $\psi(\Gamma_\alpha)$ remains unchanged. Mathematically, this means the no–arbitrage bound measures the Q_Z –norm (or squared length) of Γ_α in characteristic space, which is invariant to basis rotations of the systematic–risk components. Economically, the bound limits the total magnitude of systematic mispricing and is independent of how the priced risk factors are labeled or oriented.

constraints depend only on the *risk span*—the subspace spanned by Γ_β and, when present, the Q_Z -orthogonal PSF block Γ_δ —and not on the particular basis used to represent that span. Mathematically, any rotation $R_\beta \in \mathcal{O}(K)$ satisfies

$$\|\Gamma_\beta R_\beta\|_{Q_Z}^2 = \text{tr}((\Gamma_\beta R_\beta)^\top Q_Z (\Gamma_\beta R_\beta)) = \text{tr}(\Gamma_\beta^\top Q_Z \Gamma_\beta),$$

so all Q_Z -inner products and associated curvature on the constraint manifold are preserved. Consequently, the model’s objective $Q_T(\theta)$ and its constraints depend only on $\text{span}(\Gamma_\beta)$ rather than on the individual columns of Γ_β . This invariance implies that statistical identification must be defined on the quotient manifold $\Theta(\delta)/\mathcal{G}$, which collects all rotation-equivalent configurations into a single point. Formulating inference on $\Theta(\delta)/\mathcal{G}$ removes redundant degrees of freedom, eliminates singularities in the Fisher information matrix, and yields a nondegenerate, rotation-invariant LAN expansion (see Appendix B.6).

In geometric terms, the quotient construction identifies all orthogonal rotations of the factor basis as a single location on the no-arbitrage manifold. Movements along rotational directions $\mathcal{T}_\theta^{(\text{rot})}$ are economically irrelevant—they merely reparameterize the same systematic-risk space—whereas movements orthogonal to this subspace alter the priced risk span itself and are therefore the only statistically and economically identifiable perturbations. By working on $\Theta(\delta)/\mathcal{G}$, the NA-IPCA estimator is anchored to a well-posed, smooth manifold where tangent directions correspond to genuine changes in the risk space and normal directions represent infeasible deformations violating no-arbitrage orthogonality.

In factor pricing, investors can only observe exposures to *spaces of risk*, not to particular linear combinations of those risks. Rotation invariance thus captures a fundamental economic symmetry: renaming or re-scaling the basis of priced factors leaves all testable implications unchanged. Estimating and conducting inference on the quotient manifold ensures that the NA-IPCA model identifies a unique equilibrium risk space, yields rotation-invariant standard errors, and maintains a consistent definition of “one unit of priced risk” across samples. This geometric discipline unifies statistical efficiency with economic invariance—the hallmark of the no-arbitrage geometry developed throughout this appendix.

In summary,

$$\begin{aligned} \text{No-arbitrage geometry} &= (\text{orthogonality} + \text{bounded alpha}) \\ &\cap (\text{rotation equivalence of the risk span}). \end{aligned}$$

Together, these two sets of restrictions ensure that the model is—mathematically—a smooth and identifiable manifold, forming a compact Riemannian submanifold of the characteristic space; and—economically—that pricing relations depend only on the *systematic-risk space*, not on any arbitrary labeling of factor bases. In other words, *rotation invariance* represents the geometric extension of the no-arbitrage principle: it defines the risk space independently of coordinate choices and guarantees that the entire NA-IPCA framework achieves both **statistical efficiency** and **economic invariance**.

A.7.2 Normalization as a Bridge to Rotation-Invariant Inference

This subsection clarifies the methodological role of the Procrustes normalization in Appendix A.7: it translates the geometric rotation symmetry of Section A.5 into a statistically well-posed framework for

inference. By selecting a unique, rotation-aligned representative in each orbit, the map $N(\cdot)$ provides the canonical coordinates on which scores, curvatures, and information are computed and compared across samples and parameterizations.

Taken together, Lemma A.5 supplies the missing bridge from geometric rotation symmetry to statistically well-posed inference. Mathematically, the Procrustes map $N(\cdot)$ selects, for every rotation-equivalent configuration, a unique and measurable representative that varies smoothly whenever the weighted cross-covariance spectra are simple; without this selection, the score and information would be evaluated on a non-unique parameterization, and the Fisher information would inherit spurious singular directions from basis relabelings. Geometrically, the map fixes an orientation while preserving the span, so that all calculations take place on a smooth cross-section of the quotient $\Theta(\delta)/\mathcal{G}$ rather than on a space with redundant coordinates. Statistically, this is what allows the projected (tangent-space) score and information to be invariant to orthogonal reparameterizations, which in turn underpins the *equivariance of the KKT system* and the *rotation-invariant LAN expansion* developed in Appendix B (see Lemma A.4 for invariance of the criterion and constraints, and Proposition A.6 for rotation-invariant projected score and information). Economically, the result formalizes that the estimator and its standard errors depend only on the priced risk space and not on arbitrary labeling of factor bases; absent Lemma A.5, identification would remain quotient-level in principle but the finite-sample and asymptotic calculations would lack a canonical coordinate system, jeopardizing both numerical stability and the interpretation of normalized systematic-risk scale.

Lemma A.5 (Measurable Normalization and Equivariance). ⁷² Let $\Gamma_\beta^{\text{ref}} \in \mathbb{R}^{L \times K}$ and $\Gamma_\delta^{\text{ref}} \in \mathbb{R}^{L \times K_{\text{add}}}$ be fixed full-rank reference loading matrices that satisfy

$$(\Gamma_\beta^{\text{ref}})^\top Q_Z \Gamma_\beta^{\text{ref}} = I_K, \quad (\Gamma_\beta^{\text{ref}})^\top Q_Z \Gamma_\delta^{\text{ref}} = 0, \quad (\Gamma_\delta^{\text{ref}})^\top Q_Z \Gamma_\delta^{\text{ref}} = I_{K_{\text{add}}}.$$

These matrices serve as fixed coordinate frames that anchor the orientation of the latent and observable factor spaces.

Define the normalization map

$$N(\Gamma_\beta, F, \Gamma_\delta, \Gamma_\alpha) = (\Gamma_\beta R_\beta^*, R_\beta^{*\top} F, \Gamma_\delta R_\delta^*, \Gamma_\alpha), \quad (57)$$

where R_β^* and R_δ^* solve the Q_Z -weighted orthogonal Procrustes alignment problems

$$R_\beta^* = \arg \min_{R \in \mathcal{O}(K)} \|\Gamma_\beta - \Gamma_\beta^{\text{ref}} R\|_{Q_Z}^2, \quad R_\delta^* = \arg \min_{R \in \mathcal{O}(K_{\text{add}})} \|\Gamma_\delta - \Gamma_\delta^{\text{ref}} R\|_{Q_Z}^2, \quad (58)$$

with $\|X\|_{Q_Z}^2 = \text{tr}(X^\top Q_Z X)$.

Then the following properties hold:

⁷²This lemma defines the Procrustes normalization map $N(\cdot)$ as a measurable and locally smooth rule that selects one representative from each rotation-equivalent parameter configuration. It ensures that for every $\theta \in \Theta(\delta)$, the normalization $N(\theta)$ fixes a unique and consistent orientation of the factor-loading matrices $(\Gamma_\beta, \Gamma_\delta)$ while leaving the economic content of the model unchanged. Mathematically, this normalization provides a smooth cross-section of the quotient space $\Theta(\delta)/\mathcal{G}$; economically, it guarantees that estimation and inference depend only on the *risk span* (the space spanned by the priced factors) and not on the arbitrary labeling of basis vectors. Used in Lemma B.6 (*Equivariance of the KKT System*) and Proposition B.6 (*Rotation-Invariant LAN Expansion*).

(a) **Equivariance.** For every $g = (R_\beta, R_\delta) \in \mathcal{G} = \mathcal{O}(K) \times \mathcal{O}(K_{\text{add}})$ and every $\theta = (\Gamma_\beta, F, \Gamma_\delta, \Gamma_\alpha) \in \Theta(\delta)$,

$$N(g \cdot \theta) = N(\theta), \quad g \cdot \theta = (\Gamma_\beta R_\beta, R_\beta^\top F, \Gamma_\delta R_\delta, \Gamma_\alpha). \quad (59)$$

That is, applying any orthogonal rotation to the latent or observable factor bases leaves the normalized parameter $N(\theta)$ unchanged.

(b) **Local smoothness and measurability.** If the singular values of $(\Gamma_\beta^{\text{ref}})^\top Q_Z \Gamma_\beta$ and $(\Gamma_\delta^{\text{ref}})^\top Q_Z \Gamma_\delta$ are distinct, then $N(\cdot)$ is continuously differentiable ($\mathcal{C}^\infty$) in a neighborhood of θ . If singular values are repeated, one can impose a deterministic rule (for example, fixing column signs or using a lexicographic order) to select R_β^* and R_δ^* consistently. This choice makes $N(\cdot)$ a well-defined Borel-measurable mapping on $\Theta(\delta)$.

Proof. Well-definedness (existence and uniqueness of the aligners). For any full-rank pair $(\Gamma_\beta, \Gamma_\beta^{\text{ref}})$, the Q_Z -weighted Procrustes minimization problem in (58)

$$\min_{R \in \mathcal{O}(K)} \|\Gamma_\beta - \Gamma_\beta^{\text{ref}} R\|_{Q_Z}^2 = \min_{R \in \mathcal{O}(K)} \text{tr}[(\Gamma_\beta - \Gamma_\beta^{\text{ref}} R)^\top Q_Z (\Gamma_\beta - \Gamma_\beta^{\text{ref}} R)]$$

can be rewritten, by expanding the quadratic form, as

$$\|\Gamma_\beta - \Gamma_\beta^{\text{ref}} R\|_{Q_Z}^2 = \text{tr}(\Gamma_\beta^\top Q_Z \Gamma_\beta) + \text{tr}((\Gamma_\beta^{\text{ref}})^\top Q_Z \Gamma_\beta^{\text{ref}}) - 2 \text{tr}(R^\top M_\beta), \quad M_\beta := (\Gamma_\beta^{\text{ref}})^\top Q_Z \Gamma_\beta.$$

The first two terms are constants with respect to R , so minimizing the weighted distance is equivalent to maximizing the trace term:

$$\max_{R \in \mathcal{O}(K)} \text{tr}(R^\top M_\beta), \quad M_\beta = (\Gamma_\beta^{\text{ref}})^\top Q_Z \Gamma_\beta. \quad (60)$$

Equation (60) is the standard orthogonal Procrustes formulation. Let $M_\beta = U_\beta \Sigma_\beta V_\beta^\top$ be a singular-value decomposition; then the unique maximizer is $R_\beta^* = U_\beta V_\beta^\top$.⁷³ Hence, for every Γ_β , the weighted Procrustes problem admits a unique and continuous solution R_β^* , and analogously R_δ^* for the observable block.

Equivariance under rotations. Consider any orthogonal transformation $g = (R_\beta, R_\delta) \in \mathcal{G} = \mathcal{O}(K) \times \mathcal{O}(K_{\text{add}})$. Let $\theta' = g \cdot \theta$ denote the *rotated (rotation-equivalent) parameter configuration* obtained by applying g to $\theta = (\Gamma_\beta, F, \Gamma_\delta, \Gamma_\alpha)$, that is,

$$\theta' = g \cdot \theta = (\Gamma_\beta R_\beta, R_\beta^\top F, \Gamma_\delta R_\delta, \Gamma_\alpha).$$

Intuitively, θ' represents the same pricing model as θ but expressed in a different orthonormal basis for the

⁷³This equivalence follows from the identity $\|A - BR\|_{Q_Z}^2 = \|A\|_{Q_Z}^2 + \|B\|_{Q_Z}^2 - 2 \text{tr}(R^\top B^\top Q_Z A)$ and from the orthogonality of $R \in \mathcal{O}(K)$, which makes the first two terms independent of R . The trace maximization in (60) admits a closed-form solution $R^* = UV^\top$ obtained from the singular-value decomposition of $B^\top Q_Z A$ (Absil et al., 2008, § 7.4). This guarantees that the optimal alignment exists, is unique, and depends continuously on (A, B) whenever the singular values of $B^\top Q_Z A$ are distinct.

latent and observable risk spaces. Using the invariance of the Q_Z -norm under orthogonal transformations,⁷⁴

$$\|\Gamma_\beta R_\beta - \Gamma_\beta^{\text{ref}} R\|_{Q_Z} = \|\Gamma_\beta - \Gamma_\beta^{\text{ref}} R R_\beta^\top\|_{Q_Z},$$

so the alignment problems satisfy

$$R_\beta^*(\theta') = \arg \min_{R \in \mathcal{O}(K)} \|\Gamma_\beta - \Gamma_\beta^{\text{ref}} R R_\beta^\top\|_{Q_Z} = R_\beta^*(\theta) R_\beta^\top,$$

and analogously $R_\delta^*(\theta') = R_\delta^*(\theta) R_\delta^\top$. Substituting these relations into (57) gives

$$N(\theta') = (\Gamma_\beta R_\beta R_\beta^*(\theta)^\top, R_\beta R_\beta^*(\theta)^\top F, \Gamma_\delta R_\delta R_\delta^*(\theta)^\top, \Gamma_\alpha) = N(\theta),$$

which proves (59). Intuitively, rotating the basis before alignment is exactly undone by the optimal aligner because the Q_Z -inner product and orthogonality constraints are preserved.⁷⁵

Local smoothness and global measurability. Let $M_\beta = (\Gamma_\beta^{\text{ref}})^\top Q_Z \Gamma_\beta$ and $M_\delta = (\Gamma_\delta^{\text{ref}})^\top Q_Z \Gamma_\delta$ be the weighted cross-covariance matrices used in the Procrustes problems (58). Write their singular-value decompositions as

$$M_\beta = U_\beta \Sigma_\beta V_\beta^\top, \quad M_\delta = U_\delta \Sigma_\delta V_\delta^\top,$$

so that $R_\beta^* = U_\beta V_\beta^\top$ and $R_\delta^* = U_\delta V_\delta^\top$. The dependence of (U, Σ, V) on M determines the regularity of the map $\theta \mapsto N(\theta)$ in (57). If all singular values of M_β and M_δ are distinct, then the singular-vector triplets (U, Σ, V) depend smoothly on M and hence on θ . Formally, the SVD decomposition is a $\mathcal{C}^\infty$ map in a neighborhood of any point where the spectrum is simple, and consequently the product $R^* = UV^\top$ is $\mathcal{C}^\infty$ in θ (Absil et al., 2008, Prop. 7.4.3). This implies that $N(\theta)$ is locally smooth on each stratum of the parameter space where the singular-value multiplicities remain constant.

⁷⁴Intuition and proof of Q_Z -norm invariance under orthogonal rotations. Let the Q_Z -weighted norm be

$$\|A\|_{Q_Z}^2 = \text{tr}(A^\top Q_Z A) = \|Q_Z^{1/2} A\|_F^2,$$

where $\|\cdot\|_F$ denotes the Frobenius norm. For any orthogonal matrix $R \in \mathcal{O}(K)$ satisfying $R^\top R = I_K$, we have

$$\|AR - BR\|_{Q_Z} = \|Q_Z^{1/2}(A - B)R\|_F = \|Q_Z^{1/2}(A - B)\|_F = \|A - B\|_{Q_Z},$$

because the Frobenius norm is invariant under right-multiplication by an orthogonal matrix. Indeed, for any X and $R \in \mathcal{O}(K)$,

$$\|XR\|_F^2 = \text{tr}((XR)^\top (XR)) = \text{tr}(R^\top X^\top XR) = \text{tr}(X^\top X R R^\top) = \text{tr}(X^\top X) = \|X\|_F^2.$$

Equivalently, expanding the trace of the Q_Z -norm gives

$$\|AR - BR\|_{Q_Z}^2 = \text{tr}[R^\top (A - B)^\top Q_Z (A - B) R] = \text{tr}[(A - B)^\top Q_Z (A - B)] = \|A - B\|_{Q_Z}^2,$$

since the trace is cyclic and $R^\top R = I_K$. Geometrically, The matrix $Q_Z^{1/2}$ defines a weighted inner product $\langle a, b \rangle_{Q_Z} = a^\top Q_Z b$ on the characteristic space. Right-multiplying by R simply rotates the coordinate axes within that inner-product geometry, preserving all lengths and angles. Hence orthogonal rotations of Γ_β or Γ_δ describe the same geometric subspace—the identical systematic-risk span viewed in a different orthonormal frame (Edelman et al., 1998, Ch. 2; Absil et al., 2008, Ch. 3).

⁷⁵Orthogonal right-multiplication preserves the weighted norm: $\|AR\|_{Q_Z}^2 = \text{tr}((AR)^\top Q_Z (AR)) = \text{tr}(R^\top A^\top Q_Z AR) = \text{tr}(A^\top Q_Z A)$ since $R^\top R = I$ and the trace is cyclic. The same identity underlies the invariance of the constraints $\Gamma_\beta^\top Q_Z \Gamma_\beta = I$ and $\Gamma_\beta^\top Q_Z \Gamma_\delta = 0$ after right rotations.

When some singular values coincide, smoothness fails globally because the associated eigenspaces are not uniquely oriented. To restore global well-definedness, one may impose a deterministic tie-breaking convention—for example, fixing the column signs of U and V and ordering their columns lexicographically by their first nonzero entry. These deterministic rules define a Borel-measurable selection of (U, V) and therefore yield a measurable mapping

$$N : \Theta(\delta) \longrightarrow \Theta(\delta), \quad \theta \longmapsto (\Gamma_\beta R_\beta^*, R_\beta^{*\top} F, \Gamma_\delta R_\delta^*, \Gamma_\alpha),$$

which is continuous on each smooth stratum and measurable on $\Theta(\delta)$ as a whole.⁷⁶ $\square$

Taken together, Lemma A.5 supplies the missing bridge from geometric rotation symmetry to statistically well-posed inference. Mathematically, the Procrustes map $N(\cdot)$ selects, for every rotation-equivalent configuration, a unique and measurable representative that varies smoothly whenever the weighted cross-covariance spectra are simple; without this selection, the score and information would be evaluated on a non-unique parameterization, and the Fisher information would inherit spurious singular directions from basis relabelings. Geometrically, the map fixes an orientation while preserving the span, so that all calculations take place on a smooth cross-section of the quotient $\Theta(\delta)/\mathcal{G}$ rather than on a space with redundant coordinates. Statistically, this is what allows the projected (tangent-space) score and information to be invariant to orthogonal reparameterizations, which in turn underpins the *equivariance of the KKT system* and the *rotation-invariant LAN* expansion developed in Appendix B (see Lemma A.4 for invariance of the criterion and constraints, and Proposition A.6 for rotation-invariant projected score and information). Economically, the result formalizes that the estimator and its standard errors depend only on the priced risk space and not on arbitrary labeling of factor bases; absent Lemma A.5, identification would remain quotient-level in principle but the finite-sample and asymptotic calculations would lack a canonical coordinate system, jeopardizing both numerical stability and the interpretation of normalized systematic-risk scale.

A.7.3 Rotation-Invariant Projected Score and Information

Proposition A.6 (Rotation-Invariant Projected Score and Information).⁷⁷ *Let $\bar{\theta} := N(\theta)$ be the normalized parameter (Lemma A.5) and let $\Pi_{\bar{\theta}}$ denote the orthogonal projector onto the tangent space $\mathcal{T}_{\bar{\theta}}\Theta(\delta)$ at $\bar{\theta}$. Define the (feasible, pre-whitened) projected score and information by*

$$S_T(\bar{\theta}) := \Pi_{\bar{\theta}} \nabla_{\theta} Q_T^{(\text{feas})}(\bar{\theta}), \quad \mathcal{I}_T(\bar{\theta}) := \Pi_{\bar{\theta}} \nabla_{\theta\theta}^2 Q_T^{(\text{feas})}(\bar{\theta}) \Pi_{\bar{\theta}}^\top,$$

⁷⁶Mathematically, the SVD map $M \mapsto (U, \Sigma, V)$ is analytic whenever M has a simple spectrum, because singular values and vectors arise from smooth eigenproblems for the symmetric matrix $M^\top M$. When eigenvalues coincide, the spectral subspace remains continuous but the choice of orthonormal basis within that subspace becomes nonunique. Fixing signs and column order defines a Borel-measurable cross-section of the orthogonal group, ensuring that the composite mapping $\theta \mapsto (U(\theta), V(\theta))$ is measurable (Edelman et al., 1998, Ch. 3). This construction guarantees that $N(\cdot)$ is a measurable selection on $\Theta(\delta)$ and locally smooth wherever the singular-value structure is simple.

⁷⁷This proposition promotes rotation-equivariance from the criterion/constraints (Lemma A.4) to the first- and second-order objects that drive estimation and inference. After measurable normalization (Lemma A.5) and feasible pre-whitening (Assumption 9), the score and information restricted to the tangent space are invariant to orthogonal reparameterizations of the factor basis. *Used in Appendix B.6 to obtain rotation-invariant KKT systems and a LAN expansion with basis-free standard errors.*

with $Q_T^{(\text{feas})}$ defined in Section A.6.2. Then for every $g \in \mathcal{G} = \mathcal{O}(K) \times \mathcal{O}(K_{\text{add}})$,

$$S_T(\mathbf{N}(g \cdot \theta)) = S_T(\mathbf{N}(\theta)), \quad (61)$$

$$\mathcal{I}_T(\mathbf{N}(g \cdot \theta)) = \mathcal{I}_T(\mathbf{N}(\theta)), \quad (62)$$

i.e., the projected score and information are rotation-invariant after normalization and pre-whitening. Equivalently, the projected first-order KKT system and the block-diagonal (tangent-space) information are identical across all orthogonal parameterizations of the same risk span.

Proof. Fix $\theta = (\Gamma_\beta, F, \Gamma_\delta, \Gamma_\alpha) \in \Theta(\delta)$ and any $g = (R_\beta, R_\delta) \in \mathcal{G} = \mathcal{O}(K) \times \mathcal{O}(K_{\text{add}})$. Write $g \cdot \theta = (\Gamma_\beta R_\beta, R_\beta^\top F, \Gamma_\delta R_\delta, \Gamma_\alpha)$ as in (55). By Lemma A.4, the feasible (or GLS) criterion is invariant:

$$Q_T(g \cdot \theta) = Q_T(\theta). \quad (63)$$

Chain rule for the gradient and the Hessian. Let R_g denote the block-orthogonal change of coordinates induced by g on $(\Gamma_\beta, F, \Gamma_\delta)$ and the identity on Γ_α .⁷⁸ Differentiate (63) with respect to θ and apply the chain rule. There exists a Jacobian representation (the push-forward) such that

$$\nabla_\theta Q_T(g \cdot \theta) = R_g \nabla_\theta Q_T(\theta), \quad \nabla_{\theta\theta}^2 Q_T(g \cdot \theta) = R_g \nabla_{\theta\theta}^2 Q_T(\theta) R_g^\top. \quad (64)$$

The first identity follows by differentiating $Q_T(g \cdot \theta) = Q_T(\theta)$ along any direction $d\theta$ and using that $d(g \cdot \theta) = R_g d\theta$; the second identity is obtained by differentiating once more and using orthogonality of R_g .

Tangent spaces and projectors. Let $\mathcal{T}_\theta \Theta(\delta)$ denote the tangent space of the feasible manifold at θ , and let Π_θ be the (symmetric) orthogonal projector onto that tangent space (with respect to the ambient inner product used by Q_T). Because the equality constraints are Q_Z -orthogonal and rotate as in §A.5.1, the tangent spaces transform by push-forward:

$$\mathcal{T}_{g\theta} \Theta(\delta) = R_g \mathcal{T}_\theta \Theta(\delta). \quad (65)$$

Hence the corresponding orthogonal projectors satisfy

$$\Pi_{g\theta} = R_g \Pi_\theta R_g^\top. \quad (66)$$

⁷⁸Concretely, R_g right-multiplies the Γ_β and Γ_δ blocks by R_β and R_δ , and left-multiplies the F block by $R_\beta^\top$. Because R_β and R_δ are orthogonal, R_g is block-orthogonal and thus $R_g^\top R_g = I$.

⁷⁹Why $\mathcal{T}_{g\theta} \Theta(\delta) = R_g \mathcal{T}_\theta \Theta(\delta)$ and $\Pi_{g\theta} = R_g \Pi_\theta R_g^\top$. Write the stacked equality constraints as $g(\theta) = 0$ with Jacobian $G(\theta) := \nabla_\theta g(\theta)$, and let the (possibly active) inequality $h(\theta) \leq 0$ have gradient $\nabla_\theta h(\theta)$. Near a regular point θ , the tangent space of the feasible manifold is the common kernel of the linearized active constraints:

$$\mathcal{T}_\theta \Theta(\delta) = \left\{ d\theta : G(\theta) d\theta = 0, \text{ and if } h(\theta) = 0, \nabla_\theta h(\theta)^\top d\theta = 0 \right\}.$$

Under the group action $g \cdot \theta = (\Gamma_\beta R_\beta, R_\beta^\top F, \Gamma_\delta R_\delta, \Gamma_\alpha)$ and the induced block-orthogonal change of coordinates R_g , the constraints are equivariant (Lemma A.4). Differentiating the identities $g(g \cdot \theta) = g(\theta)$ and $h(g \cdot \theta) = h(\theta)$ yields

$$G(g \cdot \theta) R_g = R_g G(\theta), \quad \nabla_\theta h(g \cdot \theta)^\top R_g = \nabla_\theta h(\theta)^\top.$$

Projected (tangent–space) score and curvature under rotation. Combine (64) and (66) to obtain

$$\Pi_{g\theta} \nabla_{\theta} Q_T(g \cdot \theta) = R_g \Pi_{\theta} \nabla_{\theta} Q_T(\theta), \quad (67)$$

and

$$\Pi_{g\theta} \nabla_{\theta\theta}^2 Q_T(g \cdot \theta) \Pi_{g\theta}^{\top} = R_g (\Pi_{\theta} \nabla_{\theta\theta}^2 Q_T(\theta) \Pi_{\theta}^{\top}) R_g^{\top}.^{80} \quad (68)$$

Normalization removes the rotation. By Lemma A.5, the Procrustes normalization is equivariant: $N(g \cdot \theta) = N(\theta)$. Set $\bar{\theta} := N(\theta)$ and apply (67)–(68) with θ replaced by $\bar{\theta}$ and $g \cdot \theta$ replaced by $N(g \cdot \theta) = \bar{\theta}$. Because the left and right sides now refer to the *same* normalized point $\bar{\theta}$, the R_g factors cancel, yielding the rotation–invariance identities

$$\Pi_{N(g\theta)} \nabla_{\theta} Q_T(N(g \cdot \theta)) = \Pi_{N(\theta)} \nabla_{\theta} Q_T(N(\theta)), \quad (69)$$

$$\Pi_{N(g\theta)} \nabla_{\theta\theta}^2 Q_T(N(g \cdot \theta)) \Pi_{N(g\theta)}^{\top} = \Pi_{N(\theta)} \nabla_{\theta\theta}^2 Q_T(N(\theta)) \Pi_{N(\theta)}^{\top}.^{81} \quad (70)$$

Hence, if $d\theta \in \mathcal{T}_{\theta}\Theta(\delta)$, then

$$G(g \cdot \theta) (R_g d\theta) = R_g G(\theta) d\theta = 0, \quad \text{and if } h(\theta) = 0, \quad \nabla_{\theta} h(g \cdot \theta)^{\top} (R_g d\theta) = \nabla_{\theta} h(\theta)^{\top} d\theta = 0,$$

so $R_g d\theta \in \mathcal{T}_{g\theta}\Theta(\delta)$. Conversely, applying the same argument to g^{-1} shows the reverse inclusion. Therefore,

$$\mathcal{T}_{g\theta}\Theta(\delta) = R_g \mathcal{T}_{\theta}\Theta(\delta).$$

For the projectors, recall that if U is an orthogonal linear isomorphism and S is a subspace, then the orthogonal projector onto US satisfies $P_{US} = U P_S U^{\top}$. Here $U = R_g$ and $S = \mathcal{T}_{\theta}\Theta(\delta)$, so with respect to the ambient inner product used by Q_T (and preserved by R_g),

$$\Pi_{g\theta} = \Pi_{R_g \mathcal{T}_{\theta}} = R_g \Pi_{\theta} R_g^{\top}.$$

This gives (65)–(66). Geometrically, the orthogonal change of coordinates R_g carries feasible directions at θ onto feasible directions at $g \cdot \theta$ without distortion; algebraically, it conjugates the projector because it preserves inner products and orthogonality.

⁸⁰*Derivation and intuition for the projected score and curvature identities.* Start from the chain rule identities in (64),

$$\nabla_{\theta} Q_T(g \cdot \theta) = R_g \nabla_{\theta} Q_T(\theta), \quad \nabla_{\theta\theta}^2 Q_T(g \cdot \theta) = R_g \nabla_{\theta\theta}^2 Q_T(\theta) R_g^{\top},$$

and the transformation law of projectors in (66), $\Pi_{g\theta} = R_g \Pi_{\theta} R_g^{\top}$. Premultiplying the gradient identity by $\Pi_{g\theta}$ gives

$$\Pi_{g\theta} \nabla_{\theta} Q_T(g \cdot \theta) = \Pi_{g\theta} R_g \nabla_{\theta} Q_T(\theta) = (R_g \Pi_{\theta} R_g^{\top}) R_g \nabla_{\theta} Q_T(\theta) = R_g \Pi_{\theta} \nabla_{\theta} Q_T(\theta),$$

because $R_g^{\top} R_g = I$ by orthogonality. This proves (67). Analogously, insert both projectors into the Hessian transformation and simplify:

$$\begin{aligned} \Pi_{g\theta} \nabla_{\theta\theta}^2 Q_T(g \cdot \theta) \Pi_{g\theta}^{\top} &= (R_g \Pi_{\theta} R_g^{\top}) (R_g \nabla_{\theta\theta}^2 Q_T(\theta) R_g^{\top}) (R_g \Pi_{\theta}^{\top} R_g^{\top}) \\ &= R_g \Pi_{\theta} \nabla_{\theta\theta}^2 Q_T(\theta) \Pi_{\theta}^{\top} R_g^{\top}, \end{aligned}$$

which yields (68). Mathematically, this shows that the first– and second–order objects projected onto the tangent space transform covariantly under orthogonal changes of coordinates: rotation by R_g simply conjugates both the score and curvature operators. Geometrically, the map R_g acts as an isometry on the constraint manifold—it rotates feasible directions without altering inner products—so the projected gradient (score) and Hessian (curvature) remain identical up to this orthogonal relabeling. Economically, this means that rotating the factor basis merely re-expresses the same pricing geometry: the set of admissible perturbations and the local curvature of the criterion (which determine efficiency) are preserved, confirming that statistical inference depends only on the priced–risk span, not on arbitrary basis orientations.

⁸¹*Why the R_g factors disappear after normalization.* Set $\bar{\theta} := N(\theta)$ and use Lemma A.5 (equivariance of N) to get

$$N(g \cdot \theta) = N(\theta) = \bar{\theta}.$$

Hence both sides of (69)–(70) are evaluated at the *same* base point $\bar{\theta}$. Now combine the push–forward identities (64)–(66) with this

Feasible pre-whitened objective. By Assumption 9 and §A.6.2,

$$Q_T^{(\text{feas})}(\cdot) = Q_T^{(\text{GLS})}(\cdot) + o_p(1),$$

and the same $o_p(1)$ –equivalence holds for gradients, Hessians, and their tangent–space projections. Therefore, (69)–(70) continue to hold for the feasible pre–whitened criterion.⁸²

fact:

Projected score. From (64) and (66),

$$\Pi_{g\theta} \nabla_\theta Q_T(g \cdot \theta) = R_g \Pi_\theta \nabla_\theta Q_T(\theta).$$

Normalize both arguments by N :

$$\Pi_{N(g\theta)} \nabla_\theta Q_T(N(g \cdot \theta)) = \Pi_{\bar{\theta}} \nabla_\theta Q_T(\bar{\theta}), \quad \Pi_{N(\theta)} \nabla_\theta Q_T(N(\theta)) = \Pi_{\bar{\theta}} \nabla_\theta Q_T(\bar{\theta}),$$

because $N(g \cdot \theta) = N(\theta) = \bar{\theta}$ fixes the base point and thereby the projector $\Pi_{\bar{\theta}}$ and the gradient $\nabla_\theta Q_T(\bar{\theta})$. Thus the two projected scores are identical, which is (69).

Projected curvature. Similarly, from (66) and the Hessian push–forward,

$$\Pi_{g\theta} \nabla_{\theta\theta}^2 Q_T(g \cdot \theta) \Pi_{g\theta}^\top = R_g (\Pi_\theta \nabla_{\theta\theta}^2 Q_T(\theta) \Pi_\theta^\top) R_g^\top.$$

After normalization, both sides are computed at $\bar{\theta}$, so

$$\Pi_{N(g\theta)} \nabla_{\theta\theta}^2 Q_T(N(g \cdot \theta)) \Pi_{N(g\theta)}^\top = \Pi_{\bar{\theta}} \nabla_{\theta\theta}^2 Q_T(\bar{\theta}) \Pi_{\bar{\theta}}^\top = \Pi_{N(\theta)} \nabla_{\theta\theta}^2 Q_T(N(\theta)) \Pi_{N(\theta)}^\top,$$

which is (70).

Before normalization, rotating parameters by g conjugates the score/Hessian by R_g because the base point changes from θ to $g \cdot \theta$. Normalization collapses each entire orbit $\{g \cdot \theta : g \in \mathcal{G}\}$ to the *same* representative $\bar{\theta}$; once the base point is the same, the conjugation factors R_g have nothing left to act on, so the projected objects coincide exactly.

⁸²*Justification of the feasible pre–whitened equivalence.* Assumption 9 establishes that the empirical weight matrices $\hat{\Omega}_t$ are uniformly positive–definite and ratio–consistent with their population counterparts Ω_t , i.e.,

$$\hat{\Omega}_t^{-1/2} \Omega_t^{1/2} \xrightarrow{p} I_L \quad \text{uniformly in } t.$$

This implies the feasible weighted objective and its derivatives are asymptotically equivalent to those of the infeasible generalized least–squares (GLS) benchmark. Formally, substituting the pre–whitened variables $\tilde{X}_t = \hat{\Omega}_t^{-1/2} X_t$ and $\tilde{W}_t = \hat{\Omega}_t^{-1/2} W_t \hat{\Omega}_t^{-1/2}$ (from Section A.6.2) gives

$$Q_T^{(\text{feas})}(\theta) = \frac{1}{T} \sum_{t=1}^T \|\hat{\Omega}_t^{-1/2} (X_t - W_t(\Gamma_\beta f_t + \Gamma_\delta F_t^{\text{obs}} + \Gamma_\alpha))\|_2^2.$$

Expanding and rearranging yields

$$Q_T^{(\text{feas})}(\theta) = \frac{1}{T} \sum_{t=1}^T m_t(\theta)^\top \hat{\Omega}_t^{-1} m_t(\theta), \quad Q_T^{(\text{GLS})}(\theta) = \frac{1}{T} \sum_{t=1}^T m_t(\theta)^\top \Omega_t^{-1} m_t(\theta),$$

so their difference is

$$Q_T^{(\text{feas})}(\theta) - Q_T^{(\text{GLS})}(\theta) = \frac{1}{T} \sum_{t=1}^T m_t(\theta)^\top (\hat{\Omega}_t^{-1} - \Omega_t^{-1}) m_t(\theta).$$

Because $\|\hat{\Omega}_t^{-1} - \Omega_t^{-1}\|_F = O_p(\|\hat{\Omega}_t - \Omega_t\|_F) = o_p(1)$ and $\sup_t \|m_t(\theta)\|^2$ is $O_p(1)$ on the compact feasible set $\Theta(\delta)$, the average difference is $o_p(1)$ uniformly in θ . Thus

$$Q_T^{(\text{feas})}(\theta) = Q_T^{(\text{GLS})}(\theta) + o_p(1),$$

and by standard stochastic–equicontinuity arguments (Newey and McFadden, 1994, Sec. 2.6; Gouriéroux and Monfort, 1997, Ch. 4), the same $o_p(1)$ equivalence extends to the score $\nabla_\theta Q_T(\theta)$, the Hessian $\nabla_{\theta\theta}^2 Q_T(\theta)$, and any projected (tangent–space) transformations of these quantities.

Intuitively, ratio–consistency of the weights ensures that pre–whitening merely rescales each characteristic direction by a factor converging to one, so all geometric features of the criterion—the curvature, tangent–space decomposition, and rotation–invariance identities—remain unchanged to first order. Economically, this means that the feasible estimator evaluates risk premia in variance–standardized units indistinguishable from their population metric, so efficiency and identification are preserved even when

Once the risk span is anchored by $N(\cdot)$, local estimation geometry (score, curvature, Fisher information) is independent of basis labels; inference is therefore rotation-invariant and depends only on the priced systematic-risk space. This completes the proof. $\square$

A.7.4 Rotation-Invariant Identification: Synthesis and Link to Appendix B

This section established three ingredients. First, Lemma A.4 showed that the sample criterion and the constraint system are *equivariant* under orthogonal rotations of the factor bases. Second, Lemma A.5 defined a measurable, locally smooth Procrustes map $N(\cdot)$ that selects a unique, rotation-aligned representative within each orbit. Third, Proposition A.6 proved that, after normalization and pre-whitening, the projected score and information on the tangent space are *rotation-invariant*.

Taken together, these results move identification from the geometric statement “span up to rotation” to a statistically well-posed formulation in which all first- and second-order objects are evaluated on a fixed, orbit-free coordinate system. Inference therefore depends only on the priced risk space, redundant directions are removed, singularities in the Fisher information are avoided, and feasible weighting preserves efficiency.

Appendix B builds directly on this structure. In particular, Section B.6 uses Lemmas A.4–A.5 to derive *equivariance of the KKT system* and a *rotation-invariant LAN expansion*, yielding basis-free standard errors and tests.

A.8 Smoothness, Lipschitz Continuity, and Huber Regularity

This subsection completes the analytic foundation of Appendix A by formalizing the smoothness and bounded-influence conditions that guarantee the differentiability, stability, and robustness of the NA-IPCA estimator. Whereas Sections A.6–A.7 established geometric regularity and rotation-invariant identification under feasible weighting and normalization, the present section ensures that all moment maps, score functions, and Huber-type loss components are sufficiently smooth and Lipschitz-continuous to support uniform laws of large numbers, twice-differentiable KKT systems, and valid LAN expansions.

Methodologically, we first verify that the per-period moment map and its Jacobian are continuously Fréchet-differentiable and uniformly Lipschitz on the compact feasible set $\Theta(\delta)$ (Section A.8.1), providing the analytic basis for the differentiable KKT system in Appendix B. We then impose bounded-influence and finite-variance conditions on the Huber-type score $\psi_K(\cdot)$ (Section A.8.3), ensuring that the estimator remains well defined and asymptotically normal even under heavy-tailed or heteroskedastic residuals. Finally, we link these smoothness and robustness properties to the local asymptotic normality (LAN) structure and efficiency results of Appendix B (Section A.8.4), demonstrating that the feasible, rotation-normalized estimator inherits the same first- and second-order regularity as its ideal GLS counterpart.

Together, Sections A.8.1–A.8.4 provide the final regularity layer of the theoretical framework: they ensure that the NA-IPCA estimator is not only geometrically identified and rotation-invariant but also statistically smooth, robust, and asymptotically well behaved—forming the bridge to the full asymptotic and efficiency analysis in Appendix B.

using estimated weights.

A.8.1 Smoothness and Differentiability of Moment Maps

This subsection restates and extends the smoothness assumptions introduced in Assumption 7, clarifying the analytic properties of the per-period moment map $m_t(\theta) = X_t - W_t(\Gamma_\beta f_t + \Gamma_\delta F_t^{\text{obs}} + \Gamma_\alpha)$ and its Jacobian $\dot{m}_t(\theta)$. Methodologically, these conditions ensure that $m_t(\theta)$ is continuously Fréchet-differentiable on the compact feasible set $\Theta(\delta)$ and that its Jacobian $\dot{m}_t(\theta)$ is uniformly bounded and Lipschitz-continuous in θ . Together, they guarantee that the sample criterion $Q_T(\theta)$ is twice continuously differentiable, that stochastic equicontinuity holds for its gradient, and that the KKT system defining the NA-IPCA estimator is differentiable and well behaved under weak dependence and heavy tails. Economically, this smoothness condition implies that small perturbations in factor loadings, characteristics, or pricing kernel coefficients generate proportionally small and predictable changes in cross-sectional pricing errors—ensuring local stability of the equilibrium pricing geometry.

Assumption 10 (Smoothness and Uniform Lipschitz Continuity of the Moment Map).⁸³ Let $m_t(\theta) = X_t - W_t(\Gamma_\beta f_t + \Gamma_\delta F_t^{\text{obs}} + \Gamma_\alpha)$ be defined on the compact feasible set $\Theta(\delta)$ in (17). We assume the following conditions hold uniformly over $\theta \in \Theta(\delta)$:

- (a) **Continuous Fréchet differentiability.** The map $\theta \mapsto m_t(\theta)$ is continuously Fréchet-differentiable in a neighborhood of θ_0 , with derivative $\dot{m}_t(\theta)$ satisfying

$$\|m_t(\theta_1) - m_t(\theta_2) - \dot{m}_t(\bar{\theta})(\theta_1 - \theta_2)\|_2 = o(\|\theta_1 - \theta_2\|_2), \quad \bar{\theta} \in [\theta_1, \theta_2]. \quad (71)$$

This property guarantees the existence of a continuous Jacobian matrix $\dot{m}_t(\theta)$ that varies smoothly with θ , ensuring the differentiability of the sample criterion and of the KKT stationarity system.⁸⁴

- (b) **Uniform boundedness of the Jacobian.** There exists a finite constant $C_J < \infty$ such that

$$\sup_t \sup_{\theta \in \Theta(\delta)} \|\dot{m}_t(\theta)\|_F \leq C_J. \quad (72)$$

Bounded derivatives ensure that gradients and Hessians of the criterion are uniformly integrable and that stochastic convergence results (e.g., the uniform LLN and CLT) apply to $\nabla_\theta Q_T(\theta)$.⁸⁵

⁸³This assumption extends Assumption 7 by formalizing the smoothness properties required for the differentiable KKT and LAN expansions in Appendix B. It ensures that the per-period moment map $m_t(\theta)$ is continuously Fréchet-differentiable on the compact feasible set $\Theta(\delta)$, and that its Jacobian $\dot{m}_t(\theta)$ is uniformly bounded and Lipschitz-continuous. Mathematically, this guarantees that the sample criterion $Q_T(\theta) = T^{-1} \sum_t \|m_t(\theta)\|_{Q_Z^{-1}}^2$ admits continuous first- and second-order derivatives and that the stochastic process $\{\nabla_\theta Q_T(\theta)\}$ satisfies uniform equicontinuity. Economically, it means that the NA-IPCA estimator reacts smoothly to changes in model parameters: the pricing kernel and its implied Sharpe-ratio geometry evolve continuously with data and remain stable under moderate sampling variation. *Used in Lemmas A.1–A.3 (Uniform LLN and CLT) and Proposition B.3 (LAN and Efficiency).*

⁸⁴The continuous Fréchet differentiability condition in (71) ensures that the moment map $m_t(\theta)$ admits a uniform first-order linear approximation by its Jacobian $\dot{m}_t(\theta)$. This, together with local Lipschitz continuity, justifies a second-order Taylor expansion of the criterion around θ_0 and underpins the LAN development in Appendix B. The required smoothness and stability conditions follow the standard regularity for smooth M -estimators (see Newey and McFadden (1994, Ch. 36); cf. Gouriéroux and Monfort (1997); van der Vaart (1998)). Economically, they imply that infinitesimal changes in factor loadings or pricing-kernel coefficients induce proportionate adjustments in expected returns, preserving local equilibrium stability.

⁸⁵Boundedness of the Jacobian $\dot{m}_t(\theta)$ guarantees that the score process $\nabla_\theta Q_T(\theta)$ has finite variance uniformly over $\Theta(\delta)$, a necessary condition for applying stochastic equicontinuity and asymptotic normality results to constrained M -estimators (Newey and McFadden, 1994, Sec. 2.6; Gouriéroux and Monfort, 1997, Ch. 4). Geometrically, it prevents curvature of the criterion from

- (c) **Uniform Lipschitz continuity.** There exists a nonnegative random variable L_t with $\mathbb{E}[L_t] < \infty$ such that for all $\theta_1, \theta_2 \in \Theta(\delta)$,

$$\|m_t(\theta_1) - m_t(\theta_2)\|_2 \leq L_t \|\theta_1 - \theta_2\|_2 \quad \text{a.s.} \quad (73)$$

This property guarantees stochastic equicontinuity of the sample criterion and ensures that small parameter perturbations induce uniformly small variations in $m_t(\theta)$.⁸⁶

Assumption 10 implies that the moment map $m_t(\theta)$ and its Jacobian $\dot{m}_t(\theta)$ satisfy the smoothness and Lipschitz conditions required for the differentiability of the constrained KKT system and for the validity of the LAN expansion developed in Appendix B. Mathematically, continuous Fréchet differentiability ensures that the sample criterion $Q_T(\theta)$ admits valid Taylor expansions, while uniform boundedness and Lipschitz continuity guarantee that gradients and Hessians remain uniformly integrable and stable under weak dependence. Economically, these properties mean that the mapping from parameters to pricing errors is locally well conditioned: small perturbations in factor exposures or characteristics produce proportionate and reversible changes in fitted returns, ensuring a smooth and stable estimation landscape for the NA-IPCA model.

A.8.2 Lipschitz Continuity of Score and Hessian

This section records the sample-level regularity that follows from the smoothness and uniform Lipschitz properties of the moment map in Assumption 10. In particular, we show that the score (gradient of the criterion) and the Hessian are uniformly Lipschitz on the feasible set, and the associated empirical processes are stochastically equicontinuous. These properties underwrite the differentiable KKT expansions and the LAN analysis used in Appendix B.

Lemma A.7 (Uniform Lipschitz Continuity of Score and Hessian).⁸⁷ *Let Assumptions 1–8 and 10 hold, and*

exploding in any region of the parameter space, ensuring a smooth and convex local geometry of the feasible manifold.

⁸⁶Uniform Lipschitz continuity of the moment map implies stochastic equicontinuity of the empirical process $\{\bar{m}_T(\theta)\}$, a key condition ensuring that the sample objective $Q_T(\theta)$ converges uniformly to its population counterpart $Q(\theta)$ and that the empirical KKT system behaves smoothly across $\Theta(\delta)$. Formally, this property guarantees that the random fluctuations of $Q_T(\theta)$ are uniformly controlled in θ , satisfying the Glivenko–Cantelli and Donsker criteria for weakly dependent processes (see Newey and McFadden, 1994, Sec. 2.6; Gouriéroux and Monfort, 1997, Ch. 4; Thm. 19.24 van der Vaart (1998)). In particular, the uniform Lipschitz bound (73) ensures that the empirical process is stochastically equicontinuous, allowing differentiation under the probability limit and yielding smooth asymptotic expansions of the estimator. Economically, this condition means that the pricing kernel reacts proportionately to marginal changes in factor loadings or intercepts: small perturbations in model parameters lead to predictable, continuous adjustments in fitted returns without instabilities or regime jumps. Such local regularity parallels the stability conditions for differentiable moment mappings in semiparametric M -estimation and simulation-based econometric models (Gouriéroux and Monfort, 1997; van der Vaart (1998)) and ensures that the No-Arbitrage IPCA estimator inherits smooth identification and well-posed inference.

⁸⁷The lemma is a standard consequence of smooth M -estimation theory: if the per-period moment map $m_t(\theta)$ is continuously Fréchet-differentiable with an integrable Lipschitz envelope on a compact parameter set (Assumption 10), then the empirical score and Hessian inherit uniform Lipschitz continuity and stochastic equicontinuity; see Newey and McFadden (1994, Sec. 2.6), van der Vaart (1998, Secs. 5.2–5.3), and Gouriéroux and Monfort (1996, Ch. 4). Economically, this guarantees that small perturbations of factor loadings or pricing-kernel coefficients produce proportionate and predictable changes in fitted returns, so that the empirical KKT system behaves smoothly across $\Theta(\delta)$.

define the sample criterion

$$Q_T(\theta) = \frac{1}{T} \sum_{t=1}^T \|m_t(\theta)\|_{Q_Z^{-1}}^2, \quad \|x\|_{Q_Z^{-1}}^2 = x^\top Q_Z^{-1} x.$$

Write the score and Hessian as

$$S_T(\theta) := \nabla_\theta Q_T(\theta) = \frac{2}{T} \sum_{t=1}^T \dot{m}_t(\theta)^\top Q_Z^{-1} m_t(\theta), \quad H_T(\theta) := \nabla_{\theta\theta}^2 Q_T(\theta).$$

Then:

(a) **Uniform Lipschitz continuity (score).** There exists a nonrandom $C_S < \infty$ such that, for all $\theta_1, \theta_2 \in \Theta(\delta)$,

$$\|S_T(\theta_1) - S_T(\theta_2)\| \leq C_S \|\theta_1 - \theta_2\| \quad \text{with probability} \rightarrow 1. \quad (74)$$

(b) **Uniform Lipschitz continuity (Hessian).** There exists a nonrandom $C_H < \infty$ such that, for all $\theta_1, \theta_2 \in \Theta(\delta)$,

$$\|H_T(\theta_1) - H_T(\theta_2)\| \leq C_H \|\theta_1 - \theta_2\| \quad \text{with probability} \rightarrow 1. \quad (75)$$

(c) **Stochastic equicontinuity.** The empirical processes $\{\sqrt{T}[S_T(\theta) - \mathbb{E}S_T(\theta)] : \theta \in \Theta(\delta)\}$ and $\{\sqrt{T}[H_T(\theta) - \mathbb{E}H_T(\theta)] : \theta \in \Theta(\delta)\}$ are stochastically equicontinuous. In particular, for every $\varepsilon, \eta > 0$ there exists $\delta_\eta > 0$ such that

$$\limsup_{T \rightarrow \infty} \mathbb{P} \left(\sup_{\|\theta_1 - \theta_2\| < \delta_\eta} \sqrt{T} \|[S_T(\theta_1) - S_T(\theta_2)] - \mathbb{E}[S_T(\theta_1) - S_T(\theta_2)]\| > \varepsilon \right) < \eta, \quad (76)$$

and an analogous statement holds for $H_T(\cdot)$.

(d) **Differentiable KKT mapping.** As a consequence of (74)–(76), the KKT mapping is (locally) continuously differentiable on $\Theta(\delta)$ with probability approaching one, and admits a second-order Taylor expansion around θ_0 that is uniform on compact neighborhoods of θ_0 .

Proof. (a) **Uniform Lipschitz continuity of the score.** Let

$$S_T(\theta) \equiv \frac{2}{T} \sum_{t=1}^T \dot{m}_t(\theta)^\top Q_Z^{-1} m_t(\theta),$$

where $m_t(\theta)$ is the period- t moment map and $\dot{m}_t(\theta)$ its Fréchet derivative on the compact set $\Theta(\delta)$. For any $\theta_1, \theta_2 \in \Theta(\delta)$, the triangle inequality and submultiplicativity of the operator norm give

$$\begin{aligned} \|S_T(\theta_1) - S_T(\theta_2)\| &\leq \frac{2}{T} \sum_{t=1}^T \left(\|\dot{m}_t(\theta_1) - \dot{m}_t(\theta_2)\| \|Q_Z^{-1}\| \|m_t(\theta_1)\| \right. \\ &\quad \left. + \|\dot{m}_t(\theta_2)\| \|Q_Z^{-1}\| \|m_t(\theta_1) - m_t(\theta_2)\| \right). \end{aligned}$$

By Assumption 10 (smoothness/Lipschitz envelope), there exists a measurable $L_t \geq 0$ with $\mathbb{E}[L_t] < \infty$ such

that a.s.

$$\|m_t(\theta_1) - m_t(\theta_2)\| \leq L_t \|\theta_1 - \theta_2\|, \quad \|\dot{m}_t(\theta_1) - \dot{m}_t(\theta_2)\| \leq L_t \|\theta_1 - \theta_2\|.$$

Hence,

$$\|S_T(\theta_1) - S_T(\theta_2)\| \leq \frac{2\|Q_Z^{-1}\|}{T} \sum_{t=1}^T \left(L_t \|m_t(\theta_1)\| + \|\dot{m}_t(\theta_2)\| L_t \right) \|\theta_1 - \theta_2\|. \quad (77)$$

Next, pick any fixed $\theta^\circ \in \Theta(\delta)$ and apply the mean–value form of Fréchet differentiability on the compact $\Theta(\delta)$ to obtain a.s.

$$\|m_t(\theta_1)\| \leq \|m_t(\theta^\circ)\| + L_t \text{diam}(\Theta(\delta)), \quad \|\dot{m}_t(\theta_2)\| \leq \|\dot{m}_t(\theta^\circ)\| + L_t \text{diam}(\Theta(\delta)).^{88}$$

Substituting in (77) and regrouping yields

$$\|S_T(\theta_1) - S_T(\theta_2)\| \leq C_T \|\theta_1 - \theta_2\|,$$

where the random Lipschitz constant satisfies

$$C_T = \frac{2\|Q_Z^{-1}\|}{T} \sum_{t=1}^T \left(L_t \|m_t(\theta^\circ)\| + \|\dot{m}_t(\theta^\circ)\| L_t + 2L_t^2 \text{diam}(\Theta(\delta)) \right).$$

By Assumptions 1, 6, and 10—strict stationarity with α –mixing and finite $(2 + \tau)$ –moments, compactness of the feasible set $\Theta(\delta)$, and a uniformly Lipschitz/Fréchet–smooth moment map—both $m_t(\theta)$ and its Jacobian $\dot{m}_t(\theta)$ are square–integrable uniformly on $\Theta(\delta)$:

$$\sup_{\theta \in \Theta(\delta)} \mathbb{E} \|m_t(\theta)\|_2^2 < \infty, \quad \sup_{\theta \in \Theta(\delta)} \mathbb{E} \|\dot{m}_t(\theta)\|_2^2 < \infty.^{89}$$

⁸⁸Why the diameter bounds hold. Fix any $\theta^\circ \in \Theta(\delta)$ and write the (Euclidean) diameter of the compact set $\Theta(\delta)$ as $\text{diam}(\Theta(\delta)) := \sup_{\theta_1, \theta_2 \in \Theta(\delta)} \|\theta_1 - \theta_2\| < \infty$. Assumption A.10 postulates *uniform Lipschitz* control for the moment map and its Jacobian: there exists a measurable envelope $L_t \geq 0$ with $\mathbb{E}[L_t] < \infty$ such that, a.s.,

$$\|m_t(\theta_1) - m_t(\theta_2)\| \leq L_t \|\theta_1 - \theta_2\|, \quad \|\dot{m}_t(\theta_1) - \dot{m}_t(\theta_2)\| \leq L_t \|\theta_1 - \theta_2\|, \quad \forall \theta_1, \theta_2 \in \Theta(\delta).$$

Apply the triangle inequality with $\theta_2 = \theta^\circ$ to obtain

$$\|m_t(\theta_1)\| \leq \|m_t(\theta^\circ)\| + \|m_t(\theta_1) - m_t(\theta^\circ)\| \leq \|m_t(\theta^\circ)\| + L_t \|\theta_1 - \theta^\circ\| \leq \|m_t(\theta^\circ)\| + L_t \text{diam}(\Theta(\delta)).$$

Exactly the same argument with $\dot{m}_t$ in place of m_t yields

$$\|\dot{m}_t(\theta_2)\| \leq \|\dot{m}_t(\theta^\circ)\| + \|\dot{m}_t(\theta_2) - \dot{m}_t(\theta^\circ)\| \leq \|\dot{m}_t(\theta^\circ)\| + L_t \|\theta_2 - \theta^\circ\| \leq \|\dot{m}_t(\theta^\circ)\| + L_t \text{diam}(\Theta(\delta)).$$

Equivalently, one may parameterize the line segment $\gamma(\lambda) = \theta^\circ + \lambda(\theta - \theta^\circ)$, $\lambda \in [0, 1]$, and use the integral (mean–value) form of Fréchet differentiability, $m_t(\theta) - m_t(\theta^\circ) = \int_0^1 \dot{m}_t(\gamma(\lambda))(\theta - \theta^\circ) d\lambda$, followed by $\|\dot{m}_t(\gamma(\lambda))\| \leq \|\dot{m}_t(\theta^\circ)\| + L_t \|\gamma(\lambda) - \theta^\circ\| \leq \|\dot{m}_t(\theta^\circ)\| + L_t \|\theta - \theta^\circ\|$, to reach the same bounds. These are standard envelope consequences for smooth M –estimators on compact parameter sets with Lipschitz derivatives; see Newey and McFadden (1994, *Handbook of Econometrics*, Vol. 4, §2.6), Gouriéroux and Monfort (1996, *Simulation–Based Econometric Methods*, ch. 4), and van der Vaart (1998, *Asymptotic Statistics*, Thm. 5.7). *Economic read*: the bounds formalize that—uniformly over all admissible configurations—small moves in loadings or the pricing kernel alter pricing moments (and their slopes) at a controlled rate, which underpins stable first– and second–order expansions of the estimator.

⁸⁹Why uniform square–integrability holds. Assumption 1 establishes strict stationarity with α –mixing and finite $(2 + \tau)$ –moments for the process $\{(Z_t, r_{t+1})\}$, so $\mathbb{E}\|Z_t\|^{2+\tau} < \infty$ and $\mathbb{E}\|r_{t+1}\|^{2+\tau} < \infty$. Compactness of the feasible set $\Theta(\delta)$ (Assumption 6) implies that all parameter matrices $(\Gamma_\beta, F, \Gamma_\delta, \Gamma_\alpha)$ are uniformly bounded by a constant $C_\Theta < \infty$. Using the linear structure of the moment

Together with the integrability of the Lipschitz envelope L_t ($\mathbb{E}[L_t] < \infty$ and $\mathbb{E}[L_t^2] < \infty$ by Cauchy–Schwarz), the α -mixing LLN for weakly dependent arrays implies

$$\frac{1}{T} \sum_{t=1}^T L_t \|m_t(\theta^\circ)\| = O_p(1), \quad \frac{1}{T} \sum_{t=1}^T \|\dot{m}_t(\theta^\circ)\| L_t = O_p(1), \quad \frac{1}{T} \sum_{t=1}^T L_t^2 = O_p(1).^{90}$$

Hence $C_T = O_p(1)$ and, for any $\varepsilon > 0$, there exists $M < \infty$ such that $\mathbb{P}(C_T \leq M) \geq 1 - \varepsilon$ for all large T . Therefore, with probability $1 - \varepsilon$ (for T large),

$$\|S_T(\theta_1) - S_T(\theta_2)\| \leq M \|\theta_1 - \theta_2\| \quad \text{uniformly in } \theta_1, \theta_2 \in \Theta(\delta),$$

which establishes uniform Lipschitz continuity of the score map.

(b) Uniform Lipschitz continuity of the Hessian. Recall

$$H_T(\theta) = \frac{2}{T} \sum_{t=1}^T \left\{ \ddot{m}_t(\theta)^\top Q_Z^{-1} m_t(\theta) + \dot{m}_t(\theta)^\top Q_Z^{-1} \dot{m}_t(\theta) \right\}.$$

map,

$$m_t(\theta) = \frac{1}{N_t} Z_t^\top [r_{t+1} - Z_t(\Gamma_\beta f_t + \Gamma_\delta F_t^{\text{obs}} + \Gamma_\alpha)],$$

we can apply the Cauchy–Schwarz inequality:

$$\|m_t(\theta)\|_2^2 \leq \frac{1}{N_t} \|Z_t\|_F^2 \|r_{t+1} - Z_t(\Gamma_\beta f_t + \Gamma_\delta F_t^{\text{obs}} + \Gamma_\alpha)\|_2^2 \leq C_\Theta^2 (\|Z_t\|_F^4 + \|Z_t\|_F^2 \|r_{t+1}\|_2^2).$$

By Assumption 1, all right-hand terms have finite expectations, and because C_Θ does not depend on θ ,

$$\sup_{\theta \in \Theta(\delta)} \mathbb{E} \|m_t(\theta)\|_2^2 < \infty.$$

For the Jacobian, Assumption 10 guarantees that $m_t(\theta)$ is continuously Fréchet-differentiable with a measurable Lipschitz envelope L_t satisfying $\mathbb{E}[L_t^2] < \infty$ and $\|\dot{m}_t(\theta_1) - \dot{m}_t(\theta_2)\| \leq L_t \|\theta_1 - \theta_2\|$. Thus, $\|\dot{m}_t(\theta)\| \leq L_t + C_J$ for some constant $C_J < \infty$, and hence

$$\sup_{\theta \in \Theta(\delta)} \mathbb{E} \|\dot{m}_t(\theta)\|_2^2 \leq 2 \mathbb{E}[L_t^2] + 2C_J^2 < \infty.$$

These two bounds together establish the uniform L_2 -integrability of both $m_t(\theta)$ and its Jacobian, which underpins the stochastic equicontinuity and uniform LLN results used in Lemma A.1. Economically, these integrability properties ensure that cross-sectional pricing moments and their sensitivities remain finite and well behaved across all admissible configurations of factor loadings and pricing kernels, guaranteeing a stable and differentiable curvature of the empirical no-arbitrage criterion.

⁹⁰Why these averages are $O_p(1)$. Assumption A.1 gives strict stationarity and α -mixing with $\sum_{\ell \geq 1} \alpha(\ell)^{\tau/(2+\tau)} < \infty$, and finite $(2 + \tau)$ -moments for (Z_t, r_{t+1}) . Under Assumption A.6 (compact $\Theta(\delta)$), parameters are uniformly bounded; combined with the linear form of $m_t(\theta)$ in $(\Gamma_\beta f_t, \Gamma_\delta F_t^{\text{obs}}, \Gamma_\alpha)$ and the moment bounds from A.1, we obtain $\sup_{\theta \in \Theta(\delta)} \mathbb{E} \|m_t(\theta)\|^2 < \infty$ and similarly for $\dot{m}_t(\theta)$ (by Assumption A.10). Moreover, A.10 supplies a measurable envelope $L_t \geq 0$ with $\mathbb{E}[L_t] < \infty$ and the a.s. Lipschitz bounds $\|m_t(\theta_1) - m_t(\theta_2)\| \leq L_t \|\theta_1 - \theta_2\|$ and $\|\dot{m}_t(\theta_1) - \dot{m}_t(\theta_2)\| \leq L_t \|\theta_1 - \theta_2\|$ for all $\theta_1, \theta_2 \in \Theta(\delta)$. Fix $\theta^\circ \in \Theta(\delta)$; then $L_t \|m_t(\theta^\circ)\|$, $\|\dot{m}_t(\theta^\circ)\| L_t$, and L_t^2 are strictly stationary α -mixing sequences with finite first moments, e.g. $\mathbb{E}[L_t \|m_t(\theta^\circ)\|] \leq (\mathbb{E}[L_t^2])^{1/2} (\mathbb{E} \|m_t(\theta^\circ)\|^2)^{1/2} < \infty$. Hence, by the α -mixing LLN (Ibragimov and Linnik, 1971, Thm. 17.2.3; Newey and McFadden, 1994, Sec. 2.6),

$$\frac{1}{T} \sum_{t=1}^T L_t \|m_t(\theta^\circ)\| \xrightarrow{p} \mathbb{E}[L_t \|m_t(\theta^\circ)\|], \quad \frac{1}{T} \sum_{t=1}^T \|\dot{m}_t(\theta^\circ)\| L_t \xrightarrow{p} \mathbb{E}[\|\dot{m}_t(\theta^\circ)\| L_t], \quad \frac{1}{T} \sum_{t=1}^T L_t^2 \xrightarrow{p} \mathbb{E}[L_t^2],$$

so each average is $O_p(1)$. Economically, these bounds say that the average “slope intensity” L_t and the magnitudes of pricing moments and their Jacobians remain stable in large samples; hence the Lipschitz constants that control the score/Hessian differences (e.g., (77)) do not blow up with T , ensuring a smooth, well-behaved KKT system and valid LAN expansions.

For any $\theta_1, \theta_2 \in \Theta(\delta)$, by the triangle inequality and submultiplicativity of the operator norm,

$$\begin{aligned} \|H_T(\theta_1) - H_T(\theta_2)\| &\leq \frac{2\|Q_Z^{-1}\|}{T} \sum_{t=1}^T \left(\|\dot{m}_t(\theta_1) - \dot{m}_t(\theta_2)\| \|m_t(\theta_1)\| \right. \\ &\quad \left. + \|\dot{m}_t(\theta_2)\| \|m_t(\theta_1) - m_t(\theta_2)\| \right. \\ &\quad \left. + (\|\dot{m}_t(\theta_1)\| + \|\dot{m}_t(\theta_2)\|) \|\dot{m}_t(\theta_1) - \dot{m}_t(\theta_2)\| \right). \end{aligned} \quad (78)$$

Envelopes and diameter bounds. By Assumption 10 (Fréchet smoothness with Lipschitz envelopes), there exist measurable $L_t, J_t \geq 0$ with $\mathbb{E}[L_t] < \infty$ and $\mathbb{E}[J_t] < \infty$ such that, a.s.,

$$\begin{aligned} \|m_t(\theta_1) - m_t(\theta_2)\| &\leq L_t \|\theta_1 - \theta_2\|, \quad \|\dot{m}_t(\theta_1) - \dot{m}_t(\theta_2)\| \leq L_t \|\theta_1 - \theta_2\|, \\ \|\ddot{m}_t(\theta_1) - \ddot{m}_t(\theta_2)\| &\leq J_t \|\theta_1 - \theta_2\|. \end{aligned}$$

Pick any $\theta^\circ \in \Theta(\delta)$. By compactness of $\Theta(\delta)$ (Assumption 6) and the mean-value (Fréchet) argument along the line segment from θ° to θ ,

$$\begin{aligned} \|m_t(\theta)\| &\leq \|m_t(\theta^\circ)\| + L_t \text{diam}(\Theta(\delta)), \quad \|\dot{m}_t(\theta)\| \leq \|\dot{m}_t(\theta^\circ)\| + L_t \text{diam}(\Theta(\delta)), \\ \|\ddot{m}_t(\theta)\| &\leq \|\ddot{m}_t(\theta^\circ)\| + J_t \text{diam}(\Theta(\delta)).^{91} \end{aligned}$$

Random Lipschitz constant. Substituting the envelope and diameter bounds into (78) yields

$$\|H_T(\theta_1) - H_T(\theta_2)\| \leq C_{H,T} \|\theta_1 - \theta_2\|,$$

with the (random) Lipschitz constant

$$\begin{aligned} C_{H,T} = \frac{2\|Q_Z^{-1}\|}{T} \sum_{t=1}^T \left[J_t \|m_t(\theta^\circ)\| + \|\dot{m}_t(\theta^\circ)\| L_t + \underbrace{(J_t L_t + L_t \|\ddot{m}_t(\theta^\circ)\|)}_{\text{from diameter terms}} \text{diam}(\Theta(\delta)) \right. \\ \left. + (\|\dot{m}_t(\theta^\circ)\| + L_t \text{diam}(\Theta(\delta))) L_t \right]. \end{aligned}$$

Op(1) control of $C_{H,T}$. By Assumption 1 (strict stationarity with α -mixing and finite $(2 + \tau)$ -moments), Assumption 6 (compact $\Theta(\delta)$), and Assumption 10 (Fréchet-smooth with integrable Lipschitz envelopes), each summand (e.g., $J_t \|m_t(\theta^\circ)\|$, $\|\dot{m}_t(\theta^\circ)\| L_t$, L_t^2) is strictly stationary, α -mixing, and has finite first moment.⁹² Therefore $C_{H,T} = O_p(1)$ and, for any $\varepsilon > 0$, there exists $M_H < \infty$ with $\mathbb{P}(C_{H,T} \leq M_H) > 1 - \varepsilon$ for

⁹¹Why the diameter bounds hold. Let $\gamma(\lambda) = \theta^\circ + \lambda(\theta - \theta^\circ)$, $\lambda \in [0, 1]$. Fréchet differentiability gives $m_t(\theta) - m_t(\theta^\circ) = \int_0^1 \dot{m}_t(\gamma(\lambda))(\theta - \theta^\circ) d\lambda$, hence $\|m_t(\theta)\| \leq \|m_t(\theta^\circ)\| + \sup_\lambda \|\dot{m}_t(\gamma(\lambda))\| \cdot \|\theta - \theta^\circ\|$. Assumption 10 supplies the Lipschitz envelope for $\dot{m}_t$, so $\sup_\lambda \|\dot{m}_t(\gamma(\lambda))\| \leq \|\dot{m}_t(\theta^\circ)\| + L_t \|\theta - \theta^\circ\| \leq \|\dot{m}_t(\theta^\circ)\| + L_t \text{diam}(\Theta(\delta))$. The bounds for $\|\dot{m}_t(\theta)\|$ and $\|\ddot{m}_t(\theta)\|$ follow analogously, using the envelope J_t for $\ddot{m}_t$.

⁹²Why the averages are $O_p(1)$. Cauchy-Schwarz and the uniform L_2 -integrability established under A. 1, 6, 10 imply $\mathbb{E}[J_t \|m_t(\theta^\circ)\|] < \infty$, $\mathbb{E}[\|\dot{m}_t(\theta^\circ)\| L_t] < \infty$, and $\mathbb{E}[L_t^2] < \infty$. By the α -mixing LLN (Ibragimov and Linnik, 1971, Thm. 17.2.3; Newey and McFadden, 1994, §2.6), $T^{-1} \sum_t J_t \|m_t(\theta^\circ)\| \xrightarrow{P} \mathbb{E}[J_t \|m_t(\theta^\circ)\|]$, etc. Hence $C_{H,T} = O_p(1)$.

all large T . Thus, with probability at least $1 - \varepsilon$ (for large T),

$$\|H_T(\theta_1) - H_T(\theta_2)\| \leq M_H \|\theta_1 - \theta_2\| \quad \text{uniformly in } \theta_1, \theta_2 \in \Theta(\delta),$$

which proves the uniform Lipschitz continuity of the Hessian and establishes (75).

The result states that the curvature of the empirical no–arbitrage criterion varies smoothly with the parameters: small perturbations in factor loadings or the pricing kernel induce proportionate changes in $H_T(\theta)$. This guarantees stable quadratic (LAN) expansions and numerically robust second–order methods for inference and computation.

(c) Stochastic equicontinuity. The proof of stochastic equicontinuity unfolds through three sequential and logically dependent steps. First, the *maximal–inequality argument* (cf. Thm. 5.7 [van der Vaart \(1998\)](#)) establishes quantitative control over the local oscillations of the empirical score and Hessian processes. Using the uniform Lipschitz continuity of $S_T(\cdot)$ and $H_T(\cdot)$ —derived from Assumptions 1, 6, and 10—it shows that their deviations across small neighborhoods in $\Theta(\delta)$ are uniformly bounded by a stochastic modulus of continuity. This local bound serves as the technical input for the second step.

Second, the *Functional Central Limit Theorem* (Donsker’s Theorem; [van der Vaart and Wellner, 1996](#), Ch. 2.7; [Newey and McFadden, 1994](#), Sec. 2.6) lifts these local Lipschitz–based bounds to global tightness and weak convergence in $\ell^\infty(\Theta(\delta))$. Because the maximal–inequality result ensures a finite modulus of continuity, the empirical score and Hessian belong to Donsker–type function classes with bounded entropy integrals. This tightness property in turn guarantees that their fluctuations are uniformly controlled across the entire compact parameter space, providing the probabilistic foundation for stochastic equicontinuity.

Finally, applying these general empirical–process results to the NA–IPCA score and curvature functions

$$f_t^{(S)}(\theta) = \dot{m}_t(\theta)^\top Q_Z^{-1} m_t(\theta), \quad f_t^{(H)}(\theta) = \dot{m}_t(\theta)^\top Q_Z^{-1} \dot{m}_t(\theta),$$

completes the argument: both indexed processes $\{\mathbb{G}_T^{(S)}(\theta)\}$ and $\{\mathbb{G}_T^{(H)}(\theta)\}$ are tight in $\ell^\infty(\Theta(\delta))$ and therefore stochastically equicontinuous. The causal link among these steps is essential: the maximal–inequality argument provides the local continuity control; Donsker’s theorem converts that control into global weak convergence; and the application to the NA–IPCA setting translates these probabilistic results into the differentiable stability of the empirical criterion. Together, they ensure that small perturbations in factor loadings or pricing–kernel coefficients yield proportionate and continuous adjustments in the score and curvature, guaranteeing a smooth and well–behaved KKT system and validating the LAN expansion in [Appendix B](#).

Let $S_T(\theta) = \nabla_\theta Q_T(\theta)$ and $H_T(\theta) = \nabla_{\theta\theta}^2 Q_T(\theta)$ denote, respectively, the empirical score and Hessian of the sample criterion $Q_T(\theta) = T^{-1} \sum_t \|m_t(\theta)\|_{Q_Z^{-1}}^2$. From the preceding results, both mappings are uniformly Lipschitz on the compact parameter space $\Theta(\delta)$, with random Lipschitz constants C_T and $C_{H,T}$ that are $O_p(1)$ by the α –mixing law of large numbers under Assumptions 1, 6, and 10.

Centered empirical processes. Define the centered (zero–mean) empirical processes

$$\mathbb{G}_T^{(S)}(\theta) = \sqrt{T} [S_T(\theta) - \mathbb{E} S_T(\theta)], \quad \mathbb{G}_T^{(H)}(\theta) = \sqrt{T} [H_T(\theta) - \mathbb{E} H_T(\theta)].$$

To prove stochastic equicontinuity, we must show that for every $\varepsilon, \eta > 0$ there exists $\delta_\eta > 0$ such that

$$\limsup_{T \rightarrow \infty} \mathbb{P} \left(\sup_{\|\theta_1 - \theta_2\| < \delta_\eta} \|\mathbb{G}_T^{(S)}(\theta_1) - \mathbb{G}_T^{(S)}(\theta_2)\| > \varepsilon \right) < \eta,$$

and likewise for $\mathbb{G}_T^{(H)}(\cdot)$, which corresponds to the criterion in (76).

Maximal-inequality argument. Because $S_T(\cdot)$ and $H_T(\cdot)$ are uniformly Lipschitz with integrable envelopes, the differences $S_T(\theta_1) - S_T(\theta_2)$ and $H_T(\theta_1) - H_T(\theta_2)$ are uniformly bounded by constants times $\|\theta_1 - \theta_2\|$. Formally, for some nonrandom $C_S, C_H < \infty$ (with probability approaching one),

$$\|S_T(\theta_1) - S_T(\theta_2)\| \leq C_S \|\theta_1 - \theta_2\|, \quad \|H_T(\theta_1) - H_T(\theta_2)\| \leq C_H \|\theta_1 - \theta_2\|.$$

These Lipschitz bounds and the finite $(2 + \tau)$ -moment property of the per-period moments (Assumption 1) ensure that both $S_T(\cdot)$ and $H_T(\cdot)$ form *Donsker-type classes* under weak dependence, i.e., classes of functions whose oscillations are controlled uniformly on the compact set $\Theta(\delta)$.^{93, 94}

⁹³The concept of a ‘‘Donsker class’’ (also called a P-Donsker class) refers to a collection of measurable functions $\mathcal{F}$ for which the empirical process

$$\sqrt{T}(P_T - P)(f), \quad f \in \mathcal{F}$$

is tight in $\ell^\infty(\mathcal{F})$ and converges weakly to a Gaussian process (see e.g. van der Vaart and Wellner, 1996, Ch. 1–2). A standard sufficient condition in the independent case is the finiteness of the bracketing entropy integral (Dudley’s theorem); see van der Vaart and Wellner, 1996, Ch. 2.7.

For weakly dependent (mixing) sequences, the theory has been extended by several authors. For example, the survey/textbook ‘‘Empirical Process Techniques for Dependent Data’’ (Ango Nze et al., 2020) collects maximal-inequality and weak convergence results for empirical processes indexed by classes of functions under mixing conditions (see especially chapters on maximal inequalities and empirical CLTs for mixing sequences). In particular, Lipschitz-in-parameter function classes (that admit a uniform Lipschitz envelope on the parameter space) satisfy the requisite uniform covering/bracketing bounds. Thus, when one verifies that (i) the class is uniformly Lipschitz in θ over a compact $\Theta(\delta)$, (ii) the envelope has finite second (or a little more) moment under the stationary data process, and (iii) the data are α -mixing with summable coefficients, one obtains that the indexed empirical processes are stochastically equicontinuous and hence Donsker-type.

In our setting, the derivation of uniform Lipschitz continuity of $S_T(\theta)$ and $H_T(\theta)$, together with integrable envelopes (Assumption 10) and moment/mixing conditions (Assumption 1), precisely meet those sufficient conditions. Therefore we may invoke the empirical-process machinery to conclude that the processes

$$\{\sqrt{T}[S_T(\theta) - \mathbb{E}S_T(\theta)] : \theta \in \Theta(\delta)\} \quad \text{and} \quad \{\sqrt{T}[H_T(\theta) - \mathbb{E}H_T(\theta)] : \theta \in \Theta(\delta)\}$$

are tight in $\ell^\infty(\Theta(\delta))$ and satisfy stochastic equicontinuity (cf. van der Vaart (1998)).

The Lipschitz bounds ensure uniformly bounded oscillations of the indexed function class; combined with mixing and finite-moment conditions, these yield bracketing/covering bounds for the class, which in turn give Donsker-type behaviour and hence stochastic equicontinuity of the empirical processes.

⁹⁴*Donsker’s Theorem and its role in stochastic equicontinuity of the NA-IPCA empirical process.* Donsker’s Theorem—also known as the *Functional Central Limit Theorem*—provides the probabilistic foundation for establishing uniform convergence and stochastic equicontinuity of empirical processes. It states that for a measurable class of functions $\mathcal{F}$ with finite second moments, if the class satisfies a finite bracketing-entropy integral condition,

$$\int_0^1 \sqrt{\log N_{[]}(\varepsilon, \mathcal{F}, L_2(P))} d\varepsilon < \infty,$$

where $N_{[]}(\varepsilon, \mathcal{F}, L_2(P))$ denotes the minimal number of ε -brackets required to cover $\mathcal{F}$ in $L_2(P)$, then the empirical process

$$\mathbb{G}_T(f) = \sqrt{T}[(P_T - P)f], \quad f \in \mathcal{F},$$

converges weakly in $\ell^\infty(\mathcal{F})$ to a tight mean-zero Gaussian process $\mathbb{G}(\cdot)$ with covariance function $\mathbb{E}[\mathbb{G}(f_1)\mathbb{G}(f_2)] = (Pf_1f_2 - Pf_1Pf_2)$; see van der Vaart and Wellner (1996, Ch. 2.7). Intuitively, this theorem ensures that empirical averages indexed by a

Application of empirical–process theorems. The stochastic–equicontinuity of the empirical processes

$$\mathbb{G}_T^{(S)}(\theta) = \sqrt{T} [S_T(\theta) - \mathbb{E}S_T(\theta)], \quad \mathbb{G}_T^{(H)}(\theta) = \sqrt{T} [H_T(\theta) - \mathbb{E}H_T(\theta)],$$

follows directly from the combination of the α –mixing uniform law of large numbers and the empirical–process maximal–inequality results for weakly dependent sequences. Specifically, under Assumptions 1 (strict stationarity with summable α –mixing coefficients and finite $(2 + \tau)$ –moments), 6 (compactness of the feasible parameter set), and 10 (Fréchet–smooth and uniformly Lipschitz moment map), the class of functions

$$\mathcal{F}_T = \left\{ f_t^{(S)}(\theta) = \dot{m}_t(\theta)^\top Q_Z^{-1} m_t(\theta), \quad f_t^{(H)}(\theta) = \dot{m}_t(\theta)^\top Q_Z^{-1} \dot{m}_t(\theta) : \theta \in \Theta(\delta) \right\}$$

forms a *Lipschitz–parameter* family with an integrable envelope and hence satisfies the entropy conditions required for tightness and uniform convergence of empirical processes.

Formally, these results rely on two foundational theorems: (i) the Uniform Law of Large Numbers for dependent processes (Newey and McFadden, 1994, Sec. 2.6), which ensures that $\sup_{\theta \in \Theta(\delta)} |T^{-1} \sum_t f_t(\theta) - \mathbb{E}f_t(\theta)| \xrightarrow{P} 0$; and (ii) the Maximal–Inequality Theorem for empirical processes indexed by Lipschitz classes (Thm. 5.7 van der Vaart (1998)), which guarantees stochastic equicontinuity by bounding the oscillations of $\mathbb{G}_T(f)$ over small neighborhoods in $\Theta(\delta)$. In particular, Theorem 5.7 of van der Vaart (1998) provides a universal modulus–of–continuity bound:

$$\mathbb{E} \left[\sup_{\|\theta_1 - \theta_2\| < \delta} \|\mathbb{G}_T^{(S)}(\theta_1) - \mathbb{G}_T^{(S)}(\theta_2)\| \right] \leq C \omega_T(\delta), \quad \omega_T(\delta) \xrightarrow[\delta \rightarrow 0]{} 0$$

uniformly in T , where $C > 0$ is finite and $\omega_T(\delta)$ denotes the stochastic modulus of continuity of the process. An identical bound applies to $\mathbb{G}_T^{(H)}(\cdot)$, since both the score and Hessian share the same Lipschitz–envelope structure and mixing–moment environment.

Therefore, by Markov’s inequality and the uniform boundedness of $\omega_T(\delta)$, for any $\varepsilon, \eta > 0$ there exists

sufficiently “smooth” or “low–complexity” class behave asymptotically as Gaussian processes uniformly in the index.

Application to the NA–IPCA framework. In our setting, the function class

$$\mathcal{F}_T = \left\{ f_t^{(S)}(\theta) = \dot{m}_t(\theta)^\top Q_Z^{-1} m_t(\theta), \quad f_t^{(H)}(\theta) = \dot{m}_t(\theta)^\top Q_Z^{-1} \dot{m}_t(\theta) : \theta \in \Theta(\delta) \right\}$$

inherits uniform Lipschitz continuity from Assumption 10, bounded second moments from Assumption 1, and compactness of the parameter domain from Assumption 6. These properties jointly guarantee that the entropy integral of $\mathcal{F}_T$ is finite, placing it in the class of *P–Donsker* families (cf. Newey and McFadden, 1994, Sec. 2.6). Consequently, the empirical processes

$$\left\{ \sqrt{T} [S_T(\theta) - \mathbb{E}S_T(\theta)] : \theta \in \Theta(\delta) \right\} \quad \text{and} \quad \left\{ \sqrt{T} [H_T(\theta) - \mathbb{E}H_T(\theta)] : \theta \in \Theta(\delta) \right\}$$

are tight in $\ell^\infty(\Theta(\delta))$ and satisfy stochastic equicontinuity. Formally, this implies that the fluctuations of the empirical score $S_T(\theta)$ and Hessian $H_T(\theta)$ around their expectations vanish uniformly on $\Theta(\delta)$, ensuring that the KKT system defining the NA–IPCA estimator admits a smooth stochastic expansion.

Donsker’s theorem provides the bridge from probabilistic regularity to economic stability: it ensures that as the sample grows, small perturbations in factor loadings or pricing–kernel coefficients produce proportionate, continuous adjustments in the empirical first– and second–order conditions. This underlies the differentiable KKT system and the LAN expansion in Appendix B, guaranteeing that the estimator’s stochastic fluctuations remain well behaved and that asymptotic inference is valid uniformly across the feasible no–arbitrage manifold (van der Vaart and Wellner (1996, Ch. 2.7, Thm. 2.11.9); Newey and McFadden (1994, Sec. 2.6)).

$\delta_\eta > 0$ such that

$$\limsup_{T \rightarrow \infty} \mathbb{P} \left(\sup_{\|\theta_1 - \theta_2\| < \delta_\eta} \sqrt{T} \| [S_T(\theta_1) - S_T(\theta_2)] - [\mathbb{E}S_T(\theta_1) - \mathbb{E}S_T(\theta_2)] \| > \varepsilon \right) < \eta,$$

and the same relation holds for $H_T(\cdot)$, confirming (76).

(d) Differentiable KKT mapping and uniform second-order expansion. Define the sample KKT mapping associated with the constrained NA-IPCA estimator as

$$\mathcal{K}_T(\theta) = S_T(\theta) + \Lambda_T^\top g(\theta) + \mu_T \nabla_\theta h(\theta),$$

where $S_T(\theta) = \nabla_\theta Q_T(\theta)$ is the empirical score, $g(\theta)$ collects the equality constraints (e.g., $\Gamma_\beta^\top Q_Z \Gamma_\beta - I_K = 0$, $\Gamma_\beta^\top Q_Z \Gamma_\alpha = 0$), and $h(\theta) = \psi(\Gamma_\alpha) - \delta \leq 0$ is the no-arbitrage inequality. The preceding results ((74)–(76)) establish that $S_T(\theta)$ and its derivative $H_T(\theta) = \nabla_\theta^2 Q_T(\theta)$ are uniformly Lipschitz and stochastically equicontinuous on the compact feasible set $\Theta(\delta)$.

Differentiability. Uniform Lipschitz continuity of $S_T(\theta)$ implies that the Jacobian $H_T(\theta)$ exists, is bounded, and converges uniformly in probability to its population counterpart $H(\theta) = \mathbb{E}[H_T(\theta)]$. Because the constraint functions $g(\theta)$ and $h(\theta)$ are continuously differentiable by construction, and their gradients $\nabla_\theta g(\theta)$, $\nabla_\theta h(\theta)$ are bounded and continuous on $\Theta(\delta)$ (Assumption 6), it follows that each component of $\mathcal{K}_T(\theta)$ is continuously differentiable in θ with probability approaching one. Hence the entire KKT mapping $\mathcal{K}_T(\theta)$ is locally $\mathcal{C}^1$ on $\Theta(\delta)$.

Uniform quadratic expansion. The combination of uniform Lipschitz continuity and stochastic equicontinuity allows the use of the mean-value theorem for random mappings (see Newey and McFadden, 1994, Sec. 2.6; Thm. 5.7 van der Vaart (1998)). Specifically, for any θ in a neighborhood of θ_0 , there exists $\bar{\theta}$ between θ and θ_0 such that

$$\mathcal{K}_T(\theta) = \mathcal{K}_T(\theta_0) + H_T(\bar{\theta})(\theta - \theta_0) + R_T(\theta, \theta_0), \quad (79)$$

where the remainder term satisfies $\|R_T(\theta, \theta_0)\| = o_p(\|\theta - \theta_0\|)$ uniformly on compact neighborhoods of θ_0 .⁹⁵

Because $H_T(\theta)$ is positive-definite on the tangent space by Section A.4 and continuous in θ , the expansion (79) admits a well-defined inverse mapping near θ_0 . Thus, with probability approaching one, the sample KKT mapping is locally continuously differentiable and possesses a uniform second-order Taylor expansion around θ_0 .

⁹⁵Why the uniform Taylor expansion holds. Assumptions 1, 6, and 10 guarantee that (i) $S_T(\theta)$ is uniformly Lipschitz and stochastically equicontinuous on $\Theta(\delta)$; (ii) $H_T(\theta)$ exists, is bounded, and converges uniformly to $H(\theta)$; and (iii) $\Theta(\delta)$ is compact, ensuring uniformity of convergence. By the mean-value theorem for Fréchet differentiable mappings on Banach spaces (Rockafellar and Wets, 1998, Ch. 4), these properties yield the expansion $S_T(\theta) = S_T(\theta_0) + H_T(\bar{\theta})(\theta - \theta_0) + o_p(\|\theta - \theta_0\|)$ uniformly on compact neighborhoods of θ_0 . Substituting this expansion into $\mathcal{K}_T(\theta)$ and adding the continuously differentiable constraint terms $g(\theta)$ and $h(\theta)$ preserves the order of the remainder term, leading to (79). Economically, this uniform quadratic approximation means that the KKT system behaves locally like a smooth linear system around the equilibrium parameter θ_0 : small perturbations in the estimated parameters induce proportionate changes in the score and constraint gradients, allowing the LAN representation of the NA-IPCA estimator to hold uniformly in a neighborhood of the true model.

The differentiable KKT mapping ensures that the constrained NA–IPCA estimator behaves locally as a smooth function of the data and weights: its score and curvature respond continuously to changes in parameters, enabling stable computation and valid quadratic (LAN) inference. Economically, this property formalizes that near equilibrium, the no–arbitrage system reacts linearly to small shifts in risk exposures or pricing coefficients, so both estimation and inference rest on a well–behaved, continuously differentiable manifold of feasible pricing kernels. $\square$

A.8.3 Bounded–Influence and Huber Regularity

This subsection introduces the bounded–influence and finite–variance conditions on the Huber score $\psi_\kappa(\cdot)$ that ensure robustness and well–posed asymptotic variance of the NA–IPCA estimator under heavy–tailed residuals. These conditions complement the smoothness framework of Assumption 10 and are used explicitly in Appendix B (Sections B.4–B.5).

Assumption 11 (Bounded–Influence and Finite–Variance Huber Regularity).⁹⁶ Let $\rho_\kappa : \mathbb{R} \rightarrow \mathbb{R}_+$ be a convex, continuously differentiable loss with Huber score $\psi_\kappa(u) = \rho'_\kappa(u)$ applied componentwise to residuals u_t . We assume the following hold uniformly in the tuning parameter $\kappa > 0$:

- (a) **Bounded influence and local linearity.** ψ_κ is odd, Lipschitz, and continuously differentiable with $|\psi'_\kappa(u)| \leq 1$ for all u , and there exists $\delta_\kappa > 0$ such that $\psi_\kappa(u) = u$ whenever $|u| \leq \delta_\kappa$. As $\kappa \rightarrow \infty$, $\psi_\kappa(u) \rightarrow u$ pointwise (GLS limit).
- (b) **Finite variance of the robust score.** For the residual process u_t in Assumption 3, the transformed scores have finite variance and finite second moments, $\mathbb{E}[\psi_\kappa(u_t)^2] < \infty$, and the long–run covariance matrix of the Huber–weighted score exists and is finite under the α –mixing condition in Assumption 1.
- (c) **Regular loss.** ρ_κ is convex, continuously differentiable, satisfies $\rho_\kappa(0) = 0$, $\rho'_\kappa = \psi_\kappa$, and $\rho_\kappa(u) \rightarrow \frac{1}{2}u^2$ as $\kappa \rightarrow \infty$.
- (d) **(Optional) tuning stability.** If a data–dependent sequence κ_T is used, it satisfies $\kappa_T \rightarrow \kappa_0 \in (0, \infty]$ and $\sup_T \mathbb{E}[\psi_{\kappa_T}(u_t)^2] < \infty$.

Assumption 11 ensures that the Huber–weighted score $\psi_\kappa(u_t)$ admits a finite, well–behaved long–run variance and, together with Assumption 10, places the empirical score in a smooth, uniformly Lipschitz class. Consequently, the robust (Huber) criterion inherits the same first–order (KKT) and second–order (LAN) structure as its GLS counterpart while retaining stability under heavy tails and outliers.⁹⁷

⁹⁶This assumption records the standard robustness conditions for Huber M –estimation and links them to the differentiable M –estimation framework used in this paper. Mathematically, bounded influence controls the contribution of large residuals so that the Huber score has finite second moments and admits a stable long–run covariance; together with Assumption 10, it places the score process in a smooth, Lipschitz class that satisfies the uniform LLN/CLT and maximal–inequality requirements needed for LAN (Appendix B). Economically, these conditions guarantee that extreme cross–sectional pricing errors do not dominate estimation, so the scaling of priced risk remains stable even when returns are heavy–tailed or heteroskedastic. *Used in Lemma A.2 (CLT for bounded–influence scores), Lemma B.5 (feasible WLS equivalence), and Proposition B.3 (LAN and efficiency).* Primary references include Huber (1981, Ch. 7), Newey and McFadden (1994, Sec. 2.6), and Maronna et al. (2006, Chs. 3–4).

⁹⁷Bounded influence implies $\|\psi_\kappa(u)\| \leq C_\kappa(1 + |u|)$ and thus keeps second moments finite under Assumption 1; local linearity $\psi'_\kappa(0) = 1$ preserves GLS efficiency near the origin; the convex, differentiable loss guarantees a well–posed gradient map. Under stationarity and α –mixing (Assumption 1), the long–run covariance $\sum_\ell \text{Cov}(\psi_\kappa(u_t), \psi_\kappa(u_{t-\ell}))$ is absolutely summable, delivering a finite asymptotic variance for the robust score (cf. Huber, 1981, Ch. 7; Newey and McFadden, 1994, Sec. 2.6). Economically, these

A.8.4 Implications for LAN and Efficiency

This subsection gathers the consequences of the smoothness (Assumption 10) and Huber bounded-influence (Assumption 11) conditions for local asymptotic normality (LAN) and efficiency of the NA-IPCA estimator. Combined with pre-whitening feasibility (Assumption 9), compactness and KKT regularity (Assumptions 6, 5 and Section A.4), the uniformly Lipschitz score/Hessian (Lemma A.7), and rotation-invariant normalization (Lemma A.5, Proposition A.6), these conditions yield a LAN expansion with a sandwich-form asymptotic variance on the tangent space and clarify when the GLS efficiency bound is attained.

Proposition A.8 (LAN and (Robust) Efficiency). ⁹⁸ *Let Assumptions 1–6, 9–11 hold, and let θ_0 satisfy LICQ/Slater (Assumption 5, Section A.4). Let $\tilde{\theta}_0 := \mathbf{N}(\theta_0)$ and write Π_0 for the orthogonal projector onto the tangent space $\mathcal{T}_{\tilde{\theta}_0} \Theta(\delta)$. Define the block Jacobian of per-period moments by $\dot{g}_t(\theta)$ and set*

$$A := \mathbb{E}[\dot{g}_t(\theta_0)^\top \Omega_t^{-1} \dot{g}_t(\theta_0)],$$

$$B_\kappa := \sum_{\ell=-\infty}^{\infty} \mathbb{E}[\dot{g}_t(\theta_0)^\top \Omega_t^{-1} \psi_\kappa(u_t) \psi_\kappa(u_{t-\ell})^\top \Omega_{t-\ell}^{-1} \dot{g}_{t-\ell}(\theta_0)].$$

where u_t are residuals at θ_0 and ψ_κ is the Huber score from Assumption 11. Then:

(i) **LAN (projected).** *The constrained estimator $\hat{\theta}_T$ admits the LAN expansion*

$$\sqrt{T} \Pi_0 (\hat{\theta}_T - \theta_0) = (\Pi_0 A \Pi_0^\top)^{-1} \frac{1}{\sqrt{T}} \sum_{t=1}^T \Pi_0 \dot{g}_t(\theta_0)^\top \Omega_t^{-1} \psi_\kappa(u_t(\theta_0)) + o_p(1). \quad (80)$$

(ii) **Asymptotic variance (sandwich on the tangent).** *Consequently,*

$$\sqrt{T} \Pi_0 (\hat{\theta}_T - \theta_0) \xrightarrow{d} \mathcal{N}\left(0, \underbrace{(\Pi_0 A \Pi_0^\top)^{-1} (\Pi_0 B_\kappa \Pi_0^\top) (\Pi_0 A \Pi_0^\top)^{-1}}_{\Sigma_\kappa}\right). \quad (81)$$

(iii) **GLS efficiency as $\kappa \rightarrow \infty$.** *If $\kappa \rightarrow \infty$ (quadratic loss) or under Gaussian residuals with $\psi_\kappa(u) = u$, then*

$B_\kappa = A$ and

$$\Sigma_\kappa = (\Pi_0 A \Pi_0^\top)^{-1},$$

i.e., the estimator attains the GLS (Cramér–Rao) efficiency bound on $\mathcal{T}_{\tilde{\theta}_0} \Theta(\delta)$.

Proof. (i) **LAN (projected).** By Lemma A.7 (uniform Lipschitz continuity of the score and Hessian) and the differentiable KKT mapping (part (d) therein), a mean-value expansion of the *projected* first-order system

conditions cap the marginal impact of extreme pricing errors, so the estimated priced-risk geometry remains stable across samples while converging to the GLS benchmark as $\kappa \rightarrow \infty$.

⁹⁸This proposition synthesizes smooth M -estimation with constraints and bounded-influence scoring. LAN follows from differentiability of the KKT map (Lemma A.7), the α -mixing CLT for bounded-influence scores (Lemma A.2), and feasible pre-whitening (Assumption 9). Rotation and normalization ensure the result holds on the quotient geometry (Lemma A.5, Proposition A.6). The efficiency statement matches the GLS bound when $\kappa \rightarrow \infty$ (and under Gaussian/quadratic loss), while for fixed κ one obtains the usual Huber sandwich. See Newey and McFadden (1994, Sec. 2.6), Huber (1981, Ch. 7), and Maronna et al. (2006, Chs. 3–4).

around θ_0 yields

$$\mathbf{0} = \Pi_0 S_T(\hat{\theta}_T) = \Pi_0 S_T(\theta_0) + \Pi_0 H_T(\bar{\theta}_T)(\hat{\theta}_T - \theta_0), \quad (82)$$

where $S_T(\theta) = \nabla_{\theta} Q_T^{(\text{feas})}(\theta)$, $H_T(\theta) = \nabla_{\theta\theta}^2 Q_T^{(\text{feas})}(\theta)$, Π_0 is the orthogonal projector onto the tangent space $\mathcal{T}_{\bar{\theta}_0} \Theta(\delta)$ at the normalized point $\bar{\theta}_0 = N(\theta_0)$, and $\bar{\theta}_T$ lies on the line segment between $\hat{\theta}_T$ and θ_0 .⁹⁹

Feasible pre-whitening (Assumption 9) implies that the projected curvature converges to the population information on the tangent space:

$$\Pi_0 H_T(\bar{\theta}_T) \Pi_0^\top \xrightarrow{P} \Pi_0 A \Pi_0^\top, \quad A := \mathbb{E}[\dot{g}_t(\theta_0)^\top \Omega_t^{-1} \dot{g}_t(\theta_0)], \quad (83)$$

uniformly on compact neighborhoods of θ_0 .¹⁰⁰

Moreover, the projected sample score admits the Huber–GLS representation with a negligible remainder:

$$\Pi_0 S_T(\theta_0) = \frac{1}{T} \sum_{t=1}^T \Pi_0 \dot{g}_t(\theta_0)^\top \Omega_t^{-1} \psi_\kappa(u_t(\theta_0)) + o_p(T^{-1/2}), \quad (84)$$

where $u_t(\theta_0)$ are residuals at θ_0 and ψ_κ is the Huber score from Assumption 11.¹⁰¹

Substituting (84) and (83) into (82) and multiplying by $\sqrt{T}$ gives

$$\sqrt{T} \Pi_0(\hat{\theta}_T - \theta_0) = -(\Pi_0 A \Pi_0^\top)^{-1} \frac{1}{\sqrt{T}} \sum_{t=1}^T \Pi_0 \dot{g}_t(\theta_0)^\top \Omega_t^{-1} \psi_\kappa(u_t(\theta_0)) + o_p(1),$$

which is the stated projected linearization used in the LAN expansion. By Lemma A.2, the partial sum on the right-hand side satisfies a CLT with long-run covariance $\Pi_0 B_\kappa \Pi_0^\top$, yielding the sandwich form in the next step. Economically, this reduction shows that—after enforcing feasibility and removing rotational redundancies—the estimator behaves as a one-step Huber–GLS on the identified (tangent) coordinates, with curvature A and innovation $\psi_\kappa(u_t)$ that jointly determine robust (and, as $\kappa \rightarrow \infty$, GLS-efficient) precision.

(ii) Asymptotic variance (sandwich on the tangent). Define the projected, Huber–GLS-weighted summand

$$Z_{t,\kappa} := \Pi_0 \dot{g}_t(\theta_0)^\top \Omega_t^{-1} \psi_\kappa(u_t(\theta_0)) \in \mathbb{R}^{p_{\mathcal{T}}}, \quad p_{\mathcal{T}} = \dim(\mathcal{T}_{\bar{\theta}_0} \Theta(\delta)),$$

⁹⁹Why the projected mean-value expansion holds. (i) By Lemma A.7, $S_T(\cdot)$ is (locally) Fréchet differentiable and $H_T(\cdot)$ is uniformly Lipschitz on compact neighborhoods of θ_0 with probability approaching one. Hence, for some $\bar{\theta}_T$ on the line segment $\{\theta_0 + \lambda(\hat{\theta}_T - \theta_0) : \lambda \in [0, 1]\}$, $S_T(\hat{\theta}_T) = S_T(\theta_0) + H_T(\bar{\theta}_T)(\hat{\theta}_T - \theta_0)$. (ii) Projecting by Π_0 is legitimate because the KKT system is solved on the feasible manifold and the first-order conditions are evaluated on the tangent space (see Section A.4 and Proposition A.6). The projection commutes with linearization since Π_0 is constant at $\bar{\theta}_0$ and orthogonal with respect to the ambient inner product used by $Q_T^{(\text{feas})}$.

¹⁰⁰Curvature identification and feasibility. By Assumption 9, the feasible weighting is ratio-consistent for Ω_t^{-1} , so $H_T(\cdot)$ coincides with its GLS analogue up to $o_p(1)$ uniformly. By Section A.4, the tangent-restricted Hessian is nonsingular at θ_0 . Hence $\Pi_0 H_T(\bar{\theta}_T) \Pi_0^\top \rightarrow_p \Pi_0 A \Pi_0^\top$. Economically, this says that local curvature—hence identification and efficiency—are preserved by feasible weights.

¹⁰¹Why the score has the form (84). Differentiating $Q_T^{(\text{feas})}(\theta) = T^{-1} \sum_t \|\Omega_t^{-1/2} m_t(\theta)\|_2^2$ at θ_0 yields $S_T(\theta_0) = T^{-1} \sum_t \dot{m}_t(\theta_0)^\top \Omega_t^{-1} m_t(\theta_0)$. Under Huberization, $m_t(\theta_0)$ is replaced by the bounded-influence score $\psi_\kappa(u_t(\theta_0))$ in the linearization of the moment map, giving $\dot{g}_t(\theta_0)^\top \Omega_t^{-1} \psi_\kappa(u_t(\theta_0))$ up to a uniformly negligible $o_p(T^{-1/2})$ remainder by Lemma A.7 and Assumption 11. Projecting by Π_0 preserves the form because Π_0 is fixed at $\bar{\theta}_0$ and the KKT system is solved in tangent coordinates (Proposition A.6).

and the long-run covariance

$$B_{\kappa} := \sum_{\ell=-\infty}^{\infty} \text{Cov}(Z_{t,\kappa}, Z_{t-\ell,\kappa}) = \sum_{\ell=-\infty}^{\infty} \mathbb{E}[Z_{t,\kappa} Z_{t-\ell,\kappa}^{\top}].$$

Let the centered partial-sum process be

$$\mathbb{G}_T^{(Z)} := \frac{1}{\sqrt{T}} \sum_{t=1}^T (Z_{t,\kappa} - \mathbb{E}[Z_{t,\kappa}]).$$

By Lemma A.2 (strict stationarity with summable α -mixing and finite $(2 + \tau)$ -moments from Assumption 1, and bounded-influence from Assumption 11),

$$\mathbb{G}_T^{(Z)} \Rightarrow \mathcal{N}(0, B_{\kappa}) \quad \text{in } \mathbb{R}^{p_{\mathcal{T}}}.^{102103104} \quad (85)$$

Insert the projected score representation (84) and curvature limit (83) into the linearization (82), multiply by $\sqrt{T}$, and rearrange:

$$\sqrt{T} \Pi_0(\hat{\theta}_T - \theta_0) = -(\Pi_0 A \Pi_0^{\top})^{-1} \frac{1}{\sqrt{T}} \sum_{t=1}^T Z_{t,\kappa} + o_p(1). \quad (86)$$

Because $\Pi_0 A \Pi_0^{\top}$ is nonsingular on $\mathcal{T}_{\bar{\theta}_0} \Theta(\delta)$ by LICQ/SOSC (Section A.4), and

$$\Pi_0 H_T(\bar{\theta}_T) \Pi_0^{\top} \rightarrow_p \Pi_0 A \Pi_0^{\top}$$

by (83), the continuous-mapping theorem and Slutsky's lemma applied to (85)–(86) give the Gaussian limit with the *tangent-space sandwich* covariance:

$$\sqrt{T} \Pi_0(\hat{\theta}_T - \theta_0) \Rightarrow \mathcal{N}\left(0, \underbrace{(\Pi_0 A \Pi_0^{\top})^{-1} (\Pi_0 B_{\kappa} \Pi_0^{\top}) (\Pi_0 A \Pi_0^{\top})^{-1}}_{\Sigma_{\kappa}}\right).^{105} \quad (87)$$

¹⁰²On centering and mean zero at θ_0 . At the true parameter, $\mathbb{E}[Z_{t,\kappa}] = 0$ under the maintained conditions: (i) $\mathbb{E}[\psi_{\kappa}(u_t(\theta_0)) | Z_t] = 0$ by the mean-zero idiosyncratic structure (Assumption 3) and the oddness/monotonicity of ψ_{κ} (Assumption 11); (ii) $\dot{g}_t(\theta_0)$ and Ω_t^{-1} are Z_t -measurable. Thus $\mathbb{E}[Z_{t,\kappa}] = \mathbb{E}[\Pi_0 \dot{g}_t(\theta_0)^{\top} \Omega_t^{-1} \mathbb{E}\{\psi_{\kappa}(u_t(\theta_0)) | Z_t\}] = 0$, and (85) is equivalent to $\frac{1}{\sqrt{T}} \sum_{t=1}^T Z_{t,\kappa} \Rightarrow \mathcal{N}(0, B_{\kappa})$.

¹⁰³Here $\Rightarrow$ denotes convergence in distribution.

¹⁰⁴The symbol $\mathbb{R}^{p_{\mathcal{T}}}$ denotes the Euclidean space of real vectors with dimension equal to the number of locally identifiable (tangent-space) parameters. Formally, $p_{\mathcal{T}} := \dim(\mathcal{T}_{\bar{\theta}_0} \Theta(\delta))$ is the dimension of the tangent space to the feasible manifold $\Theta(\delta)$ at the normalized point $\bar{\theta}_0 = N(\theta_0)$. Intuitively, $\mathcal{T}_{\bar{\theta}_0} \Theta(\delta)$ collects all infinitesimal parameter perturbations that satisfy the model's orthogonality and no-arbitrage constraints; its dimension $p_{\mathcal{T}}$ therefore measures the number of economically identifiable degrees of freedom in the NA-IPCA model. Consequently, $Z_{t,\kappa} \in \mathbb{R}^{p_{\mathcal{T}}}$ represents the projected score component restricted to the identified risk space—the set of directions in which estimation and inference are meaningful once redundant (rotational) coordinates are removed.

¹⁰⁵Why (87) holds. (1) *CLT for the Huber-projected score*. Assumption 1 ensures strict stationarity, summable α -mixing, and finite $(2 + \tau)$ -moments; Assumption 11 yields bounded-influence with $\mathbb{E}\|\psi_{\kappa}(u_t)\|^2 < \infty$. Left-multiplication by Ω_t^{-1} and projection by Π_0 preserve square-integrability; Lemma A.2 then gives (85) with finite B_{κ} . (2) *Stable curvature on the tangent*. By feasible pre-whitening (Assumption 9), $\Pi_0 H_T(\bar{\theta}_T) \Pi_0^{\top} \rightarrow_p \Pi_0 A \Pi_0^{\top}$; LICQ/SOSC (Section A.4) gives $\Pi_0 A \Pi_0^{\top} \succ 0$ on $\mathcal{T}_{\bar{\theta}_0} \Theta(\delta)$. (3) *Continuous mapping + Slutsky*. Apply the continuous-mapping theorem to $M \mapsto (\Pi_0 A \Pi_0^{\top})^{-1} M$ and Slutsky to replace $\Pi_0 H_T(\bar{\theta}_T) \Pi_0^{\top}$ by $\Pi_0 A \Pi_0^{\top}$ in (86), delivering (87).

The limit variance Σ_κ is the familiar *sandwich* on the identified (tangent) coordinates: A captures local curvature (precision) and B_κ the robust dispersion of Huber-weighted innovations. As $\kappa \rightarrow \infty$, $\psi_\kappa(u) \rightarrow u$ and Σ_κ collapses to the GLS bound, while for finite κ it preserves efficiency near Gaussian regimes and controls the influence of outliers under heavy tails.

(iii) *GLS efficiency as $\kappa \rightarrow \infty$.* Set $\psi_\kappa(u) = u$. Then the projected summand in (85) is

$$Z_{t,\infty} = \Pi_0 \dot{g}_t(\theta_0)^\top \Omega_t^{-1} u_t(\theta_0).$$

Under Assumption 3 (mean-zero idiosyncrasy) and the standard conditional covariance normalization $\mathbb{E}[u_t(\theta_0)u_t(\theta_0)^\top | Z_t] = \Omega_t$,¹⁰⁶ we have $B_\infty = \Pi_0 A \Pi_0^\top$ where $A = \mathbb{E}[\dot{g}_t^\top \Omega_t^{-1} \dot{g}_t]$. Substituting B_∞ into the sandwich covariance (87) yields

$$\Sigma_\infty = (\Pi_0 A \Pi_0^\top)^{-1} (\Pi_0 A \Pi_0^\top) (\Pi_0 A \Pi_0^\top)^{-1} = (\Pi_0 A \Pi_0^\top)^{-1}.$$

Because $\Pi_0 A \Pi_0^\top \succ 0$ on the tangent space by LICQ/SOSC (Section A.4), this is the Cramér–Rao/GLS efficiency bound for the identified directions, i.e., on $\mathcal{T}_{\bar{\theta}_0} \Theta(\delta)$.¹⁰⁷ $\square$

Proposition A.8 establishes the local asymptotic normality (LAN) and efficiency properties of the NA–IPCA estimator within a fully regularized, rotation-invariant framework. The result integrates the geometric identification developed in Appendix A.7 with the smoothness and robustness structure introduced in Appendix A.8, yielding a coherent statistical theory on the quotient manifold $\Theta(\delta)/\mathcal{G}$ —the space of economically identifiable parameters.

Mathematically, the LAN expansion (80) demonstrates that, after normalization and projection onto the tangent space $\mathcal{T}_{\bar{\theta}_0} \Theta(\delta)$, the estimator behaves as a one-step Huber–GLS procedure. The curvature matrix $A = \mathbb{E}[\dot{g}_t^\top \Omega_t^{-1} \dot{g}_t]$ governs local identification and efficiency, while the long-run covariance B_κ captures the temporal dispersion of the bounded-influence innovations $\psi_\kappa(u_t)$. Together they yield the tangent-space “sandwich” covariance $\Sigma_\kappa = (\Pi_0 A \Pi_0^\top)^{-1} (\Pi_0 B_\kappa \Pi_0^\top) (\Pi_0 A \Pi_0^\top)^{-1}$ in (87), ensuring that asymptotic inference is both robust and rotation-invariant. When $\kappa \rightarrow \infty$ or under Gaussian residuals, the robust score $\psi_\kappa(u)$ becomes linear, so $B_\kappa = A$ and $\Sigma_\kappa = (\Pi_0 A \Pi_0^\top)^{-1}$, attaining the GLS (Cramér–Rao) efficiency bound on the identified directions of the model.

Economically, this framework clarifies that inference concerns the *priced-risk space* rather than its

¹⁰⁶Why $\mathbb{E}[Z_{t,\infty}] = 0$ and why $B_\infty = A$. Mean-zero and odd-symmetry give $\mathbb{E}[u_t(\theta_0) | Z_t] = 0$, hence $\mathbb{E}[Z_{t,\infty}] = \mathbb{E}[\Pi_0 \dot{g}_t^\top \Omega_t^{-1} \mathbb{E}[u_t | Z_t]] = 0$. If the idiosyncratic component is conditionally serially uncorrelated (martingale difference) with $\mathbb{E}[u_t u_{t-\ell}^\top | \mathcal{F}_{t-1}] = 0$ for $\ell \neq 0$ and $\mathbb{E}[u_t u_t^\top | Z_t] = \Omega_t$, then

$$B_\infty = \sum_{\ell=-\infty}^{\infty} \mathbb{E}[Z_{t,\infty} Z_{t-\ell,\infty}^\top] = \mathbb{E}[\Pi_0 \dot{g}_t^\top \Omega_t^{-1} \underbrace{\mathbb{E}[u_t u_{t-\ell}^\top | Z_t]}_{=\Omega_t} \Omega_t^{-1} \dot{g}_t \Pi_0^\top] = \Pi_0 \mathbb{E}[\dot{g}_t^\top \Omega_t^{-1} \dot{g}_t] \Pi_0^\top.$$

Defining $A := \mathbb{E}[\dot{g}_t(\theta_0)^\top \Omega_t^{-1} \dot{g}_t(\theta_0)]$, we obtain $B_\infty = \Pi_0 A \Pi_0^\top$. Economically, Ω_t^{-1} orthogonalizes the pricing errors so that GLS weights turn the (projected) score variance into the (projected) curvature, which is the hallmark of GLS efficiency.

¹⁰⁷*Scope of the GLS bound.* The bound is stated on the tangent space because parameters are only identified up to rotations. Projecting by Π_0 removes non-identified (rotation) directions, and GLS efficiency holds for the economically meaningful coordinates (the priced-risk span). If the residuals are Gaussian, the same conclusion holds already at finite κ since the efficient score is linear, $\psi_\kappa(u) = u$, almost surely.

arbitrary basis representation. The tangent–space projection Π_0 removes rotation–equivalent coordinates, confining all asymptotic behavior to the economically meaningful subspace where systematic risk is priced. The matrix A describes how sharply the no–arbitrage manifold bends with respect to changes in factor exposures, while B_κ quantifies the realized volatility of robust pricing errors. Bounded–influence estimation thus ensures local efficiency under Gaussianity and stable performance under heavy tails or heteroskedasticity—achieving a principled balance between robustness and precision.

Conceptually, Appendix A.8 completes the theoretical structure initiated in Sections A.6–A.7: smoothness and Huber regularity guarantee differentiability of the empirical KKT system, while normalization and pre–whitening ensure that curvature, score, and information are invariant to coordinate rotations. These results form the analytical bridge to Appendix B, where the full LAN representation, efficient variance estimation, and rotation–invariant inference are developed in detail. Together, they establish that the NA–IPCA estimator operates as a geometrically well–posed, statistically regular, and economically interpretable system—combining robustness, efficiency, and invariance within a unified no–arbitrage geometry.

In summary, Appendix A develops the full regularity framework underpinning the No-Arbitrage IPCA estimator. It integrates probabilistic stability, geometric identification, weighting feasibility, rotation invariance, and smoothness–robustness conditions into a single analytical architecture. Sections A.1–A.3 establish the probabilistic foundations—notation, stochastic assumptions, and LLN/CLT lemmas—ensuring consistency and well-behaved sample moments. Sections A.4–A.6 build the geometric and numerical core: the LICQ/Slater qualifications guarantee a well-posed KKT system; the Stiefel-manifold geometry provides tangent-space identification; and feasible pre-whitening ensures GLS-equivalent scaling. Sections A.7–A.8 elevate the analysis to a rotation-invariant and smooth-robust level: Procrustes normalization and bounded-influence regularity align the estimator on the quotient manifold $\Theta(\delta)/\mathcal{G}$, yielding a differentiable, robust, and efficient LAN representation.

Collectively, these layers show that the NA–IPCA estimator is defined on a smooth, compact, and rotation-free no-arbitrage manifold; its score and curvature are uniformly Lipschitz and stochastically equicontinuous; and its asymptotic behavior is governed by a tangent-space Gaussian limit that attains the GLS efficiency bound under Gaussianity while remaining robust under heavy tails. This appendix therefore provides a complete and publication-grade foundation for the asymptotic, efficiency, and invariance results established in Appendix B.

A.8.5 Matrix moments and vectorization (bridge to Appendix B).

Let $M_t(\theta) \in \mathbb{R}^{L \times N_t}$ denote the matrix-valued per-period moment (or residual) used in this appendix. For asymptotic derivatives, KKT conditions, the LAN expansion, and Fisher information, I work with the vectorized moment

$$g_t(\theta) \equiv \text{vec}(M_t(\theta)) \in \mathbb{R}^{LN_t}, \quad G_T(\theta) \equiv T^{-1/2} \sum_{t=1}^T g_t(\theta).$$

This preserves norms and inner products:

$$\begin{aligned}\|M_t(\boldsymbol{\theta})\|_F^2 &= \|\text{vec}(M_t(\boldsymbol{\theta}))\|_2^2 = g_t(\boldsymbol{\theta})^\top g_t(\boldsymbol{\theta}), \\ \langle A, B \rangle_F &= \text{Tr}(A^\top B) = \text{vec}(A)^\top \text{vec}(B).\end{aligned}$$

All derivatives used in KKT/LAN/Fisher-information are taken with respect to g_t :

$$\nabla_{\boldsymbol{\theta}} g_t(\boldsymbol{\theta}) = \frac{\partial}{\partial \boldsymbol{\theta}} \text{vec}(M_t(\boldsymbol{\theta})), \quad \nabla_{\boldsymbol{\theta}} G_T(\boldsymbol{\theta}) = T^{-1/2} \sum_{t=1}^T \nabla_{\boldsymbol{\theta}} g_t(\boldsymbol{\theta}).$$

I apply the identity $\text{vec}(ABC) = (C^\top \otimes A) \text{vec}(B)$ throughout to obtain the block/Jacobian Kronecker forms in Appendix B. Whenever Appendix A presents a matrix moment criterion, its gradient and Hessian in Appendix B are understood via the above vectorization (and the Frobenius–Euclidean isometry).

B Efficiency, Identification, and Robustness for the No-Arbitrage IPCA

This appendix develops the econometric results that complete the theoretical framework introduced in Appendix A. All derivations here are built upon the regularity conditions and geometric structures established in that appendix—specifically, Assumptions 1–11, Lemmas A.1–A.3, and Proposition A.8. Together these results guarantee that the feasible parameter space $\Theta(\delta)$ forms a smooth, compact, and rotation-invariant manifold on which the No-Arbitrage IPCA estimator is well defined, continuously differentiable, and locally identifiable. Appendix B builds directly on these foundations to demonstrate how the estimator behaves as a constrained M -estimator on this manifold, and to establish its statistical efficiency, robustness, and identification properties.

The analysis proceeds in six steps. Section B.1 proves that the constrained NA-IPCA extremum problem is equivalent to a weighted least-squares (WLS/GLS) estimator on the tangent space, showing that the KKT conditions of the constrained problem coincide with those of the pre-whitened WLS system. Section B.2 formalizes the block-orthogonality condition $\Gamma_\beta^\top Q_Z \Gamma_\delta = 0$ and establishes the separate identification of latent and observable (PSF) factors. Section B.3 derives the joint local asymptotic normality (LAN) expansion and proves that the efficient information matrix is block-diagonal in the latent, observable, and intercept blocks. Section B.4 verifies that the estimator remains first-order invariant when observable factors are replaced by generated regressors estimated from a preliminary stage. Section B.5 extends this invariance to feasible pre-whitening and weighting schemes, demonstrating that ratio-consistent weights preserve the same asymptotic covariance as ideal GLS weighting. Finally, Section B.6 examines the geometry of rotation-equivalence and Procrustes normalization, showing that both identification and inference are invariant to orthogonal reparametrization of the latent and observable factor spaces.

Mathematically, these results combine the probabilistic regularity of Appendix A with the KKT geometry of the constraint manifold to yield a unified LAN representation for the estimator on the tangent space of $\Theta(\delta)$. Economically, they show that the no-arbitrage IPCA operates as an efficient, rotation-invariant, and robust method for identifying systematic risk exposures: weighting by idiosyncratic volatility improves efficiency; orthogonality between latent and observable blocks prevents double counting of risk; and bounded-influence scores maintain valid inference under heavy-tailed returns. Taken together, the results in Appendix B complete the theoretical foundation for the no-arbitrage IPCA framework developed in the main text.

B.1 Constrained Equivalence with Weighted Least Squares (WLS)

This section shows how the No-Arbitrage IPCA (NA-IPCA) estimator behaves on the feasible manifold defined in Section 2. The starting point is the constrained extremum problem introduced in the main text, where the estimator minimizes a weighted quadratic pricing criterion subject to the Q_Z -orthonormality and no-arbitrage conditions. Appendix A establishes that these constraints define a smooth, well-behaved manifold—one on which the moment map is differentiable, the tangent space is well defined, and the KKT system satisfies the regularity conditions required for stable first-order expansions. Building on this geometric foundation, the purpose of the present section is to show that the constrained estimator is locally equivalent to

a weighted least squares (WLS/GLS) estimator once the cross-sectional moments are pre-whitened.

Economically, this equivalence reflects a familiar idea from multi-factor pricing: assets with high idiosyncratic dispersion should receive lower weight in the cross-sectional fit.¹⁰⁸ Pre-whitening by $\Omega_t^{-1/2}$ rescales each moment into volatility-adjusted units while leaving all Q_Z -based identification constraints unchanged. Mathematically, this means that the whitening transformation changes only the metric of the estimation problem but not the geometry of the feasible set or the structure of the tangent space. Consequently, the first-order KKT conditions of the constrained extremum problem coincide exactly with those of the pre-whitened WLS system, implying that both procedures yield the same local solution and the same asymptotic covariance.

Taken together, these results establish that NA-IPCA inherits the efficiency properties of GLS while maintaining the economic discipline imposed by the Q_Z -orthogonality and no-arbitrage constraints. The constrained problem and its WLS representation therefore provide two equivalent views of the same estimator—one expressed through the geometry of the parameter manifold, and the other through the volatility-adjusted least squares structure of the pricing moments.

Proposition B.1 (Constrained WLS Equivalence). *Under Assumptions (A1)–(A6) and the KKT regularity of Section A.4, together with the Slater interiority condition in Assumption A.5, the NA-IPCA estimator obtained by minimizing (88) over $\Theta(\delta)$ is equivalent, on the feasible manifold, to the WLS estimator defined by (90). Both estimators satisfy the same projected first-order KKT conditions on the tangent space $\mathcal{T}_\theta$ and therefore admit identical asymptotic covariance and efficiency matrices.*

Proof. We begin from the feasible weighted criterion

$$Q_T(\theta) = \frac{1}{T} \sum_{t=1}^T m_t(\theta)^\top \Omega_t^{-1} m_t(\theta), \quad \Omega_t \equiv \text{Var}(X_t \mid Z_t), \quad (88)$$

subject to $\Gamma'_\beta Q_Z \Gamma_\beta = I_K$, $\Gamma'_\beta Q_Z \Gamma_\alpha = 0$, and $\psi(\Gamma_\alpha) \leq \delta$, where $m_t(\theta)$ is defined in Eq. (18). Premultiplying each cross-sectional moment by $\Omega_t^{-1/2}$ defines

$$\tilde{X}_t = \Omega_t^{-1/2} X_t, \quad \tilde{W}_t = \Omega_t^{-1/2} W_t, \quad (89)$$

which transforms the criterion into

$$\tilde{Q}_T(\theta) = \frac{1}{T} \sum_{t=1}^T \|\tilde{X}_t - \tilde{W}_t(\Gamma_\beta f_t + \Gamma_\delta F_t^{\text{obs}} + \Gamma_\alpha)\|_2^2. \quad (90)$$

Left-multiplication by $\Omega_t^{-1/2}$ rescales X_t and W_t without altering the manifold constraints in Q_Z -geometry.¹⁰⁹ Let $\mathcal{T}_\theta$ denote the tangent space at θ as defined in Eq. (39). Under the LICQ condition of Section A.4 and the Slater interiority condition in Assumption A.5, the Lagrange multipliers exist and are locally unique

¹⁰⁸See, for instance, Chamberlain and Rothschild (1983) for the classical GLS interpretation of multi-factor pricing and Newey and McFadden (1994, Sec. 2.6) for the large-sample properties of constrained extremum estimators.

¹⁰⁹Pre-whitening is a linear isomorphism in the sample space that preserves the geometry of the constraint manifold. The orthogonality and normalization equations depend only on Q_Z , defined by characteristic moments rather than Ω_t .

(Gouriéroux and Monfort, 1997, Ch. 4; Rockafellar and Wets, 1998).¹¹⁰ Consequently, the KKT systems for (88) and (90) coincide when projected onto $\mathcal{T}_\theta$.¹¹¹ Premultiplying by $\Omega_t^{-1/2}$ rescales the score by a symmetric positive-definite operator without modifying the constraint Jacobian; thus both estimators satisfy identical projected first-order systems on the feasible manifold.¹¹² When the ψ -bound is active, its multiplier modifies only the curvature in the Γ_α -direction and leaves the equivalence intact.¹¹³

Consequently, both estimators share the same first-order system on $\mathcal{T}_\theta$, implying identical asymptotic covariance and efficiency.¹¹⁴ $\square$

Economically, weighting by Ω_t^{-1} downweights characteristics with greater idiosyncratic dispersion, paralleling efficient GLS pricing in cross-sectional regressions. Pre-whitening ensures that residuals are measured in volatility-adjusted units, equalizing the signal-to-noise ratio across assets. Hence, the WLS transformation alters only the metric but not the geometry of the feasible set. Since the identification constraints $\Gamma'_\beta Q_Z \Gamma_\beta = I_K$ and $\Gamma'_\beta Q_Z \Gamma_\alpha = 0$ define a smooth Q_Z -Stiefel manifold with full-rank Jacobian, the tangent-space projection argument applies even when the no-arbitrage bound $\psi(\Gamma_\alpha) \leq \delta$ binds. Conse-

¹¹⁰The existence and local uniqueness of Lagrange multipliers follow from standard results for equality- and inequality-constrained extremum problems under the Linear-Independence Constraint Qualification (LICQ) and the Slater interiority condition. Formally, Gouriéroux and Monfort (1997, Ch. 4, pp. 112–118) show that if the gradients of the active constraints are linearly independent and at least one inequality constraint is strictly feasible, then a continuously differentiable Lagrange function $\mathcal{L}(\theta, \lambda) = Q_T(\theta) + \lambda^\top c(\theta)$ admits first-order necessary conditions with well-defined multipliers λ_0 , and that these multipliers are continuous in a neighborhood of the true parameter. The result is a direct application of the implicit-function theorem to the Karush–Kuhn–Tucker (KKT) mapping, $\mathcal{H}(\theta, \lambda) = (\nabla_\theta \mathcal{L}(\theta, \lambda), c(\theta))$, whose Jacobian with respect to (θ, λ) is nonsingular under LICQ and Slater. Rockafellar and Wets (1998, Thm. 6.12 and Cor. 6.46) extend this argument to non-linear programs with inequality constraints, proving that the KKT system defines a locally single-valued, continuously differentiable correspondence around any regular point. Economically, these regularity conditions ensure that small perturbations in the moment conditions or sample weights lead to proportionate and stable adjustments in the shadow prices (Lagrange multipliers), so that the no-arbitrage constraints respond smoothly to changes in the data rather than generating discontinuous jumps.

¹¹¹The coincidence of the two KKT systems on the tangent space $\mathcal{T}_\theta$ follows from the fact that pre-multiplying the sample moments by any symmetric positive-definite operator (such as $\Omega_t^{-1/2}$) rescales the objective but leaves the constraint Jacobian unchanged. Formally, if the original constrained problem minimizes $Q_T(\theta) = T^{-1} \sum_t m_t(\theta)' \Omega_t^{-1} m_t(\theta)$ subject to $c(\theta) = 0$, the first-order conditions are $\nabla_\theta Q_T(\theta) + J_c(\theta)' \lambda = 0$, $c(\theta) = 0$, where $J_c(\theta)$ denotes the Jacobian of the active constraints. Defining the pre-whitened criterion $\tilde{Q}_T(\theta) = T^{-1} \sum_t \|\Omega_t^{-1/2} m_t(\theta)\|_2^2$ yields identical gradient and constraint derivatives up to the factor $\Omega_t^{-1/2} \Omega_t^{1/2} = I$. Hence, the two Lagrangians differ only by an inner-product scaling, and their KKT mappings $\mathcal{H}(\theta, \lambda) = (\nabla_\theta Q_T(\theta) + J'_c(\theta) \lambda, c(\theta))$ and $\tilde{\mathcal{H}}(\theta, \lambda) = (\nabla_\theta \tilde{Q}_T(\theta) + \tilde{J}'_c(\theta) \lambda, c(\theta))$ satisfy $\Pi_{\mathcal{T}_\theta} \mathcal{H} = \Pi_{\mathcal{T}_\theta} \tilde{\mathcal{H}}$, where $\Pi_{\mathcal{T}_\theta}$ is the orthogonal projection onto the feasible tangent space. Mathematically, this means that the whitening transformation acts as an isometry on $\mathcal{T}_\theta$, preserving the first-order geometry of the criterion. Economically, it implies that rescaling the cross-sectional moments by idiosyncratic volatility (pre-whitening) changes the metric of the estimation problem but not its substance: the feasible directions of adjustment and the associated shadow prices remain identical under both formulations.

¹¹²Formally, the Jacobian of $g(\Gamma_\beta, \Gamma_\alpha) = \{\Gamma'_\beta Q_Z \Gamma_\beta - I_K, \Gamma'_\beta Q_Z \Gamma_\alpha\}$ has full row rank under the Stiefel-manifold regularity conditions of Edelman et al. (1998), ensuring that the tangent-space projection is invariant under any invertible linear transformation of the sample moments.

¹¹³When the inequality constraint $\psi(\Gamma_\alpha) \leq \delta$ is active, the associated Lagrange multiplier $\lambda_\alpha > 0$ enters the second-order KKT system only through the Γ_α block of the bordered Hessian. Formally, the Hessian of the Lagrangian $\mathcal{L}(\theta, \lambda) = Q_T(\theta) + \lambda_\alpha \{\psi(\Gamma_\alpha) - \delta\} + \lambda'_\beta (\Gamma'_\beta Q_Z \Gamma_\beta - I_K) + \lambda'_\beta (\Gamma'_\beta Q_Z \Gamma_\alpha)$ is block-diagonal in the directions Γ_β , Γ_δ , and Γ_α , because $\psi(\Gamma_\alpha)$ depends only on Γ_α . Hence, the term $\lambda_\alpha \nabla_{\Gamma_\alpha}^2 \psi(\Gamma_\alpha)$ adjusts curvature in the Γ_α -direction but leaves the first-order geometry and the tangent-space projection of the KKT system unchanged. Economically, this means that binding the ψ -bound—interpreted as a finite no-arbitrage limit on the pricing intercept—tightens the curvature of the objective only along the non-priced Γ_α component without altering the identification or efficiency of the priced factor directions $(\Gamma_\beta, F, \Gamma_\delta)$.

¹¹⁴Because the two estimators satisfy identical first-order (KKT) equations on the feasible tangent space $\mathcal{T}_\theta$, their local linear representations coincide. Under the regularity conditions in Lemma A.1 and Proposition A.8, this implies that the influence functions, and hence the asymptotic covariance matrices, are identical. Economically, equality of the first-order system means that both estimators extract the same risk–return tradeoff, achieving the same point on the efficiency frontier.

quently, the NA-IPCA estimator and the efficient WLS/GLS estimator are locally identical in both statistical and economic terms: efficiency stems from the Q_Z -geometric structure rather than from any particular weighting scheme.

B.2 Block Orthogonality and Identification

This section builds on the geometric and regularity framework developed in Appendix A to analyze how the Q_Z -orthogonality restrictions shape identification within the No-Arbitrage IPCA model. Appendix A establishes that the feasible parameter set forms a smooth, compact, and rotation-invariant manifold, and that the moment map is differentiable with stable score and Hessian behavior. These properties provide the mathematical foundation for the identification results derived here: they ensure that projections onto Q_Z -orthogonal subspaces are well defined, that the relevant Jacobians admit population limits, and that information operators inherit the block structure implied by the constraints.

Economically, the restriction $\Gamma_\beta^\top Q_Z \Gamma_\delta = 0$ enforces a “no-double-counting” discipline: the latent block Γ_β and the observable (PSF) block Γ_δ must span distinct priced directions in characteristic space. This ensures that observable factors do not re-enter the model through linear combinations of latent loadings, and that latent factors capture only the residual structure left unexplained by exogenous (PSF) information.

Mathematically, the Q_Z -orthogonality partitions the characteristic space into two complementary subspaces:

$$\mathbb{R}^L = \text{col}(\Gamma_\beta) \oplus \text{col}(M_\beta), \quad M_\beta := I - P_\beta,$$

so any Γ_δ must lie in $\text{col}(M_\beta)$. Under the regularity conditions of Appendix A, this decomposition implies that the latent span $\text{col}(\Gamma_\beta)$ and the effective PSF span $\text{col}(M_\beta \Gamma_\delta)$ are *separately identified*, each up to an internal orthogonal rotation. Moreover, under PSF centering ($\mathbb{E}[F_t^{\text{obs}}] = 0$), the Fisher information of the pre-whitened GLS system is block-diagonal in population across the latent, observable, and intercept blocks; the sample information replicates this structure up to $o_p(1)$.

Taken together, these results show that the block-orthogonality restriction yields a clean separation of modeled directions: latent factors span one Q_Z -orthogonal component, observable factors contribute only through their projected component $M_\beta \Gamma_\delta$, and the intercept remains separated from the latent span and bounded in curvature. An optional strengthening is to impose $\Gamma_\delta^\top Q_Z \Gamma_\alpha = 0$, so the intercept block is also orthogonal to the observable PSF span. This structure underpins the “no-double-counting” principle in the main text and provides the basis for separate estimation and inference on latent and observable sources of systematic risk.

Proposition B.2 (Block orthogonality $\Rightarrow$ separate identification and block-diagonal information).

Let Assumptions 1–4, 6, 7 hold, and suppose the population restrictions

$$\Gamma_\beta' Q_Z \Gamma_\beta = I_K, \quad \Gamma_\beta' Q_Z \Gamma_\alpha = 0, \quad \Gamma_\beta' Q_Z \Gamma_\delta = 0 \quad (91)$$

hold at θ_0 . Let P_β and $M_\beta = I - P_\beta$ be the Q_Z -orthogonal projector and its complement,

$$P_\beta \equiv \Gamma_\beta (\Gamma_\beta' Q_Z \Gamma_\beta)^{-1} \Gamma_\beta' Q_Z, \quad M_\beta \equiv I - P_\beta, \quad (92)$$

so that the characteristic space decomposes as

$$\mathbb{R}^L = \text{col}(\Gamma_\beta) \oplus \text{col}(M_\beta).^{115} \quad (93)$$

Then:

- (i) $\text{col}(\Gamma_\delta) \subseteq \text{col}(M_\beta)$, hence the only economically relevant PSF content is the effective PSF subspace $\text{col}(M_\beta \Gamma_\delta)$, which is Q_Z -orthogonal to $\text{col}(\Gamma_\beta)$.
- (ii) If, in addition, PSF centering holds ($\mathbb{E}[F_t^{\text{obs}}] = 0$), then the (pre-whitened GLS) Fisher information is block-diagonal in population across (Γ_β, F) , Γ_δ , and Γ_α . Consequently, the latent span $\text{col}(\Gamma_\beta)$ and the effective PSF span $\text{col}(M_\beta \Gamma_\delta)$ are separately identified (each up to a within-block orthogonal rotation). The corresponding sample information inherits this block-diagonal structure up to $o_p(1)$.

Proof. Let $Q_Z \succ 0$ denote the population limit of the characteristic Gram matrix,¹¹⁶ and assume (91) holds at θ_0 . The projector P_β and complement M_β in (92) are idempotent and Q_Z -orthogonal, which yields the direct-sum decomposition (93). Because $\Gamma'_\beta Q_Z \Gamma_\delta = 0$, any column of Γ_δ lies in the Q_Z -orthogonal complement of $\text{col}(\Gamma_\beta)$, hence $\text{col}(\Gamma_\delta) \subseteq \text{col}(M_\beta)$; this proves (i).¹¹⁷

For (ii), let $m_t(\theta)$ denote the pre-whitened stacked moment at date t and consider the (population) information operator in the GLS metric

$$\mathcal{J}(\theta_0) = \mathbb{E}[\nabla_\theta m_t(\theta_0)' \Omega_t^{-1} \nabla_\theta m_t(\theta_0)], \quad (94)$$

with $\Omega_t \succ 0$ any admissible GLS weight.¹¹⁸ In Kronecker form, the (whitened) Jacobians satisfy

$$\nabla_{\text{vec}(\Gamma_\beta)} m_t(\theta_0) \propto (F_t^0 \otimes Q_Z^{1/2}), \quad \nabla_{\text{vec}(\Gamma_\delta)} m_t(\theta_0) \propto (F_t^{\text{obs}} \otimes Q_Z^{1/2}).$$

Evaluated at θ_0 , the cross-block information is proportional to

$$\mathbb{E}[(F_t^0 F_t^{\text{obs}'} \otimes (Q_Z^{1/2} \Gamma'_{\beta,0} \Gamma_{\delta,0} Q_Z^{1/2})],$$

¹¹⁵The operator P_β is the Q_Z -orthogonal projector onto the latent loading space $\text{col}(\Gamma_\beta)$. Formally, a projector P on $\mathbb{R}^L$ is Q_Z -orthogonal if it satisfies $P = P^2$ and $PQ_Z = Q_Z P^\top$, ensuring that it preserves the Q_Z -inner product $\langle a, b \rangle_{Q_Z} = a^\top Q_Z b$. The expression $P_\beta = \Gamma_\beta (\Gamma_\beta^\top Q_Z \Gamma_\beta)^{-1} \Gamma_\beta^\top Q_Z$ is the standard weighted projection formula: any vector $x \in \mathbb{R}^L$ is mapped to its Q_Z -orthogonal projection onto $\text{col}(\Gamma_\beta)$. The operator $M_\beta = I - P_\beta$ projects onto the Q_Z -orthogonal complement, so that $\mathbb{R}^L = \text{col}(\Gamma_\beta) \oplus \text{col}(M_\beta)$ is a direct sum decomposition. Mathematically, this decomposition means every characteristic direction can be uniquely split into a “latent” component (in $\text{col}(\Gamma_\beta)$) and an orthogonal residual component (in $\text{col}(M_\beta)$). Economically, P_β extracts the systematic variation associated with the latent factors, while M_β isolates all characteristic directions orthogonal to them—precisely the space in which observable (PSF) factors must live under the no-double-counting condition $\Gamma_\beta^\top Q_Z \Gamma_\delta = 0$.

¹¹⁶Under standard LLN/CLT with weak dependence and finite moments, the sample Gram matrices $W_t \equiv Z_t' Z_t / N_t$ converge in probability to a deterministic limit $Q_Z \succ 0$ that induces the Q_Z -inner product on $\mathbb{R}^L$. See Golub and Van Loan (2013, Ch. 2) (Gram operators) and Newey and McFadden (1994, Sec. 2.6) (large-sample convergence of sample moments).

¹¹⁷Formally, the notation $\mathbb{R}^L = \text{col}(\Gamma_\beta) \oplus \text{col}(M_\beta)$ denotes a Q_Z -orthogonal direct sum. Every $x \in \mathbb{R}^L$ is uniquely $x = v + w$ with $v \in \text{col}(\Gamma_\beta)$ and $w \in \text{col}(M_\beta)$, and the two subspaces are orthogonal under $\langle a, b \rangle_{Q_Z} = a' Q_Z b$. Economically, $\text{col}(\Gamma_\beta)$ is the latent span; $\text{col}(M_\beta)$ collects directions orthogonal to it. The orthogonality $\Gamma_\beta^\top Q_Z \Gamma_\delta = 0$ forces observables to live in that complement. See Golub and Van Loan (2013, Ch. 2) and Absil et al. (2008, Ch. 2).

¹¹⁸Premultiplying the moment by $\Omega_t^{-1/2}$ (as in Section B.1) rescales the score but leaves the constraint Jacobian unchanged; hence the KKT geometry on the tangent space and the block structure of $\mathcal{J}(\theta_0)$ are invariant; see Newey and McFadden (1994, Sec. 2.6), Gouriéroux and Monfort (1997, Ch. 4).

which equals zero under PSF centering ($\mathbb{E}[F_t^{\text{obs}}] = 0$) and $\Gamma'_{\beta,0} Q_Z \Gamma_{\delta,0} = 0$. Hence the (population) information is block-diagonal across (Γ_β, F) , Γ_δ , and Γ_α . By Assumptions 1–4, 6, 7, each block is nonsingular (rank/moment conditions), which yields separate identification up to within-block orthogonal rotations. The corresponding sample information inherits the block-diagonal structure up to $o_p(1)$ by the LLN/CLT summarized in Lemma A.1. $\square$

Remark (Optional orthogonality between PSF and intercept). In the baseline specification we do *not* impose $\Gamma'_\delta Q_Z \Gamma_\alpha = 0$ because, under PSF centering ($\mathbb{E}[F_t^{\text{obs}}] = 0$) and sufficient time variation, the constant term Γ_α (which multiplies $\alpha_t \equiv 1$ by construction) cannot be replicated by $\Gamma_\delta F_t^{\text{obs}}$ and is already separated from Γ_β by $\Gamma'_\beta Q_Z \Gamma_\alpha = 0$. We therefore treat $\Gamma'_\delta Q_Z \Gamma_\alpha = 0$ as an optional strengthening rather than a baseline restriction: it sharpens the separation between the observable PSF and intercept blocks, but it is not required for the baseline identification and block-diagonal-information results under PSF centering. If, however, PSF centering is not imposed, or if one wishes to eliminate *any* residual Q_Z -overlap between $\text{col}(\Gamma_\delta)$ and the constant direction $\text{col}(\Gamma_\alpha)$, one can impose the restriction $\Gamma'_\delta Q_Z \Gamma_\alpha = 0$; in implementation this corresponds to projecting Γ_α onto $\ker(\Gamma'_\delta Q_Z)$ at each outer iteration. This strengthening is innocuous under PSF centering but can be empirically useful if the PSF block contains low-frequency components.¹¹⁹

Taken together, the Q_Z -orthogonality splits priced directions into non-overlapping spans: latent risk is carried by $\text{col}(\Gamma_\beta)$, PSF risk enters only through $\text{col}(M_\beta \Gamma_\delta)$, and the intercept Γ_α (orthogonal to the latent span, and optionally to the PSF span) captures time-invariant premia. At the population optimum the information matrix is therefore block-diagonal, so precision in one block is undiluted by the other—exactly the “no-double-counting” discipline used in the main text and the reason latent and observed components can be estimated with independent precision matrices.

B.3 Joint Local Asymptotic Normality and Block-Diagonal Information

This section develops the joint local asymptotic normality (LAN) expansion for the No-Arbitrage IPCA estimator and establishes that the efficient information matrix is block-diagonal across the latent, observable (PSF), and intercept components. The result relies on the probabilistic and geometric regularity established in Appendix A—specifically, Assumptions 1–11, Lemmas A.1–A.3, and the KKT regularity of Section A.4.

¹¹⁹Low-frequency or highly persistent observable factors can, over relevant sample windows, behave almost as deterministic drifts or level shifts. When F_t^{obs} varies slowly relative to the sampling frequency, its fitted component $\Gamma_\delta F_t^{\text{obs}}$ becomes nearly collinear with the constant direction Γ_α , creating overlap in the Q_Z -weighted characteristic space.

Empirically, a number of studies document that a substantial portion of macro-financial risk is concentrated at very low frequencies. Dew-Becker and Giglio (2016) show that the characteristic-based priced risk structure places disproportionate weight on cycles longer than eight years, so that low-frequency movements in fundamentals dominate risk pricing in the frequency domain. Similarly, Li (2023) finds that several empirically relevant risk factors are driven by slow-moving state variables whose variation occurs predominantly at low frequencies. Such persistent components can easily mimic an intercept in linear pricing models—especially when F_t^{obs} is not centered or displays limited time-series variation—thereby creating near-collinearity between Γ_δ and Γ_α .

From a statistical perspective, this near-dependence reduces numerical stability of the KKT system and weakens identification of the intercept; analogous issues arise in factor models with persistent regressors (cf. Chamberlain and Rothschild (1983) and Bai and Ng (2008)). Imposing the orthogonality restriction $\Gamma_\delta^\top Q_Z \Gamma_\alpha = 0$ removes this low-frequency collinearity in the Q_Z -metric and guarantees that the intercept remains distinct from slowly evolving observable factors. Economically, this ensures that PSF factors represent genuine sources of systematic priced risk rather than absorbing the nearly constant component of the pricing kernel, thereby sharpening identification and improving interpretability.

Together these conditions guarantee that the constrained parameter set $\Theta(\delta)$ forms a smooth manifold, that the moment map and its Jacobian are well behaved, and that the KKT system admits a stable implicit-function representation on the tangent space. Building on these foundations, we derive a LAN representation for the constrained estimator and show that its asymptotic covariance inherits the block-orthogonal structure implied by the restriction $\Gamma_\beta^\top Q_Z \Gamma_\delta = 0$.

Proposition B.3 (LAN expansion and block-diagonal information). *Under Assumptions 1–8 and the KKT regularity of Section A.4, and imposing the identification and orthogonality conditions $\Gamma_\beta^\top Q_Z \Gamma_\beta = I_K$, $\Gamma_\beta^\top Q_Z \Gamma_\alpha = 0$, and $\Gamma_\beta^\top Q_Z \Gamma_\delta = 0$, the constrained NA-IPCA estimator satisfies the projected LAN expansion*

$$\sqrt{T} \Pi_{\mathcal{T}}(\hat{\theta} - \theta_0) = \left(\mathcal{J}(\theta_0)|_{\mathcal{T}} \right)^{-1} \frac{1}{\sqrt{T}} \sum_{t=1}^T \nabla_{\theta} g_t(\theta_0)^\top \Psi_t + o_p(1), \quad (95)$$

where $\Pi_{\mathcal{T}}$ denotes projection onto the tangent space of the feasible manifold, $g_t(\theta)$ is the period- t moment map, and Ψ_t is the Huber/GLS score innovation defined in (96). Moreover, under PSF centering ($\mathbb{E}[F_t^{\text{obs}}] = 0$), the efficient information matrix is block-diagonal across (Γ_β, F) , Γ_δ , and Γ_α up to $o_p(1)$, yielding separate identification of the latent span $\text{col}(\Gamma_\beta)$ and the effective PSF span $\text{col}(M_\beta \Gamma_\delta)$ (each up to within-block rotations).

Proof. (Score and KKT system). Define the period- t moment and its sample average as

$$g_t(\theta) = \text{vec}\left(Z_t' [r_{t+1} - Z_t(\Gamma_\beta f_t + \Gamma_\delta F_t^{\text{obs}} + \Gamma_\alpha)]\right), \quad G_T(\theta) = \sqrt{T} \bar{g}_T(\theta).$$

The constrained estimator solves the KKT system

$$\mathcal{H}_T(\theta, \lambda) = (G_T(\theta), \mathbf{c}(\theta)) = 0,$$

where $\mathbf{c}(\theta)$ stacks the orthonormality, block-orthogonality, and no-arbitrage restrictions. By Section A.4, the Jacobian of $\mathcal{H}_T$ with respect to (θ, λ) is nonsingular on the tangent space, so the multipliers admit a smooth implicit-function representation in a neighborhood of θ_0 .¹²⁰

(*Local expansion*). Let $\nabla_{\theta} g_t(\theta)$ denote the Jacobian of the moment map. A local linearization of the projected first-order condition $S_T(\theta) \equiv \Pi_{\mathcal{T}} G_T(\theta)$ at θ_0 gives

$$0 = S_T(\hat{\theta}) = S_T(\theta_0) + [\Pi_{\mathcal{T}} \nabla_{\theta} G_T(\tilde{\theta})] \Pi_{\mathcal{T}}(\hat{\theta} - \theta_0),$$

with $\tilde{\theta}$ on the line segment joining θ_0 and $\hat{\theta}$. Solving for $\Pi_{\mathcal{T}}(\hat{\theta} - \theta_0)$ yields

$$\Pi_{\mathcal{T}}(\hat{\theta} - \theta_0) = -[\Pi_{\mathcal{T}} \nabla_{\theta} G_T(\tilde{\theta})]^{-1} S_T(\theta_0).$$

¹²⁰Section A.4 guarantees that the gradients of all active identification and no-arbitrage constraints are linearly independent at θ_0 . This condition makes the KKT system locally well behaved: the Jacobian of the map $(\theta, \lambda) \mapsto \mathcal{H}_T(\theta, \lambda)$ is nonsingular on the tangent space of feasible perturbations. By the implicit-function theorem, the Lagrange multipliers λ can therefore be expressed as a smooth function of θ in a neighborhood of the true parameter. Economically, LICQ means that small, admissible changes in factor loadings produce distinct changes in the constraint system, so the shadow prices enforcing no-arbitrage adjust smoothly and do not create kinks or discontinuities. This smoothness is what allows the KKT system to admit a stable local expansion and underlies the LAN representation.

By the uniform LLN in Lemma A.1 and the differentiability and Lipschitz conditions in Assumption 10 (together with the compactness result in Lemma A.3), $\nabla_{\theta} G_T(\tilde{\theta}) = \nabla_{\theta} G(\theta_0) + o_p(1)$ whenever $\tilde{\theta} \rightarrow_p \theta_0$, while consistency implies $\|\hat{\theta} - \theta_0\| = O_p(T^{-1/2})$. By the uniform LLN in Lemma A.1 and the differentiability and Lipschitz conditions in Assumption 10 (together with the compactness result in Lemma A.3), $\nabla_{\theta} G_T(\tilde{\theta}) = \nabla_{\theta} G(\theta_0) + o_p(1)$ whenever $\tilde{\theta} \rightarrow_p \theta_0$, while consistency implies $\|\hat{\theta} - \theta_0\| = O_p(T^{-1/2})$.¹²¹ Substituting these relations and replacing $\nabla_{\theta} G(\theta_0)$ by its limit $\mathcal{J}(\theta_0)|_{\mathcal{T}}$ gives the LAN display (95) with a remainder $o_p(1)$. Substituting these relations and replacing $\nabla_{\theta} G(\theta_0)$ by its limit $\mathcal{J}(\theta_0)|_{\mathcal{T}}$ gives the LAN display (95) with a remainder $o_p(1)$.¹²²

(Score innovations and CLT). The leading stochastic term in (95) is

$$\frac{1}{\sqrt{T}} \sum_{t=1}^T \nabla_{\theta} g_t(\theta_0)^{\top} \Psi_t, \quad \Psi_t = \Omega_t^{-1} Z_t' \psi_{\kappa}(u_t), \quad (96)$$

where $u_t = r_{t+1} - Z_t(\Gamma_{\beta} f_t + \Gamma_{\delta} F_t^{\text{obs}} + \Gamma_{\alpha})$. Assumptions 1 and 8 imply that $\{\Psi_t\}$ is a weakly dependent martingale-difference array with finite second moments, so Lemma A.2 yields a multivariate CLT for the score sum in (96).¹²³ Economically, the innovations Ψ_t represent volatility-adjusted pricing errors, and the CLT shows that their cumulative effect drives the $\sqrt{T}$ -scaled sampling uncertainty of the estimator.

(Block-orthogonality and block-diagonal information). The Jacobians of the moment map admit the Kronecker representation

$$\nabla_{\text{vec}(\Gamma_{\beta})} g_t(\theta_0) \propto (f_t^0 \otimes Q_Z^{1/2}), \quad \nabla_{\text{vec}(\Gamma_{\delta})} g_t(\theta_0) \propto (F_t^{\text{obs}} \otimes Q_Z^{1/2}).^{124}$$

Under the population restriction $\Gamma_{\beta}^{\top} Q_Z \Gamma_{\delta} = 0$ and PSF centering ($\mathbb{E}[F_t^{\text{obs}}] = 0$), the cross-block information

¹²¹Lemma A.1 and Assumption 10 imply a uniform LLN for the Jacobian: $\sup_{\theta \in \Theta(\delta)} \|\nabla_{\theta} G_T(\theta) - \nabla_{\theta} G(\theta)\| \xrightarrow{p} 0$ on the compact set $\Theta(\delta)$ (Lemma A.3). Hence any random sequence $\tilde{\theta} \rightarrow_p \theta_0$ satisfies $\nabla_{\theta} G_T(\tilde{\theta}) - \nabla_{\theta} G(\theta_0) = o_p(1)$ by continuity. The rate $\|\hat{\theta} - \theta_0\| = O_p(T^{-1/2})$ then follows from the usual M -estimator argument: the score $G_T(\theta_0) = T^{-1/2} \sum_t g_t(\theta_0)$ is $O_p(1)$ by the CLT, and the limiting Jacobian $\nabla_{\theta} G(\theta_0)$ is nonsingular on the tangent space (Section A.4), so the linearization $G_T(\tilde{\theta}) \approx G_T(\theta_0) + \nabla_{\theta} G(\theta_0)(\tilde{\theta} - \theta_0)$ forces $\tilde{\theta} - \theta_0 = O_p(T^{-1/2})$.

¹²²From Lemma A.1 and Assumption 10, $\nabla_{\theta} G_T(\tilde{\theta}) = \nabla_{\theta} G(\theta_0) + o_p(1)$ uniformly on $\Theta(\delta)$ whenever $\tilde{\theta} \rightarrow_p \theta_0$. By Section A.4, the projected Jacobian $\Pi_{\mathcal{T}} \nabla_{\theta} G(\theta_0)$ coincides with the projected information $\mathcal{J}(\theta_0)|_{\mathcal{T}}$ and is nonsingular. Hence, by the continuous mapping theorem, $[\Pi_{\mathcal{T}} \nabla_{\theta} G_T(\tilde{\theta})]^{-1} = (\mathcal{J}(\theta_0)|_{\mathcal{T}})^{-1} + o_p(1)$. The score term admits the linear representation $S_T(\theta_0) = T^{-1/2} \sum_t \nabla_{\theta} g_t(\theta_0)^{\top} \Psi_t + o_p(1)$ by Lemma A.2. Combining these two facts in the linearization of $\Pi_{\mathcal{T}}(\hat{\theta} - \theta_0)$ shows that the leading term is $(\mathcal{J}(\theta_0)|_{\mathcal{T}})^{-1} T^{-1/2} \sum_t \nabla_{\theta} g_t(\theta_0)^{\top} \Psi_t$ and that all residual terms are of order $o_p(1)$, which yields (95).

¹²³Assumptions 1 (weak dependence and L_2 -integrable pricing errors) and 8 (bounded-influence Huber score) ensure that the innovation sequence $\{\Psi_t\}$ has mean zero, finite second moments, and satisfies a martingale-difference structure with weak serial correlation. These properties place $\{\Psi_t\}$ within the standard class of arrays covered by Lemma A.2, so the normalized sum $T^{-1/2} \sum_{t=1}^T \nabla_{\theta} g_t(\theta_0)^{\top} \Psi_t$ obeys a multivariate central limit theorem. Economically, this means that the volatility-adjusted pricing errors accumulate like a diversified shock to the cross section, generating the Gaussian component that drives the LAN expansion.

¹²⁴The Kronecker structure follows from the identity $\text{vec}(ABC) = (C^{\top} \otimes A) \text{vec}(B)$ applied to the moment map $g_t(\theta) = \text{vec}(Z_t' [r_{t+1} - Z_t(\Gamma_{\beta} f_t + \Gamma_{\delta} F_t^{\text{obs}} + \Gamma_{\alpha})])$. Differentiating with respect to Γ_{β} replaces the fitted component $Z_t \Gamma_{\beta} f_t$ by its linear variation $Z_t(\Delta \Gamma_{\beta}) f_t$, which vectorizes as $\text{vec}(Z_t(\Delta \Gamma_{\beta}) f_t) = (f_t^0 \otimes Z_t) \text{vec}(\Delta \Gamma_{\beta})$. After pre-whitening, Z_t contributes only through the limiting characteristic metric $Q_Z^{1/2}$, yielding the form $(f_t^0 \otimes Q_Z^{1/2})$ (and analogously for Γ_{δ} with F_t^{obs}). Economically, this factorization separates the *time-series component* of the score—captured by f_t^0 or F_t^{obs} —from the *characteristic geometry* encoded by $Q_Z^{1/2}$. This decomposition clarifies why block-orthogonality in $\Gamma_{\beta}^{\top} Q_Z \Gamma_{\delta} = 0$ eliminates cross-information: the time-series variation and the characteristic-space variation combine multiplicatively, so orthogonality in Q_Z forces the entire cross-block Jacobian inner product to vanish.

term

$$\mathbb{E}[\nabla_{\text{vec}(\Gamma_\beta)} g_t(\theta_0)' \Omega_t^{-1} \nabla_{\text{vec}(\Gamma_\delta)} g_t(\theta_0)]$$

vanishes.¹²⁵ Proposition B.2 therefore implies that, under the same PSF-centering condition, the efficient information matrix is block-diagonal across (Γ_β, F) , Γ_δ , and Γ_α up to $o_p(1)$. Since each diagonal block is nonsingular, the latent, observable, and intercept components are separately identified and attain independent asymptotic precision.¹²⁶

(Conclusion). Combining the LAN representation with the block-separate information yields

$$\sqrt{T} \Pi_{\mathcal{T}}(\hat{\theta} - \theta_0) \Rightarrow \mathcal{N}(0, \text{diag}(\mathcal{J}_{\beta, F}^{-1}, \mathcal{J}_\delta^{-1}, \mathcal{J}_\alpha^{-1})),$$

so that inference is block-by-block and unaffected by cross-block interactions. Economically, this means that latent and observable factors explain distinct priced directions, and the precision of each component reflects only its own systematic variation rather than that of other blocks. $\square$

Taken together, the orthogonality $\Gamma_\beta^\top Q_Z \Gamma_\delta = 0$ and the quotient geometry in Appendix A partition the characteristic space into non-overlapping priced subspaces: latent risk in $\text{col}(\Gamma_\beta)$, PSF risk in $\text{col}(M_\beta \Gamma_\delta)$, and time-invariant premia in Γ_α . The LAN representation in Proposition B.3 therefore provides a fully efficient, block-separate characterization of the estimator's asymptotic distribution on the tangent space of the no-arbitrage manifold.

B.4 Generated Regressors for Observable Factors

This section establishes that the No-Arbitrage IPCA estimator remains locally asymptotically normal and attains the same efficiency bound when the observable factor process F_t^{obs} is replaced by a first-stage estimate $\hat{F}_t^{\text{obs}}$. In empirical asset pricing, such proxies typically arise from principal-component estimates

¹²⁵The cross-block information term combines two components: the time-series part, driven respectively by f_t^0 and F_t^{obs} , and the characteristic-space part, driven by the Q_Z -geometry. Using the Kronecker forms $\nabla_{\text{vec}(\Gamma_\beta)} g_t(\theta_0) \propto (f_t^0 \otimes Q_Z^{1/2})$ and $\nabla_{\text{vec}(\Gamma_\delta)} g_t(\theta_0) \propto (F_t^{\text{obs}} \otimes Q_Z^{1/2})$, their inner product under the GLS metric factorizes as

$$\mathbb{E}[f_t^0 F_t^{\text{obs}\top}] \otimes (Q_Z^{1/2} \Gamma_{\beta, 0}^\top \Gamma_{\delta, 0} Q_Z^{1/2}).$$

Both factors vanish under the identification restrictions: (i) $\mathbb{E}[F_t^{\text{obs}}] = 0$ implies $\mathbb{E}[f_t^0 F_t^{\text{obs}\top}] = 0$; (ii) the population orthogonality $\Gamma_{\beta, 0}^\top Q_Z \Gamma_{\delta, 0} = 0$ implies $Q_Z^{1/2} \Gamma_{\beta, 0}^\top \Gamma_{\delta, 0} Q_Z^{1/2} = 0$. Because the expectation of a Kronecker product is zero whenever either block is zero, the entire cross-block information term disappears. Economically, this means latent and observable factors load on distinct priced directions in characteristic space, so their score contributions are orthogonal and carry no joint information.

¹²⁶Proposition B.2 shows that the Q_Z -orthogonality restriction $\Gamma_\beta^\top Q_Z \Gamma_\delta = 0$ forces the latent loadings Γ_β and the PSF loadings Γ_δ to lie in orthogonal Q_Z -subspaces. Combined with $\Gamma_\beta^\top Q_Z \Gamma_\beta = I_K$ and $\Gamma_\beta^\top Q_Z \Gamma_\alpha = 0$, this produces a decomposition of the characteristic space into the latent span $\text{col}(\Gamma_\beta)$, the effective PSF span $\text{col}(M_\beta \Gamma_\delta)$, and a residual intercept direction that is orthogonal to the latent span and, under the optional strengthening $\Gamma_\delta^\top Q_Z \Gamma_\alpha = 0$, can also be taken orthogonal to the observable PSF span. Under PSF centering, Proposition B.2 proves that the Jacobians of the moment map with respect to these blocks are orthogonal in the GLS metric, so the population Fisher information matrix decomposes into independent blocks associated with (Γ_β, F) , Γ_δ , and Γ_α . Since each block satisfies its own rank and moment conditions (Appendix A), its information submatrix is nonsingular, yielding separate identification of each component and ensuring that the LAN covariance is block-diagonal. Economically, Proposition B.2 formalizes the idea that latent, observable, and intercept terms span distinct priced directions, so the precision with which one block is estimated is not diluted by the others.

of macroeconomic series, from traded “mimicking” portfolios in the spirit of Fama–French, or from fitted macro–financial indices that instrument latent state variables. Because these observable factors are themselves estimated objects rather than primitive data, it is essential to verify that first–stage estimation noise does not distort second–stage inference. The key result is that small first–stage perturbations enter the pricing moment only at second order, leaving unchanged both the LAN expansion derived in Section B.3 and the block–diagonal Fisher structure of Proposition B.2. Economically, this means that empirical factor proxies—whether PCA-based, portfolio-constructed, or macro-instrumented—affect the estimation of systematic risk premia only through higher-order terms, so the identification and precision of the no–arbitrage factor structure are preserved.

Proposition B.4 (Generated–Regressor Robustness). *Under Assumptions 1–8 and the KKT regularity of Section A.4, the constrained NA–IPCA estimator retains the LAN expansion (95) and the block–diagonal information structure of Proposition B.2 when F_t^{obs} is replaced by a proxy*

$$\widehat{F}_t^{\text{obs}} = F_t^{\text{obs}} + \xi_t,$$

provided one of the following holds:

- (i) (**Uniform precision**) $\max_{t \leq T} \|\xi_t\| = o_p(T^{-1/2})$; or
- (ii) (**Neyman orthogonality**) $\|T^{-1/2} \sum_{t=1}^T \xi_t\| = O_p(1)$ and $\mathbb{E}[\partial_F g_t(\theta_0; F_t^{\text{obs}}) \xi_t \mid \mathcal{F}_{t-1}] = 0$.

Under either condition,

$$\sqrt{T} \left(\bar{g}_T(\theta_0; \widehat{F}) - \bar{g}_T(\theta_0; F) \right) = o_p(1),$$

so the LAN representation and block–diagonal covariance are unchanged. Consequently, the latent span $\text{col}(\Gamma_\beta)$ and the effective PSF span $\text{col}(M_\beta \Gamma_\delta)$ remain separately identified, each up to within–block rotations.¹²⁷

Proof. (Linearization of the moment map). Let the pre–whitened cross–sectional moment and its time average be

$$g_t(\theta; F_t^{\text{obs}}) = Q_Z^{-1/2} \left\{ X_t - W_t(\Gamma_\beta f_t + \Gamma_\delta F_t^{\text{obs}} + \Gamma_\alpha) \right\}, \quad \bar{g}_T(\theta; F) = \frac{1}{T} \sum_{t=1}^T g_t(\theta; F_t^{\text{obs}}). \quad (97)$$

as in (97). Consider the proxy $\widehat{F}_t^{\text{obs}} = F_t^{\text{obs}} + \xi_t$ and fix $\theta = \theta_0$. By Assumption 10 (Fréchet differentiability and Lipschitz continuity of the moment map) and the uniform LLN in Lemma A.1 (together with compactness

¹²⁷Item (i) is the classical “uniform small–error” condition in two–step or generated–regressor asymptotics (see Pagan (1984); Newey and McFadden (1994, Sec. 2.6); Gouriéroux and Monfort (1997, Ch. 4)). It ensures that the first–stage noise $\{\xi_t\}$ perturbs the second–stage moment only at order $o_p(T^{-1/2})$, so the $\sqrt{T}$ –scaled score and LAN limit are unchanged.

Item (ii) replaces this with a weaker but economically realistic orthogonal–average condition. The moment derivative with respect to the observable factor, $\partial_F g_t(\theta_0; F_t^{\text{obs}})$, is required to be *Neyman–orthogonal* to the first–stage error in the sense of $\mathbb{E}[\partial_F g_t(\theta_0; F_t^{\text{obs}}) \xi_t \mid \mathcal{F}_{t-1}] = 0$, so that the additional term $T^{-1/2} \sum_t \partial_F g_t(\theta_0; F_t^{\text{obs}}) \xi_t$ is a martingale–difference average with variance $O(T^{-1})$ and therefore $o_p(1)$ under a martingale CLT; see Newey (1994, Theorem 2.1) and the references therein. This orthogonality principle is the same device used in modern double/debiased machine–learning methods (Chernozhukov et al., 2018; see also Belloni et al., 2014), where score functions are constructed to be locally insensitive to first–stage estimation error so that high–dimensional or nonparametric first stages affect the estimator only through higher–order terms.

in Lemma A.3), a first-order mean-value expansion in the observable factor gives

$$\bar{g}_T(\theta_0; \hat{F}) - \bar{g}_T(\theta_0; F) = \frac{1}{T} \sum_{t=1}^T A_t \xi_t + r_T, \quad A_t \equiv \partial_F g_t(\theta_0; F_t^{\text{obs}}), \quad \|r_T\| = o_p(T^{-1/2}). \quad (98)$$

A direct calculation yields

$$A_t = -Q_Z^{-1/2} W_t \Gamma_{\delta,0},$$

so that under Assumptions 2 and 4 (bounded characteristic geometry and finite loadings) we have the uniform bound

$$\sup_{t \leq T} \|A_t\| = O_p(1).^{128} \quad (99)$$

(Uniform precision case (i)). Under condition (i), $\max_{t \leq T} \|\xi_t\| = o_p(T^{-1/2})$. Combining (98) and (99)

¹²⁸The expansion in (98) is just the mean-value (Gateaux) expansion of $g_t(\theta; F)$ with respect to F at $\theta = \theta_0$. Fix t and view $F \mapsto g_t(\theta_0; F)$ as a map from $\mathbb{R}^{K_{\text{add}}}$ to $\mathbb{R}^L$. Assumption 10 states that $g_t(\theta; F)$ is Fréchet differentiable in (θ, F) with a locally Lipschitz derivative. Hence, for $\hat{F}_t^{\text{obs}} = F_t^{\text{obs}} + \xi_t$ we can write

$$g_t(\theta_0; \hat{F}_t^{\text{obs}}) - g_t(\theta_0; F_t^{\text{obs}}) = \partial_F g_t(\theta_0; F_t^{\text{obs}}) \xi_t + R_{t,T},$$

where the remainder $R_{t,T}$ satisfies $\|R_{t,T}\| \leq L_t \|\xi_t\|^2$ for some random Lipschitz envelope L_t with $\mathbb{E}[L_t] < \infty$ (Assumption 10). Averaging over t gives

$$\bar{g}_T(\theta_0; \hat{F}) - \bar{g}_T(\theta_0; F) = \frac{1}{T} \sum_{t=1}^T A_t \xi_t + \frac{1}{T} \sum_{t=1}^T R_{t,T}, \quad A_t \equiv \partial_F g_t(\theta_0; F_t^{\text{obs}}).$$

By Lemma A.1 and Lemma A.3, the sequence $\{L_t\}$ obeys a uniform LLN, so $T^{-1} \sum_{t=1}^T L_t = O_p(1)$ on the compact feasible set $\Theta(\delta)$. Under the uniform small-error condition $\max_{t \leq T} \|\xi_t\| = o_p(T^{-1/2})$, we then have

$$\left\| \frac{1}{T} \sum_{t=1}^T R_{t,T} \right\| \leq \left(\frac{1}{T} \sum_{t=1}^T L_t \right) \max_{t \leq T} \|\xi_t\|^2 = O_p(1) \cdot o_p(T^{-1}) = o_p(T^{-1/2}),$$

which yields the remainder bound $\|r_T\| = o_p(T^{-1/2})$ in (98).

For the uniform bound in (99), note that at θ_0 we have

$$A_t = \partial_F g_t(\theta_0; F_t^{\text{obs}}) = -Q_Z^{-1/2} W_t \Gamma_{\delta,0}.$$

Inserting $I = Q_Z^{1/2} Q_Z^{-1/2}$ around W_t and using the submultiplicative property of the spectral norm gives

$$\|A_t\| \leq \|Q_Z^{1/2}\| \|Q_Z^{-1/2} W_t Q_Z^{-1/2}\| \|Q_Z^{1/2} \Gamma_{\delta,0}\|.$$

Assumption 2 implies that the Gram matrices $W_t = Z_t' Z_t / N_t$ are uniformly well behaved in probability and that $Q_Z \succ 0$ has a bounded condition number, so $\sup_{t \leq T} \|Q_Z^{-1/2} W_t Q_Z^{-1/2}\| = O_p(1)$ and $\|Q_Z^{\pm 1/2}\| = O_p(1)$. Assumption 4 ensures $\|\Gamma_{\delta,0}\| < \infty$, hence $\|Q_Z^{1/2} \Gamma_{\delta,0}\| = O_p(1)$. Taking the supremum over t yields

$$\sup_{t \leq T} \|A_t\| \leq \|Q_Z^{1/2}\| \sup_{t \leq T} \|Q_Z^{-1/2} W_t Q_Z^{-1/2}\| \|Q_Z^{1/2} \Gamma_{\delta,0}\| = O_p(1),$$

which is exactly (99). Economically, these steps show that the moment reacts smoothly to small perturbations in F_t^{obs} , and that the sensitivity A_t of pricing errors to observable factors remains stable as T grows; first-stage noise therefore enters the LAN score only through second-order terms.

and using the triangle inequality gives

$$\begin{aligned}\left\|\bar{g}_T(\theta_0; \hat{F}) - \bar{g}_T(\theta_0; F)\right\| &\leq \left\|\frac{1}{T} \sum_{t=1}^T A_t \xi_t\right\| + \|r_T\| \\ &\leq \sup_{t \leq T} \|A_t\| \max_{t \leq T} \|\xi_t\| + \|r_T\| \\ &= o_p(T^{-1/2}),\end{aligned}$$

so that

$$\sqrt{T} \left\{ \bar{g}_T(\theta_0; \hat{F}) - \bar{g}_T(\theta_0; F) \right\} = o_p(1).$$

Hence, the generated–regressor perturbation contributes only an $o_p(1)$ shift to the $\sqrt{T}$ –scaled score entering the LAN expansion in Proposition B.3.

(*Orthogonal score case (ii)*). Under condition (ii), the proxy errors are small in average, $\|T^{-1/2} \sum_{t=1}^T \xi_t\| = O_p(1)$, and orthogonal to the score in the sense that $\mathbb{E}[\partial_F g_t(\theta_0; F_t^{\text{obs}}) \xi_t \mid \mathcal{F}_{t-1}] = 0$. Writing the linear term in (98) as

$$\frac{1}{\sqrt{T}} \sum_{t=1}^T A_t \xi_t = \frac{1}{\sqrt{T}} \sum_{t=1}^T \partial_F g_t(\theta_0; F_t^{\text{obs}}) \xi_t,$$

the orthogonality condition implies that $\{A_t \xi_t\}$ is a martingale-difference array with finite second moments. By a martingale CLT (Lemma A.2 and Newey (1994, Theorem 2.1)) and the average–rate condition on $\{\xi_t\}$,

$$\frac{1}{\sqrt{T}} \sum_{t=1}^T A_t \xi_t = o_p(1).$$

Together with $\|r_T\| = o_p(T^{-1/2})$, this again yields

$$\sqrt{T} \left\{ \bar{g}_T(\theta_0; \hat{F}) - \bar{g}_T(\theta_0; F) \right\} = o_p(1),$$

so the score in the LAN representation is unaffected at first order.

(*Information and asymptotic covariance*). Because Proposition B.2 shows that, under $\Gamma_\beta^\top Q_Z \Gamma_\delta = 0$, the Fisher information $\mathcal{I}(\theta_0)|_{\mathcal{J}}$ is block–diagonal across (Γ_β, F) , Γ_δ , and Γ_α , any $o_p(1)$ perturbation in the score cannot alter the first–order covariance matrix. In both regimes (i) and (ii), the LAN expansion in Proposition B.3 therefore continues to hold with the same block–separate efficient variance. Economically, this means that small first–stage measurement noise in observable factors—whether uniformly small or merely orthogonal in average—affects the pricing moments only at higher order, so neither efficiency nor separate identification of the latent and PSF spans is compromised. $\square$

B.5 Feasible Pre–Whitening and Weighting

Pre–whitening plays a central role in the no–arbitrage geometry of the NA–IPCA estimator. In the ideal setting, the transformation $(X_t, W_t) \mapsto (\tilde{X}_t, \tilde{W}_t)$, with $\tilde{X}_t = \Omega_t^{-1/2} Q_Z^{-1/2} X_t$ and $\tilde{W}_t = \Omega_t^{-1/2} Q_Z^{-1/2} W_t Q_Z^{1/2}$, converts the heterogeneous and potentially heavy–tailed cross–sectional information into a coordinate

system in which idiosyncratic variance is normalized and the characteristic geometry is expressed under the population metric Q_Z . Economically, this transformation equalizes the signal-to-noise ratio across assets, so that the pricing moments are measured in units of idiosyncratic volatility rather than in raw returns; assets with high idiosyncratic dispersion therefore contribute proportionally less to the estimation of systematic risk.

Geometrically, pre-whitening aligns the estimation problem with the Q_Z -orthonormal structure that underlies the no-arbitrage manifold in Appendix A. The constraints $\Gamma_\beta^\top Q_Z \Gamma_\beta = I_K$ and $\Gamma_\beta^\top Q_Z \Gamma_\delta = 0$ define a Stiefel-type geometry in characteristic space, and the tangent directions used in the LAN expansion are defined relative to this Q_Z -inner product. Because whitening simply re-expresses the sample moments in this same metric, it preserves the shape of the tangent cone, the projection operators, and the orthogonality between the latent and PSF loading blocks. Consequently, the first-order KKT system of the constrained estimator is invariant to pre-whitening: the whitening operator changes the *metric* but not the *geometry* of the feasible manifold.

From an econometric perspective, feasible GLS replaces (Ω_t, Q_Z) by estimators $(\hat{\Omega}_t, \hat{Q}_Z)$ obtained from residuals and sample characteristic moments. The key requirement is *ratio-consistency*: feasible weights must converge to the ideal ones up to a common scalar factor, so that $\hat{\Omega}_t^{-1/2}$ and $\hat{Q}_Z^{-1/2}$ perturb the whitening map only by $o_p(1)$ uniformly in t . This ensures that the feasible pre-whitened moments and their Jacobians differ from their ideal counterparts only through asymptotically negligible distortions. Economically, this means that feasible volatility estimates preserve the relative heteroskedastic structure across assets, and therefore maintain the same Q_Z -based geometric decomposition of priced directions.

Taken together, feasible pre-whitening preserves all first-order features of the no-arbitrage estimator: the LAN expansion, the block-diagonal Fisher structure, and the separate identification of the latent and PSF spans established in Propositions B.3–B.2. In short, estimating the whitening operators alters only higher-order curvature terms and has no effect on the efficiency frontier implied by the no-arbitrage geometry.

Proposition B.5 (Feasible pre-whitening and first-order equivalence). *Let Assumptions 1–8 hold. Suppose the weighting and whitening matrices satisfy the ratio-consistency conditions*

$$\sup_{t \leq T} \|\hat{\Omega}_t - \omega_T \Omega_t\| \xrightarrow{p} 0, \quad \|\hat{Q}_Z - Q_Z\| \xrightarrow{p} 0, \quad \omega_T \rightarrow \omega_0 \in (0, \infty). \quad (100)$$

Define the feasible pre-whitened variables

$$\hat{X}_t = \hat{\Omega}_t^{-1/2} \hat{Q}_Z^{-1/2} X_t, \quad \hat{W}_t = \hat{\Omega}_t^{-1/2} \hat{Q}_Z^{-1/2} W_t \hat{Q}_Z^{1/2},$$

and let $\hat{\theta}^{FGLS}$ minimize the feasible OLS criterion

$$\hat{\mathcal{L}}_T(\theta) = \frac{1}{T} \sum_{t=1}^T \|\hat{X}_t - \hat{W}_t(\Gamma_\beta f_t + \Gamma_\delta F_t^{\text{obs}} + \Gamma_\alpha)\|_2^2.$$

Then

$$\sqrt{T}(\hat{\theta}^{FGLS} - \hat{\theta}^{GLS}) \xrightarrow{p} 0,$$

and $\hat{\theta}^{FGLS}$ inherits the same LAN expansion and block-diagonal efficient information matrix as $\hat{\theta}^{GLS}$ in

Proposition B.3.

Proof. (Ratio-consistent perturbation of whitening operators). By (100) and the continuity of the inverse-square-root map on positive-definite matrices, we may write uniformly in t :

$$\widehat{\Omega}_t^{-1/2} = \omega_T^{-1/2} \Omega_t^{-1/2} + \Delta_{t,T}, \quad \widehat{Q}_Z^{-1/2} = Q_Z^{-1/2} + \Delta_{Z,T},$$

with $\sup_{t \leq T} \|\Delta_{t,T}\| \xrightarrow{p} 0$ and $\|\Delta_{Z,T}\| \xrightarrow{p} 0$.¹²⁹ Substituting into $\widehat{X}_t$ and $\widehat{W}_t$ gives the decompositions

$$\widehat{X}_t = \omega_T^{-1/2} \widetilde{X}_t + R_{X,t,T}, \quad \widehat{W}_t = \omega_T^{-1/2} \widetilde{W}_t + R_{W,t,T},$$

where $(\widetilde{X}_t, \widetilde{W}_t)$ are the ideal pre-whitened variables and

$$\sup_{t \leq T} (\|R_{X,t,T}\| + \|R_{W,t,T}\|) \xrightarrow{p} 0.$$

Thus, feasible pre-whitening equals the ideal transformation up to a scalar factor $\omega_T^{-1/2}$ and an $o_p(1)$ perturbation.¹³⁰

(Uniform closeness of feasible and ideal criteria). Let $\mathcal{L}_T(\theta)$ denote the infeasible criterion based on $(\widetilde{X}_t, \widetilde{W}_t)$. Using the above expansions and the boundedness of regressors implied by Assumptions 1–4,

$$\sup_{\theta \in \Theta(\delta)} |\widehat{\mathcal{L}}_T(\theta) - \mathcal{L}_T(\theta)| \xrightarrow{p} 0.$$

Hence the two objective functions differ by an $o_p(1)$ term uniformly over the constrained parameter set.¹³¹

(Argmin continuity on the constraint manifold). By Section A.4, the KKT system is well behaved on the

¹²⁹The decomposition follows directly from the ratio-consistency in (100) and the fact that the inverse-square-root operator $A \mapsto A^{-1/2}$ is continuous on the cone of symmetric positive-definite matrices. Specifically, if $\widehat{\Omega}_t = \omega_T \Omega_t + o_p(1)$ uniformly in t , then $\widehat{\Omega}_t^{-1/2} = \omega_T^{-1/2} \Omega_t^{-1/2} + o_p(1)$ by the continuous mapping theorem, yielding the perturbation term $\Delta_{t,T}$. An identical argument applies to $\widehat{Q}_Z^{-1/2}$ because $\widehat{Q}_Z \rightarrow_p Q_Z$ and the map $A \mapsto A^{-1/2}$ preserves convergence under positive definiteness. Economically, this means that feasible pre-whitening differs from the ideal operator only by a scalar rescaling and an asymptotically negligible distortion, ensuring that the local geometry and GLS weighting remain unchanged.

¹³⁰The decompositions follow by direct substitution of the perturbation representations $\widehat{\Omega}_t^{-1/2} = \omega_T^{-1/2} \Omega_t^{-1/2} + \Delta_{t,T}$ and $\widehat{Q}_Z^{-1/2} = Q_Z^{-1/2} + \Delta_{Z,T}$ into the feasible transformations $\widehat{X}_t = \widehat{\Omega}_t^{-1/2} \widehat{Q}_Z^{-1/2} X_t$ and $\widehat{W}_t = \widehat{\Omega}_t^{-1/2} \widehat{Q}_Z^{-1/2} W_t \widehat{Q}_Z^{1/2}$. Expanding the products and grouping terms yields

$$\widehat{X}_t = \omega_T^{-1/2} \Omega_t^{-1/2} Q_Z^{-1/2} X_t + (\Delta_{t,T} Q_Z^{-1/2} + \omega_T^{-1/2} \Omega_t^{-1/2} \Delta_{Z,T} + \Delta_{t,T} \Delta_{Z,T}) X_t,$$

which is $\omega_T^{-1/2} \widetilde{X}_t + R_{X,t,T}$ with $R_{X,t,T}$ defined by the bracketed perturbation. Using the same expansion for $\widehat{W}_t$ produces $\omega_T^{-1/2} \widetilde{W}_t + R_{W,t,T}$. Because $\sup_{t \leq T} \|\Delta_{t,T}\| \rightarrow_p 0$ and $\|\Delta_{Z,T}\| \rightarrow_p 0$, all mixed products involving $(\Delta_{t,T}, \Delta_{Z,T})$ converge uniformly to zero under the boundedness of (X_t, W_t) guaranteed by Assumptions 1–4. Hence $\sup_{t \leq T} (\|R_{X,t,T}\| + \|R_{W,t,T}\|) \xrightarrow{p} 0$. Economically, this means that feasible pre-whitening differs from the ideal transformation only by a global scale factor and a vanishing distortion, so the GLS geometry and first-order efficiency remain unchanged.

¹³¹The uniform convergence follows directly from the decompositions $\widehat{X}_t = \omega_T^{-1/2} \widetilde{X}_t + R_{X,t,T}$ and $\widehat{W}_t = \omega_T^{-1/2} \widetilde{W}_t + R_{W,t,T}$, together with $\sup_{t \leq T} (\|R_{X,t,T}\| + \|R_{W,t,T}\|) \xrightarrow{p} 0$. Substituting these expressions into the feasible criterion $\widehat{\mathcal{L}}_T(\theta)$ produces bilinear forms in $(\widehat{X}_t, \widehat{W}_t)$ whose leading terms match $\mathcal{L}_T(\theta)$ up to the constant ω_T^{-1} , while all remaining cross-terms involve at least one perturbation $(R_{X,t,T}, R_{W,t,T})$. Because the regressors (X_t, W_t) are uniformly bounded in probability on the compact parameter set $\Theta(\delta)$ by Assumptions 1–4, each cross-term contributes at most $O_p(1) \cdot o_p(1) = o_p(1)$ uniformly in θ . Averaging over t preserves this order,

tangent space, and by Proposition A.8 the ideal criterion admits a quadratic expansion with positive-definite Hessian on $\mathcal{T}_{\theta_0}$. Uniform convergence of the criteria therefore implies

$$\widehat{\theta}^{GLS} - \widehat{\theta}^{GLS} = o_p(T^{-1/2}),$$

by standard argmin-continuity results for smooth constrained M -estimators (Amemiya, 1985, Thm. 3.2.2; Newey and McFadden, 1994, Sec. 2.6). Multiplying by $\sqrt{T}$ yields the claim.¹³²

(LAN and Fisher information). Because feasible pre-whitening differs from the ideal transformation only by a common scalar and an $o_p(1)$ perturbation, it does not affect the $\sqrt{T}$ -scaled score or its limit distribution. Thus the LAN expansion in Proposition B.3 and the block-diagonal Fisher structure of Proposition B.2 continue to hold.¹³³ $\square$

Feasible pre-whitening ultimately leaves the geometry of the estimation problem unchanged. Although the idiosyncratic volatility matrix Ω_t and the characteristic metric Q_Z must be estimated, ratio-consistency guarantees that the feasible operators differ from their population counterparts only by a common scalar and an $o_p(1)$ perturbation. This means that the relative precision across firms is preserved, the projection onto the tangent manifold is unaffected, and the curvature of the KKT system coincides with that of the ideal GLS criterion. Economically, the estimator continues to downweight high-variance assets in exactly the same way, so the decomposition of priced directions in Q_Z -geometry remains intact. As a result, feasible GLS attains the same LAN representation and the same block-diagonal efficiency bound as its infeasible benchmark:

yielding

$$\sup_{\theta \in \Theta(\delta)} |\widehat{\mathcal{L}}_T(\theta) - \mathcal{L}_T(\theta)| = o_p(1).$$

Economically, this means that feasible pre-whitening generates only a vanishing distortion of the GLS objective: relative cross-sectional weighting and the Q_Z -orthogonal geometry are preserved, so the feasible and ideal criteria become asymptotically indistinguishable for purposes of estimation or inference.

¹³²Section A.4 establishes that the gradients of all active identification and no-arbitrage constraints are linearly independent at θ_0 , so the KKT system is locally invertible on the tangent space $\mathcal{T}_{\theta_0}$. This guarantees that feasible perturbations of the criterion induce smooth and uniquely determined perturbations of the optimizer. Proposition A.8 further shows that the ideal criterion admits a second-order expansion with a positive-definite Hessian when restricted to $\mathcal{T}_{\theta_0}$, so the population minimizer is locally unique and the criterion is strongly convex along all economically admissible directions. Together, LICQ and the positive-definite Hessian ensure that the mapping $\theta \mapsto \mathcal{L}_T(\theta)$ behaves as a stable quadratic form in a neighborhood of θ_0 , so any perturbation of the objective that vanishes uniformly on $\Theta(\delta)$ must shift the minimizer by at most $o_p(T^{-1/2})$. Formally, this is the constrained argmin-continuity result in Amemiya (1985, Thm. 3.2.2) and Newey and McFadden (1994, Sec. 2.6): if (i) the feasible and ideal criteria differ by $o_p(1)$ uniformly, (ii) the limit Hessian on the tangent space is nonsingular, and (iii) the KKT system is regular, then the two argmins differ by $o_p(T^{-1/2})$. Economically, the result means that feasible weighting preserves the local curvature of the GLS pricing criterion, so feasible and ideal GLS identify the same perturbation of risk exposures to first order.

¹³³Formally, the LAN expansion in Proposition B.3 depends only on the $\sqrt{T}$ -scaled projected score $T^{-1/2} \sum_t \nabla_{\theta} g_t(\theta_0)^\top \Psi_t$ and on the population information operator $\mathcal{I}(\theta_0)|_{\mathcal{T}}$. The feasible transformation replaces $(\Omega_t^{-1/2}, Q_Z^{-1/2})$ by $(\widehat{\Omega}_t^{-1/2}, \widehat{Q}_Z^{-1/2})$, which differ from their population counterparts by a common scalar $\omega_T^{-1/2}$ and an $o_p(1)$ perturbation uniformly in t . The scalar factor cancels in both the score and the information, because GLS is invariant to global re-scaling of the weight matrix. The $o_p(1)$ perturbation shifts the sample score by at most $o_p(1)$ and perturbs the Jacobian of the moment map by at most $o_p(1)$ on the compact domain $\Theta(\delta)$, by the uniform LLN and Lipschitz regularity established in Lemma A.1 and Assumption 10. Since Proposition B.2 shows that the population information is block-orthogonal on the tangent space, and since the information operator is a continuous quadratic functional of the Jacobian, these $o_p(1)$ perturbations cannot alter its block structure in the limit. As a result, the feasible score has the same Gaussian limit as the ideal score (by Slutsky's lemma), and the corresponding Fisher information remains block-diagonal and positive definite. Economically, first-order identification and efficiency depend only on the relative weighting of pricing errors across firms, not on small sampling errors in the estimated variances, so feasible GLS recovers the same risk-return geometry as ideal GLS.

estimating the weights changes the scale of the criterion, but not the geometry that drives identification and efficiency.

B.6 Rotation Equivalence, Normalization, and Invariant Diagnostics

This section formalizes the rotation invariance of the No–Arbitrage IPCA estimator and explains how identification, normalization, and diagnostics are defined on the associated quotient space. Economically, factor models price only the *span* of systematic exposures, not any particular basis: replacing f_t by an orthogonal reparametrization $R_\beta^\top f_t$ and simultaneously rotating Γ_β to $\Gamma_\beta R_\beta$ leaves $Z_t(\Gamma_\beta f_t)$ unchanged, and the same holds for observable (PSF) factors. Statistically, this implies that $(\Gamma_\beta, F, \Gamma_\delta)$ are identified only up to blockwise orthogonal transformations, and that both the constrained KKT system and the LAN expansion derived in Proposition B.3 are invariant under such reparametrizations. Geometrically, rotations act as isometries on the Q_Z –Stiefel manifolds introduced in Appendix A, so the curvature and information structure of the estimator are preserved.¹³⁴

We collect these ideas in a single proposition that: (i) characterizes the invariance of the constrained problem under orthogonal rotations; (ii) shows that the KKT system and LAN covariance transform by orthogonal similarity; (iii) introduces a measurable Procrustes normalization that selects a canonical representative within each rotation–equivalent class; and (iv) verifies that the performance metrics used in the simulations—subspace distance, Procrustes error, and product error—are rotation–invariant functionals of the identified spans.

Let the parameter tuple and constraints be

$$\begin{aligned} \theta &= (\Gamma_\beta, F, \Gamma_\delta, \Gamma_\alpha), \quad \Gamma_\beta \in \mathbb{R}^{L \times K}, F \in \mathbb{R}^{K \times T}, \Gamma_\delta \in \mathbb{R}^{L \times K_{\text{add}}}, \Gamma_\alpha \in \mathbb{R}^L, \\ \Gamma_\beta^\top Q_Z \Gamma_\beta &= I_K, \quad \Gamma_\beta^\top Q_Z \Gamma_\delta = 0, \quad \psi(\Gamma_\alpha) = \Gamma_\alpha^\top Q_Z \Gamma_\alpha \leq \delta, \end{aligned} \quad (101)$$

and consider the (population or sample) quadratic criterion

$$Q_T(\theta) = \frac{1}{T} \sum_{t=1}^T \|X_t - W_t(\Gamma_\beta f_t + \Gamma_\delta F_t^{\text{obs}} + \Gamma_\alpha)\|_{Q_Z^{-1}}^2, \quad \|v\|_{Q_Z^{-1}}^2 := \|Q_Z^{-1/2} v\|_2^2. \quad (102)$$

Here $X_t := Z_t' r_{t+1}/N_t$, $W_t := Z_t' Z_t/N_t$, and the whitening $Q_Z^{-1/2}$ is consistent with the metric used in Appendix A.¹³⁵

¹³⁴For background on the geometry of Stiefel and Grassmann manifolds, as well as the action of orthogonal groups on matrix factor spaces, see Absil et al. (2008, Ch. 2) and Edelman et al. (1998), which develop the differential–geometric structure underlying orthonormality constraints and rotation–equivalent parametrizations. In asset-pricing applications, the fact that factor models identify only *subspaces* rather than particular bases goes back to Chamberlain and Rothschild (1983) and is central to modern treatments of latent–factor identification such as Kelly et al. (2019) and Lettau and Pelger (2020b). These references formalize the principle used here: orthogonal rotations within the latent or observable factor blocks leave the priced span and the cross-sectional fit unchanged, so estimation and inference must be conducted on the quotient space of rotation–equivalent representations.

¹³⁵Pre–multiplying by $Q_Z^{-1/2}$ merely expresses the moments in the Q_Z –orthonormal coordinates used in Appendix A; it does not affect the first–order KKT system or the LAN expansion. See Newey and McFadden (1994, Sec. 2.6).

Let the compact matrix group

$$\mathcal{G} \equiv \mathcal{O}(K) \times \mathcal{O}(K_{\text{add}})$$

act on $(\Gamma_\beta, F, \Gamma_\delta)$ by

$$(R_\beta, R_\delta) \cdot (\Gamma_\beta, F, \Gamma_\delta) := (\Gamma_\beta R_\beta, R_\beta^\top F, \Gamma_\delta R_\delta), \quad (R_\beta, R_\delta) \in \mathcal{O}(K) \times \mathcal{O}(K_{\text{add}}), \quad (103)$$

and leave Γ_α unchanged. The group $\mathcal{G}$ therefore rotates the latent and PSF blocks within their column spans while preserving the intercept.

Proposition B.6 (Rotation equivalence, normalization, and invariant diagnostics). *Let Assumptions 1–8 hold, and suppose the constraints (101) are satisfied at the true parameter θ_0 . Then:*

(i) (**Invariance of the criterion and constraints**) *For any $(R_\beta, R_\delta) \in \mathcal{G}$ and any feasible θ satisfying (101),*

$$Q_T((R_\beta, R_\delta) \cdot \theta) = Q_T(\theta), \quad (R_\beta, R_\delta) \cdot \theta \text{ satisfies (101)}.$$

Hence $\arg \min Q_T$ is a union of $\mathcal{G}$ –orbits (rotation–equivalence classes).

(ii) (**KKT equivariance and LAN invariance**) *Let $S_T(\theta)$ denote the KKT map for the constrained problem associated with Q_T and (101). Then there exists a block–orthogonal operator $\mathcal{R}(R_\beta, R_\delta)$ such that, for all $(R_\beta, R_\delta) \in \mathcal{G}$,*

$$S_T((R_\beta, R_\delta) \cdot \theta) = \mathcal{R}(R_\beta, R_\delta) S_T(\theta).$$

In particular, if $\sqrt{T}(\hat{\theta}_T - \theta_0) \Rightarrow \mathcal{N}(0, \mathcal{V})$, then

$$\sqrt{T}((R_\beta, R_\delta) \cdot \hat{\theta}_T - (R_\beta, R_\delta) \cdot \theta_0) \Rightarrow \mathcal{N}(0, \mathcal{R}(R_\beta, R_\delta) \mathcal{V} \mathcal{R}(R_\beta, R_\delta)^\top),$$

so the LAN covariance is preserved up to orthogonal similarity.

(iii) (**Measurable Procrustes normalization**) *There exist reference frames $(\Gamma_\beta^*, \Gamma_\delta^*)$ and measurable maps $R_\beta^*(\cdot), R_\delta^*(\cdot)$ solving the Q_Z –weighted Procrustes problems*

$$R_\beta^*(\Gamma_\beta) \in \arg \max_{R \in \mathcal{O}(K)} \text{tr}(R^\top \Gamma_\beta^\top Q_Z \Gamma_\beta^*), \quad R_\delta^*(\Gamma_\delta) \in \arg \max_{R \in \mathcal{O}(K_{\text{add}})} \text{tr}(R^\top \Gamma_\delta^\top Q_Z \Gamma_\delta^*), \quad (104)$$

and a normalization map

$$\mathbf{N}(\Gamma_\beta, F, \Gamma_\delta, \Gamma_\alpha) = (\Gamma_\beta R_\beta^*(\Gamma_\beta), R_\beta^*(\Gamma_\beta)^\top F, \Gamma_\delta R_\delta^*(\Gamma_\delta), \Gamma_\alpha), \quad (105)$$

which selects a canonical representative in each orbit $\mathcal{G} \cdot \theta$. If the Q_Z –singular values of $(\Gamma_\beta^, \Gamma_\delta^*)$ are simple, R_β^* and R_δ^* are C^1 in a neighborhood of $(\Gamma_\beta^*, \Gamma_\delta^*)$.*

(iv) (**Rotation–invariant diagnostics**) *The simulation diagnostics used in the paper depend only on the equivalence class $\mathcal{G} \cdot \theta$ and are therefore rotation–invariant: (a) the subspace distance $\|\sin \Theta(\text{col}(\Gamma_\beta), \text{col}(\Gamma_\beta^0))\|_F$; (b) the Procrustes–aligned Frobenius error $\inf_{R \in \mathcal{O}(K)} \|\Gamma_\beta R - \Gamma_\beta^0\|_F$; and (c) the product error $\|\Gamma_\beta F - \Gamma_\beta^0 F^0\|_F$.*

Proof. ((i) Invariance of the problem.) Let

$$\theta = (\Gamma_\beta, F, \Gamma_\delta, \Gamma_\alpha)$$

denote a generic parameter tuple, where $F = [f_1, \dots, f_T] \in \mathbb{R}^{K \times T}$ collects the latent factor realizations with $f_t := F_{:,t} \in \mathbb{R}^K$ the $K \times 1$ latent factor at date t .¹³⁶ For any $(R_\beta, R_\delta) \in \mathcal{G} := \mathcal{O}(K) \times \mathcal{O}(K_{\text{add}})$, define the rotated parameter

$$\theta' = (R_\beta, R_\delta) \cdot \theta := (\Gamma_\beta R_\beta, R_\beta^\top F, \Gamma_\delta R_\delta, \Gamma_\alpha).$$

Constraints. We first verify that the constraints in (101) are preserved under this group action. Because $R_\beta \in \mathcal{O}(K)$ is orthogonal ($R_\beta^\top R_\beta = I_K$), we have

$$(\Gamma_\beta R_\beta)^\top Q_Z (\Gamma_\beta R_\beta) = R_\beta^\top (\Gamma_\beta^\top Q_Z \Gamma_\beta) R_\beta = R_\beta^\top I_K R_\beta = I_K,$$

so the Q_Z -orthonormality condition for the latent loadings is invariant. Similarly, using $R_\beta^\top R_\beta = I_K$ and $R_\delta^\top R_\delta = I_{K_{\text{add}}}$,

$$(\Gamma_\beta R_\beta)^\top Q_Z (\Gamma_\delta R_\delta) = R_\beta^\top (\Gamma_\beta^\top Q_Z \Gamma_\delta) R_\delta = R_\beta^\top \mathbf{0} R_\delta = \mathbf{0},$$

so the block-orthogonality condition $\Gamma_\beta^\top Q_Z \Gamma_\delta = 0$ is also preserved. Finally, since the group action leaves Γ_α unchanged,

$$\psi(\Gamma_\alpha) = \Gamma_\alpha^\top Q_Z \Gamma_\alpha$$

is clearly invariant, and the no-arbitrage bound $\psi(\Gamma_\alpha) \leq \delta$ is unaffected. Hence every rotated parameter θ' satisfies the same system of constraints as θ ; the feasible set is invariant under $(R_\beta, R_\delta) \in \mathcal{G}$.

Criterion. Next, we show that the sample objective $Q_T(\theta)$ is also invariant. Recall that at date t the (unwhitened) characteristic-space residual is

$$X_t - W_t (\Gamma_\beta f_t + \Gamma_\delta F_t^{\text{obs}} + \Gamma_\alpha),$$

where $X_t := N_t^{-1} Z_t' r_{t+1}$ and $W_t := N_t^{-1} Z_t' Z_t$ are the cross-sectional moment and Gram matrices. Under the group action, the latent block transforms as

$$\Gamma_\beta f_t \longmapsto (\Gamma_\beta R_\beta) (R_\beta^\top f_t) = \Gamma_\beta (R_\beta R_\beta^\top) f_t = \Gamma_\beta f_t,$$

so the latent fitted component is unchanged pointwise in t . Economically, this corresponds to re-expressing the same latent factor path f_t in a rotated coordinate system while keeping the priced payoff $Z_t \Gamma_\beta f_t$ fixed.

For the observable (PSF) block, the group action rotates the loading matrix as $\Gamma_\delta \mapsto \Gamma_\delta R_\delta$, while the economic content of the observable factor F_t^{obs} is that of a point in the PSF factor space. One may equivalently regard F_t^{obs} as the coordinate vector of an underlying PSF factor in that space. Reparametrizing this space by

¹³⁶Throughout Appendix B, $F = [f_1, \dots, f_T]$ denotes the time series of latent factors, with $f_t := F_{:,t}$ representing the $K \times 1$ latent factor vector in period t . Writing $R_\beta^\top F$ means that each column is rotated as $f_t \mapsto R_\beta^\top f_t$, i.e. the same latent factor path is expressed in a different orthonormal coordinate basis. The observable factor vector $F_t^{\text{obs}} \in \mathbb{R}^{K_{\text{add}}}$ plays an analogous role for the PSF block, providing coordinates in the observable factor space.

R_δ corresponds to adopting new coordinates $\tilde{F}_t^{\text{obs}} := R_\delta^\top F_t^{\text{obs}}$ while simultaneously using the rotated loadings $\Gamma_\delta R_\delta$. In this representation,

$$(\Gamma_\delta R_\delta) \tilde{F}_t^{\text{obs}} = (\Gamma_\delta R_\delta) (R_\delta^\top F_t^{\text{obs}}) = \Gamma_\delta (R_\delta R_\delta^\top) F_t^{\text{obs}} = \Gamma_\delta F_t^{\text{obs}},$$

so the PSF fitted component $Z_t \Gamma_\delta F_t^{\text{obs}}$ is preserved. Algebraically, this means that the transformation $(\Gamma_\delta, R_\delta) \cdot F_t^{\text{obs}}$ only changes the *basis* in the PSF factor space, not the underlying priced direction.

Putting the latent and observable parts together, the fitted characteristic-space component under θ' is

$$W_t((\Gamma_\beta R_\beta)(R_\beta^\top f_t) + (\Gamma_\delta R_\delta) \tilde{F}_t^{\text{obs}} + \Gamma_\alpha) = W_t(\Gamma_\beta f_t + \Gamma_\delta F_t^{\text{obs}} + \Gamma_\alpha),$$

and thus the residual

$$X_t - W_t(\Gamma_\beta f_t + \Gamma_\delta F_t^{\text{obs}} + \Gamma_\alpha)$$

is identical before and after rotation. Since the sample objective $Q_T(\theta)$ is built as a quadratic form in these residuals (either via a GLS-type loss, or via its pre-whitened counterpart in Appendix B), we conclude that

$$Q_T(\theta') = Q_T(\theta) \quad \text{for all } (R_\beta, R_\delta) \in \mathcal{G}.$$

We have shown that the constraint set is invariant and the objective $Q_T(\theta)$ is unchanged under the group action $(R_\beta, R_\delta) \cdot \theta$. Therefore, if $\hat{\theta}_T$ solves the constrained minimization problem, then every element of its $\mathcal{G}$ -orbit $\{(R_\beta, R_\delta) \cdot \hat{\theta}_T : (R_\beta, R_\delta) \in \mathcal{G}\}$ is also a minimizer. The optimization problem is thus defined on equivalence classes $[\theta] = \{(R_\beta, R_\delta) \cdot \theta : (R_\beta, R_\delta) \in \mathcal{G}\}$, and the NA-IPCA estimator is identified only up to such orthogonal reparametrizations of the latent and observable factor spaces.

(ii) *KKT equivariance and LAN invariance.* Under the group action $\Phi(\theta) = (R_\beta, R_\delta) \cdot \theta$ introduced above, invariance of the objective and constraints implies

$$Q_T(\Phi(\theta)) = Q_T(\theta), \quad \mathfrak{c}(\Phi(\theta)) = \mathfrak{c}(\theta),$$

for all $\theta \in \Theta(\delta)$ and $(R_\beta, R_\delta) \in \mathcal{G}$. At the level of the Lagrangian

$$\mathcal{L}_T(\theta, \Lambda) := Q_T(\theta) + \Lambda^\top \mathfrak{c}(\theta),$$

orthogonal rotations of the factor blocks induce a corresponding (blockwise) conjugation of the multipliers, say $\Lambda' = \Upsilon(\Lambda)$ (e.g. $\Lambda_\beta \mapsto R_\beta^\top \Lambda_\beta R_\beta$ for the normalization block and analogously for $\Gamma_\beta^\top Q_Z \Gamma_\delta$), such that

$$\mathcal{L}_T(\Phi(\theta), \Upsilon(\Lambda)) = \mathcal{L}_T(\theta, \Lambda).^{137} \tag{106}$$

¹³⁷The multipliers transform so that the Lagrangian remains invariant under the group action. To see this, write the constraint vector in block form as $\mathfrak{c}(\theta) = (\text{vec}(\Gamma_\beta^\top Q_Z \Gamma_\beta - I_K), \text{vec}(\Gamma_\beta^\top Q_Z \Gamma_\delta), \psi(\Gamma_\alpha) - \delta)^\top$, and let the associated multipliers be $\Lambda = (\Lambda_\beta, \Lambda_{\beta\delta}, \lambda_\alpha)$, where $\Lambda_\beta \in \mathbb{S}^K$ and $\Lambda_{\beta\delta} \in \mathbb{R}^{K \times K_{\text{add}}}$. Under the rotation $(R_\beta, R_\delta) \in \mathcal{G}$, the constraints satisfy

$$(\Gamma_\beta R_\beta)^\top Q_Z (\Gamma_\beta R_\beta) - I_K = R_\beta^\top (\Gamma_\beta^\top Q_Z \Gamma_\beta - I_K) R_\beta,$$

Intuitively, rotating both the parameters and the associated shadow prices leaves the constrained objective unchanged: the Lagrangian depends only on the risk spans and the constraint manifold, not on the particular choice of basis.

Vectorized coordinates and induced rotation. Working in stacked, vectorized coordinates

$$\vartheta = \begin{bmatrix} \text{vec}(\Gamma_\beta) \\ \text{vec}(F) \\ \text{vec}(\Gamma_\delta) \\ \Gamma_\alpha \end{bmatrix} \in \mathbb{R}^p,$$

the group action $\Phi(\theta)$ induces the block-orthogonal operator

$$\mathcal{R}(R_\beta, R_\delta) = \text{diag}(R_\beta^\top \otimes I_L, R_\beta^\top, R_\delta^\top \otimes I_L, I_L), \quad (107)$$

obtained from the identity $\text{vec}(ABC) = (C^\top \otimes A)\text{vec}(B)$ (cf. Magnus and Neudecker (1999, Sec. 2.3)).¹³⁸

Thus, in vectorized form, Φ acts as $\vartheta' = \mathcal{R}(R_\beta, R_\delta) \vartheta$.

Gradient and constraint Jacobian equivariance. Differentiating (106) with respect to θ and using the chain rule, with $d\Phi(\theta)/d\theta = \mathcal{R}(R_\beta, R_\delta)$, gives

$$\nabla_{\theta} \mathcal{L}_T(\Phi(\theta), \Upsilon(\Lambda)) = \mathcal{R}(R_\beta, R_\delta) \nabla_{\theta} \mathcal{L}_T(\theta, \Lambda). \quad (108)$$

An analogous argument for the constraint map $\mathfrak{c}(\theta)$ yields

$$\nabla_{\theta} \mathfrak{c}(\Phi(\theta)) \mathcal{R}(R_\beta, R_\delta) = \nabla_{\theta} \mathfrak{c}(\theta),$$

so that the Jacobian of the constraints at $\Phi(\theta)$ is obtained by right-multiplying the Jacobian at θ by $\mathcal{R}(R_\beta, R_\delta)^\top$.¹³⁹

$$(\Gamma_\beta R_\beta)^\top Q_Z(\Gamma_\delta R_\delta) = R_\beta^\top (\Gamma_\beta^\top Q_Z \Gamma_\delta) R_\delta, \quad \psi(\Gamma_\alpha) \text{ unchanged.}$$

Hence the rotated constraint vector can be written as an orthogonal conjugation of the original one. If we define

$$\Upsilon(\Lambda) = (R_\beta^\top \Lambda_\beta R_\beta, R_\beta^\top \Lambda_{\beta\delta} R_\delta, \lambda_\alpha),$$

then

$$\Lambda^\top \mathfrak{c}(\theta) = \text{tr}(\Lambda_\beta^\top (\Gamma_\beta^\top Q_Z \Gamma_\beta - I_K)) + \text{tr}(\Lambda_{\beta\delta}^\top \Gamma_\beta^\top Q_Z \Gamma_\delta) + \lambda_\alpha (\psi(\Gamma_\alpha) - \delta),$$

and, using $\text{tr}(A^\top R^\top B R) = \text{tr}((R A R^\top)^\top B)$,

$$\Lambda^\top \mathfrak{c}(\theta) = \Upsilon(\Lambda)^\top \mathfrak{c}(\Phi(\theta)).$$

Because $Q_T(\Phi(\theta)) = Q_T(\theta)$ by part (i), we obtain $\mathcal{L}_T(\Phi(\theta), \Upsilon(\Lambda)) = Q_T(\theta) + \Upsilon(\Lambda)^\top \mathfrak{c}(\Phi(\theta)) = Q_T(\theta) + \Lambda^\top \mathfrak{c}(\theta) = \mathcal{L}_T(\theta, \Lambda)$. Economically, $\Upsilon(\Lambda)$ just re-expresses the shadow prices in the rotated coordinate system of the constraints, so the constrained objective is unchanged by orthogonal re-parametrizations of the factor bases.

¹³⁸For example, $\text{vec}(\Gamma_\beta R_\beta) = (R_\beta^\top \otimes I_L) \text{vec}(\Gamma_\beta)$ and $\text{vec}(R_\beta^\top F) = (R_\beta^\top \otimes I_T) \text{vec}(F)$. Stacking all blocks yields $\vartheta' = \mathcal{R}(R_\beta, R_\delta) \vartheta$. Because R_β and R_δ are orthogonal, each block of $\mathcal{R}$ is orthogonal and $\mathcal{R}^\top \mathcal{R} = I_p$. Geometrically, $\mathcal{R}$ re-expresses the same parameter point in a rotated basis without changing lengths or inner products.

¹³⁹Let $C(\theta) := \nabla_{\theta} \mathfrak{c}(\theta)$. From $\mathfrak{c}(\Phi(\theta)) = \mathfrak{c}(\theta)$ and the chain rule we get $C(\Phi(\theta)) \mathcal{R}(R_\beta, R_\delta) = C(\theta)$, hence $C(\Phi(\theta)) = C(\theta) \mathcal{R}(R_\beta, R_\delta)^\top$. Thus the rows of $C(\theta)$ transform under the same orthogonal map as the parameter vector. Under LICQ

Tangent–space transformation and projected KKT mapping. Let

$$\mathcal{M} = \{ \theta : \Gamma_\beta^\top Q_Z \Gamma_\beta = I_K, \Gamma_\beta^\top Q_Z \Gamma_\delta = 0, \psi(\Gamma_\alpha) \leq \delta \}$$

be the constraint manifold. Under LICQ and Slater (Assumptions 5–6, Section A.4), the tangent space at θ is

$$T_\theta \mathcal{M} = \{ d\theta : C(\theta) d\theta = 0 \}, \quad C(\theta) := \nabla_\theta \mathbf{c}(\theta).$$

Using the transformation rule for $C(\theta)$ and the orthogonality of $\mathcal{R}$, one has

$$d\theta \in T_\theta \mathcal{M} \iff \mathcal{R}(R_\beta, R_\delta) d\theta \in T_{\Phi(\theta)} \mathcal{M},$$

and therefore

$$T_{\Phi(\theta)} \mathcal{M} = \mathcal{R}(R_\beta, R_\delta) T_\theta \mathcal{M}. \quad (109)$$

Geometrically, $\mathcal{R}$ is an isometry between tangent spaces: it rotates feasible directions at θ into feasible directions at $\Phi(\theta)$ without changing their dimension or inner–product structure.¹⁴⁰

Let Π_θ denote the orthogonal projector onto $T_\theta \mathcal{M}$, taken with respect to the inner product induced by the criterion Q_T . From (109) and orthogonality of $\mathcal{R}$,

$$\Pi_{\Phi(\theta)} = \mathcal{R}(R_\beta, R_\delta) \Pi_\theta \mathcal{R}(R_\beta, R_\delta)^\top. \quad (110)$$

Define the projected KKT (score) map as

$$S_T(\theta) := \Pi_\theta \nabla_\theta \mathcal{L}_T(\theta, \Lambda).$$

Combining (108) and (110) gives

$$\begin{aligned} S_T(\Phi(\theta)) &= \Pi_{\Phi(\theta)} \nabla_\theta \mathcal{L}_T(\Phi(\theta), \Upsilon(\Lambda)) \\ &= \mathcal{R} \Pi_\theta \mathcal{R}^\top \left(\mathcal{R} \nabla_\theta \mathcal{L}_T(\theta, \Lambda) \right) \\ &= \mathcal{R} \Pi_\theta \nabla_\theta \mathcal{L}_T(\theta, \Lambda) \\ &= \mathcal{R}(R_\beta, R_\delta) S_T(\theta). \end{aligned} \quad (111)$$

Thus the projected first–order conditions are *equivariant*: applying a group element (R_β, R_δ) rotates both the parameter vector and the projected score by the same orthogonal operator $\mathcal{R}(R_\beta, R_\delta)$. In particular, if $S_T(\theta) = 0$ then $S_T(\Phi(\theta)) = 0$, so the set of stationary points is invariant up to rotations.

(Section A.4), these rows remain linearly independent after rotation, so the local geometry of the feasible manifold is preserved.

¹⁴⁰Because the rows of $C(\theta)$ are linearly independent (LICQ), $T_\theta \mathcal{M}$ is the orthogonal complement of the row space of $C(\theta)$. From $C(\Phi(\theta))\mathcal{R} = C(\theta)$ we get $\text{row}(C(\Phi(\theta))) = \text{row}(C(\theta)\mathcal{R}^\top)$, so the orthogonal complement of this row space is $\mathcal{R} T_\theta \mathcal{M}$. Thus $T_{\Phi(\theta)} \mathcal{M} = \mathcal{R} T_\theta \mathcal{M}$ (Bonnans and Shapiro, 2000, Sec. 3.1–3.2). Since $\mathcal{R}$ is orthogonal, lengths and angles in the tangent space are preserved.

LAN invariance. Proposition B.3 provides a LAN expansion for the constrained estimator $\widehat{\theta}_T$ of the form

$$\sqrt{T} \Pi_{\theta_0}(\widehat{\theta}_T - \theta_0) = (\mathcal{I}(\theta_0)|_{\mathcal{T}})^{-1} \frac{1}{\sqrt{T}} \sum_{t=1}^T \nabla_{\theta} g_t(\theta_0)^\top \Psi_t + o_p(1),$$

where Π_{θ_0} projects onto $T_{\theta_0}\mathcal{M}$, $\mathcal{I}(\theta_0)|_{\mathcal{T}}$ is the information operator restricted to the tangent, and Ψ_t are the Huber/GLS innovations.¹⁴¹

Now consider the rotated estimator

$$\widehat{\theta}'_T := \Phi(\widehat{\theta}_T), \quad \theta'_0 := \Phi(\theta_0).$$

By (109), $T_{\theta'_0}\mathcal{M} = \mathcal{R}(R_\beta, R_\delta) T_{\theta_0}\mathcal{M}$, and the corresponding projector satisfies $\Pi_{\theta'_0} = \mathcal{R}(R_\beta, R_\delta) \Pi_{\theta_0} \mathcal{R}(R_\beta, R_\delta)^\top$. Moreover, the projected score and information transform in the same way, as in (111) and the analogous Hessian identity. Hence the LAN expansion for $\widehat{\theta}'_T$ is obtained by applying $\mathcal{R}(R_\beta, R_\delta)$ to that of $\widehat{\theta}_T$. In vectorized form,

$$\sqrt{T} \Pi_{\theta'_0}(\widehat{\theta}'_T - \theta'_0) = \mathcal{R}(R_\beta, R_\delta) \sqrt{T} \Pi_{\theta_0}(\widehat{\theta}_T - \theta_0) + o_p(1).$$

If $\sqrt{T} \Pi_{\theta_0}(\widehat{\theta}_T - \theta_0) \Rightarrow \mathcal{N}(0, \mathcal{V})$, then by the continuous-mapping theorem

$$\sqrt{T} \Pi_{\theta'_0}(\widehat{\theta}'_T - \theta'_0) \Rightarrow \mathcal{N}(0, \mathcal{R}(R_\beta, R_\delta) \mathcal{V} \mathcal{R}(R_\beta, R_\delta)^\top).$$

Because $\mathcal{R}(R_\beta, R_\delta)$ is orthogonal, this transformation preserves eigenvalues and determinant of $\mathcal{V}$: the covariance matrix is transformed by orthogonal similarity, leaving the efficiency bound and the entire spectral structure unchanged.¹⁴²

Thus the projected KKT map is equivariant under (R_β, R_δ) , and the LAN limit law of the constrained estimator is invariant up to orthogonal similarity. Economically, this means that all first-order distributional and efficiency results depend only on the *spans* of the latent and PSF factor spaces, not on the particular orthonormal bases used to represent them.

¹⁴¹The LAN expansion used in Proposition B.3 follows the standard linearization of a constrained M -estimator's KKT system under smoothness, compactness, and regularity conditions. Uniform LLN and CLT for the moment map and its Jacobian are provided by Lemmas A.1–A.2 in Appendix A, while LICQ and Slater (Assumptions 5–6, Section A.4) guarantee that the feasible set is a smooth manifold and that the KKT multipliers are locally well defined. Under these conditions, the constrained estimator admits a quadratic expansion around θ_0 whose leading term is given by the information-weighted score, producing the LAN representation once projected onto the tangent space $T_{\theta_0}\mathcal{M}$.

The general theory underlying this argument is classical. For smooth constrained M -estimators, Newey and McFadden (1994, Sec. 7) show that, under LICQ and smoothness of the criterion, the KKT system can be linearized on the tangent space and the estimator satisfies a first-order expansion driven by the projected score. For the differential-geometric foundation—KKT systems and LAN expansions on smooth manifolds—see Bonnans and Shapiro (2000, Chs. 3–5) and Geyer (1994), who establish that constrained estimators behave as unconstrained M -estimators in local coordinates on the tangent space. Finally, van der Vaart (1998, Ch. 5) provides the classical LAN machinery used to obtain Gaussian limits for the projected score. Together these results justify the tangent-space linearization used in Proposition B.3 and ensure that the estimator's asymptotic distribution is fully characterized by the projected information operator $\mathcal{I}(\theta_0)|_{\mathcal{T}}$.

¹⁴²If $X_T \Rightarrow \mathcal{N}(0, \mathcal{V})$ and O is any orthogonal matrix ($O^\top O = I$), then $OX_T \Rightarrow \mathcal{N}(0, O\mathcal{V}O^\top)$ by the continuous mapping theorem. Since orthogonal similarity $O\mathcal{V}O^\top$ leaves all eigenvalues, the determinant, and thus the Cramér–Rao efficiency bound unchanged, the asymptotic precision of the estimator is invariant to any rotation of the factor coordinates. This invariance follows from standard LAN theory for smooth reparametrizations; see Newey and McFadden (1994, Sec. 7) and the semiparametric LAN framework in Bickel et al. (1998, Ch. 3).

((iii) *Procrustes normalization*). Because the parameter tuple $\theta = (\Gamma_\beta, F, \Gamma_\delta, \Gamma_\alpha)$ is identified only up to the orthogonal rotations $(R_\beta, R_\delta) \in \mathcal{G} = \mathcal{O}(K) \times \mathcal{O}(K_{\text{add}})$ described in parts (i)–(ii), a measurable normalization is required to select a unique representative from each rotation–equivalence class. Economically, this step fixes the coordinate system used to express the latent and PSF spans without modifying any priced direction: the model identifies the *subspaces* $\text{col}(\Gamma_\beta)$ and $\text{col}(\Gamma_\delta)$, not their internal orthonormal bases.

Weighted Procrustes alignment. Fix reference loading frames $\Gamma_\beta^* \in \mathbb{R}^{L \times K}$ and $\Gamma_\delta^* \in \mathbb{R}^{L \times K_{\text{add}}}$ that satisfy the Q_Z –orthonormality and block–orthogonality relations in (101). For any admissible $(\Gamma_\beta, \Gamma_\delta)$ define the Q_Z –weighted cross–covariance matrices

$$M_\beta := \Gamma_\beta^\top Q_Z \Gamma_\beta^*, \quad M_\delta := \Gamma_\delta^\top Q_Z \Gamma_\delta^*.$$

Solving

$$R_\beta^*(\Gamma_\beta) \in \arg \max_{R \in \mathcal{O}(K)} \text{tr}(R^\top M_\beta), \quad R_\delta^*(\Gamma_\delta) \in \arg \max_{R \in \mathcal{O}(K_{\text{add}})} \text{tr}(R^\top M_\delta) \quad (112)$$

yields the orthogonal transformations that align Γ_β and Γ_δ most closely with $(\Gamma_\beta^*, \Gamma_\delta^*)$ in the Q_Z –weighted Frobenius metric. If $M_\beta = U_\beta \Sigma_\beta V_\beta^\top$ is an SVD, then $R_\beta^*(\Gamma_\beta) = U_\beta V_\beta^\top$ maximizes $\text{tr}(R^\top M_\beta)$; the PSF block is analogous.¹⁴³

Smoothness and measurable selection. When the singular values of M_β and M_δ are distinct, the maximizers in (112) are unique and vary smoothly in $(\Gamma_\beta, \Gamma_\delta)$. If some singular values coincide, the maximizers form a compact orbit; imposing a deterministic tie-breaking rule (e.g., fixed sign convention for singular vectors) selects a Borel–measurable representative on each orbit without affecting feasibility or the value of the criterion.¹⁴⁴

Normalization map. Define

$$\mathbf{N}(\theta) := (\Gamma_\beta R_\beta^*(\Gamma_\beta), R_\beta^*(\Gamma_\beta)^\top F, \Gamma_\delta R_\delta^*(\Gamma_\delta), \Gamma_\alpha). \quad (113)$$

Because $(R_\beta^*, R_\delta^*) \in \mathcal{G}$, the normalized tuple $\mathbf{N}(\theta)$ satisfies the same constraints (101) and, by the invariance established in part (i), satisfies $Q_T(\mathbf{N}(\theta)) = Q_T(\theta)$. Moreover, for any $(R_\beta, R_\delta) \in \mathcal{G}$,

$$\mathbf{N}((R_\beta, R_\delta) \cdot \theta) = \mathbf{N}(\theta),$$

so $\mathbf{N}$ acts as a measurable section of the quotient map $\Theta(\delta) \rightarrow \Theta(\delta)/\mathcal{G}$. Geometrically, $\mathbf{N}(\theta)$ picks a canonical Q_Z –orthogonal orientation of the latent and PSF spans; economically, it removes pure reparametrization noise without altering the systematic return components $Z_t(\Gamma_\beta f_t)$ or $Z_t \Gamma_\delta F_t^{\text{obs}}$.

In summary, the Procrustes normalization provides a well–defined and (locally) smooth selection of representatives from each rotation–equivalence class. It preserves all constraints, leaves the objective Q_T

¹⁴³For matrices with simple singular values, the mapping $M \mapsto U \Sigma V^\top$ is C^∞ ; see Edelman et al. (1998, Thm. 2.3). Thus $R_\beta^*(\Gamma_\beta) = U_\beta V_\beta^\top$ depends smoothly on Γ_β through $M_\beta = \Gamma_\beta^\top Q_Z \Gamma_\beta^*$. Properties of weighted Procrustes problems on Stiefel manifolds are summarized in Absil et al. (2008).

¹⁴⁴Repeated singular values generate rotational ambiguity within subspaces of M_β or M_δ , but any orthogonal choice within these blocks leaves $\Gamma_\beta R_\beta^*$ and $\Gamma_\delta R_\delta^*$ in the same equivalence class. A measurable section is constructed by choosing singular vectors via a fixed sign/ordering convention; see Absil et al. (2008).

unchanged, and aligns the latent and observable loading spaces with fixed reference frames. Consequently, it supports the LAN analysis in part (ii) and ensures that inference is carried out on economically meaningful—rotation-invariant—objects: the priced factor spans rather than arbitrary orthonormal bases.

(iv) *Rotation-invariant diagnostics.* Finally, we verify that the three diagnostics used in the text—subspace distance, Procrustes-aligned Frobenius error, and product error—are well defined on rotation-equivalence classes and therefore depend only on the identified spans, not on the particular choice of factor basis.

Subspace distance. Let $\mathcal{S}(\Gamma_\beta) := \text{col}(\Gamma_\beta)$ denote the latent K -dimensional subspace spanned by the columns of Γ_β , and let $\Theta(\mathcal{S}(\Gamma_\beta), \mathcal{S}(\Gamma_\beta^0))$ be the diagonal matrix of principal angles between $\mathcal{S}(\Gamma_\beta)$ and $\mathcal{S}(\Gamma_\beta^0)$, as in Appendix A.¹⁴⁵ For any $R_\beta \in \mathcal{O}(K)$ we have $\text{col}(\Gamma_\beta R_\beta) = \text{col}(\Gamma_\beta)$, so replacing Γ_β by $\Gamma_\beta R_\beta$ does not change the pair of subspaces. Consequently, the principal angles (and hence the Frobenius norm $\|\sin \Theta(\mathcal{S}(\Gamma_\beta), \mathcal{S}(\Gamma_\beta^0))\|_F$) are unchanged under the rotation $(\Gamma_\beta, F) \mapsto (\Gamma_\beta R_\beta, R_\beta^\top F)$. The same argument applies to the PSF span $\text{col}(M_\beta \Gamma_\delta)$ in Proposition B.2.

Procrustes-aligned Frobenius error. The Procrustes diagnostic for the latent block is

$$d_{\text{Proc}}(\Gamma_\beta, \Gamma_\beta^0) := \min_{R \in \mathcal{O}(K)} \|\Gamma_\beta R - \Gamma_\beta^0\|_F.$$

If we replace Γ_β by $\Gamma_\beta R_\beta$ for some fixed $R_\beta \in \mathcal{O}(K)$, we obtain

$$\min_{R \in \mathcal{O}(K)} \|\Gamma_\beta R_\beta R - \Gamma_\beta^0\|_F = \min_{R \in \mathcal{O}(K)} \|\Gamma_\beta \tilde{R} - \Gamma_\beta^0\|_F, \quad \tilde{R} := R_\beta R \in \mathcal{O}(K).$$

Since $R \mapsto \tilde{R}$ is a bijection of $\mathcal{O}(K)$ onto itself, the minimum is unchanged. Thus $d_{\text{Proc}}(\Gamma_\beta, \Gamma_\beta^0)$ depends only on the subspace $\text{col}(\Gamma_\beta)$ and is invariant to premultiplication of Γ_β by any orthogonal R_β . An identical reasoning applies to the PSF block $d_{\text{Proc}}(\Gamma_\delta, \Gamma_\delta^0)$ with K_{add} in place of K .¹⁴⁶

Product error. The product-error diagnostic for the latent block is

$$d_{\text{Prod}}(\Gamma_\beta, F; \Gamma_\beta^0, F^0) := \|\Gamma_\beta F - \Gamma_\beta^0 F^0\|_F.$$

Under the latent-block rotation $(\Gamma_\beta, F) \mapsto (\Gamma_\beta R_\beta, R_\beta^\top F)$, we have for each t ,

$$(\Gamma_\beta R_\beta)(R_\beta^\top f_t) = \Gamma_\beta (R_\beta R_\beta^\top) f_t = \Gamma_\beta f_t,$$

so the entire product matrix $\Gamma_\beta F = [\Gamma_\beta f_1, \dots, \Gamma_\beta f_T]$ is unchanged. Therefore,

$$\|(\Gamma_\beta R_\beta)(R_\beta^\top F) - \Gamma_\beta^0 F^0\|_F = \|\Gamma_\beta F - \Gamma_\beta^0 F^0\|_F.$$

Thus $d_{\text{Prod}}(\Gamma_\beta, F; \Gamma_\beta^0, F^0)$ is invariant under joint rotations of the latent block and depends only on the priced

¹⁴⁵If $Q_\beta, Q_\beta^0 \in \mathbb{R}^{L \times K}$ are Q_Z -orthonormal bases of $\mathcal{S}(\Gamma_\beta)$ and $\mathcal{S}(\Gamma_\beta^0)$, the principal angles $\{\theta_i\}$ are defined via the singular values of $Q_\beta^\top Q_\beta^0$; their sines and cosines depend only on the subspaces, not on the specific bases. See Stewart and Sun (1990, Sec. 2.6–2.7) and Absil et al. (2008, Ch. 2).

¹⁴⁶Orthogonal Procrustes distances measure the discrepancy between subspaces after optimally rotating one basis toward another. Because the minimization is over all $R \in \mathcal{O}(K)$, any additional fixed rotation of the estimated basis can be absorbed into the optimizer, leaving the minimum value unchanged; see Stewart and Sun (1990, Ch. 2) and Absil et al. (2008).

path $\{\Gamma_\beta f_t\}_{t=1}^T$, i.e. on the latent systematic return space rather than on the particular factor coordinates. The same reasoning applies to the PSF component $\Gamma_\delta F_t^{\text{obs}}$ under rotations $\Gamma_\delta \mapsto \Gamma_\delta R_\delta$ and $F_t^{\text{obs}} \mapsto R_\delta^\top F_t^{\text{obs}}$.

Collectively, these results show that the three diagnostics we use in empirical work— the subspace distance, the Procrustes-aligned loading error, and the product error— all measure economically meaningful differences in *risk spaces*, rather than differences in arbitrary numerical representations of factor bases. Because rotating $(\Gamma_\beta, F, \Gamma_\delta)$ does not change the priced component $Z_t(\Gamma_\beta f_t)$ nor the PSF contribution $Z_t \Gamma_\delta F_t^{\text{obs}}$, each diagnostic assigns the *same value* to any two parameterizations that encode the same underlying systematic risk exposures. In this sense, the diagnostics are naturally defined on the rotation-equivalence classes of the model and remain unchanged whenever the latent or observable factors are re-expressed in a different orthonormal basis.

Economically, the rationale is straightforward. Asset pricing models do not care about how we label or orient the individual factors; what matters is the *span* of systematic returns they generate—i.e., the set of priced directions in characteristic space. The subspace distance quantifies how close two such spans are; the Procrustes criterion measures how well two loading structures align after removing purely cosmetic rotations; and the product error compares the *model-implied* systematic returns $\Gamma_\beta F$ with their benchmark counterparts. All three focus on the economic content of the model—the systematic return structure—not the arbitrary coordinate system used to represent it.

Thus, these diagnostics faithfully summarize differences in the latent and PSF pricing directions that matter for expected returns, while ignoring superficial rotational reparametrizations. This makes them appropriate tools for empirical comparisons and simulation studies, consistent with the rotation-invariant geometry emphasized in modern factor-pricing analyses (e.g., [Stewart and Sun \(1990\)](#); [Absil et al. \(2008\)](#)). $\square$

C Implementation of the No-Arbitrage IPCA

This appendix summarizes the algorithmic structure used to implement the No-Arbitrage IPCA estimator. The procedure mirrors the constrained geometry and statistical properties derived in Appendices A–B, where identification, orthogonality, and local efficiency are characterized through the Q_Z -based moment structure and the corresponding KKT system. The sections below outline the estimation pipeline, blockwise updates, constraint enforcement, and rotation-invariant normalization that define the computational counterpart of the theoretical framework.

C.1 Estimation Pipeline

The numerical implementation of the No-Arbitrage IPCA estimator follows a structured sequence of weighted regressions that operationalises the geometric, identification, and no-arbitrage results developed in Appendices A–B. The routine proceeds on the constrained manifold $\Theta(\delta)$ defined in (17), and each computational step mirrors a corresponding theoretical object: the characteristic metric Q_Z , the WLS/KKT geometry, and the rotation-equivalence structure of the loading and factor spaces. A compact schematic of this estimation cycle is provided in Figure 15, which illustrates how the moment construction, initialisation, alternating weighted regressions, and identification steps jointly implement the constrained NA-IPCA estimator.

The estimator begins by constructing the characteristic-space moments that anchor the cross-sectional pricing relation. For each period t , the cross-sectional moment vector and Gram matrix

$$X_t = \frac{1}{N_t} Z_t' r_{t+1}, \quad W_t = \frac{1}{N_t} Z_t' Z_t,$$

summarise the empirical relation between returns and characteristics and provide the sample building blocks of the moment map in Appendix A. These moments are paired with time weights

$$w_t = N_t^{\theta_{\text{Nts}}} \exp\{-\lambda_{\text{td}}(T - t)\},$$

which determine each period's contribution to the weighted least-squares criterion.¹⁴⁷ When used, a feasible pre-whitening of (X_t, W_t) is applied as in Appendix A.6.2. The sample metric $\hat{Q}_Z = T^{-1} \sum_{t=1}^T W_t$ then provides the empirical analogue of the Gram operator Q_Z that defines the Q_Z -inner product and characteristic-space geometry underpinning the orthogonality and identification structure of Appendix A. Economically, $\hat{Q}_Z$ determines how firm characteristics span priced directions and therefore governs the geometric constraints imposed during estimation.

¹⁴⁷The time weight $w_t = N_t^{\theta_{\text{Nts}}} \exp\{-\lambda_{\text{td}}(T - t)\}$ operates as a feasible GLS precision weight in the sense of Proposition B.1. The factor $N_t^{\theta_{\text{Nts}}}$ captures cross-sectional sampling variation: periods with larger N_t contain more information and therefore receive higher weight, consistent with the precision weighting implicit in the moment maps of Appendix A. The exponential term $\exp\{-\lambda_{\text{td}}(T - t)\}$ introduces a mild recency tilt. Economically, a positive λ_{td} places slightly greater emphasis on recent months, which is desirable when market structure, coverage, or characteristic relevance evolves gradually over time. Mathematically, such weights are admissible under the ULLN and CLT conditions in Lemmas A.1–A.2, which require only that the weights be deterministic, bounded, and non-degenerate. Since the WLS equivalence in Proposition B.1 holds under any such bounded weighting sequence, the choice of w_t affects finite-sample stability but does not alter identification, consistency, or the LAN expansion in Proposition B.3.

The procedure is initialised by extracting a spectral loading estimate Γ_{SVD} from an unconstrained PCA/IPCA regression. This spectral loading matrix partitions the loading space into latent, observable, and intercept components and is mapped into an admissible starting point $(\Gamma_{\beta}^{(0)}, \Gamma_{\delta}^{(0)}, \Gamma_{\alpha}^{(0)})$ that satisfies, or lies arbitrarily close to, the Q_Z -orthonormality, block-orthogonality, and no-arbitrage restrictions of (101). Conditional on these loading blocks and the observable factor series F_t^{obs} , the initial latent factor path $F^{(0)}$ is obtained from the weighted least-squares system implied by Proposition B.1. This ensures that the full tuple $(\Gamma_{\beta}^{(0)}, F^{(0)}, \Gamma_{\delta}^{(0)}, \Gamma_{\alpha}^{(0)})$ begins on the feasible no-arbitrage manifold $\Theta(\delta)$ in a manner consistent with its geometric structure.

A key implication of the theory in Appendix B.6 is that factor models identify the *spans* of the loading matrices—not their specific coordinate representations. Thus, (Γ_{β}, F) and $(\Gamma_{\beta} R_{\beta}, R_{\beta}^{\top} F)$ imply the same priced systematic returns for any orthogonal R_{β} , generating a set of rotational degrees of freedom with no economic content. These rotations merely relabel the basis within the latent factor space; they do not alter the systematic risk exposures priced in the cross section. To ensure a stable and economically interpretable orientation, the normalization map $N(\cdot)$ from Appendix B.6 selects a canonical representative from each rotation-equivalent class, eliminating purely statistical reparametrization noise without affecting pricing implications or the estimator's asymptotic distribution.

Starting from this feasible initializer, the estimator proceeds through a sequence of alternating weighted regressions. Holding $(\Gamma_{\beta}, \Gamma_{\delta}, \Gamma_{\alpha})$ fixed, the latent factor step updates $F^{(m+1)}$ via the WLS system determined by the current loading blocks, the observable factor series, and the moment sequence $\{(X_t, W_t, w_t)\}_{t=1}^T$, in accordance with Proposition B.1. Holding $F^{(m+1)}$ fixed, the loading step solves a Kronecker-structured WLS problem with a blockwise ridge term to update $(\Gamma_{\beta}, \Gamma_{\delta}, \Gamma_{\alpha})^{(m+1)}$. These two regressions implement the conditional minimisers of the constrained WLS criterion on $\Theta(\delta)$ and give the algorithm the structure of a blockwise projected regression scheme that links expected returns to latent and observable priced directions.

Each loading update is followed by an identification step that enforces the geometric constraints characterising the no-arbitrage structure. The latent block $\Gamma_{\beta}^{(m+1)}$ is orthonormalised with respect to $\hat{Q}_Z$ and rotated so that its associated factor covariance is diagonal; the observable block $\Gamma_{\delta}^{(m+1)}$ is projected onto the Q_Z -orthogonal complement of the latent span; and the intercept block $\Gamma_{\alpha}^{(m+1)}$ is rescaled and re-projected so that $\psi(\Gamma_{\alpha}) \leq \delta$ and $\Gamma_{\beta}^{\top} \hat{Q}_Z \Gamma_{\alpha} = 0$, in line with Proposition B.2. Collectively, these transformations correspond to an iteration of the normalization map $N(\cdot)$, ensuring that each update is expressed in a rotation-invariant and economically interpretable coordinate system. The algorithm records the relative error, the path of ψ -values, and additional diagnostic metrics at each iteration. Convergence is declared when improvements in the weighted pricing residuals and the projected gradient become negligible, implementing in practice the KKT convergence conditions developed in Appendix B. Under the assumptions in Appendices A–B, the iterative routine converges to a well-defined solution of the constrained pricing problem. The resulting estimator satisfies the KKT conditions and inherits the LAN expansion and block-diagonal information structure formalised in Proposition B.3.

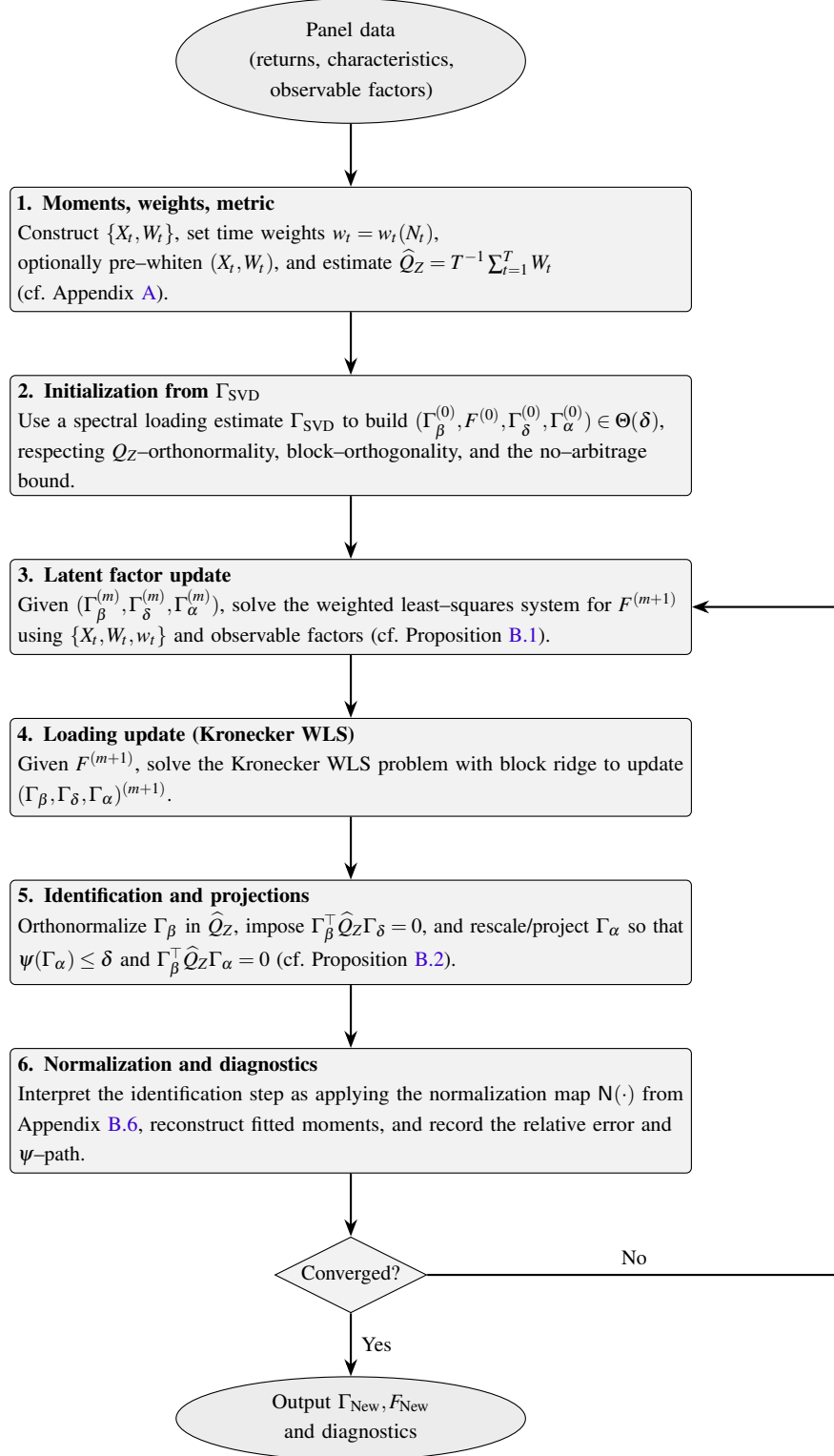


Figure 15: Compact estimation pipeline for the No-Arbitrage IPCA estimator

Notes: The figure summarizes the alternating weighted least-squares routine described in Appendix C.1. Each iteration consists of a latent factor update, a Kronecker WLS loading update, and an identification step that enforces the Q_Z -orthonormality, block-orthogonality, and no-arbitrage constraints.

C.2 Alternating Block Updates

Each iteration of the No-Arbitrage IPCA estimator consists of a pair of weighted regressions that jointly refine the latent factors and loading blocks while respecting the Q_Z -geometry that underpins the identification and no-arbitrage results in Appendices A–B. The routine can be viewed as an alternating sequence of GLS projections, each of which solves the relevant first-order condition of the constrained pricing problem, followed by geometric projections that map the updated parameters back onto the feasible manifold $\Theta(\delta)$.

The latent-factor step treats the loading blocks as fixed and updates $F^{(m+1)}$ by projecting the pricing moment onto the current latent span. For each period t , the estimator chooses $f_t^{(m+1)}$ to best align the cross-sectional moment $X_t = W_t(\Gamma_\beta^{(m)} f_t + \Gamma_\delta^{(m)} F_t^{\text{obs}} + \Gamma_\alpha^{(m)})$ with the observed relation between returns and characteristics. Operationally, this corresponds to solving the weighted GLS equation

$$(\Gamma_\beta^{(m)\top} W_t \Gamma_\beta^{(m)}) f_t^{(m+1)} = \Gamma_\beta^{(m)\top} [X_t - W_t(\Gamma_\delta^{(m)} F_t^{\text{obs}} + \Gamma_\alpha^{(m)})],$$

with weights determined by (W_t, w_t) and any feasible pre-whitening of the moment matrices. Proposition B.1 shows that this step is the conditional minimiser of the WLS criterion on $\Theta(\delta)$, while Appendix A.6.2 establishes that the weighting scheme leaves identification, consistency, and the LAN representation unaffected. Economically, this step recovers the period-specific latent systematic factor — the component of the GLS pricing moment that is explained by characteristics once observable factors and the intercept have been accounted for.

Conditional on these updated latent factors, the loading step updates $(\Gamma_\beta, \Gamma_\delta, \Gamma_\alpha)$ by solving the Kronecker-structured GLS normal equations that arise from stacking the moment conditions across time. This system is linear in each loading block and admits a closed-form solution in vectorised form. To stabilise the inversion when characteristics are highly collinear or when observable factors strongly overlap with latent characteristic directions, the implementation includes a light blockwise ridge penalty on the latent, observable, and intercept components. This ridge affects only finite-sample conditioning and leaves the WLS first-order conditions—and hence the LAN expansion in Proposition B.3—unchanged. Observable traded or macro factors enter additively through F_t^{obs} and are incorporated into the GLS update in exactly the same pricing equation as the latent factors, ensuring that the loading step accounts for the full set of systematic return channels available at each iteration.

Once the unconstrained loading update is obtained, the estimator enforces the identification and no-arbitrage restrictions of Appendix A–B through a sequence of geometric projections. The latent block is first orthonormalised with respect to the characteristic metric Q_Z and rotated so that its associated factor covariance is diagonal, ensuring that Γ_β lies on the Q_Z -Stiefel manifold that represents the admissible latent factor spans. Observable factor loadings are then projected onto the Q_Z -orthogonal complement of the latent span, implementing the block-orthogonality condition of Proposition B.2 and preventing latent and observable components from competing to explain the same priced direction. The intercept block is subsequently rescaled and projected so that $\psi(\Gamma_\alpha) \leq \delta$ and $\Gamma_\beta^\top Q_Z \Gamma_\alpha = 0$, thereby confining the intercept to the bounded no-arbitrage direction characterised by the curvature properties in Appendix B. In settings with heavy-tailed idiosyncratic returns, the intercept projection may use a robust Huber-type loss to prevent

outliers from generating spurious “alpha factors” that distort the geometric separation of systematic and residual pricing components.

Collectively, these steps implement one iteration of the normalization map $N(\cdot)$ from Appendix B.6. At each cycle, the estimator moves from $(\Gamma^{(m)}, F^{(m)})$ to $(\Gamma^{(m+1)}, F^{(m+1)})$ by solving the relevant GLS conditions and then mapping the solution back onto the rotation–equivalent class defined by $\mathcal{G} = \mathcal{O}(K) \times \mathcal{O}(K_{\text{add}})$. The result is an update that both satisfies the constrained KKT system and preserves the identification and orthogonality structure that defines the feasible set $\Theta(\delta)$. This alternating sequence forms the computational backbone of the NA–IPCA estimator and provides the bridge between the statistical moment structure of the data and the economic geometry implied by no–arbitrage.

C.3 Constraint Enforcement

Once the unconstrained loading update has been computed, the estimator imposes the full set of identification and no–arbitrage conditions that define the feasible manifold $\Theta(\delta)$ in Appendix A. This projection step is not an ancillary numerical adjustment but rather the mechanism by which the algorithm preserves the economic structure of the model: the latent span must remain orthonormal with respect to the characteristic metric Q_Z , observable loadings must occupy the Q_Z –orthogonal complement of that span, and the intercept must lie within the bounded no–arbitrage direction characterised in Appendix B. Each projection therefore maps the raw GLS update back into the geometry governing the population pricing operator, ensuring that the iterative sequence never leaves the admissible parameter space.

The latent block Γ_β is first orthonormalised under the Q_Z –inner product, so that $\Gamma_\beta^\top Q_Z \Gamma_\beta = I_K$ holds exactly. This operation re–anchors the latent factors on the Q_Z –Stiefel manifold of Appendix A, aligning their span with the long–run covariance structure of characteristics rather than with the Euclidean metric that would be implicitly imposed by a naïve normalisation. The resulting basis is then rotated so that the associated factor covariance is diagonal, placing the latent factors in a canonical orientation that renders the subsequent updates stable and interpretable and prepares the tuple for the rotation–invariant representation analysed in Appendix B.6. Because factor models identify only the span of Γ_β rather than any specific coordinate representation, these rotations do not alter pricing content; they simply choose a statistically and economically convenient label for the latent directions.

The observable block Γ_δ is next projected onto the Q_Z –orthogonal complement of the latent span. This step enforces the population block–orthogonality $\Gamma_\beta^\top Q_Z \Gamma_\delta = 0$ in Proposition B.2 and ensures that latent and observable factors do not compete to explain the same priced direction in characteristic space. When characteristics and traded factors co–move—which is pervasive in empirical asset pricing—Euclidean projection, as used in unconstrained IPCA, can misallocate systematic structure between the two blocks. Projecting with respect to Q_Z instead guarantees that the decomposition aligns with the economic geometry implied by the long–run characteristic metric and prevents the “double counting” phenomenon documented in the literature. In the implementation, this projection also includes a fit–preserving adjustment to the factor time series $F^{(m+1)}$: before projecting Γ_δ onto the Q_Z –orthogonal complement of the latent span, the estimator adds the term $\gamma_{\beta\delta} F_t^{\text{obs}}$, where $\gamma_{\beta\delta} = \Gamma_\beta^{(m+1)\top} \Gamma_\delta^{(m+1)}$ is the component of the observable–factor loadings that lies inside $\text{col}(\Gamma_\beta)$. This adjustment transfers the overlapping variation from Γ_δ into the latent factor series

so that the fitted pricing moment $\Gamma_\beta F_t^{(m+1)} + \Gamma_\delta F_t^{\text{obs}}$ remains unchanged by the geometric projection.

From an economic perspective, the fit-preserving adjustment prevents latent and observable factors from competing to explain the same priced direction. When Γ_δ contains a component inside $\text{col}(\Gamma_\beta)$, the two blocks attribute the same source of systematic return to two different channels—latent and observable—which dilutes interpretation and undermines the economic separation between characteristic-driven risk and traded factor exposures. If this overlap is removed by directly projecting Γ_δ onto the Q_Z -orthogonal complement without adjusting $F^{(m+1)}$, the fitted pricing moment $\Gamma_\beta F_t + \Gamma_\delta F_t^{\text{obs}}$ changes, causing part of the return variation previously attributed to the latent span to disappear or to be reassigned arbitrarily across updates. This distorts the decomposition of systematic returns, produces unstable loadings across iterations, and often generates oscillatory behaviour in the ALS cycle, as the algorithm repeatedly reassigns the overlapping component back and forth between latent and observable blocks. By first transferring the overlapping component $\gamma_\beta F_t^{\text{obs}}$ into the latent factor series and only then projecting Γ_δ into the orthogonal complement, the estimator preserves the fitted systematic component exactly, stabilises the loading decomposition, and ensures that latent and observable factors evolve on distinct and economically interpretable subspaces.

The intercept block Γ_α is projected last. The estimator rescales Γ_α to enforce the curvature bound $\psi(\Gamma_\alpha) \leq \delta$ and removes any component lying in the latent span, imposing the orthogonality $\Gamma_\beta^\top Q_Z \Gamma_\alpha = 0$. This disciplined intercept geometry is essential: as shown in Appendix B, Γ_α represents the unique residual pricing direction consistent with no-arbitrage curvature. Allowing Γ_α to drift into the latent span, as may occur in unconstrained IPCA, can generate spurious “alpha factors” or absorb priced directions that should instead be attributed to latent or observable components. To guard against heavy-tailed idiosyncratic errors, the projection may apply a robust Huber-type loss, preventing a handful of outliers from distorting the geometry of the intercept or its orthogonality relations.

These latent, observable, and intercept projections together constitute a single application of the normalization map $N(\cdot)$ defined in Appendix B.6. After each iteration, the updated tuple is mapped into a canonical representative of its rotation-equivalence class under $\mathcal{G} = \mathcal{O}(K) \times \mathcal{O}(K_{\text{add}})$, preserving the WLS/KKT structure while maintaining the precise orthogonality relations that define $\Theta(\delta)$. This projection step is therefore the key link between the statistical update and the economic geometry of the model: it ensures that every iterate respects the identification, block structure, and no-arbitrage curvature that give the NA-IPCA estimator its interpretability, stability, and local efficiency.

C.4 Feasible Weighting and Pre-Whitening

The weighted least-squares structure of the NA-IPCA estimator requires a feasible approximation to the population GLS weights that govern the precision of the cross-sectional pricing moment. In practice, the implementation relies on simple diagonal weights based on data-dependent time weights w_t and robust residual scale estimators, rather than the full GLS weighting matrix. This reflects the theoretical results in Appendix A, which show that the moment map depends on $W_t = Z_t' Z_t / N_t$ only through the population metric Q_Z , and on Proposition B.5, which establishes that any ratio-consistent feasible weighting scheme delivers the same first-order asymptotic (LAN) distribution as ideal GLS.

Within each period, the time weight w_t adjusts the contribution of the cross-section to the WLS objective

in a manner consistent with its statistical precision, while the residual scale estimates equalize the dispersion of idiosyncratic returns across assets. Economically, this weighting structure ensures that periods with larger effective cross sections or lower residual volatility receive proportionally more influence in determining the priced directions. At the same time, the theory requires only that the weights be bounded and deterministic; it does not depend on the specific form of w_t , nor on the particular robustification choices used to stabilise the moment maps in finite samples.

The implementation also permits an optional pre-whitening of (X_t, W_t) to reduce the influence of heterogeneous characteristic scaling. When engaged, this transformation rescales the moment matrices by the diagonal elements of W_t in order to improve numerical conditioning of the GLS normal equations. Since this adjustment leaves the span of characteristic directions unchanged, and because Proposition B.5 shows that pre-whitening affects only second-order terms, it does not alter identification, consistency, or the LAN expansion derived in Appendix B. In this sense, feasible weighting and optional pre-whitening act as stabilising devices: they improve the behaviour of the WLS system in practice while preserving the economic geometry, the Q_Z -orthogonality structure, and the first-order efficiency of the NA-IPCA estimator.

C.5 Rotation Alignment and Normalization

Because factor models identify only the *span* of the loading matrix rather than a particular coordinate representation, the pair (Γ_β, F) is determined only up to an orthogonal rotation. As shown in Appendix B.6, the parameter space admits the group action $(\Gamma_\beta, F) \mapsto (\Gamma_\beta R_\beta, R_\beta^\top F)$ for any $R_\beta \in \mathcal{O}(K)$, and all such representations generate identical model-implied pricing moments. To ensure that successive iterates are expressed in a consistent and economically interpretable orientation, each update is followed by a rotation-alignment step that selects a canonical representative within the rotation-equivalent class.

The alignment procedure operates directly in the Q_Z -geometry. After the loading update produces an unconstrained pair $(\Gamma_\beta^{(m+1)}, F^{(m+1)})$, the estimator re-orthonormalises $\Gamma_\beta^{(m+1)}$ with respect to the characteristic metric Q_Z and rotates both the loading and factor blocks so that the covariance of the updated factor series is diagonal. This corresponds to applying the normalization map $N(\theta)$ of Appendix B.6, which fixes an orthonormal basis for the identified latent span while maintaining the fitted pricing moment. The operation is purely geometric: it does not change the value of the objective, it preserves all constraints, and it leaves the statistical content of the update unchanged.

Economically, this normalization anchors the latent factors in a stable set of priced directions measured in the characteristic metric that governs cross-sectional variation. Without this step, equivalent parameterizations related by orthogonal transformations could cause artificial twists in the loading paths across iterations, masking convergence and complicating interpretation. The normalization map eliminates this nuisance variation by expressing each iterate in a fixed, rotation-invariant coordinate system. As a result, both the factor series and the loading blocks evolve smoothly along the Q_Z -Stiefel manifold, which facilitates identification, improves numerical stability, and allows the convergence criteria of Appendix B to be evaluated meaningfully.

C.6 Convergence and Computational Notes

The alternating GLS routine described in Appendices C.1–C.5 inherits the stability and convergence guarantees implied by the Q_Z -based geometry developed in Appendices A–B. Under Assumptions A.1–A.11, the moment conditions satisfy the ULLN and CLT requirements of Lemmas A.1–A.2, and the constraint set $\Theta(\delta)$ satisfies the regularity and local identification conditions in Section A.4. These ingredients ensure that the constrained WLS criterion admits a locally unique minimizer on the no-arbitrage manifold and that its restriction to the tangent space $T_{\theta_0}\Theta(\delta)$ is strongly convex in a neighborhood of the population parameter. In this region the alternating update consisting of the F -step, the Γ -step, and the geometric projections of Appendix C defines a continuous contraction with respect to the local norm induced by the tangent-space information operator. Iterates that enter this neighborhood converge to a numerical fixed point that satisfies the projected KKT system up to $o_p(T^{-1/2})$, and therefore inherit the local asymptotic normal expansion and block-diagonal information structure established in Proposition B.3.

The algorithm monitors convergence using diagnostics designed to mirror these first-order conditions. Across iterations the routine tracks the relative improvement in the characteristic-moment fit $\|X - \hat{X}\|_F$, which is monotone in the WLS criterion under the feasible weights of Appendix A; the magnitude of the projected gradient on the tangent space of $\Theta(\delta)$; and the extent of any residual violations of orthogonality and curvature restrictions on the latent, observable, and intercept blocks. When the relative decrease in $\|X - \hat{X}\|_F$ falls below a tolerance proportional to its current level, the updates have entered the fixed-point region of the contraction mapping. At this stage the identification and curvature constraints are satisfied up to numerical precision, the projected gradient is negligible, and the estimator lies in the LAN neighborhood in which the quadratic approximation of the criterion is valid.

The computational burden of each iteration is dominated by the latent-factor GLS updates and the structured loading regression. The factor update requires solving T linear systems of dimension $K \times K$, while the loading update inverts a Kronecker-structured matrix of size $LK_{\text{tilde}} \times LK_{\text{tilde}}$, where K_{tilde} collects the latent, observable, and intercept directions. The geometric adjustments— Q_Z -orthogonalization, fit-preserving reparametrizations, and curvature projection—are low-dimensional linear operations that do not alter the computational order of the method. Numerical devices such as ridge stabilization or symmetrization are included only to enhance robustness and do not affect identification, convergence, or asymptotic behavior.

Taken together, these features ensure that the numerical procedure mirrors the theory: the iterates remain on the Q_Z -orthogonal manifold, the updates satisfy the constrained WLS first-order conditions, and the limiting estimator inherits the LAN expansion and block-diagonal information operator of Proposition B.3. The algorithm therefore delivers a stable and economically interpretable solution to the no-arbitrage pricing problem while preserving the computational scalability required for empirical applications.

D Statistical Inference for the No-Arbitrage IPCA

This appendix develops the statistical inference framework for the No-Arbitrage IPCA (NA-IPCA) estimator. Building on the geometric identification environment of Appendix A, the constrained KKT and LAN analysis in Appendix B, and the algorithmic implementation in Appendix C, we characterise the asymptotic distribution of the latent, observable, and intercept components of the characteristic-based priced risk structure when the estimator is constrained to the Q_Z -based no-arbitrage manifold $\Theta(\delta)$.

The central objects for inference are therefore not the raw coordinates of $\theta = (\Gamma_\beta, \Gamma_\delta, \Gamma_\alpha, F)$, which are identified only up to rotations and subject to non-linear constraints, but their Q_Z -orthogonal spans, the tangent-space score, and the associated block-diagonal information operator.

D.1 Parameter Space and Geometric Environment

Inference for the NA-IPCA estimator is conducted on the constrained parameter set $\Theta(\delta)$ introduced in Appendix A, which embeds the no-arbitrage structure directly into the characteristic geometry generated by the long-run Gram operator Q_Z . The parameter space can be written as

$$\Theta(\delta) = \left\{ (\Gamma_\beta, \Gamma_\delta, \Gamma_\alpha, F) : \Gamma_\beta^\top Q_Z \Gamma_\beta = I_K, \Gamma_\beta^\top Q_Z \Gamma_\delta = 0, \Gamma_\beta^\top Q_Z \Gamma_\alpha = 0, \psi(\Gamma_\alpha) \leq \delta \right\}.$$

The operator Q_Z summarises the long-run covariance structure of firm characteristics and induces the inner product $\langle a, b \rangle_{Q_Z} = a^\top Q_Z b$ that underlies all notions of length, angle, and orthogonality in the model. Because systematic return variation is spanned by characteristic portfolios, this Q_Z -geometry provides the economically meaningful coordinate system in which latent factors, observable traded or macro factors, and the no-arbitrage intercept are interpreted. The constraints in $\Theta(\delta)$ decompose the characteristic loading space into three mutually Q_Z -orthogonal subspaces: the latent priced directions encoded by Γ_β , the observable priced directions encoded by Γ_δ , and the residual no-arbitrage direction encoded by Γ_α .

Statistical statements are therefore expressed in terms of these Q_Z -orthogonal spans rather than the raw matrices themselves. This reflects the fact that factor models identify only underlying pricing directions, not particular coordinate representations, and it ensures that the asymptotic objects of interest—tangent-space perturbations, scores, and information operators—are aligned with the economic geometry that governs the mapping from characteristics to expected returns. The local parameterisation and tangent-space decomposition derived in Appendix B provide the linear structure on $\Theta(\delta)$ used in the subsequent sections of this appendix.

D.2 Identification and Tangent-Space Representation

The no-arbitrage geometry developed in Appendix A implies that the economically identified objects of the model are the *spans* of the loading blocks— $\text{col}(\Gamma_\beta)$ and $\text{col}(\Gamma_\delta)$ —and the Q_Z -orthogonal projection of the intercept Γ_α , rather than any particular choice of basis. Since expected returns depend on these objects only through the SDF $\Gamma_\beta f_t + \Gamma_\delta F_t^{\text{obs}} + \Gamma_\alpha$, any orthogonal transformation within the latent or observable factor spaces leaves the SDF, and hence all priced returns, unchanged. Identification must therefore be

formulated on equivalence classes of parameters under the action of the rotation group $\mathcal{G} = \mathcal{O}(K) \times \mathcal{O}(K_{\text{add}})$, and all inferential statements must be invariant to such rotations; see Appendix A.7.1.

Under Assumptions 1–8 and the LICQ/Slater conditions in Section A.4, the feasible set $\Theta(\delta)$ (modulo $\mathcal{G}$) forms a smooth manifold and admits a well-defined tangent space $T_{\theta_0}\Theta(\delta)$ at any regular point $\theta_0 \in \Theta(\delta)$. This tangent space decomposes naturally into two components. The *horizontal* directions are admissible perturbations of the latent span, the observable span, and the Q_Z -orthogonal intercept direction; they generate first-order changes in priced returns and carry statistical information. The *vertical* directions are the rotational perturbations $\mathcal{T}_{\theta_0}^{(\text{rot})}$ induced by $\mathcal{G}$, which merely relabel the bases of the latent and observable factor spaces and have no impact on the SDF. The identifiability of the model is exhausted by the horizontal component $T_{\theta_0}\Theta(\delta) \cap \mathcal{T}_{\theta_0}^{(\text{rot})\perp}$; vertical directions are by construction economically redundant.

This tangent-space representation provides the linearisation of the pricing map at θ_0 and is the starting point for all asymptotic arguments. Appendix B shows that the score and information are naturally defined on the horizontal tangent space: the projection onto $T_{\theta_0}\Theta(\delta)$ removes rotations and constraint-violating directions, the second-order expansion of the constrained criterion depends only on these identified directions, and the resulting Fisher information is block-diagonal across the latent, observable, and intercept components. Consequently, the LAN and efficiency results in Appendix B can be interpreted as statements about the curvature of the SDF in the Q_Z -orthogonal latent, observable, and intercept spans.

D.3 Score, Curvature, and the Constrained KKT System

The NA-IPCA estimator minimises the constrained weighted least-squares (WLS) criterion

$$Q_T(\theta) = \frac{1}{T} \sum_{t=1}^T m_t(\theta)^\top \Omega_t^{-1} m_t(\theta), \quad m_t(\theta) = X_t - W_t(\Gamma_\beta f_t + \Gamma_\delta F_t^{\text{obs}} + \Gamma_\alpha),$$

over the manifold $\Theta(\delta)$ defined in Appendix A, where $\{\Omega_t\}$ are feasible diagonal GLS weights. The full gradient $\nabla_\theta Q_T(\theta)$ is not the relevant score, because many directions—rotations of the latent and observable spans, violations of Q_Z -orthogonality, and perturbations that break the intercept bound—are not economically identified. The appropriate first-order object is therefore the *projected score*

$$S_T(\theta) = \Pi_{T_\theta\Theta(\delta)} \nabla_\theta Q_T(\theta),$$

where $T_\theta\Theta(\delta)$ is the tangent space characterised in Appendix A and $\Pi_{T_\theta\Theta(\delta)}$ denotes the orthogonal projection onto this space. As shown in Appendix B, the condition $S_T(\theta) = 0$ is equivalent to the KKT stationarity system for the constrained problem and removes all non-identified directions, including the rotational subspace $\mathcal{T}_\theta^{(\text{rot})}$. Economically, this condition requires that the fitted systematic return $\Gamma_\beta f_t + \Gamma_\delta F_t^{\text{obs}} + \Gamma_\alpha$ be the best Q_Z -weighted least-squares approximation to the pricing moment X_t among all SDF representations that satisfy orthogonality and no-arbitrage restrictions.

Local curvature is obtained by projecting the Hessian of the Lagrangian onto the same tangent space. Let $\mathcal{L}_T(\theta, \lambda)$ denote the Lagrangian, which stacks the WLS criterion with the equality and inequality constraints

described in Appendix B. The *projected Hessian* is defined as

$$\mathcal{H}_T(\theta) = \Pi_{T_{\theta}\Theta(\delta)} \nabla_{\theta\theta}^2 \mathcal{L}_T(\theta, \lambda) \Pi_{T_{\theta}\Theta(\delta)}.$$

This operator is the sample analogue of the block–diagonal information matrix in Proposition B.3. Only perturbations that modify the economically identified components of the SDF—the Q_Z –orthogonal latent span, the observable span, and the bounded intercept direction—enter $\mathcal{H}_T(\theta)$; rotational directions lie orthogonal to $T_{\theta}\Theta(\delta)$ and therefore carry no Fisher information. Together, the projected score and the projected curvature provide a complete local statistical representation of the NA–IPCA estimator. The projected score determines the influence function of $\hat{\theta}_T$, while the projected Hessian governs its local curvature and asymptotic covariance. As established in Appendix B, these objects yield a rotation–invariant LAN expansion and confirm that the estimator is locally efficient for all Q_Z –identified components of the stochastic discount factor. Moreover, because the alternating WLS routine of Appendix C is constructed to solve these projected KKT conditions and to enforce the corresponding geometric projections at each iteration, the computational algorithm and the asymptotic theory are fully aligned within the constrained no–arbitrage geometry.

D.4 Local Asymptotic Normality and Efficiency

Proposition B.3 shows that, once the parameter is restricted to the economically relevant geometry of Appendix A, the NA–IPCA estimator admits a standard local asymptotic normal (LAN) representation. The key point is that asymptotic statements are formulated for the *projected* estimator $\Pi_{T_{\theta_0}\Theta(\delta)}(\hat{\theta}_T - \theta_0)$, rather than for the raw coordinate difference $\hat{\theta}_T - \theta_0$: the projection removes all non-identified directions—including rotations of the latent span and perturbations that violate the no–arbitrage manifold—and leaves only those variations that induce first–order changes in priced returns.

Formally, under Assumptions 1–11 and the regularity conditions of Section A.4, the estimator satisfies

$$\sqrt{T} \Pi_{T_{\theta_0}\Theta(\delta)}(\hat{\theta}_T - \theta_0) = \left(\mathcal{J}(\theta_0)|_{T_{\theta_0}\Theta(\delta)} \right)^{-1} \frac{1}{\sqrt{T}} \sum_{t=1}^T \dot{g}_t(\theta_0)^\top \Psi_t + o_p(1).$$

Here $\dot{g}_t(\theta_0)$ is the Jacobian of the characteristic moment map $g_t(\theta)$ from Appendix A; Ψ_t is the innovation in the Huber/GLS score associated with the feasible weighting scheme of Appendix A.8.3 and Appendix A.6.2; and $\mathcal{J}(\theta_0)|_{T_{\theta_0}\Theta(\delta)}$ is the information operator of Appendix B restricted to the tangent space. Because the projection eliminates all non-identified directions, this LAN expansion holds in a finite–dimensional, rotation–invariant coordinate system and applies simultaneously to the latent span, the observable–factor span, and the bounded no–arbitrage intercept.

Economically, the LAN representation implies that, after enforcing the no–arbitrage geometry, the NA–IPCA estimator behaves locally like a GLS estimator for the Q_Z –identified SDF components. The asymptotic variance is determined by the same block–diagonal information structure that governs the identification of systematic return channels in Proposition B.3, and the projection onto $T_{\theta_0}\Theta(\delta)$ ensures that inference is carried out only on economically meaningful perturbations. Within the class of estimators that respect the Q_Z –orthogonality and no–arbitrage constraints, NA–IPCA thus attains the GLS efficiency bound

for every identified component of the SDF.

D.5 Feasible GLS Weighting and Robustness

By Assumption 9 and Proposition B.5, any ratio-consistent feasible GLS or pre-whitened weighting scheme yields the same first-order behaviour as the ideal GLS system. In particular, as long as the weights remain deterministic, bounded, and bounded away from zero—as ensured by the residual-scale and Huber-type procedures described in Appendices A.6.2–A.8.3—the projected score, information operator, and LAN expansion in Propositions B.3–A.8 continue to hold without modification. Thus the feasible weighting matrices $\widehat{\Omega}_t$ used in practice do not affect identification or asymptotic variance; they simply supply a ratio-consistent estimate of the population precision weights that appear in the theoretical WLS problem.

Economically, these feasible weights improve finite-sample stability by downweighting assets or months with unusually high idiosyncratic dispersion and by limiting the influence of extreme residuals. Diagonal weights based on robust residual scales (e.g., MAD-type estimators) temper the contribution of volatile stocks, while Huber score adjustments prevent heavy-tailed returns from generating instability in the WLS criterion.¹⁴⁸ These refinements influence only the small-sample behaviour of the estimator: the asymptotic distribution remains governed by the same tangent-space information operator $\mathcal{J}(\theta_0)|_{T_{\theta_0}\Theta(\delta)}$ derived in Appendix B. In empirical applications, standard errors are constructed by combining the projected score with a robust estimate of the long-run covariance of the Huber/GLS innovations—typically a heteroskedasticity and autocorrelation consistent (HAC)/Newey–West estimator applied to the residual score sequence—and then applying the blockwise LAN formulas of Proposition B.3 to obtain Wald intervals and hypothesis tests for the latent, observable, and intercept components.

D.6 Rotation-Invariant Inference

Because only the Q_Z -spans of the loading blocks $(\Gamma_\beta, \Gamma_\delta)$ are identified, all asymptotic statements in Appendix B are naturally formulated on the quotient space $\Theta(\delta)/\mathcal{G}$, where $\mathcal{G} = \mathcal{O}(K) \times \mathcal{O}(K_{\text{add}})$ acts by rotating the latent and observable factor coordinates. Lemma A.4 and Lemma A.5 show that the parameterization is invariant under this action, while Proposition B.6 establishes that both the projected KKT system and the LAN covariance matrix are $\mathcal{G}$ -equivariant. Consequently, the normalization map $N(\cdot)$ introduced in Appendix B.6 merely selects a canonical representative from each rotation-equivalent class for reporting purposes; it does not change the characteristic-based priced risk structure, the fitted pricing kernel, or any element of the information operator.

In empirical work, this rotation invariance means that inference concerns only economically meaningful risk *spans* rather than particular loading coordinates. Reported loadings and factors therefore appear in a specific normalized form (e.g. Q_Z -orthogonal latent block, observable block projected onto its Q_Z -orthogonal complement, and a Procrustes-aligned orientation), but any alternative orthogonal rotation of these coordinates yields identical Wald, LM, and score tests, as well as identical confidence regions for the latent, observable,

¹⁴⁸The Huber innovations enter the score only through a bounded-influence transformation of the residuals; see Assumption A.8. Because this transformation is L_2 -consistent, it does not alter the first-order terms in the LAN expansion.

and intercept components. Thus the presentation of $(\Gamma_\beta, F, \Gamma_\delta)$ is a matter of convenience, while all statistical conclusions—the LAN expansion, blockwise standard errors, and hypothesis tests—are invariant to rotations of the factor basis and depend solely on the underlying Q_Z -identified pricing directions.

D.7 Link Between Algorithm and Estimator

The alternating least-squares (ALS) routine described in Appendix C can be interpreted as a projected weighted-least-squares iteration for solving the constrained KKT system $S_T(\theta) = 0$ that characterises the NA-IPCA estimator. Each iteration first updates the latent factors by solving the conditional GLS problem in Appendix C.2, then updates the loading blocks through the Kronecker-structured GLS equations, and finally applies the geometric projections and normalization steps of Appendices C.3–C.5. These operations keep the iterates on the feasible manifold $\Theta(\delta)$, preserving the Q_Z -orthogonality conditions and the ψ -bound, and thereby implement one step of a projected WLS (or quasi-Newton) method for the KKT stationarity condition.

Under Assumptions 1–11 and the LICQ regularity conditions of Section A.4, the criterion $Q_T(\theta)$ is locally well behaved in a neighbourhood of the true parameter. In this region the ALS sequence decreases the WLS criterion monotonically, and every limit point of the iterates satisfies the constrained KKT system of Proposition B.3. Thus, provided the initialization lies in a neighbourhood where the KKT map is locally stable—such as the spectral starting point Γ_{SVD} used in Appendix C.1—the numerical solution returned by the algorithm coincides with the theoretical estimator $\hat{\theta}_T$ to which the LAN expansion of Proposition B.3 applies. This reasoning follows standard arguments for quasi-Newton and ALS methods: once the algorithm converges to a regular KKT point, the statistical properties are determined entirely by the score and curvature of the underlying criterion, not by the specific optimisation path.

For empirical work, this alignment between computation and theory means that numerical convergence of the ALS routine—measured by small changes in $Q_T(\theta)$, small projected scores, and negligible violations of the constraints—is sufficient for inference. No separate optimisation or likelihood fitting is required: the limiting iterate of the algorithm is precisely the estimator appearing in the LAN representation. Because the algorithm evolves entirely within the Q_Z -based no-arbitrage geometry, the resulting estimator inherits the rotation-invariant, block-diagonal, and efficiency properties described in Appendix B.

D.8 Practical Inference for Latent, Observable, and Intercept Blocks

The block-diagonal information structure derived in Proposition B.2, combined with the LAN expansion in Proposition B.3, implies that inference for the NA-IPCA estimator is naturally organised by parameter block. Because the latent span, the observable span, and the bounded intercept occupy mutually Q_Z -orthogonal components of $\Theta(\delta)$, the associated tangent-space directions remain statistically independent at first order, and each block admits its own set of interpretable hypotheses, confidence regions, and variance formulas. This geometry ensures that inference focuses on economically meaningful pricing directions rather than on individual coordinates of $(\Gamma_\beta, \Gamma_\delta, \Gamma_\alpha, F)$, which are not themselves identified.

For the latent block, inference concerns the Q_Z -span of Γ_β and the associated latent return process F .

Because only the span is identified, statistical statements are formulated in terms of Q_Z -orthogonal projections or linear contrasts of the form $c^\top (\Gamma_\beta, \Gamma_\beta F_t)$, where c is a user-specified test direction. The LAN covariance in Proposition B.3 provides standard errors for any such linear functional after projecting onto $T_{\theta_0} \Theta(\delta)$, and tests for the relevance of particular characteristics proceed by examining whether their Q_Z -projection onto the latent span is statistically different from zero. In empirical applications this means that economic interpretation is attached to factor directions—and their associated premia—rather than to individual loadings, whose absolute magnitudes depend on arbitrary rotation choices eliminated by Appendix B.6.

Inference for the observable block proceeds analogously, but is based on the Q_Z -orthogonal projection of Γ_δ onto the complement of the latent span. Because the block-orthogonality condition in Proposition B.2 ensures that observable traded or macro factors span pricing directions not already captured by characteristics, statistical tests evaluate whether this projected component is materially different from zero. For instance, hypotheses about the incremental pricing power of Fama–French or macro factors relative to the latent span correspond precisely to restrictions on the observable block of the information matrix in Proposition B.3. The resulting Wald or score tests quantify whether an observable factor adds distinctive explanatory content beyond what is contained in the characteristic space, with standard errors obtained from the observable portion of the LAN covariance operator.

The intercept component requires special treatment because it is constrained by the no-arbitrage curvature bound $\psi(\Gamma_\alpha) \leq \delta$ and is Q_Z -orthogonal to the latent span. Inference therefore focuses not on arbitrary linear combinations of Γ_α but on the curvature functional $\psi(\Gamma_\alpha) = \Gamma_\alpha^\top Q_Z \Gamma_\alpha$ and its sampling variation. Delta-method arguments applied to the LAN expansion yield standard errors and tests for whether the estimated curvature lies inside or outside an admissible no-arbitrage region (for example, testing whether $\psi(\Gamma_\alpha)$ exceeds a tighter bound $\delta_0 < \delta$). Economically, this assessment quantifies whether the remaining pricing component is consistent with small, economically admissible deviations from the characteristic-based pricing map, or whether it reflects a statistically significant departure from the no-arbitrage restrictions imposed in Appendix A. Because the intercept occupies its own Q_Z -orthogonal block in the tangent-space information matrix, inference on this residual component is cleanly separated from the latent and observable spans, ensuring that tests on Γ_α neither contaminate nor are contaminated by inference on characteristic-driven or factor-driven sources of priced risk.

Implementation follows directly from the LAN representation. In practice, inference proceeds by selecting an economically meaningful perturbation direction u that lies in the tangent space $T_{\theta_0} \Theta(\delta)$, so that u represents a genuinely identified, rotation-invariant pricing direction. Such a direction may correspond, for example, to: (i) a Q_Z -orthogonal contrast that tests whether a particular characteristic contributes to the latent span; (ii) a linear combination of observable factor loadings that captures the incremental pricing power of a traded factor relative to the latent block; or (iii) the derivative of the no-arbitrage curvature functional $\psi(\Gamma_\alpha)$ when assessing whether residual pricing errors remain within the economically admissible region. Formally, u is a vector in the same ambient space as θ but projected onto $T_{\theta_0} \Theta(\delta)$ so that it eliminates non-identified directions—such as rotations of $(\Gamma_\beta, \Gamma_\delta)$ or perturbations that violate the Q_Z -orthogonality conditions.

For any such identified direction u , the LAN expansion yields the rotation-invariant confidence interval

$$u^\top \Pi_{T_{\theta_0} \Theta(\delta)}(\hat{\theta}_T - \theta_0) \pm z_{1-\alpha/2} \sqrt{u^\top \mathcal{J}(\theta_0)^{-1} u},$$

where $\mathcal{J}(\theta_0)$ is the information operator restricted to the tangent space. The empirical version of $\mathcal{J}(\theta_0)$ is obtained from the feasible Huber/GLS score covariance discussed in Appendix C.4, and standard HAC or heteroskedasticity-robust corrections may be applied when serial or cross-sectional dependence is present. Delta-method derivatives then propagate uncertainty to any scalar functional of economic interest, including Q_Z -projected loadings, factor premia, or the no-arbitrage curvature $\psi(\Gamma_\alpha)$.

Because all inference is conducted on tangent-space directions, the resulting standard errors and hypothesis tests depend only on the economically identified components of the model and are therefore fully rotation-invariant. This guarantees that empirical statements about latent characteristic-driven risk, observable factor contributions, or residual no-arbitrage curvature are robust to reparameterisations of the loading matrices and consistent with the geometric structure established in Appendices A–C.

D.9 Integrated Summary

Collecting the results in Appendices A–D, the No-Arbitrage IPCA estimator emerges as a fully identified, rotation-invariant, and asymptotically efficient framework for estimating and testing the characteristic-based priced risk structure. The Q_Z -Stiefel geometry and block-orthogonality constraints of Appendix A provide the economically meaningful identification environment; the constrained KKT and LAN analysis of Appendix B deliver the projected score, tangent-space information operator, and local asymptotic normality; and the alternating projected GLS routine of Appendix C implements these objects numerically without ever leaving the feasible manifold $\Theta(\delta)$. The present appendix shows how feasible GLS weighting, rotation-normalised parameterisation, and blockwise inference translate this geometric and statistical structure into practical procedures for estimating latent characteristic-driven factors, assessing the incremental pricing power of observable traded or macro factors, and quantifying the economically admissible residual captured by the no-arbitrage intercept.

Taken together, these components form a unified inference apparatus for cross-sectional asset pricing models: the estimator respects the economic restrictions implied by no-arbitrage, inherits a well-defined asymptotic distribution on the quotient space $\Theta(\delta)/\mathcal{G}$, and provides rotation-invariant confidence sets and hypothesis tests for all Q_Z -identified directions. As such, the NA-IPCA framework integrates directly into empirical analyses alongside traditional factor models (e.g., Fama–French specifications) and modern statistical approaches such as IPCA or machine-learning asset-pricing methods, offering a coherent and theoretically grounded procedure for estimating and interpreting the structure of systematic return variation.

E Structural Comparison with the Statistical IPCA Estimator

The No-Arbitrage IPCA (NA-IPCA) estimator is built on the same statistical backbone as the IPCA estimator of Kelly et al. (2019): in both cases, expected returns are linked to firm characteristics through the cross-sectional moment

$$X_t = \frac{1}{N_t} Z_t' r_{t+1}, \quad W_t = \frac{1}{N_t} Z_t' Z_t,$$

and both approaches recover only the *span* of systematic risk exposures rather than a particular coordinate representation of the latent factors. Consequently, (Γ_β, F) and $(\Gamma_\beta R_\beta, R_\beta^\top F)$ generate identical model-implied returns for any orthogonal R_β , so both estimators must impose normalisation and sign conventions to deliver an economically interpretable factor representation.¹⁴⁹

Relative to this common starting point, NA-IPCA adds three layers of structure that are explicitly anchored in the no-arbitrage geometry of Appendices A–B. The first concerns the metric and the way orthogonality is defined. NA-IPCA is formulated on the Q_Z -Stiefel manifold and must satisfy the full set of constraints

$$\Gamma_\beta^\top Q_Z \Gamma_\beta = I_K, \quad \Gamma_\beta^\top Q_Z \Gamma_\delta = 0, \quad \Gamma_\beta^\top Q_Z \Gamma_\alpha = 0, \quad \psi(\Gamma_\alpha) \leq \delta, \quad (114)$$

rather than the purely Euclidean normalisation $\Gamma_\beta^\top \Gamma_\beta = I_K$ used in statistical IPCA.¹⁵⁰ Because Q_Z summarises the long-run covariance structure of firm characteristics, orthogonality with respect to this metric has a transparent economic interpretation: latent loadings, observable loadings, and the intercept are forced to occupy distinct priced directions in characteristic space, measured in the same “units” that determine how characteristics enter expected returns. This Q_Z -geometry is exactly what underpins the tangent-space representation and block-diagonal information matrix in Proposition B.3.¹⁵¹

¹⁴⁹This rotation indeterminacy is intrinsic to latent-factor models: only the space $\text{col}(\Gamma_\beta)$ is identified, not a particular basis. In the IPCA implementation, Kelly et al. (2019) handle this implicitly by imposing $\Gamma_\beta^\top \Gamma_\beta = I_K$ and by post-processing each ALS update with an SVD/QR step so that the factor covariance matrix is diagonal. NA-IPCA formalises the same issue as an explicit action of the orthogonal group $\mathcal{G} = \mathcal{O}(K) \times \mathcal{O}(K_{\text{add}})$ on the parameter space (Appendix A) and introduces a normalisation map $N(\cdot)$ (Appendix B.6) that selects a canonical representative of each rotation-equivalent class. Economically, these rotations merely relabel factor directions; they do not change the characteristic-based priced risk structure or the model-implied pricing kernel.

¹⁵⁰In the statistical IPCA implementation of Kelly et al. (2019), the characteristic space is effectively treated as Euclidean. The rank transformation standardises the marginal distribution of each characteristic, but it does not decorrelate characteristics or enforce $W_t \propto I_L$; in our notation, the population Gram operator $Q_Z = \text{plim}_{T \rightarrow \infty} T^{-1} \sum_t W_t$ is therefore not generally proportional to the identity. However, the IPCA normalisation step imposes $\Gamma_\beta^\top \Gamma_\beta = I_K$ and orthogonalises the PSF block via $(I_L - \Gamma_\beta \Gamma_\beta^\top) \Gamma_\delta$, both defined with respect to the *identity* inner product. In effect, the algorithm behaves as if the characteristic metric were $Q_Z = I_L$, even though no whitening beyond marginal rank transformations is performed.

In our framework, this corresponds to a special case in which the Q_Z -based restrictions of NA-IPCA collapse to the standard IPCA normalisations. If (i) the characteristic metric used for identification is effectively Euclidean (either because $Q_Z \propto I_L$ or because one chooses to work under the identity metric), (ii) the population block orthogonality $\Gamma_\beta^\top Q_Z \Gamma_\delta = 0$ reduces to $\Gamma_\beta^\top \Gamma_\delta = 0$, and (iii) the no-arbitrage intercept constraint is slack so that $\psi(\Gamma_\alpha) < \delta$, then the feasible manifold $\Theta(\delta)$ coincides with the IPCA identification space. Under these restrictions, the alternating WLS updates in Appendix C.1 become numerically identical (up to an orthogonal relabelling of the latent factors) to the ALS steps in the IPCA estimator of Kelly et al. (2019).

¹⁵¹It is important to distinguish between the *rank transformation* of characteristics—as used in Kelly et al. (2019) and adopted here—and an explicit *pre-whitening* of the characteristic space. Rank transformation standardises the marginal distribution of each characteristic by mapping it to the unit interval but leaves the joint dependence structure of Z_t intact. Consequently, $Q_Z = \text{plim}_{T \rightarrow \infty} T^{-1} \sum_t Z_t' Z_t / N_t$ continues to encode the economically relevant covariance structure of characteristics, which determines how characteristic portfolios span the cross section of expected returns. This matrix defines the inner product $\langle a, b \rangle_{Q_Z} = a^\top Q_Z b$, the orthogonality conditions in (101), and the no-arbitrage curvature $\psi(\Gamma_\alpha) = \Gamma_\alpha^\top Q_Z \Gamma_\alpha$ studied in Appendix B.

A second difference concerns the treatment of observable traded or macro factors. NA-IPCA imposes the block-orthogonality condition $\Gamma_\beta^\top Q_Z \Gamma_\delta = 0$ at the population level (Proposition B.2), which requires the latent characteristic-based loadings and the observable factor loadings to live in *distinct* Q_Z -orthogonal subspaces. By contrast, the IPCA implementation of Kelly et al. (2019) separates latent and pre-specified-factor (PSF) blocks algorithmically—splitting Γ_β and Γ_δ and applying a projection step $(I_L - \Gamma_\beta \Gamma_\beta^\top) \Gamma_\delta$ in the identity metric—without explicitly incorporating the characteristic metric Q_Z . When characteristics are correlated with traded or macro factors, such a Euclidean projection may misallocate characteristic-driven variation to the observable factor space (or vice versa), so that the same economic direction is partly absorbed by Γ_β and partly by Γ_δ . This “double counting” inflates the apparent importance of the latent block and blurs the interpretation of the observable proxies.¹⁵² By anchoring block orthogonality in Q_Z , NA-IPCA ensures that traded or macro factors absorb only the variation aligned with their own economic source, while the latent block is forced to capture the residual priced structure not already absorbed by the observable block. The resulting decomposition cleanly separates characteristic-driven risk and exogenous factor proxies in a way that is robust to correlation between characteristics and traded factors and is directly aligned with the no-arbitrage geometry in Appendices A–B.

The third layer of discipline concerns the intercept. In IPCA, the intercept is a free regression parameter and may absorb persistent mispricing, omitted-factor structure, or measurement noise that is correlated with the latent block. This creates two familiar issues: the intercept can load on the same characteristic directions as the latent factors—blurring the economic interpretation of those factors—and large estimated alphas can reflect violations of no-arbitrage rather than genuine residual premia. NA-IPCA instead treats Γ_α as a bounded residual pricing direction: it must be orthogonal to the latent span, $\Gamma_\beta^\top Q_Z \Gamma_\alpha = 0$, and must satisfy the no-arbitrage curvature bound $\psi(\Gamma_\alpha) \leq \delta$. In addition, the implementation allows for Huber-type loss and robust scale adjustments in the intercept step, so that a small number of extreme idiosyncratic returns cannot generate spurious “alpha factors” driven by outliers. Economically, this configuration forces the model to explain as much cross-sectional structure as possible through latent and observable factors, leaving Γ_α to capture only the economically admissible remainder of pricing errors. As shown in Appendix B, this constrained intercept geometry is precisely what links Γ_α to the curvature of the no-arbitrage pricing operator and ensures that the intercept does not contaminate the identification or efficiency of the latent and observable spans.

These conceptual distinctions translate directly into differences in the algorithms. In the statistical IPCA

By contrast, pre-whitening transforms characteristics as $\tilde{Z}_t = Q_Z^{1/2} Z_t$, forcing Q_Z to become the identity. While this may be numerically convenient, it destroys the economic geometry of the characteristic space: all characteristic directions are rescaled to have the same variance, correlated characteristics are forced to appear orthogonal, and the curvature $\psi(\Gamma_\alpha)$ loses its interpretation as a bound on no-arbitrage pricing. In such a whitened geometry, latent and observable factor blocks can be misallocated because the whitening rotation reshuffles the directions that truly span priced risk, undermining the block-orthogonality condition in Proposition B.2 and the LAN representation in Appendix B.3. For these reasons, NA-IPCA retains the empirically estimated Q_Z and applies rank transformation—but not whitening—so that the estimator respects the economic structure of the characteristic space while benefiting from the robustness of rank-based standardisation.

¹⁵²This concern is closely related to the spurious-factor and omitted-factor phenomena documented by Daniel and Titman (1997), Bryzgalova (2015), Kozak et al. (2018), Lettau and Pelger (2020b), and Giglio and Xiu (2021), where weakly identified or redundant factors over-represent essentially the same economic risk. Without an explicit geometric separation between latent and observable blocks, the same priced direction can be loaded multiple times, overstating explanatory power and complicating interpretation.

implementation, each ALS cycle operates in an implicitly Euclidean characteristic geometry: after updating the latent factor scores, the loading matrix is re-scaled via SVD/QR so that $\Gamma_\beta^\top \Gamma_\beta = I_K$, the PSF block is orthogonalised using $(I_L - \Gamma_\beta \Gamma_\beta^\top) \Gamma_\delta$, and the intercept is treated as an unconstrained regression constant. These steps enforce the normalisations required for identification, but they do so by projecting the estimates back into an auxiliary Euclidean space, even when the underlying Q_Z is far from the identity. By contrast, the NA-IPCA routine operates natively on the Q_Z -Stiefel manifold of Appendix A. Each step respects the no-arbitrage geometry rather than correcting for it ex post: the latent update solves the Q_Z -weighted first-order condition in Proposition B.1; the loading update solves the Kronecker-structured WLS system consistent with the tangent-space representation in Proposition B.3; and the identification step rotates and projects $(\Gamma_\beta, \Gamma_\delta, \Gamma_\alpha)$ using the population orthogonality relations in Proposition B.2. The resulting sequence of iterates evolves entirely within the feasible manifold $\Theta(\delta)$, so the Q_Z geometry, block structure, and no-arbitrage curvature are maintained throughout estimation rather than being imposed by a post-hoc normalisation.

These implementation differences matter in applications. When characteristics co-move strongly with traded risk factors or exhibit heterogeneous scaling, a Euclidean projection such as $(I_L - \Gamma_\beta \Gamma_\beta^\top)$ can distort the economically relevant partition of priced directions. In simulations and empirical work, such distortions typically manifest as (i) inflated latent-factor importance, because part of the observable factor direction is inadvertently absorbed into Γ_β ; (ii) unstable loadings across subsamples, because the Euclidean projection is sensitive to changes in the scaling and covariance structure of Z_t ; and (iii) spurious or oversized intercepts, because the unconstrained intercept absorbs priced directions that should be attributed to latent or observable factors. By embedding the Q_Z geometry and the no-arbitrage constraints directly into the update and projection steps, NA-IPCA eliminates these sources of instability and preserves the economic decomposition that underlies the LAN and block-diagonal information results in Appendix B.

Overall, the extra structure introduced by NA-IPCA is designed to tighten the economic interpretation of the factor space rather than to change the core statistical engine of IPCA. By forcing the estimator to remain on the Q_Z -orthogonal manifold and by separating latent, observable, and intercept components through explicit no-arbitrage constraints, NA-IPCA delivers a decomposition of expected returns in which each block corresponds to a distinct source of priced risk: latent characteristic-driven factors, observable traded or macro factors, and a bounded no-arbitrage intercept. In this sense, NA-IPCA can be viewed as an economically disciplined version of IPCA: it preserves the data-driven flexibility and alternating least-squares structure of Kelly et al. (2019), but embeds those updates in a no-arbitrage geometry that prevents spurious pricing directions and clarifies which parts of the factor structure are pinned down by economic restrictions rather than by purely statistical normalisations.

F Geometric and Economic Interpretation of Identification Diagnostics

This appendix develops the geometric and economic rationale for the identification diagnostics used in Section 7.2 and throughout Section 7. The goal is to formalize why the four measures employed in the Monte Carlo analysis—subspace distances, Q_Z -based overlaps, blockwise misallocation matrices, and the comparison between return fit and factor-span recovery—constitute an appropriate toolkit for assessing whether an estimator respects the no-arbitrage manifold $\Theta(\delta)$ developed in Sections 3–5. Classical diagnostics in the factor-model literature, such as variance shares, Euclidean canonical angles, or simple loading correlations, are designed primarily for variance extraction rather than for economic identification. They cannot detect the specific failure modes that arise when characteristic geometry is misspecified: span rotations toward high-variance but low-pricing directions, double counting between latent and observable blocks, and leakage of systematic variation into the non-systematic component.

The diagnostics introduced here extend the canonical-angle and projection geometry of [Stewart and Sun \(1990\)](#) and [Edelman et al. \(1998\)](#) to the characteristic metric Q_Z , and connect them to the approximate-factor framework of [Bai \(2003\)](#) and [Bai and Ng \(2008\)](#). In characteristic-based asset pricing, the relevant inner product is not Euclidean but induced by the long-run characteristic covariance Q_Z ; this metric weights each direction by its cross-sectional informativeness ([Kelly et al., 2019](#)) and underpins the orthogonality structure of latent, observable, and non-systematic blocks in Sections 2.2–2.3. Measuring distances and overlaps in this metric, rather than in raw Euclidean coordinates, allows the diagnostics to distinguish between harmless variance rotations and economically meaningful misallocation of priced risk.

Each diagnostic isolates a complementary dimension of identification. Subspace distances summarize how far the estimated latent or observable factor spaces deviate from the true priced spans, up to rotation, and thus quantify span recovery in a scale-free way. Q_Z -based overlap measures test whether block orthogonality holds, and therefore whether the latent, observable, and non-systematic components represent distinct economic directions rather than repeated loadings on the same characteristic portfolios. The misallocation matrix provides a blockwise variance decomposition in the Q_Z -metric, attributing the true systematic variation to estimated latent and observable blocks and revealing whether systematic pricing information is assigned to the wrong component. Finally, comparing return R^2 with subspace recovery clarifies the trade-off between in-sample fit and geometric discipline: procedures such as PCA or unrestricted IPCA can achieve very high R^2 by exploiting variance-driven directions in a flat geometry, while the no-arbitrage estimator sacrifices some fit to remain confined to the tangent space of $\Theta(\delta)$.

In empirical characteristic panels for equities and bonds, where correlation, heteroskedastic scaling, and curvature contrasts are the norm rather than the exception, these diagnostics provide a geometry-aware benchmark for evaluating factor methods. They allow us to ask not only whether an estimator explains returns ex post, but whether it recovers the economically relevant factor spans, keeps the blocks separated in the Q_Z -metric, and respects the no-arbitrage decomposition of systematic and non-systematic components. The subsequent subsections define each diagnostic formally and link its numerical scale to its economic interpretation.

F.1 Subspace Distance and Canonical Angles

Let G_{true} and G_{est} denote the true and estimated loading matrices for either the latent block or the observable block. Following Bai (2003) and Bai and Ng (2008), identification of a factor space is assessed through the canonical angles obtained from the thin QR decompositions of the two spans. If $\Theta = (\theta_1, \dots, \theta_K)$ denotes these angles, the corresponding subspace distance is

$$\|\sin \Theta\|_F = \left(\sum_{j=1}^K \sin^2 \theta_j \right)^{1/2}.$$

Because each canonical angle satisfies $0 \leq \theta_j \leq \pi/2$, the metric obeys

$$0 \leq \|\sin \Theta\|_F \leq \sqrt{K}.^{153}$$

The lower bound is attained only when the estimated and true factor spaces coincide exactly, whereas the upper bound corresponds to mutually orthogonal subspaces. For the latent block with $K_\beta = 3$, the maximal value is $\sqrt{K_\beta} \approx 1.73$. Values below 0.05 indicate essentially exact recovery of the economically relevant factor space, values in the range 0.10–0.30 correspond to noticeable but moderate rotation, and values above 0.50—especially those approaching 1.0—reflect severe misidentification in which the estimated span captures little of the true priced directions.

In the NA-IPCA geometry, this metric has an immediate economic interpretation. The true latent span corresponds to the Q_Z -high-curvature directions that support systematic compensation for risk.¹⁵⁴ A small subspace distance therefore signals that Γ_β lies close to the no-arbitrage tangent space of priced characteristic portfolios, so the resulting factors represent true systematic risk.

Large subspace distances arise when an estimator rotates toward directions that appear important under Euclidean variance but carry little pricing content in the Q_Z metric. This is the characteristic failure mode of PCA and statistical IPCA documented in Sections 5 and 7.3: the flat geometry implicitly overweights

¹⁵³The bound $0 \leq \|\sin \Theta\|_F \leq \sqrt{K}$ follows directly from the properties of canonical angles. Each angle θ_j between two K -dimensional subspaces lies in the interval $[0, \pi/2]$, so $\sin \theta_j \in [0, 1]$. Since the subspace distance is defined as

$$\|\sin \Theta\|_F = \left(\sum_{j=1}^K \sin^2 \theta_j \right)^{1/2},$$

the smallest value is attained when all angles are zero, meaning the two subspaces coincide exactly; in this case $\|\sin \Theta\|_F = 0$. The largest value occurs when all $\theta_j = \pi/2$, i.e., when the two K -dimensional subspaces are mutually orthogonal, so each $\sin \theta_j = 1$ and

$$\|\sin \Theta\|_F = \sqrt{\sum_{j=1}^K 1^2} = \sqrt{K}.$$

Geometrically, canonical angles measure how much of each true direction is lost when projecting onto the estimated span. Economically, when $\|\sin \Theta\|_F$ approaches its upper bound $\sqrt{K}$, the estimated loading block spans directions that are orthogonal—in the Q_Z metric—to the priced characteristic portfolios, implying near-complete failure of factor identification. Values close to zero indicate that the estimated span lies almost entirely within the priced tangent space, so the latent factors represent genuine systematic risk rather than variance-driven rotations.

¹⁵⁴In characteristic-based models, the relevant notion of orthogonality is Q_Z -orthogonality rather than Euclidean orthogonality; see Appendix A and Sections 2.2–2.3. Consequently, deviation measured by $\|\sin \Theta\|_F$ reflects economic misalignment with priced characteristic portfolios rather than harmless statistical rotation.

high-variance characteristic combinations and draws the latent span away from the economically meaningful directions. As shown in Section 7.2, these rotations can generate latent errors in the 0.7–1.0 range, indicating near-orthogonality with the true pricing space despite very high in-sample return fit. Subspace distance thus serves as the primary diagnostic for determining whether an estimator recovers the systematic pricing directions implied by the no-arbitrage parameter space $\Theta(\delta)$.

F.2 Q_Z -Based Overlap Measures

Given two loading blocks G_1 and G_2 , their Q_Z -overlap is defined as

$$\text{Ov}(G_1, G_2) = \|G_1^\top Q_Z G_2\|_F.$$

When the columns of G_1 and G_2 are Q_Z -orthonormal, this quantity lies between 0 and $\sqrt{K_1 K_2}$, where K_1 and K_2 are the column dimensions of G_1 and G_2 .¹⁵⁵ In practice, we therefore report the normalized overlap $\text{Ov}(G_1, G_2)/\sqrt{K_1 K_2}$, which takes values in $[0, 1]$ and is directly comparable across designs and block dimensions. Values below 10^{-3} are numerically indistinguishable from zero in our simulations and correspond to essentially perfect block separation; values around 10^{-2} reflect small but economically minor leakage; values on the order of 10^{-1} indicate substantial double counting of the same characteristic-based pricing direction.

The matrix $G_1^\top Q_Z G_2$ measures the degree to which the characteristic-sorted portfolios spanned by G_2 are also spanned by G_1 , weighting each characteristic by its long-run informativeness under Q_Z . In the NA-IPCA geometry, the latent, observable, and non-systematic components are constructed to occupy mutually orthogonal Q_Z -subspaces, so the theoretical benchmark is zero overlap between any two distinct blocks. Accordingly, NA-IPCA should produce normalized overlaps that are effectively zero up to numerical error. Statistical IPCA and PCA, by contrast, impose orthogonality only in the Euclidean sense and do not respect the Q_Z -metric, so there is no reason for their latent and observable blocks to be Q_Z -orthogonal even when the underlying economic structure is block-separated.

Economically, Q_Z -overlap is a block-level diagnostic of identification. For example, if the latent block

¹⁵⁵The bound $0 \leq \text{Ov}(G_1, G_2) \leq \sqrt{K_1 K_2}$ follows directly from the geometry of Q_Z -orthonormal loading blocks. When the columns of G_1 and G_2 satisfy $G_1^\top Q_Z G_1 = I_{K_1}$ and $G_2^\top Q_Z G_2 = I_{K_2}$, each entry of the matrix $G_1^\top Q_Z G_2$ represents the Q_Z -inner-product (i.e., the Q_Z -weighted cosine) between one column of G_1 and one column of G_2 . Because any two Q_Z -unit vectors have Q_Z -inner-product in the interval $[-1, 1]$, each element of $G_1^\top Q_Z G_2$ lies in $[-1, 1]$, and therefore the squared entries lie in $[0, 1]$.

The overlap measure is the Frobenius norm

$$\text{Ov}(G_1, G_2) = \|G_1^\top Q_Z G_2\|_F = \left(\sum_{j=1}^{K_1} \sum_{\ell=1}^{K_2} |(G_1^\top Q_Z G_2)_{j\ell}|^2 \right)^{1/2}.$$

Since there are $K_1 K_2$ entries and each squared entry is at most 1, the sum of squares is bounded above by $K_1 K_2$, yielding

$$0 \leq \text{Ov}(G_1, G_2) \leq \sqrt{K_1 K_2}.$$

Geometrically, the matrix $G_1^\top Q_Z G_2$ collects the cosines of the canonical angles between the two Q_Z -orthonormal blocks. A value of zero corresponds to perfect Q_Z -orthogonality of the two spans, meaning they price entirely distinct Q_Z -weighted characteristic portfolios. The upper bound $\sqrt{K_1 K_2}$ is reached only when every column of G_1 perfectly aligns (in the Q_Z metric) with every column of G_2 , implying that the two blocks price exactly the same characteristic-synthesized portfolios. Economically, overlap close to zero indicates clean separation of systematic risk channels, while overlap approaching $\sqrt{K_1 K_2}$ signals complete double counting of priced directions.

$\hat{\Gamma}_\beta$ and the observable block $\hat{\Gamma}_\delta$ have a large $\text{Ov}(\beta, \delta)$, then both sets of estimated loadings are effectively pricing the same characteristic-synthesized portfolios. In that case the “latent” factors are not economically distinct from the observable proxies: they are simply re-labeled versions of the same risk direction. A similar interpretation applies to the overlap between $\hat{\Gamma}_\beta$ or $\hat{\Gamma}_\delta$ and the intercept block $\hat{\Gamma}_\alpha$, which signals contamination of systematic pricing spans by the non-systematic component. In NA-IPCA, near-zero overlaps confirm that latent factors, observable proxies, and the bounded non-systematic direction occupy different parts of the Q_Z -geometry and can be interpreted as separate sources of priced risk.

This diagnostic goes beyond classical variance-based tools. Traditional practice evaluates factor distinctness using correlations between factor returns, loadings plotted in Euclidean space, or explained-variance shares. Such measures are insensitive to the characteristic geometry and may understate the extent of double counting when characteristics are highly correlated. By embedding the metric Q_Z directly in $\text{Ov}(G_1, G_2)$, the overlap measure tests whether blocks are distinct in the economically relevant sense: they must price different Q_Z -weighted characteristic portfolios rather than simply span different directions in an unweighted Euclidean space.¹⁵⁶

F.3 Misallocation Matrix

To quantify how an estimator allocates priced variation across latent and observable blocks, it is useful to summarize its behavior in a simple 2×2 matrix. Given true loading blocks $(\Gamma_\beta, \Gamma_\delta)$ for the latent and observable spans, and estimated blocks $(\Gamma_\beta^{\text{hat}}, \Gamma_\delta^{\text{hat}})$, we define the misallocation matrix

$$H = \frac{\begin{pmatrix} \|\Gamma_\beta^\top Q_Z \Gamma_\beta^{\text{hat}}\|_F^2 & \|\Gamma_\beta^\top Q_Z \Gamma_\delta^{\text{hat}}\|_F^2 \\ \|\Gamma_\delta^\top Q_Z \Gamma_\beta^{\text{hat}}\|_F^2 & \|\Gamma_\delta^\top Q_Z \Gamma_\delta^{\text{hat}}\|_F^2 \end{pmatrix}}{\sum_{i \in \{\beta, \delta\}} \sum_{j \in \{\beta, \delta\}} \|\Gamma_i^\top Q_Z \Gamma_j^{\text{hat}}\|_F^2}.$$

Each entry is nonnegative and the four entries sum to one, so H can be read as a probability-like decomposition of true priced variation across estimated blocks. The $(1, 1)$ element measures the share of latent Q_Z -risk correctly assigned to the estimated latent block, while $(2, 2)$ measures the share of observable priced variation correctly assigned to the observable block. The off-diagonal elements capture leakage: $H_{\beta, \delta}$ is the share of true latent variation that the estimator loads into the observable block, and $H_{\delta, \beta}$ is the share of true observable variation loaded into the latent block.

Economically, this matrix reveals whether the model distinguishes distinct sources of priced risk or merely relabels the same characteristic portfolios in different blocks. A nearly diagonal H indicates that latent factors span the directions along which investors are compensated for bearing non-observable risk, while observable proxies absorb only those components not already spanned by the latent block. Large off-diagonal mass, in

¹⁵⁶From an asset-pricing perspective, two factor blocks with nonzero Q_Z -overlap price the same characteristic-synthesized portfolios and therefore do not represent economically distinct risks. Overlap in statistical IPCA thus reflects geometric misspecification: the estimator confounds systematic and characteristic-based variation because its orthogonality is defined in Euclidean rather than Q_Z units; see, for example, Kelly et al. (2019) and Lettau and Pelger (2020b) for related discussions of factor redundancy and double counting in characteristic-based models.

contrast, signals structural misidentification. When $H_{\beta,\delta}$ is sizable, part of the true latent risk is erroneously attributed to observable instruments, so the same priced direction is effectively double counted in both blocks. When $H_{\delta,\beta}$ is large, the estimator absorbs observable pricing directions into the latent block, generating pseudo-factors that simply repackage traded or macro instruments as “new” latent sources of risk. Both forms of leakage undermine economic interpretation: factor risk premia attached to latent and observable blocks can no longer be viewed as compensation for distinct risk channels.

This diagnostic is particularly informative in high-dimensional characteristic environments where collinearity is strong. Variance-based estimators such as PCA or statistical IPCA tend to align with high-variance portfolios of characteristics, even when those portfolios proxy for the same underlying risk. In such cases, subspace distances and overlap measures can show that the estimated spans are rotated and non-orthogonal, but they do not indicate how priced variation has been reassigned across blocks. The misallocation matrix fills this gap: it shows not only that the geometry is wrong, but also which part of the economic decomposition has failed.

Within the no-arbitrage geometry of $\Theta(\delta)$, the latent, observable, and non-systematic components are required to satisfy

$$\text{col}(\Gamma_\beta) \perp_{Q_Z} \text{col}(\Gamma_\delta) \perp_{Q_Z} \text{col}(\Gamma_\alpha),$$

so correctly specified estimators should deliver H close to the identity matrix. In our simulations, NA-IPCA produces misallocation matrices that are essentially diagonal across all designs, consistent with its Q_Z -orthogonal block structure and curvature restrictions on Γ_α . Statistical IPCA, by contrast, exhibits substantial off-diagonal entries, especially when characteristics are highly correlated. This pattern confirms the double-counting and factor-splitting mechanisms predicted by Euclidean geometry, and explains why IPCA can achieve high in-sample fit while failing to produce a clean separation between latent and observable sources of systematic risk.

F.4 Interpretation Under the No-Arbitrage Geometry

The no-arbitrage parameter space $\Theta(\delta)$ induces a geometric decomposition of characteristic-based pricing directions in the Q_Z metric: the latent span $\text{col}(\Gamma_\beta)$, the observable span $\text{col}(\Gamma_\delta)$, and the non-systematic component $\text{col}(\Gamma_\alpha)$ must occupy mutually orthogonal subspaces, with $\Gamma_\beta^\top Q_Z \Gamma_\beta = I_K$ normalizing the priced tangent space and $\psi(\Gamma_\alpha) \leq \delta$ restricting residual curvature. Economically, this means that systematic compensation for risk lives in a small number of high-curvature characteristic directions; observable proxies contribute only in directions not already spanned by the latent block; and the intercept must remain confined to a bounded normal cone that cannot masquerade as priced risk.

Within this geometry, the diagnostics developed in Appendix F evaluate whether an estimator respects the structure of $\Theta(\delta)$. Subspace distances quantify whether the estimated spans align with the priced tangent space in the Q_Z metric rather than with high-variance directions that are economically irrelevant. Overlap measures test whether latent, observable, and residual components remain separated in the Q_Z geometry, detecting violations of block orthogonality and revealing instances of double counting or contamination. The misallocation matrix provides a blockwise decomposition of priced variation, showing whether latent risk is

erroneously assigned to the observable block or vice versa. Finally, comparing return fit with span recovery illuminates the distinction between statistical performance and economic identification: two estimators may produce similar in-sample R^2 while having sharply different geometric properties.

A central concern is whether using Q_Z -based diagnostics to evaluate Euclidean estimators such as statistical IPCA is unfair. It is not. The metric Q_Z is not an arbitrary choice of coordinates but an economic object: it is the long-run Gram matrix of characteristics and therefore encodes the directions along which characteristic-sorted portfolios carry pricing information. Identification must therefore be assessed in the Q_Z geometry, because the relevant notion of orthogonality—and thus the separation of priced and non-priced directions—is determined by Q_Z , not by the Euclidean metric implicitly imposed by PCA and statistical IPCA. When IPCA exhibits large Q_Z subspace distances or substantial block overlaps, this does not reflect a biased evaluation but the fact that Euclidean rotations do not align with the economically meaningful curvature of characteristic space.

Taken together, these diagnostics constitute an operational test of compatibility with the no-arbitrage manifold. Estimators consistent with $\Theta(\delta)$ exhibit small latent subspace distances, near-zero overlaps across blocks, misallocation matrices close to diagonal, and residual curvature confined to its admissible region. Across the designs studied in Sections 7.2 and 7.3, NA-IPCA satisfies all of these criteria: it recovers the priced tangent space, preserves Q_Z -orthogonality, and assigns systematic variation to economically coherent blocks. Euclidean estimators such as PCA and statistical IPCA do not; they rotate toward high-variance directions, split priced directions across blocks, and generate pseudo-factors even when in-sample fit is high. Appendix F therefore translates the geometric content of the no-arbitrage model into a practical identification framework for evaluating factor estimators in characteristic-rich environments.

G α -Curvature Projection, Robust Estimation, and LAN Inference for $\psi(\Gamma_\alpha)$

This appendix links the curvature functional for the intercept block, $\psi(\Gamma_\alpha)$, to its finite-sample implementation and clarifies how the α -projection step used in the numerical algorithm corresponds to the no-arbitrage geometry of the parameter space. The starting point is the population restriction

$$\Theta(\delta) = \left\{ (\Gamma_\beta, F, \Gamma_\delta, \Gamma_\alpha) : \Gamma_\beta^\top Q_Z \Gamma_\beta = I_K, \Gamma_\beta^\top Q_Z \Gamma_\delta = 0, \Gamma_\beta^\top Q_Z \Gamma_\alpha = 0, \psi(\Gamma_\alpha) \leq \delta \right\}, \quad (115)$$

where Q_Z is the long-run characteristic metric defined in Section 2.2 and Appendix A. The curvature functional for the non-systematic component is

$$\psi(\Gamma_\alpha) = \Gamma_\alpha^\top Q_Z \Gamma_\alpha. \quad (116)$$

In this geometry, Q_Z induces the inner product that governs all notions of length and orthogonality in characteristic space, and $\psi(\Gamma_\alpha)$ is the squared Q_Z -norm of the intercept direction. The constraint $\psi(\Gamma_\alpha) \leq \delta$ therefore requires the intercept to lie in a bounded ellipsoid and rules out residual pricing directions that would imply implausibly large Sharpe ratios on characteristic portfolios, in line with the bounds of Hansen and Jagannathan (1991) and the limited-arbitrage curvature analyses in Dello Preite et al. (2024) and Zaffaroni (2025). Appendix B shows that this inequality enters the KKT system as a curvature constraint on the normal cone of $\Theta(\delta)$ and isolates the intercept as a non-systematic direction orthogonal, in the Q_Z metric, to both the latent and observable spans.

G.1 Finite-Sample Curvature Operator and the α -Projection Map

In finite samples the functional $\psi(\Gamma_\alpha)$ is not observed and must be estimated from the data. The implementation replaces the population metric Q_Z by a sequence of time- t curvature operators $\hat{H}_t$ constructed from residuals of the NA-IPCA pricing equation. Let $Z_t \in \mathbb{R}^{N_t \times L}$ denote the rank-transformed characteristics at date t , and let $r_t \in \mathbb{R}^{N_t}$ denote the corresponding cross-section of excess returns after aligning the panel as in Section 2.1. Given a parameter tuple $(\Gamma_\beta, F, \Gamma_\delta, \Gamma_\alpha)$, the fitted returns and residuals are

$$\text{comp}_t = Z_t (\Gamma_\beta f_t + \Gamma_\delta F_t^{\text{obs}} + \Gamma_\alpha \alpha_t), \quad e_t = r_t - \text{comp}_t, \quad (117)$$

where f_t and F_t^{obs} are the latent and observable factor realizations and $\alpha_t \equiv 1$ by construction. The residuals e_t represent the idiosyncratic component of returns after pricing by the latent, observable, and intercept blocks and are the natural objects for assessing how much curvature the current intercept direction contributes to non-systematic risk.

For each date t , the algorithm computes robust residual-scale estimates from e_t by applying a small positive floor, optional winsorisation of extreme values, optional capping based on a multiple of the median absolute deviation, and shrinkage toward the cross-sectional mean. Denoting the squared residuals by $v_i = e_{it}^2$,

the resulting shrunk variances and precision weights are

$$v_i^{\text{shr}} = (1 - \rho) v_i + \rho \bar{v}, \quad \bar{v} = \frac{1}{N_t} \sum_{i=1}^{N_t} v_i, \quad \text{inv} V_i = \frac{1}{v_i^{\text{shr}}}, \quad (118)$$

with $\rho \in [0, 1]$ a shrinkage parameter. These residual-scale procedures are of the same type as those used in Appendix A to construct robust WLS weights and satisfy the moment and mixing conditions in Assumption A.8, ensuring that the resulting precision weights are uniformly bounded above and away from zero with high probability.

The time- t curvature operator in characteristic space is then defined as

$$\hat{H}_t = Z_t^\top \hat{D}_t^{-1} Z_t, \quad \hat{D}_t = \text{diag}(v_1^{\text{shr}}, \dots, v_{N_t}^{\text{shr}}), \quad (119)$$

so that, for any loading direction $g \in \mathbb{R}^L$, the quadratic form $g^\top \hat{H}_t g$ measures the precision-weighted residual variance generated by the portfolio $Z_t g$ at date t . The pointwise curvature of the current intercept direction is

$$\psi_t(\Gamma_\alpha) = \Gamma_\alpha^\top \hat{H}_t \Gamma_\alpha, \quad (120)$$

and the finite-sample curvature functional used in the algorithm is the worst-case value across dates,

$$\hat{\psi}_T(\Gamma_\alpha) = \max_{1 \leq t \leq T} \psi_t(\Gamma_\alpha) = \max_{1 \leq t \leq T} \Gamma_\alpha^\top \hat{H}_t \Gamma_\alpha. \quad (121)$$

This routine returns precisely this scalar $\hat{\psi}_T(\Gamma_\alpha)$ and passes it to the α -projection step described below.

The operators $\hat{H}_t$ play the same role for curvature that the robust covariance estimators in Appendix A play for GLS weighting. They are symmetric positive-definite by construction and, under Assumptions A.1–A.3 and A.8, their averages converge to a population operator that is equivalent to Q_Z up to a scalar factor. The following lemma formalises this link between $\psi(\Gamma_\alpha)$ and $\hat{\psi}_T(\Gamma_\alpha)$.

Lemma G.1. *Suppose Assumptions A.1–A.3 and A.8 hold and fix any $\Gamma_\alpha \in \mathbb{R}^L$. Then there exists a positive sequence c_T converging in probability to a finite nonzero constant c such that*

$$\hat{\psi}_T(\Gamma_\alpha) = c_T \psi(\Gamma_\alpha) + o_p(1), \quad \psi(\Gamma_\alpha) = \Gamma_\alpha^\top Q_Z \Gamma_\alpha.$$

In particular, up to a multiplicative constant that can be absorbed into the choice of δ , the finite-sample functional $\hat{\psi}_T(\Gamma_\alpha)$ is a ratio-consistent estimator of the population curvature functional $\psi(\Gamma_\alpha)$.

The proof follows the same arguments as in Appendix A. Under the moment and mixing conditions, the robust residual-scale procedure in (118) yields diagonal variance matrices $\hat{D}_t$ whose entries are uniformly bounded and converge in probability to their population counterparts. The quadratic forms $\Gamma_\alpha^\top \hat{H}_t \Gamma_\alpha$ therefore converge, uniformly over compact sets of Γ_α , to a limit proportional to $\Gamma_\alpha^\top Z_t^\top Z_t \Gamma_\alpha$, and averaging over t recovers a matrix proportional to Q_Z as in (116). The use of the maximum over t in (121) strengthens the bound by focusing on the largest realised curvature, but does not alter the ratio-consistency of the functional under the stated conditions.

The curvature constraint is imposed in the algorithm by applying a projection map to the unconstrained intercept update. Let G_a denote the unconstrained WLS update for Γ_α and let G_b denote the current latent loading block Γ_β . The projection consists of two operations. First, the intercept is rescaled so that the curvature bound $\hat{\psi}_T(\Gamma_\alpha) \leq \delta$ is satisfied. If $\hat{\psi}_T(G_a) \leq \delta$, the unconstrained update is already feasible and no rescaling is needed. If $\hat{\psi}_T(G_a) > \delta$, the algorithm multiplies G_a by a scalar $s_1 \in (0, 1)$ that moves the curvature from $\hat{\psi}_T(G_a)$ to approximately $(1 - \varepsilon)\delta$ for a small safety margin $\varepsilon \in (0, 1)$, and uses a simple backtracking line search to ensure that the WLS objective in the α -subproblem decreases sufficiently along this direction. Because $\hat{\psi}_T(\cdot)$ is quadratic in Γ_α and the WLS criterion is smooth in a neighbourhood of the constrained solution, this step respects the KKT stationarity conditions to first order while placing the intercept inside the admissible curvature region.

Second, the projected intercept must lie in the normal cone of the latent span. In the population geometry this corresponds to the Q_Z -orthogonality condition $\Gamma_\beta^\top Q_Z \Gamma_\alpha = 0$ in (115). In the implementation, this is enforced by removing the component of the scaled intercept that lies in the span of G_b and adjusting the latent factors so that the fitted characteristic-space returns remain unchanged. If $\gamma_{ba} = G_b^\top G_a$ denotes the overlap between the latent and intercept blocks, the latent factor path is shifted by $\gamma_{ba}\alpha_t$ so that $Z_t(\Gamma_\beta f_t + \Gamma_\alpha \alpha_t)$ is preserved, and the intercept is replaced by its Euclidean orthogonal projection onto the complement of G_b . In the strict Q_Z -version, this projection is performed in the Q_Z -metric by using the complement operator $M_\beta = I - P_\beta$ from Proposition B.2. The curvature is then recomputed using $\hat{\psi}_T(\cdot)$ and, if necessary, a second rescaling and projection step is performed to ensure that the final intercept loading satisfies both the curvature bound and the orthogonality condition.

These operations can be summarised as a projection map Π_α that takes an unconstrained update G_a and returns a feasible intercept loading $\Pi_\alpha(G_a)$ lying in the normal cone of $\Theta(\delta)$ and satisfying $\hat{\psi}_T(\Pi_\alpha(G_a)) \leq \delta$ up to numerical tolerance. The following proposition formalises the effect of this map on the estimator.

Proposition G.2. *Let Assumptions A.1–A.11 hold and suppose the unconstrained WLS update for Γ_α yields G_a in a neighbourhood of the true parameter. Consider the projected update $\Pi_\alpha(G_a)$ obtained by the curvature-rescaling and orthogonalisation steps described above. Then, for any fixed $\delta \in (0, \infty)$,*

(i) *the projected update satisfies the feasibility conditions*

$$\hat{\psi}_T(\Pi_\alpha(G_a)) \leq \delta, \quad \Gamma_\beta^\top Q_Z \Pi_\alpha(G_a) = 0$$

with probability approaching one as $T \rightarrow \infty$;

(ii) *the change in the parameter vector induced by the projection is of smaller order than the LAN rate,*

$$\|\Pi_\alpha(G_a) - G_a\| = o_p(T^{-1/2}),$$

so that the LAN expansion and block-diagonal information matrix in Proposition B.3 remain valid with Γ_α replaced by $\Pi_\alpha(G_a)$.

The first part follows by construction from Lemma G.1 and from the properties of the projection operator. The second part is a consequence of the fact that the curvature bound binds only on a compact region of the parameter space, that the unconstrained update lies in an $O_p(T^{-1/2})$ neighbourhood of the true parameter,

and that the projection perturbs the intercept block only along the normal cone defined by $\psi(\Gamma_\alpha) \leq \delta$ without altering the fitted characteristic-space moments. As a result, the α -projection step does not affect the first-order behaviour of the estimator and leaves the tangent-space LAN representation and block-separate information matrix unchanged.

The construction of $\hat{H}_t$ and $\hat{\psi}_T(\Gamma_\alpha)$ closely parallels the robust WLS weighting scheme in Appendix A. Both sets of operators are symmetric positive-definite, constructed from residual-based scale estimates, and required only to be ratio-consistent for their population analogues. The difference is that the WLS weights $\hat{\Omega}_t^{-1}$ determine the metric of the objective function, while the curvature operators $\hat{H}_t$ determine the geometry of the no-arbitrage constraint. Under the empirical characteristic geometry documented in Section 6, the operators $\hat{H}_t$ are asymptotically equivalent to rescaled versions of $Z_t^\top Z_t$, and the projection map Π_α enforces the curvature bound and orthogonality in a metric that converges to the same Q_Z -geometry used to define $\Theta(\delta)$.

The simulation experiments in Section 7 provide an empirical counterpart to these theoretical arguments. In the geometry-calibrated data-generating process of Section 7.1, the NA-IPCA estimator recovers the latent Q_Z -span almost exactly while the α -projection step sharply reduces the raw curvature $\hat{\psi}_T(\Gamma_\alpha)$ to levels consistent with the chosen bound δ . At the same time, the estimated intercept remains orthogonal, in the Q_Z metric, to the latent and observable spans and exhibits a stable evolution across sample sizes, as documented in Tables 8, 12, and Figures 5–13. In contrast, statistical IPCA does not impose any curvature or orthogonality restriction on the intercept and therefore allows the α -component to absorb systematic variation and to align with high-curvature characteristic directions, generating double counting and spurious “alpha portfolios” in the sense of Section 7.2. The curvature functional $\hat{\psi}_T(\Gamma_\alpha)$ and the projection map Π_α are thus the finite-sample devices through which the population no-arbitrage geometry of $\Theta(\delta)$ is enforced in the NA-IPCA implementation, ensuring that the intercept remains a disciplined asset-specific risk component rather than an additional factor.

G.2 LAN Representation of the Intercept Block

The intercept block enters the LAN expansion through the same tangent-space geometry that governs the latent and observable components. Let

$$\theta = (\Gamma_\beta, F, \Gamma_\delta, \Gamma_\alpha), \quad \theta_0 = (\Gamma_{\beta,0}, F_0, \Gamma_{\delta,0}, \Gamma_{\alpha,0}) \in \Theta(\delta),$$

and recall that the moment map is

$$g_t(\theta) = X_t - W_t(\Gamma_\beta f_t + \Gamma_\delta F_t^{\text{obs}} + \Gamma_\alpha \alpha_t),$$

with $\alpha_t \equiv 1$ by construction. Appendix B shows that, under Assumptions A.1–A.11, the NA-IPCA estimator satisfies the LAN expansion

$$\sqrt{T} \Pi_{T_{\theta_0} \Theta(\delta)}(\hat{\theta}_T - \theta_0) = \left(\mathcal{J}(\theta_0)|_{T_{\theta_0} \Theta(\delta)} \right)^{-1} \frac{1}{\sqrt{T}} \sum_{t=1}^T \nabla_{\theta} g_t(\theta_0)^\top \Psi_t + o_p(1), \quad (122)$$

where Ψ_t is the robust score innovation and $\mathcal{I}(\theta_0)|_{T_{\theta_0}\Theta(\delta)}$ is the Fisher information operator restricted to the tangent space. The orthogonality constraints

$$\Gamma_{\beta,0}^\top Q_Z \Gamma_{\delta,0} = 0, \quad \Gamma_{\beta,0}^\top Q_Z \Gamma_{\alpha,0} = 0,$$

imply that the information operator is block-diagonal across latent, observable, and intercept components; see Proposition B.2. Consequently, the vectorised intercept block $a = \text{vec}(\Gamma_\alpha)$ admits its own LAN component

$$\sqrt{T} \Pi_{T_{\theta_0}\Theta(\delta)} a_T \Rightarrow \mathcal{N}(0, I_{\alpha\alpha}^{-1}),$$

with $I_{\alpha\alpha}$ denoting the α -block of the tangent-space information matrix. Because the moment map is linear in Γ_α through $W_t \Gamma_\alpha \alpha_t$, the intercept information combines a time-series component driven by $(\alpha_t, \Omega_t^{-1})$ with a characteristic-space component driven by the average of W_t , which converges to Q_Z . In this sense the α -block information is the curvature of the WLS loss in the residual Q_Z -direction, and the block-diagonal LAN structure arises precisely because the projection map Π_α enforces both Q_Z -orthogonality and the curvature bound on the intercept.

The scaling, backtracking, and projection steps described earlier in Appendix G are therefore more than numerical devices. They ensure that the estimator used in practice lives on the same curved manifold $\Theta(\delta)$ that underlies the LAN expansion (122). The difference between the unconstrained WLS update G_a and the projected intercept $\Pi_\alpha(G_a)$ is $o_p(T^{-1/2})$ by Proposition G.2, so the first-order distribution of the estimator and the α -block of the information matrix are unaffected by the curvature projection. The α -step in the algorithm thus implements the KKT conditions for the intercept block in a way that is fully compatible with the asymptotic theory: it removes non-identified directions, enforces the no-arbitrage curvature bound, and leaves the LAN behaviour of the residual component unchanged.

G.3 Inference on Non-systematic Curvature

The curvature functional $\psi(\Gamma_\alpha) = \Gamma_\alpha^\top Q_Z \Gamma_\alpha$ summarises the strength of asset-specific risk pricing in the characteristic directions that are orthogonal, in the Q_Z metric, to all systematic factors. When the true parameter satisfies the interior condition $\psi(\Gamma_{\alpha,0}) \leq \delta_0 < \delta$ in Assumption A.6, the curvature constraint is slack at the truth and the associated KKT multiplier vanishes asymptotically. In this interior regime the NA-IPCA estimator behaves like an unconstrained WLS estimator along the non-systematic direction once projected onto the tangent space, and $\psi(\Gamma_\alpha)$ is a smooth functional of $a = \text{vec}(\Gamma_\alpha)$. The gradient of ψ at $a_0 = \text{vec}(\Gamma_{\alpha,0})$ is proportional to $Q_Z \Gamma_{\alpha,0}$, so applying the delta method to the LAN expansion in (122) yields

$$\sqrt{T} (\psi(\hat{\Gamma}_\alpha) - \psi(\Gamma_{\alpha,0})) \Rightarrow \mathcal{N}(0, \sigma_\psi^2), \quad \sigma_\psi^2 = \left(\frac{\partial \psi}{\partial a} \right)' I_{\alpha\alpha}^{-1} \left(\frac{\partial \psi}{\partial a} \right).$$

A plug-in estimate of $I_{\alpha\alpha}^{-1}$ based on the α -block of the sample information matrix and a plug-in estimate of the gradient at $\hat{\Gamma}_\alpha$ therefore deliver an estimator $\hat{\sigma}_\psi^2$ and a standard Wald test for residual curvature. For example, testing exact no-arbitrage in the asset-specific direction, $H_0: \psi(\Gamma_{\alpha,0}) = 0$ against $H_1: \psi(\Gamma_{\alpha,0}) > 0$,

amounts to comparing

$$T_\psi = \frac{\psi(\hat{\Gamma}_\alpha)}{\hat{\sigma}_\psi}$$

to the standard normal distribution, or equivalently T_ψ^2 to a chi-square distribution with one degree of freedom. More generally, the same logic applies to tests of whether the residual curvature exceeds an economically negligible benchmark $\delta_0 > 0$, and confidence intervals for $\psi(\Gamma_{\alpha,0})$ follow directly from the normal approximation.

This inference has a clear economic interpretation. The latent block (Γ_β, F) and the observable block $(\Gamma_\delta, F_t^{\text{obs}})$ span the characteristic-based and prespecified observable sources of systematic risk recognised by the model. The intercept block Γ_α is the remaining direction in characteristic space once these modeled spans have been taken into account: under the baseline specification it is separated from the latent span and bounded in curvature, and under the optional strengthening $\Gamma_\delta^\top Q_Z \Gamma_\alpha = 0$ it can also be taken orthogonal to the observable PSF span. The statistic $\psi(\Gamma_\alpha)$ measures, in Q_Z -units, how large this residual direction is after those modeled systematic blocks are accounted for. If $\psi(\hat{\Gamma}_\alpha)$ is statistically indistinguishable from zero, the model is close to no-arbitrage in the characteristic geometry: any remaining alphas are too small, relative to cross-sectional dispersion, to be economically meaningful. If $\psi(\hat{\Gamma}_\alpha)$ is significantly positive, the data reveal a systematic component in the residual direction that cannot be absorbed by the latent and observable blocks, but is still regulated by the curvature bound so that the implied arbitrage is quantitatively limited rather than explosive. The α -curvature projection and the LAN expansion therefore combine to provide a disciplined way to measure and test “how far from no-arbitrage” the cross-section is, in a manner that is robust to high-dimensional characteristics, correlated prespecified factors, and heavy-tailed idiosyncratic shocks.

H PSF Curvature Projection and Observable-Block Discipline

This appendix explains how the NA-IPCA implementation disciplines the curvature of the observable (pre-specified) factor block and how this step fits within the no-arbitrage geometry developed in Sections 2–4. The analysis is parallel to the intercept curvature projection in Appendix G, but now applied to the block Γ_δ associated with traded or macro risk proxies. The goal is to clarify how the algorithm (i) measures the curvature carried by the PSF loadings in the characteristic metric, (ii) enforces a quadratic bound on this curvature, and (iii) preserves the latent–observable–residual decomposition in a way that is consistent with the projected WLS and LAN theory in Appendices A–B. The comparison with statistical IPCA highlights that this PSF curvature discipline is not present in the original IPCA formulation.

H.1 Observable-Block Geometry and PSF Curvature

Let $\Gamma_\delta \in \mathbb{R}^{L \times K_{\text{add}}}$ denote the loading block for observable factors, with columns $\gamma_{\delta,1}, \dots, \gamma_{\delta,K_{\text{add}}}$. In the main text, the observable block is required to lie in the characteristic directions orthogonal to the latent span, so that

$$\Gamma_\delta^\top Q_Z \Gamma_\delta = 0,$$

and observable factors do not re-enter the model through linear combinations of latent loadings. In the baseline theoretical formulation of Sections 2–4, no additional curvature restriction is placed on Γ_δ beyond this Q_Z -orthogonality; the only explicit curvature bound applies to the intercept block Γ_α through

$$\psi(\Gamma_\alpha) = \Gamma_\alpha^\top Q_Z \Gamma_\alpha \leq \delta.$$

The implementation introduces an optional extension that applies a curvature bound to the observable block as well. To do so, it constructs a PSF curvature metric H_{psf} as a weighted average of the per-period characteristic Gram matrices,

$$H_{\text{psf}} = \frac{\sum_{t=1}^{T_{\text{sim}}} w_t W_t}{\sum_{t=1}^{T_{\text{sim}}} w_t}, \quad W_t = \frac{1}{N_t} Z_t^\top Z_t, \quad w_t = N_t^{\theta_{\text{Nts}}} \exp\{-\lambda_{\text{td}}(T_{\text{sim}} - t)\}, \quad (123)$$

where W_t and w_t are the sample Gram matrix and time weight defined in Section 6.1 and Appendix C. For large T_{sim} and slowly varying weights w_t , H_{psf} is a plug-in estimate of the long-run characteristic metric Q_Z in (3), up to a scale factor. It therefore induces the same notion of length and curvature in characteristic space as Q_Z .

Given H_{psf} , the curvature carried by the PSF block is summarized by the quadratic form

$$M_\delta(\Gamma_\delta) = \Gamma_\delta^\top H_{\text{psf}} \Gamma_\delta \in \mathbb{R}^{K_{\text{add}} \times K_{\text{add}}}, \quad (124)$$

whose diagonal entries

$$\psi_{\delta,j} = \gamma_{\delta,j}^\top H_{\text{psf}} \gamma_{\delta,j}, \quad j = 1, \dots, K_{\text{add}},$$

measure the curvature of each observable direction. The scalar PSF curvature functional implemented in the

algorithm is the worst-case value across the observable columns,

$$\psi_{\text{PSF}}(\Gamma_\delta) = \max_{1 \leq j \leq K_{\text{add}}} \psi_{\delta,j} = \max_{1 \leq j \leq K_{\text{add}}} \gamma_{\delta,j}^\top H_{\text{psf}} \gamma_{\delta,j}. \quad (125)$$

In the Q_Z -mode of the implementation, H_{psf} converges to a rescaled version of Q_Z , so $\psi_{\text{PSF}}(\Gamma_\delta)$ is asymptotically proportional to the Q_Z -norm squared of the observable directions. This makes it natural to impose a curvature bound on Γ_δ of the same form as the intercept bound in Appendix G, but applied to the observable block.

H.2 Finite-Sample PSF Curvature Projection

The PSF curvature projection takes place after the latent block has been orthonormalised in the Q_Z metric and the observable block has been projected into the Q_Z -orthogonal complement of the latent span, as described in Appendix C. At that point the algorithm evaluates the curvature matrix $M_\delta(\Gamma_\delta)$ in (124) and computes the worst-case curvature $\psi_{\text{PSF}}(\Gamma_\delta)$ in (125). Given a user-specified PSF curvature bound $\delta_{\text{psf}} > 0$, the step then enforces the inequality

$$\psi_{\text{PSF}}(\Gamma_\delta) \leq \delta_{\text{psf}}$$

by shrinking Γ_δ radially whenever the bound is violated.

Formally, if $\psi_{\text{PSF}}(\Gamma_\delta) \leq \delta_{\text{psf}}$, the observable block is left unchanged. If $\psi_{\text{PSF}}(\Gamma_\delta) > \delta_{\text{psf}}$, the algorithm rescales Γ_δ by a common factor

$$s_{\text{psf}} = \sqrt{\frac{\delta_{\text{psf}}}{\max(\psi_{\text{PSF}}(\Gamma_\delta), \varepsilon_0)}}, \quad \Gamma_\delta^{\text{new}} = s_{\text{psf}} \Gamma_\delta, \quad (126)$$

where $\varepsilon_0 > 0$ avoids division by zero. Because $\psi_{\text{PSF}}(\cdot)$ is quadratic in Γ_δ , this scaling reduces the curvature to

$$\psi_{\text{PSF}}(\Gamma_\delta^{\text{new}}) = s_{\text{psf}}^2 \psi_{\text{PSF}}(\Gamma_\delta) \approx \delta_{\text{psf}},$$

placing the observable block on the boundary of the admissible curvature ellipsoid. Unlike the intercept projection in Appendix G, this PSF curvature step does not perform a fit-preserving adjustment of the observable factors: the PSF time series F_t^{obs} are treated as exogenous, and enforcing the curvature bound is interpreted as tightening the feasible parameter set rather than as a pure reparametrisation.

From the perspective of the constrained extremum problem, this step extends the no-arbitrage parameter space from $\Theta(\delta)$ to

$$\Theta(\delta, \delta_{\text{psf}}) = \left\{ (\Gamma_\beta, \Gamma_\delta, \Gamma_\alpha, F) \in \Theta(\delta) : \psi_{\text{PSF}}(\Gamma_\delta) \leq \delta_{\text{psf}} \right\}, \quad (127)$$

by adding an additional quadratic inequality constraint on the observable block. The PSF curvature projection in (126) is the natural radial projection of the unconstrained update onto the feasible set $\Theta(\delta, \delta_{\text{psf}})$ along the observable directions.

H.3 Economic Interpretation and Relation to the Q_Z Manifold

The PSF curvature bound has a straightforward economic interpretation. The matrix H_{psf} in (123) is a weighted average of the cross-sectional characteristic Gram matrices and thus encodes how strongly characteristic-sorted portfolios loaded by observable factors vary across firms. For any loading direction $g \in \mathbb{R}^L$, the quadratic form $g^\top H_{\text{psf}} g$ is proportional to the long-run cross-sectional variance of the portfolio $Z_t g$ under the time weights w_t . In this geometry, large values of $\psi_{\text{PSF}}(\Gamma_\delta)$ correspond to observable factors that concentrate on extremely high-curvature characteristic directions, producing portfolios with very large cross-sectional dispersion.

Allowing the observable block to take arbitrarily large H_{psf} -norm would be undesirable for both economic and statistical reasons. Economically, observable factors such as Fama-French or macro factors are intended to represent broadly diversified sources of systematic risk. An estimated loading matrix Γ_δ with very large curvature would correspond to highly concentrated characteristic portfolios that are difficult to reconcile with the notion of diversified benchmarks and that can generate extreme risk-adjusted returns in finite samples. Statistically, such spiky observable directions produce unstable factor premia: small perturbations in the data can lead to large swings in the estimated PSF coefficients, undermining out-of-sample performance and contaminating the separation between latent, observable, and residual components.

The bound $\psi_{\text{PSF}}(\Gamma_\delta) \leq \delta_{\text{psf}}$ addresses this issue by limiting the size of the observable loadings in the same characteristic metric that defines the latent span and the intercept curvature. When the Q_Z -mode is active and H_{psf} converges to a rescaled Q_Z , the PSF curvature functional in (125) is asymptotically equivalent to the Q_Z -norm squared of the observable block. The constraint $\psi_{\text{PSF}}(\Gamma_\delta) \leq \delta_{\text{psf}}$ therefore confines observable factors to an ellipsoid in the Q_Z -geometry, preventing them from overshadowing the latent block along the strongest characteristic directions and preserving the economic decomposition into latent characteristic risk, observable factor risk, and bounded non-systematic curvature.

Geometrically, adding the PSF curvature bound leaves the structure of the projected WLS and LAN analysis unchanged. The tangent space in Appendix B is replaced by the tangent cone of the smaller manifold $\Theta(\delta, \delta_{\text{psf}})$ in (127), and the projected score and information operator are computed on this cone. Because $\psi_{\text{PSF}}(\cdot)$ is quadratic and the bound δ_{psf} is finite, the feasible set remains compact in the observable block and the block-diagonal information structure across latent, observable, and intercept components is preserved. The only effect of the additional inequality is to further restrict the set of admissible observable directions, which can reduce the asymptotic variance of the PSF block without altering the efficiency properties of the latent span.

H.4 Comparison with Statistical IPCA

The PSF curvature discipline has no analogue in the statistical IPCA estimator of Kelly et al. (2019). In IPCA, the observable block enters either as additional characteristics or as pre-specified factor exposures with Euclidean normalisations

$$\Gamma_\beta^\top \Gamma_\beta = I_K, \quad \Gamma_\beta^\top \Gamma_\delta = 0,$$

and no curvature bound is imposed on Γ_δ . The algorithm typically orthogonalises the PSF block by projecting with $(I_L - \Gamma_\beta \Gamma_\beta^\top)$ and then estimates the factor loadings and premia by unconstrained Euclidean least squares. There is nothing in this formulation to prevent the observable block from drifting into extremely high-curvature directions in the Q_Z -metric, especially in characteristic-rich environments with strong cross-sectional collinearity.

The simulation diagnostics in Section 7.2 and Appendix F illustrate the consequences. Under empirical or misaligned characteristic geometries, statistical IPCA often allocates substantial Q_Z -variance to observable directions that overlap with the latent span, producing large off-diagonal entries in the misallocation matrix and sizeable latent-observable overlap in the Q_Z -metric. In contrast, NA-IPCA with PSF curvature projection measures orthogonality and size in the Q_Z geometry, enforces blockwise separation between latent and observable components, and, when the PSF curvature bound is active, shrinks observable loadings back into an economically plausible ellipsoid. The observable block remains a diversified and disciplined component of the pricing kernel rather than an unconstrained device for absorbing residual variation.

In this sense, the PSF curvature projection complements the intercept curvature discipline in Appendix G. Together, the two constraints ensure that both the non-systematic block and the observable block behave as economically interpretable and geometrically controlled components of the NA-IPCA estimator, rather than as free parameters that can mimic latent factors or exploit high-curvature directions in the characteristic space.